

# MEASURING AND MITIGATING SOLUTION MODE COLLAPSE IN RLVR

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## ABSTRACT

A language model (LM) can usually answer the same question in more than one way, but reinforcement learning with verifiable rewards (RLVR) is indifferent to which correct answer a model produces. A solution will earn the same reward whether it is the thousandth copy of a familiar answer or one the model has never produced before. Yet, there is potential value in having the model retain multiple correct solutions as it is trained. For instance, multiple modes may give users a choice and provide problem-solving strategies that improve overall model performance. Here, we measure solution diversity through RLVR training using MODEBENCH, a set of multi-solution tasks, where the verifier returns both correctness and mode discovered. We find that RLVR post-training concentrates probability onto fewer correct modes even as accuracy holds or improves, and moreover, that frontier models are already highly concentrated. We then introduce Re:Max, which uses the verifier to index a memory containing one verified example per discovered mode and rehearses those modes uniformly. A solution found once is, therefore, practiced as often as one found repeatedly. Across three model scales, two RL objectives, and harder task constructions, Re:Max improves both how often a policy succeeds and how many different ways it can succeed.

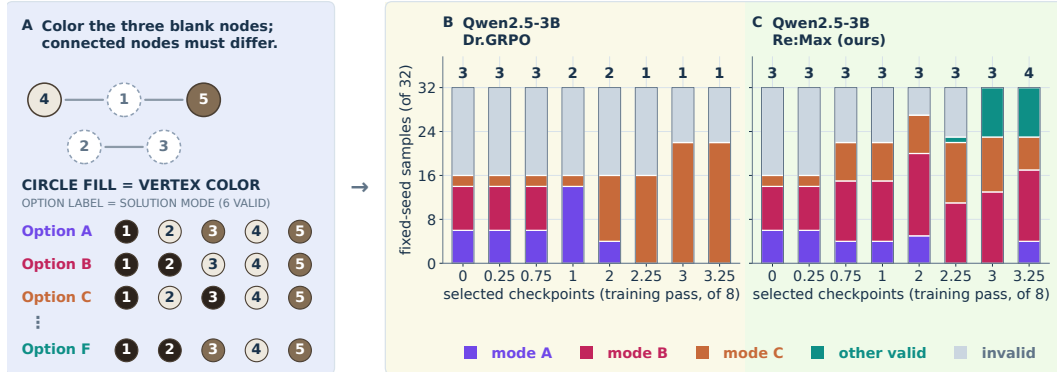


Figure 1: **Motivation.** Consider the Graph coloring problem shown in A (left), with six valid solutions. We evaluate Qwen2.5-3B across training, tracking the number and diversity of correct solutions. At each training pass, we show 32 sampled questions, eight from each of four matched runs. Training with Dr. GRPO (B) improves accuracy and leads to a single solution dominating the output. In contrast, Re:Max (C) increases both accuracy and diversity of correct solutions.

## 1 INTRODUCTION

Training and evaluation of reinforcement learning with verifiable rewards (RLVR) algorithms are driven entirely by model accuracy. However, what makes models useful at deployment depends on having different candidates to choose from, i.e. test-time search and reranking (Brown et al., 2024; Snell et al., 2024; Cobbe et al., 2021; Wang et al., 2023). This raises a question: as a model gets better at finding a correct answer during RLVR, *what happens to other correct candidates?*



ModeBench dataset and models



Code repository

Links anonymized for review.

RLVR trains against an executable check (Guo et al., 2025), but its binary reward does not distinguish a solution the model has produced a thousand times from one it has never produced before. Nothing in the objective asks the model to maintain diversity. Fig. 1 shows what this can look like on a graph-coloring problem with six valid answers: under Dr.GRPO (Liu et al., 2025), the policy becomes more accurate while its correct outputs concentrate onto a single mode.

Mode collapse, or narrowing of the output space, has been observed in alignment (Kirk et al., 2024; Cui et al., 2025), creative generation (Doshi & Hauser, 2024), brainstorming and ideation (Anderson et al., 2024), public argument (Kim et al., 2026), and LM-assisted writing (Abdulhai et al., 2026). Quantifying this narrowing requires deciding whether two outputs are actually the same answer. Token-level diversity (Li et al., 2016; Zhu et al., 2018) counts two spellings of one solution as variety, while a judge that groups answers by meaning uses a second model to evaluate whether a given response is distinct (Farquhar et al., 2024; Zheng et al., 2023).

Here, we investigate the prevalence and impact of mode collapse in problem-solving tasks under RLVR post-training. Our contributions are threefold:

1. **MODEBENCH**: We contribute a benchmark for empirically studying mode collapse in tasks allowing for multiple correct solutions, ranging from code evaluation to constrained planning. **MODEBENCH** provides both output correctness and maintains a unique identity for each available modal solution to a given problem. The benchmark spans five executable domains, each with five difficulty levels (Sec. 2).
2. We conduct comprehensive experiments on **MODEBENCH** across 4 open-weight model families and 7 frontier models. Across open-weight models and at 3 model scales, the number of solution modes collapses in every domain under both Dr.GRPO and GRPO. Further, in inference, all frontier models we measured reached fewer than 1.5 effective modes out of up to 355 unique solutions (Sec. 5.1 and App. N.1).
3. We propose *mode replay*, which stores one verified exemplar per mode and rehearses all modes equally. Added to Dr.GRPO (Liu et al., 2025), it raises accuracy and pairwise correct mode diversity across 3 model scales and all 5 **MODEBENCH** domains. We outperform 8 other RLVR baselines. Added to MaxRL (Tajwar et al., 2026) (which we dub as Re:Max), it adds diversity and accuracy across **MODEBENCH** difficulty levels (Sections 3 and 5.2–5.3).

Overall, our results quantify the severity of mode collapse and suggest that incorporating mode replay as part of RLVR post-training can lead to both increased diversity of solutions and improved model performance.

## 2 MODEBENCH: MEASURING SUCCESSFUL ALTERNATIVES

Building a benchmark to study solution model collapse is challenging for two reasons. First, each problem must allow for several valid discrete solutions. Second, given a model’s response, we must be able to determine which solution mode was reached. For example, in Fig. 2D, we show algebra problems in our benchmark. In this example, we show that the same algebraic solution can be reached through two different series of steps.

Ensuring these two properties hold is domain-specific. Fig. 2 illustrates a subset of solution modes across the considered domains; e.g., in **Countdown** and **MathIR**, modes mean the model reached

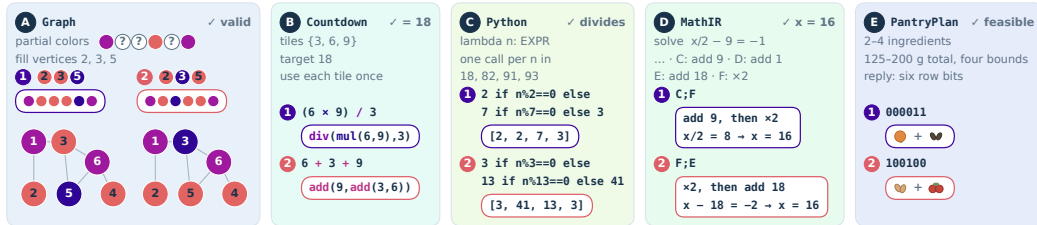


Figure 2: **MODEBENCH** comprises questions with many acceptable solutions across domains. (1) and (2) identify different solution modes across the five domains, which include Graph coloring, a normalized computation tree, executable Python code, algebraic problems and constraint optimization.

the same answer by a different route, whereas in Graph and PantryPlan, modes mean a different valid answer altogether. We provide the number of unique possible solutions per domain in Table 3. Refer to App. B and App. C for per-domain rules and prompts for dataset generation.

For each domain, we provide a verifier that evaluates whether a valid solution exists within the model’s response, as well as the mode a solution belongs to. This was built by creating in advance a comprehensive set of potential solutions to the problem. Precisely, for a prompt  $x$  with specification  $s_x$ , the validator maps an output  $y$  either to failure  $\perp$  or to a key  $V(y, s_x) \in \mathcal{C}_x$ . Two accepted outputs are the same mode if and only if their keys agree. The full set of valid keys  $\mathcal{C}_x$  is never revealed, neither its members nor its size. A policy, and the memory that stores its successes, only ever sees keys for answers the policy actually produced.

## 2.1 SUCCESS AND VERIFIED-MODE DIVERSITY

To measure performance on MODEBENCH, we must consider two distinct properties: how often the output is accurate, and, given that it is right, which of the available right answers has been given. A benchmark reporting only accuracy cannot capture the failure we study.

The obvious way to measure modal diversity is to count distinct correct answers in  $K$  samples. We define  $P$  as the chance a sample is correct and  $q_c$  be how that correct probability is divided among the valid answers:

$$\underbrace{\Pr[D_K > 0]}_{\text{pass@K}} = 1 - (1 - P)^K, \quad \underbrace{\mathbb{E}[D_K]}_{\text{distinct@K}} = \sum_c [1 - (1 - Pq_c)^K], \quad (1)$$

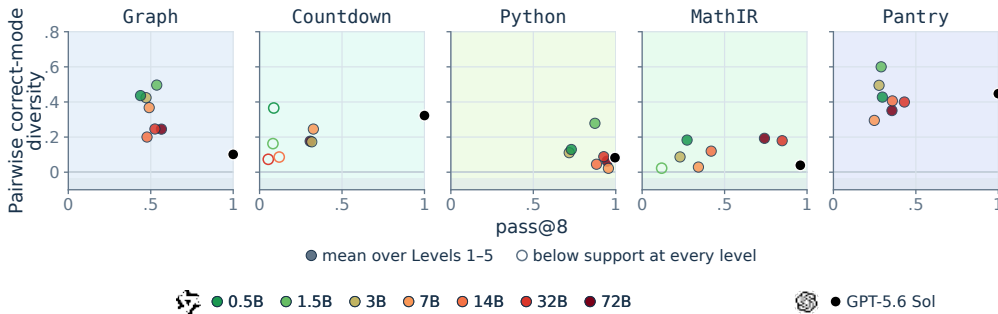
where  $D_K$  counts the distinct correct answers. However, both metrics increase with  $P$ , and thus a policy that is right more often may have higher `pass@K` and `distinct@K` while producing less diverse solutions.

We therefore propose an alternative metric, denoted as pairwise correct-mode diversity:

$$\text{PCMD} = 1 - \sum_c q_c^2 = \Pr[Z_1 \neq Z_2 \mid \text{both responses verified}], \quad (2)$$

measuring the joint probability of different solution modes given two correct responses. A PCMD of zero thus implies every correct answer belongs to the same solution mode, and an even spread over  $m$  yields a PCMD of  $1 - 1/m$ , e.g., if answers spread evenly across two possible solution modes, then the PCMD is  $1/2$ . We note that when infinite solution modes are possible, the maximum PCMD converges to 1, and that when the model outputs no correct solutions, we cannot calculate the metric. We only calculate PCMD when a model produces a minimum number of correct solutions; App. A.2 gives that threshold.

Fig. 3 shows how PCMD captures diversity of responses independently of accuracy. GPT-5.6 Sol has near-perfect `pass@8` for all five tasks considered, yet much weaker models in the Qwen2.5-Instruct family have higher PCMD. For the context of this work, we define a *solution mode collapse* as a decline in PCMD during RLVR.



**Figure 3: Higher performance is not associated with increased solution mode diversity.** We show the relationship between a performance-related metric (`pass@8`;  $x$ -axis) and our proposed solution mode diversity metric (PCMD;  $y$ -axis) for GPT-5.6 Sol (in black), and Qwen2.5-Instruct models of various sizes (where parameter count is indicated by color). Note that accuracy and model size are not correlated with solution mode diversity.

## 2.2 HARDER TASKS WITH THE SAME NOTION OF A MODE

We construct MODEBENCH so that tasks can be made more or less challenging, which helps prevent  $\text{pass@8}$  accuracy saturation. This allows us to explore the impact of problem difficulty on mode collapse. For each domain, we include 5 independent difficulty levels, e.g. for our Graph domain we allow more hidden vertices, and for Countdown we give the model more operands. Each difficulty level holds constant (1) our definition of mode, (2) our approximate distribution of modes, (3) our verifier constant, and (4) the sample size of 384 train and 128 test problems. To ensure (2) holds, and moreover, that the problems we sample are in fact more challenging, we fit each level with consideration to  $\text{pass@1}$  and  $\text{pass@8}$  of a reference model, e.g. Level 4 was tuned such that a Qwen2.5-Instruct-7B model has the same  $\text{pass@1}$  and  $\text{pass@8}$  scores as a Qwen2.5-Instruct-0.5B on Level 1.

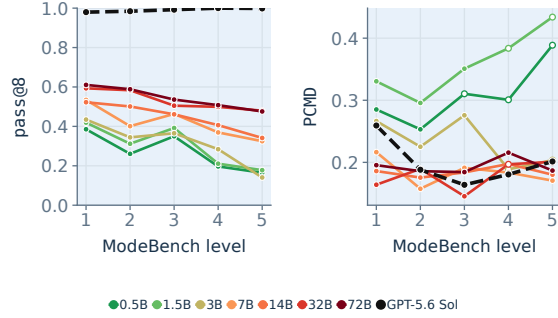


Figure 4: **Harder problems of the same domain exhibit fewer solution modes.** When we ask the Qwen2.5-Instruct family and GPT-5.6 Sol increasingly difficult questions from harder levels of MODEBENCH,  $\text{pass@8}$  declined and so too does PCMD. For our benchmark, as problems get harder, solution mode collapse worsens.

LMs with a higher parameter count generally have higher  $\text{pass@8}$  and can solve more problems. Yet, in Fig. 4, for our harder problems, we find that solution mode diversity declines, even as our underlying mode count per level remains stable. This is most noticeable when frontier models are evaluated. Even though correctness is saturated, we are able to use PCMD to isolate solution mode collapse (App. N.1), and this phenomena happens across MODEBENCH. For example, frontier models answer .991 of questions from Level 1-3 correctly under normalized grading, but their mean PCMD is only .232, below the .570 of the most diverse sub-2B model, which answers .273 of its prompts.

## 2.3 CONCENTRATION IN FRONTIER MODELS

A common fix for solution mode collapse is to increase the sampling budget, as repeated sampling can substantially increase solution coverage at test time (Brown et al., 2024; Yue et al., 2025). But, across MODEBENCH, we find that more samples do not recover the complete set of modes. For example, all 1,024 GPT-5.6 Sol draws on the 128 Level-3 Graph problems verify, yet eight draws yield just 1.16 distinct colorings, and 95.1% of correct pairs repeat one (App. N.1). Fig. 5 follows MODEBENCH at three levels out to 512 draws per problem, and the curves flatten far below the modal set: an LM that has settled on two colorings does not find the rest even after being asked five hundred times.

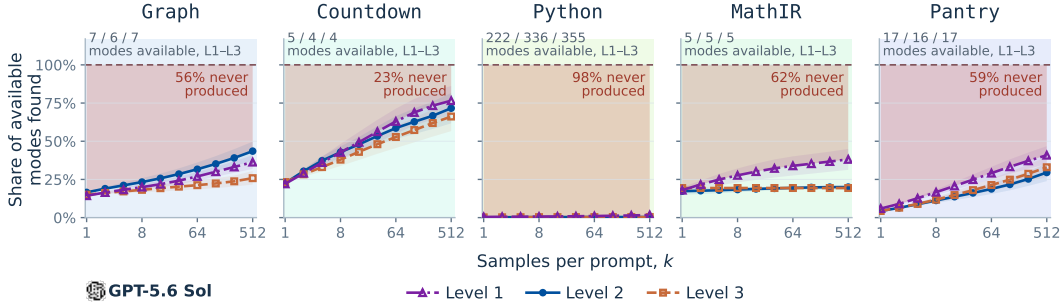


Figure 5: **Asking the model more times may not recover all available modes.** Our other frontier measurements ask each question eight times, so we test directly whether a larger budget recovers the missing modes. GPT-5.6 Sol, with medium reasoning on, answers 32 problems per domain and level with 512 responses each. Curves are distinct verified modes in random  $k$ -subsets as a share of the modes those problems admit, and the shaded band is what no level ever produced. Shading is the pointwise 95% problem bootstrap. Printed counts are the enumerated support those problems admit, in Level 1/2/3 order, and every panel shares one scale (App. K.3).

Python is the extreme case. Its problems admit several hundred distinct return vectors on average, and five hundred draws surface between one and four solutions, under two percent of all possible solutions, and under one percent at Levels 2 and 3. Each doubling of the sampling budget buys a fraction of one mode, against gaps of several modes everywhere and several hundred in Python. This problem becomes worse as we increase problem difficulty. All 7 frontier models tested narrow in PCMD from Level 1 to Level 3 while staying near-perfect on correctness (Fig. 15, App. N.1).

We similarly check our local models over larger samplings, and we find quadrupling to  $K = 32$  draws per problem moves control PCMD by at most .010, and MathIR at  $K = 64$  draws is as concentrated as at  $K = 8$  (Apps. J and K), while disabling reasoning lowers correctness and verified-mode counts in every deployment tested, though its PCMD effect differs by deployment (Table 59, App. O.6).

We do, however, note that cheap inference interventions do exist. For example, we find that because different models collapse onto different answers, mixing responses from two independent models covers more verified solution modes than either alone (App. L.2). Additionally, raising inference temperature gives more solution modes too, but at the cost of model accuracy (Fig. 39).

### 3 RE:MAX: LEARN FROM NEW SAMPLES AND REMEMBER SUCCESSES

#### 3.1 THEORETICAL INTUITION FOR MODE COLLAPSE

Solution mode collapse follows from two properties of the objective: 1) the binary verifier assigns the same reward to every correct answer, so the objective provides no incentive to preserve one correct mode over another; and 2) an on-policy update can reinforce only the modes that appear in its fresh rollout samples, which are drawn according to the policy’s current preferences. Together, these properties create a positive feedback loop, where a mode with even a small early advantage is sampled more often, receives more reinforcement, and therefore becomes even more likely to be sampled in subsequent updates. Figure 6 illustrates this process.

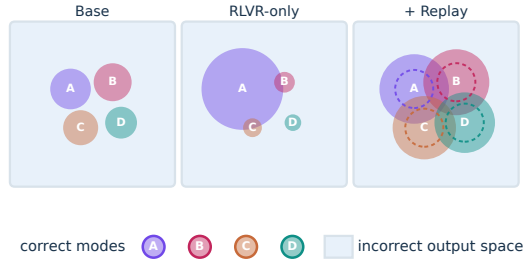


Figure 6: **RLVR early "winner" (A) becomes the predominant way to solve.** Under base model, we see that before RLVR, many modes exist. RLVR-only training concentrates on one mode. Our replay approach keeps every buffered mode & leaves room for more.

We formalize this intuition in three theoretical results, developed in detail in the appendices. First, Theorem P.4 shows that, under an independent-logit parametrization and Euclidean expected-gradient flow, this positive feedback leads to a “winner-takes-all” limit: a unique initial leader among correct modes eventually dominates, as Sinha et al. (2026) and Lochab et al. (2026) also establish. Second, App. P.14 shows that this result depends on the interaction between binary rewards and optimization geometry; binary rewards alone do not force this concentration. Third, Lemma P.39 characterizes a limitation of standard RLVR; LMs post-trained with RLVR can amplify rare modes when sampled, but both the chance of sampling a mode and its expected absolute logit-gradient contribution vanish as its probability approaches zero. When the applied advantages sum to zero within the group, a mode absent from the current batch has no gradient.

#### 3.2 REPLAYING DISTINCT VERIFIED SUCCESSES

Our theoretical analysis suggests that, to achieve diversity in inference we need to maintain our solution modes during training. We address this challenge drawing inspiration from classic RL approaches towards memory management through replay (Lin, 1992; Mnih et al., 2013; Rolnick et al., 2019). Once a successful solution mode has been found and stored, replay allows us to continue training on it whether or not the current policy rediscovers it. Under a fixed-buffer analysis, a constant replay from the existing set of unique found modes keeps every buffered mode above a positive probability floor (Theorem P.22).

For each training prompt  $x$ , we maintain an initially empty, bounded buffer  $\mathcal{B}_x$ . The first time a verified mode appears, we store one exemplar  $e_c$  that realizes it (subsequent rediscoveries do not modify the buffer). A deterministic schedule revisits nonempty buffer throughout training. For an exemplar of token length  $|e_c|$ , the replay loss is

$$L_{\text{replay}}(x) = -\frac{1}{|\mathcal{B}_x|} \sum_{c \in \mathcal{B}_x} \frac{\log \pi_{\theta}(e_c | x)}{|e_c|}. \quad (3)$$

Uniform weighting breaks the link between discovery frequency, how often a mode appears in fresh rollouts, and rehearsal frequency, how often a stored mode is replayed. Once a mode enters the buffer, it is rehearsed as often as every other retained mode, so frequently sampled modes receive no additional replay.

Adding this loss to the fresh update from standard MaxRL (Tajwar et al., 2026) gives our primary method, Re:Max. But, this principle is modular, and can be applied to any RLVR algorithm. For example, we also evaluate with Dr.GRPO (Liu et al., 2025) resulting in the algorithm, Re:Dr. Solution modes discovered during training do not enter the on-policy reward group, and contribute only through the separate replay loss. This separation allows one to vary the RLVR objective and the replay mechanism independently. Finally, our replay buffers contain only responses to training prompts; held-out evaluation therefore measures changes in the policy itself rather than retrieval or recitation from the buffer (App. D).

Fig. 7 illustrates how replay changes learning dynamics. In the standard RLVR setting, a policy learns only from modes present in the current sampling, which often leads previously discovered modes to disappear. Re:Max adds an additional incentive for the policy to continuously generate previously found solution modes that yield correct solutions. We note that this is fundamentally different from approaches that encourage diversity by modifying the on-policy learning signal to favor rare or diverse outcomes (Song et al., 2025; Li et al., 2026a), as we do not use mode identity to modify the task reward or advantage.

## 4 EXPERIMENTAL DESIGN

In our main experiment, we examine whether replay improves correctness and solution-mode diversity across different RLVR objectives. We compare the impact of our replay component against both Dr.GRPO (Liu et al., 2025) and MaxRL (Tajwar et al., 2026). First, on all five MODEBENCH Level 1 train/test sets we carry out five seeds of post-training using Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B. We train for eight epochs with  $G = 16$  new rollouts. We note that replay does not increase this sampling budget. Instead, it performs an additional training update using at most  $M = 16$  previously collected exemplars from the replay buffer for each visited prompt (Table 6). Second, we conduct the same experiment on MODEBENCH Levels 2 and 3 using only a Qwen2.5-0.5B. Third, we compare our method with other approaches, including frequency-weighted replay, UCPO (Lochab et al., 2026), RLEP (Zhang et al., 2025), GAPO (Anschel et al., 2025), SetPO (Li et al., 2026a), and fixed Semantic-MaxEnt under a specific setting (Qwen2.5-0.5B, five seed, level 1).

We additionally evaluate 7 hosted models: GPT-5.6 Sol (OpenAI, 2026b), GPT-5.4 (OpenAI, 2026a), Grok 4.3 (xAI, 2026), Kimi K3 (Moonshot AI, 2026), DeepSeek V4 Pro (DeepSeek-AI, 2026), Claude

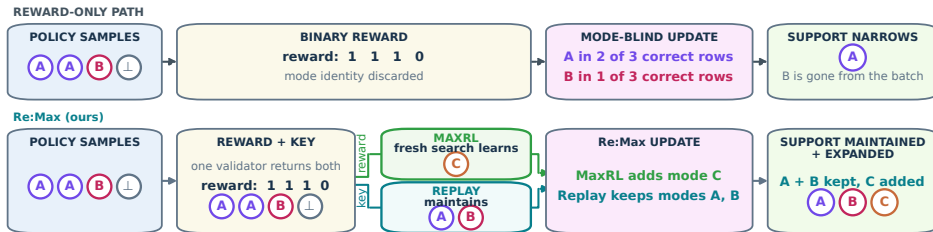


Figure 7: **Replay preserves successes that on-policy learning may stop seeing.** MaxRL learns from the policy’s current samples, while replay stores one verified exemplar per discovered mode and revisits those exemplars with equal weight. Together, these two learning paths allows Re:Max to preserve previously discovered solution modes while continuing to explore for new ones.

Table 1: **Compared objectives under Qwen2.5-0.5B, MODEBENCH Level 1.** Each method is compared against the same task performed under Dr.GRPO; avg weights the five domains equally within seed (5-seeds). Bold marks a column’s best. An em-dash means that there are few more-than-one solution pairs to evaluate, and a superscript gives the paired seed count where it falls below five. Red-to-green shading scales to each column’s largest magnitude.

|                        | $\Delta$ pass@8 |               |               |               |               |               | $\Delta$ PCMD             |               |                         |               |                         |               |
|------------------------|-----------------|---------------|---------------|---------------|---------------|---------------|---------------------------|---------------|-------------------------|---------------|-------------------------|---------------|
|                        | Count           |               |               | Pantry        |               |               | Count                     |               |                         | Pantry        |                         |               |
|                        | Graph           | down          | Python        | MathIR        | Plan          | Avg           | Graph                     | down          | Python                  | MathIR        | Plan                    | Avg           |
| Re:Dr ( <i>ours</i> )  | <b>+ .646</b>   | <b>+ .191</b> | <b>+ .356</b> | <b>+ .285</b> | <b>+ .207</b> | <b>+ .337</b> | <b>+ .559</b>             | <b>+ .480</b> | <b>+ .082</b>           | <b>+ .017</b> | <b>+ .334</b>           | <b>+ .294</b> |
| Re:Max ( <i>ours</i> ) | <b>+ .621</b>   | <b>+ .185</b> | <b>+ .509</b> | <b>+ .277</b> | <b>+ .199</b> | <b>+ .358</b> | <b>+ .525</b>             | <b>+ .504</b> | <b>+ .249</b>           | <b>+ .005</b> | <b>+ .317</b>           | <b>+ .320</b> |
| GAPO                   | <b>+ .175</b>   | <b>+ .011</b> | <b>+ .009</b> | <b>− .054</b> | <b>+ .279</b> | <b>+ .084</b> | <b>+ .131</b>             | <b>+ .340</b> | <b>.000</b>             | <b>.000</b>   | <b>+ .653</b>           | <b>+ .225</b> |
| SetPO                  | <b>+ .064</b>   | <b>− .050</b> | <b>+ .497</b> | <b>+ .051</b> | <b>+ .154</b> | <b>+ .143</b> | <b>+ .039</b>             | <b>+ .007</b> | <b>+ .013</b>           | <b>+ .003</b> | <b>+ .266</b>           | <b>+ .066</b> |
| MaxRL (no replay)      | <b>+ .213</b>   | <b>+ .101</b> | <b>.000</b>   | <b>− .066</b> | <b>+ .009</b> | <b>+ .051</b> | <b>+ .086</b>             | <b>+ .015</b> | <b>.000</b>             | <b>.000</b>   | <b>.000</b>             | <b>+ .020</b> |
| RLEP-Dr                | <b>+ .098</b>   | <b>+ .103</b> | <b>.000</b>   | <b>+ .126</b> | <b>.000</b>   | <b>+ .066</b> | <b>+ .035</b>             | <b>+ .012</b> | <b>.000</b>             | <b>− .001</b> | <b>.000</b>             | <b>+ .009</b> |
| UCPO                   | <b>− .003</b>   | <b>+ .133</b> | <b>.000</b>   | <b>+ .031</b> | <b>+ .053</b> | <b>+ .043</b> | <b>− .003<sup>2</sup></b> | <b>+ .019</b> | <b>.000<sup>2</sup></b> | <b>− .001</b> | <b>.000<sup>2</sup></b> | <b>+ .009</b> |
| Semantic-MaxEnt        | <b>+ .002</b>   | <b>+ .141</b> | <b>.000</b>   | <b>− .121</b> | <b>+ .201</b> | <b>+ .045</b> | <b>+ .004</b>             | <b>+ .020</b> | <b>.000</b>             | <b>+ .001</b> | <b>—</b>                | <b>+ .006</b> |
| GRPO                   | <b>+ .037</b>   | <b>+ .159</b> | <b>.000</b>   | <b>− .052</b> | <b>+ .023</b> | <b>+ .034</b> | <b>+ .008</b>             | <b>+ .001</b> | <b>.000</b>             | <b>+ .001</b> | <b>.000</b>             | <b>+ .002</b> |
| Untrained checkpoint   | <b>+ .132</b>   | <b>− .383</b> | <b>− .172</b> | <b>− .292</b> | <b>+ .380</b> | <b>− .067</b> | <b>+ .108</b>             | <b>—</b>      | <b>—</b>                | <b>+ .027</b> | <b>+ .591</b>           | <b>+ .242</b> |

Opus 5 (Anthropic, 2026b), and Claude Opus 4.8 (Anthropic, 2026a) on MODEBENCH Levels 1–3, collecting eight responses per prompt and grading them through a fixed formatting normalizer so that presentation differences are not counted as verified alternatives (App. O). We carry out the same inference-only evaluation Level 1–5 across the available scales and benchmark levels of SmolLM2 (Ben Allal et al., 2025), Qwen2.5 (Qwen et al., 2024), Falcon3 (Falcon-LLM Team, 2024), and OLMo-2 (Team OLMo et al., 2024). These experiments are inference-only; no replay (or training) is applied.

## 5 RESULTS

Our results are three-fold. First, we document solution-mode collapse under RLVR on MODEBENCH (Section 5.1); Second, we find that replay reverses this collapse without sacrificing correctness (Section 5.2), outperforming 8 existing RLVR approaches. Third, we show that replay gains persist across difficulty levels (Section 5.3), KL-chaining, and domain changes.

### 5.1 RLVR CONCENTRATES SOLUTION MODES

**RLVR replaces diverse correct attempts with the same repeated answers.** We compare each policy’s solution-mode distribution before and after training using PCMD. Across Dr.GRPO, GRPO, and MaxRL results, we find fewer solution modes after RLVR post-training in 14 out of 15 domains for Qwen2.5-3B (Fig. 8), with the same pattern appearing across model scales and difficulty levels (Fig. 27). The starkest case is Python, where Qwen2.5-3B goes from solving 73 of 128 held-out prompts in multiple ways to as few as 22 after Dr.GRPO post-training.

**As outputs concentrate, rollouts under RLVR lose training signal.** Dr.GRPO and MaxRL learn from a rollout group only when it contains a mix of correct and incorrect responses. Thus, we audit every rollout group during training to track how often they have provided a non-zero learning signal. We find that Re:Max systematically increases the percentage of roll-

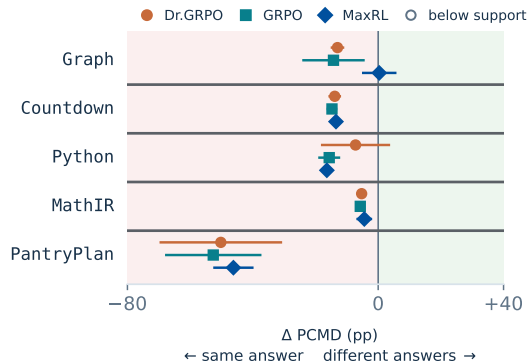


Figure 8: **Training reduces solution mode diversity.** We compare PCMD before and after RLVR on Qwen2.5-3B, over MODEBENCH Level 1 test set, under three objectives, five domains, five seeds. The figure shows the change in PCMD from before to after. Marks are means over training seeds with nominal 95% intervals. In 14 of 15 comparisons, we find fewer solution modes after training. See Fig. 27 for other model scales.

out groups with a learning signal (App. H.4); e.g., on Python, 18.5% of groups have successes and failures under replay, compared to only 1.5% without.

## 5.2 REPLAY IMPROVES BOTH ACCURACY AND DIVERSITY

**Replay improves accuracy and diversity across RLVR algorithms and model scales.** We add our proposed verified-mode replay mechanism to Dr.GRPO and MaxRL and compare each against its matched no-replay control across five MODEBENCH domains and three model scales. Across these comparisons, we find that Re:Max and Re:Dr. (ours) shift the outcome toward both higher pass@8 and greater solution-mode diversity (Fig. 9; Tab. 8).

**Mode diversity comes from retaining modes; balancing makes it stronger.** A single-slot buffer, rehearsing one verified success per prompt and never seeing a second key, already recovers 58% of Re:Dr’s diversity effect (App. Q.4). To isolate balancing itself, we compare uniform replay with a frequency-weighted variant that rehearses modes in proportion to their discovery rate. Holding buffer construction, advantage scheduling, and loss update unchanged, uniform weighting preserves greater verified diversity, most clearly in Graph, Countdown, and PantryPlan: in Graph it recovers roughly 0.6 additional modes per prompt while raising pass@8 by 6.6 points. Across domains, these gains come without an aggregate difference in correctness (App. F.7, App. Fig. 25).

**Alternative diversity mechanisms produce isolated rather than consistent gains.** We compare Re:Max and Re:Dr(ours) against UCPO (Lochab et al., 2026), RLEP (Zhang et al., 2025), GAPO (Anschel et al., 2025), SetPO (Li et al., 2026a), and fixed Semantic-MaxEnt (Tab. 1, App. F.5). Some of these methods produce substantial gains in individual domains, e.g., GAPO reaches  $2.9\times$  its control’s effective number of verified modes in PantryPlan, while SetPO raises pass@8 by 147% in Python. Yet these gains are domain-specific, while Re:Max improves both metrics across domains, raising pass@8 by 90% relative to Dr.GRPO and reaching  $1.9\times$  its effective number of verified modes (App. P.12).

## 5.3 REPLAY REMAINS EFFECTIVE BEYOND THE BASE TRAINING SETTING

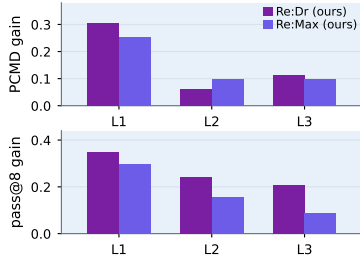


Figure 10: **Replay gains persist across difficulty levels.** Bars show the gain from adding replay to Dr.GRPO (Re:Dr) and MaxRL (Re:Max) at each MODEBENCH level, averaged across domains for Qwen2.5-0.5B. Replay improves both PCMD (top) and pass@8 (bottom) at all three levels.

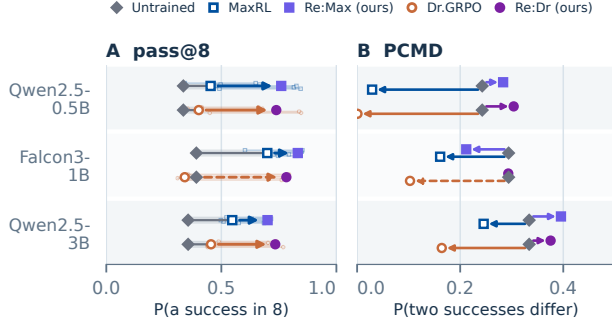


Figure 9: **Replay leads to higher accuracy and solution mode diversity.** Diamonds are initial LMs; open/filled markers are training without/with replay. (A) averages all five domains within seed; faint paths show seeds. (B) draws both arrows from the untrained model, one to the standard RLVR objective alone, one to it with replay, over the domains where every arm clears the support bar. App. Fig. 18 gives results for each domain and scale.

**Replay’s gains persist on harder task constructions.** We repeat the replay-versus-control comparison within each of the first three MODEBENCH levels, holding the level and RLVR objective fixed. Re:Dr. and Re:Max (ours) improves average pass@8 and PCMD under both Dr.GRPO and MaxRL at every level (Fig. 10). At Level 2, for instance, Graph rises from 1.04 to 1.55 effective verified modes under Dr.GRPO, and from 1.13 to 1.48 under MaxRL. Gains vary across domains, but remain positive in aggregate even at Level 3 (App. F.3).

**Replay leaves more usable alternative correct solutions when the task changes.** We ask whether Re:Max / Re:Dr has a downstream impact on inference. Ideally, replay should make a model more

able to give a correct answer when the original most-common answer is no longer available. To test, we take a portfolio of eight generated answers, remove one option from the task—e.g. an ingredient in `PantryPlan` or a color in `Graph`—and re-check under the modified task. Across all 5 domains, Re:Max / Re:Dr (ours) are more likely to give a valid alternate answer. When no answer is valid, Re:Max / Re:Dr find a new valid answer in fewer additional attempts than RLVR or tuned-temperature sampling (Apps. M.3, L.1).

**Reference KL and replay preserve different sources of diversity.** One explanation for the diversity gains of Re:Max / Re:Dr (ours) is that the model retains the diversity of its base. We test this by anchoring the policy to its base: we sweep the coefficient of a KL penalty to the initial policy and compare its correctness–diversity curve with Re:Max / Re:Dr (ours). The anchor is limited by the diversity already present in its reference, whereas Re:Max/ Re:Dr (ours) is limited by the modes its buffer discovers. Averaged over the three domains with a reportable initial PCMD, matching diversity with a KL anchor costs model correctness, although in `Python` and `Pantry` the reference is already broad enough for the anchor to win (Fig. 11). We further show analytically that diversity depends on whether all modes were contained within the initial policy or discovered during training (App. P.11).

## 6 RELATED WORK

**Measuring solution diversity under alignment RLHF** and reasoning RL can reduce output diversity, entropy, and high-budget coverage (Kirk et al., 2024; Dang et al., 2025; Cui et al., 2025; Yue et al., 2025). Measuring this collapse, however, requires defining when two correct answers differ meaningfully. Semantic clustering infers it from language (Farquhar et al., 2024), which is general but depends on a judge. MODEBENCH instead defines solution identity through the same executable check that assigns reward, so two outputs differ exactly when they execute differently: a narrower but cheaper measure, shared across training and evaluation. Preserving alternatives matters because sampling-based selection relies on generating multiple viable proposals (Brown et al., 2024; Snell et al., 2024; Cobbe et al., 2021; Lightman et al., 2024).

**Expanding diversity.** Outcome bonuses, set prediction, distribution matching, and uniform-correct objectives all reshape the learning signal (Song et al., 2025; Chen et al., 2025; Li et al., 2025; 2026a;b; Lochab et al., 2026), while MaxRL combines objectives across sampling budgets (Tajwar et al., 2026). Yet existing approaches still depend on what the current policy samples—precisely where our analysis finds that solution diversity disappears. Memory can break this dependence: standard replay (Lin, 1992; Rolnick et al., 2019), self-imitation learning (Oh et al., 2018), and RLEP (Zhang et al., 2025) all revisit past successes in proportion to how often they occur. Instead, we homogeneously balance replay across solution modes, an idea related to explicitly preserving behavioral diversity in quality-diversity methods (Mouret & Clune, 2015; Ecoffet et al., 2021).

## 7 CONCLUSION

We contribute MODEBENCH, a benchmark for studying solution mode diversity in RLVR, and Re:Max, a replay-based method that preserves discovered solution modes during RLVR. Across objectives, model scales, and task difficulties, replay retains solution diversity while improving correctness. Diversity may itself be a target for optimization: future work can move beyond preserving discovered modes towards training models to produce desired distributions over ways to solve a problem.

**Limitations.** Replay retains only what sampling finds, and mode count says nothing about the “worth” of additional solutions. Our equivalences are executable by construction, so future work is needed to evaluate the benefit and feasibility of replay in more open-ended solution spaces.

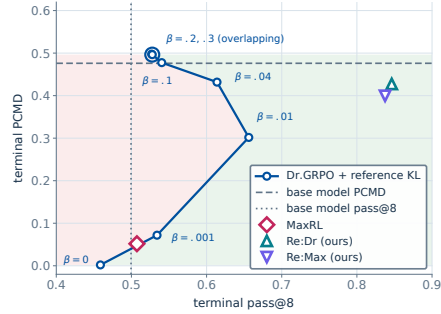


Figure 11: **Replay and KL anchoring preserve different diversity.** Means over the 3 domains with reportable initial PCMD, Qwen2.5-0.5B. The anchor curve sweeps  $\beta$ ; larger  $\beta$  preserves more diversity at lower pass@8. Red/green marks lower/higher pass@8 than KL at equal measured diversity (App. P.12).

AI USE STATEMENT

Generative AI assisted manuscript editing, mathematical analysis, and supporting code. The authors reviewed and verified the resulting content and take responsibility for the final manuscript.

ETHICS STATEMENT

The benchmark and experiment-specific data are synthetic; this study collects no human-subject or personal data. More verified modes are not inherently better: a distinct execution may be risky or costly even when it passes the validator, and PCMD does not measure downstream value. Systems that evaluate or optimize breadth should report correctness, invalidity, and application-specific costs alongside mode counts.

REPRODUCIBILITY STATEMENT

Every reported number is computed from a versioned result file. App. S records protocols, hashes, validations, and regeneration commands, and links the public dataset and model archive; App. B records split identities. The AI Use Statement above discloses generative-AI assistance.

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## APPENDIX ORGANIZATION

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## NOTATION

Table 2 summarizes the notation used for the benchmark, replay algorithm, and theoretical analysis. Prompt and policy indices are omitted when their meaning is clear from context.

Table 2: **Notation.** Symbols reused across the benchmark, the algorithm, and the idealized analysis.

| Symbol                       | Meaning                                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------------------|
| <i>Benchmark and metrics</i> |                                                                                                               |
| $V(y, s_x)$                  | Validator output for response $y$ and specification $s_x$ : a canonical execution key, or $\perp$ for failure |
| $\mathcal{C}_x$              | Set of valid solution keys for prompt $x$ , unavailable to the learner                                        |
| $m$                          | Number of correct modes a prompt admits, $m =  \mathcal{C}_x $                                                |
| $P$                          | Probability that a sampled response is verified correct                                                       |
| $q_c$                        | Conditional probability of key $c$ given correctness                                                          |
| $Z_i$                        | Canonical key of the $i$ th draw, or $\perp$                                                                  |
| $C(q)$                       | Conditional collision $\sum_c q_c^2$ ; the concentration measure                                              |
| PCMD                         | Success-conditional diversity $1 - C(q)$ ; the reported diversity axis                                        |
| $N_{\text{eff}}^{(2)}$       | Effective number of verified modes, $1/(1 - \text{PCMD})$                                                     |
| $K$                          | Number of draws per evaluation group ( $K = 8$ for the main comparisons)                                      |
| $n_c, n_+$                   | Number of responses verified on key $c$ , and total verified responses, $n_+ = \sum_c n_c$                    |
| $D_K, F_K$                   | Distinct verified keys, and correct fraction, in $K$ draws                                                    |
| $\bar{D}_K$                  | Expected distinct verified keys, $\mathbb{E}[D_K] = \text{distinct}@K$                                        |
| $B$                          | Extra verified modes, $\text{distinct}@K - \text{pass}@K$                                                     |
| <i>Algorithm</i>             |                                                                                                               |
| $\mathcal{B}_x$              | Replay buffer for prompt $x$ , with capacity $M = 16$ keys                                                    |
| $k$                          | Number of buffered keys, $k =  \mathcal{B}_x  \leq M$ ; in $\text{pass}@k$ , the number of evaluation draws   |
| $P_{\mathcal{B}}$            | Total probability mass of buffered keys                                                                       |
| $e_c$                        | Stored exemplar realizing key $c$ ; $ e_c $ its token length                                                  |
| $\ell_c$                     | Mean response-token log probability of the exemplar $e_c$                                                     |
| $G$                          | Fresh rollout group size ( $G = 16$ in training)                                                              |
| $R_i, R$                     | Binary reward of response $i$ , and group success count $\sum_i R_i$                                          |
| $A_i$                        | Fresh task advantage of response $i$                                                                          |
| $\sigma$                     | Round-robin replay cursor                                                                                     |
| $\eta, \kappa$               | Scale and clip constants of the Semantic-MaxEnt comparator                                                    |
| <i>Idealized analysis</i>    |                                                                                                               |
| $\mathcal{A}$                | Category set, correct modes $\mathcal{C}$ and incorrect $\mathcal{N}$                                         |
| $c_G(P), \Psi$               | Mean-flow correctness coefficient and its potential                                                           |
| $\rho$                       | Effective replay dose in the mean flow; $\lambda_{\text{replay}}$ its implemented weight                      |
| $\tau$                       | Effective optimization time in the mean flow                                                                  |
| $C_T$                        | Energy constant in the retention bound                                                                        |
| $\mu, \beta$                 | Reference policy and its KL coefficient in (A4')                                                              |
| $D$                          | Reverse divergence $\text{KL}(p \parallel \mu)$ to that reference                                             |

## Part I. Scope and measurement

### A METRIC AND OBJECTIVE DETAILS

Correctness and verified-mode diversity are distinct properties of a policy. For a prompt  $x$  with reference specification  $s_x$ , the validator maps a response to either failure or a canonical key:

$$V(\cdot, s_x) : \mathcal{Y} \rightarrow \mathcal{C}_x \cup \{\perp\}.$$

Here  $\mathcal{C}_x$  is the set of valid solution keys and  $\perp$  denotes failure. The learner observes a key only for a response it generates; neither the full set  $\mathcal{C}_x$  nor its size is available during training.

For  $Y \sim \pi_\theta(\cdot | x)$ , define the probability of a correct response and the conditional distribution over the solution key of a verified correct response:

$$P_\theta^+(x) = \Pr[V(Y, s_x) \neq \perp], \quad p_\theta^+(c | x) = \Pr[V(Y, s_x) = c | V(Y, s_x) \neq \perp].$$

We use the abbreviations  $P$  and  $q_c$  from the main text when the policy and prompt are clear. The conditional distribution is defined only when  $P > 0$ . If  $P = 0$ , `pass@K`, `distinct@K`, and `mean@K` are all zero, whereas PCMD is undefined.

We use *mode collapse* to describe increasing concentration over correct solution keys, particularly when correctness remains stable or improves. Transferring probability from a less probable correct key to a more probable one decreases  $\text{PCMD} = 1 - \sum_c q_c^2$  (Lemma A.1). A decrease in PCMD alone does not establish that such transfers relate the distributions, or that any key has zero probability. Finite samples cannot establish that a mode is absent from the policy.

Estimating PCMD requires two verified responses per prompt. We distinguish estimates with at least 30 eligible prompts from those with less support; paired training comparisons require at least 20 eligible evaluation prompts in each arm (App. A.2).

#### A.1 MEASURING SUCCESS AND DIVERSITY

For a given prompt, let  $n_c$  be the number of responses verified on key  $c$  and  $n_+ = \sum_c n_c$  the total number of verified responses. When  $n_+ \geq 2$ , we estimate PCMD as the fraction of verified response pairs with different keys:

$$\widehat{\text{PCMD}} = 1 - \frac{\sum_c \binom{n_c}{2}}{\binom{n_+}{2}} = 1 - \frac{\sum_c n_c(n_c - 1)}{n_+(n_+ - 1)}, \quad (4)$$

For independent responses from a fixed policy, this estimator is unbiased conditional on any verified count  $n_+ \geq 2$ . Given that count, the verified keys are independent draws from  $q$ , so each pair agrees with probability  $\sum_c q_c^2$ . The denominator therefore uses the verified count, rather than the total sampling budget. This argument does not establish unbiasedness after additional selection on observed outcomes.

Local evaluations pool four groups of eight responses per prompt; hosted evaluations use eight responses per prompt. Pooling independent responses from the same policy increases the number of verified pairs available to estimate  $q$ . We report the equally weighted mean of prompt-level estimates where  $n_+ \geq 2$ , with an across-prompt standard error. Pooling pairs across prompts would instead weight each prompt by its number of verified pairs, giving greater influence to prompts the policy solves more often.

Some local evaluations reuse sampling streams across groups (App. H.2). The resulting repeated responses increase measured collision and reduce estimated PCMD, violating the independence assumption above. A separate evaluation uses 64 responses per prompt with independent sampling streams for `Graph` and `PantryPlan` (App. H.1). In all six model-scale-domain comparisons, the paired `Re:Dr` minus `Dr:GRPO` PCMD gain is larger in this evaluation by .122–.208 (mean .160), and each confidence interval for the independently sampled gain excludes zero. Thus, the replay advantage persists under independent sampling in these comparisons. The main Level-1 comparisons also use 32 disjoint sampling streams (Table 61). Estimates from evaluations with overlapping streams describe diversity within the sampled response pools; they do not have the independent-sampling interpretation.

Two transformations aid interpretation:  $\text{rarefied-distinct@2} = 1 + \text{PCMD}$  is the expected number of distinct modes in two independent verified draws, and  $N_{\text{eff}}^{(2)} = 1/(1 - \text{PCMD})$  is the effective number of verified modes. Both are increasing functions of PCMD and order policies identically for a given prompt.

**Relation to standard diversity indices.** PCMD is the Gini–Simpson index applied to the conditional distribution over correct solution keys. It is the complement of Simpson’s concentration measure (Simpson, 1949), and Eq. (4) is its standard unbiased finite-sample estimator under independent sampling. The effective mode count  $N_{\text{eff}}^{(2)}$  is the inverse Simpson index, or Hill number of order two, which expresses diversity as an equivalent number of equally probable classes (Hill, 1973; Jost, 2006). Our notation identifies the population to which these established indices are applied: verified responses grouped by canonical execution key. They measure diversity among correct solutions.

For  $K$  independent samples  $Y_{1:K}$ , let

$$D_K = |\{V(Y_i, s_x) : i \leq K\} \setminus \{\perp\}|, \quad F_K = K^{-1} \sum_i \mathbf{1}[V(Y_i, s_x) \neq \perp].$$

MODEBENCH reports three sampling averages with different units.  $\text{distinct@K} = \mathbb{E}[D_K]$  is an expected count in  $[0, K]$  of unique verified keys;  $\text{pass@K} = \Pr[D_K > 0]$  and  $\text{mean@K} = \mathbb{E}[F_K]$  are probabilities in  $[0, 1]$ .

**Lemma A.1** (Success, sampled diversity, and PCMD). *Fix a prompt, a finite or countable valid-key set, and  $K \geq 1$  independent draws from the same fixed sampling law. Let  $\mu_c = \Pr[V(Y, s_x) = c]$  and  $P = \sum_c \mu_c$ . Then*

$$\text{pass@K} = 1 - (1 - P)^K, \quad \bar{D}_K := \text{distinct@K} = \sum_c [1 - (1 - \mu_c)^K]. \quad (5)$$

*For a finite valid-key set of size  $m \geq 1$  and  $P > 0$ , write  $q_c = \mu_c/P$ . At fixed  $P$ , concentration lowers  $\bar{D}_K$ : if  $q$  majorizes  $q'$  (its decreasingly sorted partial sums are at least those of  $q'$ ), then  $\bar{D}_K(P, q) \leq \bar{D}_K(P, q')$ . In particular,*

$$1 - (1 - P)^K \leq \bar{D}_K(P, q) \leq m[1 - (1 - P/m)^K]. \quad (6)$$

*For  $K \geq 2$ , the lower bound is attained only by concentration on one key, and the upper bound only by uniform  $q$ . For  $K = 1$ ,  $\bar{D}_1 = P$  for every  $q$ . At fixed  $q$ , increasing  $P$  cannot decrease  $\bar{D}_K$ . By contrast  $\text{PCMD}(q) = 1 - \sum_c q_c^2$  does not involve  $P$ : it is unchanged when correctness rises or falls at a fixed allocation. Concentration lowers it: if  $q$  majorizes  $q'$  then  $\text{PCMD}(q) \leq \text{PCMD}(q')$ , strictly unless  $q'$  is a permutation of  $q$ . It runs from 0 at a point mass to  $1 - 1/m$  at the uniform allocation, so its ceiling is set by the number of valid keys rather than by correctness.*

*Proof.* All  $K$  draws fail with probability  $(1 - P)^K$ . For each key  $c$ , its appearance indicator has expectation  $1 - (1 - \mu_c)^K$ ; summing indicators gives the second identity, also for countably many keys by nonnegativity. For fixed  $P > 0$  and  $K \geq 2$ , the function  $f(v) = 1 - (1 - Pv)^K$  is strictly concave on  $[0, 1]$ . Transferring mass from a larger coordinate to a smaller one without reversing their order increases  $\sum_c f(q_c)$ ; successive such transfers give the majorization assertion. The point mass and uniform vector are its extreme cases, giving the bounds and equality conditions. Finally,  $\partial \bar{D}_K / \partial P = K \sum_c q_c (1 - Pq_c)^{K-1} \geq 0$ , which proves monotonicity in correctness. For the final claim,  $\sum_c q_c^2$  is symmetric and convex, so the same transfer strictly decreases it whenever the two coordinates differ; PCMD is one minus that sum and no term in it involves  $P$ . The point mass and the uniform vector again give its extremes, by the same succession of transfers that gave the bounds on  $\bar{D}_K$ .  $\square$

These identities hold separately for each prompt. Because the expressions are nonlinear, substituting correctness averaged over prompts generally does not recover the average metric. In both the training and hosted-model studies,  $\text{pass@K}$  depends on total correct probability, whereas  $\text{distinct@K}$  also depends on how that probability is distributed over solution keys.

Raw  $\text{distinct@K}$  measures the number of modes observed at a given sampling budget; it does not estimate the total number of modes with positive probability. To define a uniform-sampling reference,

consider a finite space of  $N$  actions, of which  $N_c$  map to valid key  $c$ . Uniform independent action sampling yields  $\sum_c [1 - (1 - N_c/N)^K]$  expected verified keys. When the  $m$  valid actions have distinct keys, this reduces to  $m[1 - (1 - 1/N)^K]$ , as in the finite-action references reported in Table 22. Open-ended program and derivation spaces have no canonical uniform sampling distribution.

## A.2 ENDPOINTS AND THE REPORTED DIVERSITY AXIS

At the final checkpoint, we report `pass@K`, the probability of at least one correct response, and `distinct@K`, the expected number of distinct verified keys. We measure diversity among correct responses with PCMD, which depends only on their conditional distribution  $q$  (Lemma A.1). By comparison, `distinct@K` increases with correctness  $P$  even when  $q$  is unchanged (Eq. (5)). Across the 375 evaluations of untrained models, the correlation between `distinct@8` and `pass@8` is .936. The corresponding correlation for PCMD is -.332 among the 225 evaluations with sufficient verified responses. These correlations describe different populations because some evaluations have too few correct responses to report PCMD.

Fig. 14 illustrates this distinction. For models with at least 7B parameters, Python `pass@8` is at least .988 at Levels 1–3, while PCMD never exceeds .044. A high probability of finding a correct answer therefore coexists with repeated sampling of the same solution mode.

**Support thresholds.** The prompt-level PCMD estimator requires at least two verified responses. The fraction of eligible prompts has correlation .988 with `pass@8`; comparisons can therefore differ in both diversity and the population on which it is estimated. Each reported estimate includes its eligible prompt count. Filled markers denote at least 30 eligible prompts; hollow markers denote 5–29 and include standard errors. Estimates based on fewer than five prompts are omitted. Of the 375 evaluations of untrained models, 225 meet the 30-prompt threshold. Missing estimates indicate insufficient observations to measure diversity.

For paired training comparisons, each arm must have at least 20 eligible evaluation prompts. The eligible subsets may differ between arms, so a difference in their means can reflect both policy differences and changes in which prompts are included. The independent-sampling evaluation in App. H.1 instead compares the two conditions on jointly eligible prompts. These paired contrasts need not equal differences between means computed on each arm’s separately eligible subset.

PCMD equals one minus the conditional-collision U-statistic in App. H.2. The supplementary measure `distinct@K - pass@K` =  $\mathbb{E}[D_K - \mathbf{1}[D_K > 0]]$  counts expected verified modes beyond the first. Like PCMD, it requires no knowledge of the full valid-key set  $\mathcal{C}_x$ , but it also depends on the number of successful draws. Dividing `distinct@8` by eight rescales the metric without removing this dependence on correctness.

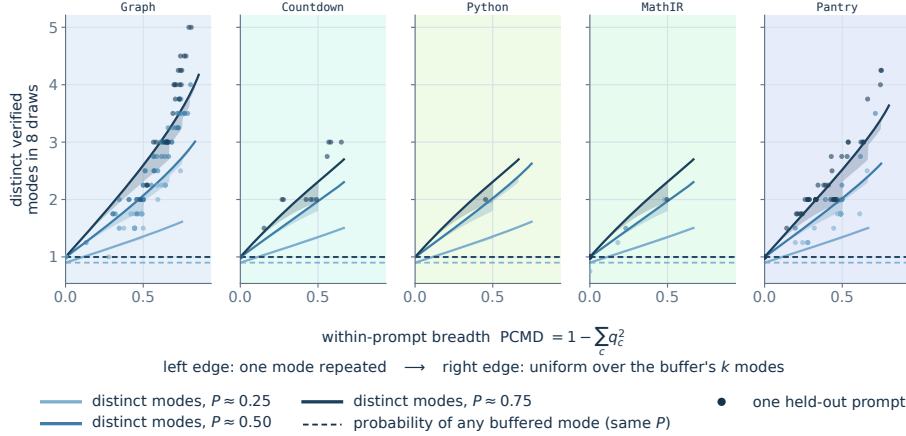
## A.3 WHY BALANCE WITHIN THE REPLAY BUFFER

Uniform sampling over buffered keys assigns equal replay frequency to each stored mode, regardless of how often it was discovered. To examine this objective, consider an equally weighted categorical loss without length normalization (Fig. 6). The response-level objective used in training is given in App. D; the probability of its stored exemplar can differ from the total probability of all responses with the same key. For  $k = |\mathcal{B}_x|$  buffered keys, let  $p(c)$  denote each key’s probability,  $P_B$  their total probability mass, and  $U_k$  the uniform distribution over these keys. Then

$$-\frac{1}{k} \sum_{c \in \mathcal{B}_x} \log p(c) = -\log P_B + \log k + \text{KL}(U_k \parallel p(c \mid c \in \mathcal{B}_x)). \quad (7)$$

Minimizing this loss favors both greater total probability on buffered keys and a more uniform conditional distribution within the buffer. These two terms have distinct effects. The probability of sampling at least one buffered key in  $K$  independent draws is  $1 - (1 - P_B)^K$ , which depends only on total buffered mass. The expected number of distinct buffered keys is  $\sum_{c \in \mathcal{B}_x} [1 - (1 - p(c))^K]$ . For  $K \geq 2$  and  $P_B > 0$ , strict concavity makes this expectation largest when probability is allocated uniformly over the  $k$  keys.

Thus, at fixed buffered mass, uniform allocation maximizes sampled diversity without changing the probability of reaching any buffered key. Fig. 12 illustrates this distinction: the dashed curves show the probability of at least one buffered key, and the solid curves show the expected number of distinct keys. Their difference is the expected number of modes beyond the first. Uniform allocation also maximizes the minimum conditional probability of any buffered key. These properties characterize the categorical objective; they do not imply that every combined training update improves both correctness and diversity.



**Figure 12: Uniform allocation maximizes sampled diversity at fixed buffered probability mass.** The horizontal axis traces allocations from a single mode to a uniform distribution over  $k$  buffered keys. Dashed curves show  $1 - (1 - P)^8$ , the probability of sampling at least one buffered key when their total mass is  $P$ . Solid curves show the expected number of distinct buffered keys for  $P \in \{.25, .50, .75\}$ ; bands span a range of mode counts based on the observed prompts and are analytic ranges rather than confidence intervals. Points show held-out Level-1 prompts, which have no replay buffer, evaluated at the final Re:Dr and Re:Max checkpoints of Qwen2.5-0.5B, using one matched seed and grouping prompts by their verified response fraction. Their horizontal coordinates use empirical key frequencies in  $1 - \sum_c q_c^2$ , rather than the unbiased pair estimator. Both coordinates use the same sampled responses, so the scatter illustrates agreement with the analytic relationship rather than an independent prediction of diversity on new responses.

These properties describe the categorical objective on training-prompt buffers. Generalization is measured on held-out prompts, whose responses never enter the buffers. The surrogate alone does not establish that the implemented response-level objective increases held-out PCMD.

## Part II. Benchmark and method

### B BENCHMARK AND EVALUATION PROTOCOL

MODEBENCH uses a single executable validator to determine correctness and solution identity, through the map  $V(\cdot, s_x)$  defined in App. A. Invalid outputs receive reward zero and no key; valid outputs receive reward one and a canonical execution key. The valid-key catalogue and its size are unavailable to the learner and are not used to schedule replay or determine when training stops. Table 3 summarizes the five domains and the response variants that share a key. The datasets are available in the public release; the training comparisons in this paper cover Levels 1–3.

Table 3: **Executable solution identities in the five MODEBENCH domains.** The validator determines both correctness and identity, while the catalogue of valid modes is unavailable during training. *Valid modes* reports the mean and range of certified catalogue sizes over the Level-1 evaluation split. These catalogues need not include every accepted output: *Countdown*, for example, accepts valid keys outside its enumeration. *Reached* reports mean distinct verified modes from one hosted deployment using 512 draws on 32 prompts per domain (Fig. 5). These prompts were selected from the evaluation split independently of model outputs. The two columns therefore summarize the full split and a subset, respectively. *MathIR*’s catalogue includes longer derivations; limited observed coverage does not establish whether these routes are inaccessible or simply too rare to observe at the given budget. Higher levels have different constructions and mode-count distributions (App. B.4). The full prompts and model-specific chat templates appear in App. C.

| Domain         | Generated answer / check                          | Executed key       | Equivalent responses                 | Valid modes | Reached |
|----------------|---------------------------------------------------|--------------------|--------------------------------------|-------------|---------|
| Graph coloring | Complete assignment; check fixed colors and edges | Color vector       | Spacing only                         | 6.27        | 2.53    |
|                |                                                   |                    |                                      | 4–12        |         |
| Countdown      | Parse operands; execute exactly                   | Normalized AST     | Commutative order                    | 4.52        | 3.47    |
|                |                                                   |                    |                                      | 2–8         |         |
| Python factors | Execute restricted lambda externally              | Return vector      | Programs with the same return vector | 229.44      | 3.75    |
|                |                                                   |                    |                                      | 16–1,512    |         |
| MathIR         | Execute bounded action program                    | State trajectory   | Programs with the same states        | 5.00        | 1.91    |
|                |                                                   |                    |                                      | 5           |         |
| PantryPlan     | Exact ingredient allocation; check all bounds     | Ingredient support | Quantity and ordering aliases        | 18.19       | 6.91    |
|                |                                                   |                    |                                      | 8–40        |         |

#### B.1 SOLUTION MODE DEFINITIONS AND CONCENTRATION BY DOMAIN

MODEBENCH defines solution modes through three kinds of executable identity. In *Countdown* and *MathIR*, distinct modes are different routes to the same answer. In *Python*, they are different return vectors on the public test inputs. In *Graph* and *PantryPlan*, they are different valid answers. The response budgets also differ (Table 6): a *PantryPlan* response is a short plan, while a *Countdown* response is a derivation. These keys identify verified solutions, not latent reasoning processes.

**Sensitivity to the chosen equivalence.** The canonical key determines which responses count as distinct solutions. A coarser, task-relevant equivalence can therefore change the measured diversity gain. App. L.3 tests one such equivalence per domain. Most contrasts persist, but the replay gain for Level-3 *Python* with the original wording falls from .419 to zero when  $d$  and  $n/d$  are treated as the same unordered factor pair. Among the five domains, *Python* is the most sensitive to this choice. Evidence that the gain extends beyond the original key definitions comes from the other domains. Training uses the original canonical keys throughout; the coarser analysis measures sensitivity to alternative definitions of solution identity.

**Answer-only training prompts.** The training prompts request only the final answer (App. C), with response budgets chosen accordingly. Restricting outputs to executable answers allows the validator to determine solution identity directly, without a learned judge or an embedding-similarity threshold. The finite solution catalogues also provide the reference values in Table 22.

For example, uniform sampling over the 64 ingredient-selection bit masks in `PantryPlan` reaches more verified modes than any trained policy evaluated here.

The training experiments do not measure diversity in long reasoning traces. The separate hosted-model evaluations include reasoning-enabled deployments and also exhibit concentration (App. N.1), but these inference-only observations do not establish how training changes reasoning behavior.

Across RLVR-only arms, the mean initial-to-terminal increase in conditional collision is +0.202 for route-keyed domains over 98 seed-blocks, +0.201 for the behavior-keyed domain over 15, and +0.404 for answer-keyed domains over 106. All three grouped means increase, with the largest increase in the answer-keyed group. These comparisons describe concentration under each definition; differences between domains do not isolate the effect of choosing a particular key.

`Countdown`’s mean collision increase is +0.383 over 46 seed-blocks, compared with +0.230 in `Graph`. Thus concentration also occurs among different derivations of a common answer. `MathIR` has a smaller increase, +0.042. Its certified catalogue includes longer derivations that are absent from the observed outputs (App. D.5). Their absence at the evaluated budgets does not establish that those modes are unreachable.

**Explicit-reasoning settings and observed solution counts.** App. O.6 compares five deployments on the same 480 prompts, with eight responses per prompt and an 8,192-token output cap in both conditions. Disabling explicit reasoning through the provider interface lowers overall mean `distinct@8` in all five deployments: `GPT-5.6 Sol` changes from 1.62 to 0.99 and `GPT-5.4` from 1.85 to 1.18. These counts include unsolved prompts and depend on both correctness and allocation among verified solution modes. Changes in success-conditional `PCMD` differ across deployments, so lower raw counts do not establish a common increase in conditional concentration. The requested setting does not establish the absence of hidden computation, and collection times and provider defaults can also differ. This comparison therefore describes solution distributions under the tested configurations, without identifying an isolated effect of internal deliberation.

## B.2 DATASET SPLITS AND VALIDATION

Each domain has disjoint training and evaluation splits (Table 6). `PantryPlan` and Level 2 also include separate development splits (App. B.4). For every `Python` problem, execution verifies at least two functions with distinct valid return vectors. For `MathIR`, each catalogue route is checked with the same interpreter used to evaluate model outputs.

Table 4: **Dataset sizes and certified mode counts.** `Python` and `MathIR` use disjoint training and evaluation splits. Catalogue ranges refer to the training split; Table 3 gives evaluation-split counts.

|                                    | <b>Python factors</b> | <b>MathIR action menu</b> |
|------------------------------------|-----------------------|---------------------------|
| Training problems                  | 384                   | 384                       |
| Evaluation problems                | 128                   | 128                       |
| Certified modes per prompt (train) | 16–3,600              | 5                         |

**PantryPlan.** `PantryPlan` contains 384 training, 64 development, and 128 evaluation prompts across four problem families. The number of feasible ingredient supports ranges from 8–45 in training and 8–40 in evaluation, with an evaluation mean of 18.19. The validator checks inventory, serving increments, mass, nutrient bounds, and dietary restrictions using exact arithmetic. The solution key records which ingredients are used. These checks establish satisfaction of the stated constraints; they do not assess taste, preparation feasibility, or food safety.

## B.3 PYTHON EXECUTION AND VALIDATION

After removing surrounding whitespace, the validator parses each candidate as a `Python` expression, with limits of 240 characters and 64 abstract syntax tree (AST) nodes. The accepted form is `lambda n: EXPR`, with one ordinary argument named `n` and no default value, positional-only, keyword, or variadic arguments. Integer literals have absolute value at most 10,000. Permitted expressions use `n`, addition, subtraction, multiplication, floor division, remainder, integer comparisons, Boolean `and/or`, unary plus/minus/`not`, and conditionals. The validator rejects Boolean and non-integer literals, calls, imports, attributes, comprehensions, containers, subscripts, assignment, loops, and other identifiers.

The task specification contains  $J$  distinct integers in  $[4, 1000]$ , with  $2 \leq J \leq 8$ . Every Level-1 problem uses 4 inputs drawn from  $[6, 96]$  and admits at least two valid return vectors.

**External execution.** Candidate expressions run in an isolated Python process with an empty `__builtins__` environment. The process repeats the syntax checks and evaluates the function on the fixed inputs, subject to a 0.25-second execution limit and a one-second response deadline. Timeouts, execution errors, malformed results, non-integer outputs, and failed divisor checks produce  $\perp$  and zero reward. Only a validated integer return vector receives a key.

**Behavioral identity.** For each of the  $J$  fixed inputs  $n_j$ , the returned  $d_j$  must be an integer but not a Boolean, satisfy  $1 < d_j < n_j$ , and divide  $n_j$  exactly. All checks must pass. The key is the ordered return vector  $(d_1, \dots, d_J)$ ; two programs share a key exactly when these vectors agree. The exact support count  $\prod_j |\{d : 1 < d < n_j, d \mid n_j\}|$  is used only for dataset validation.

#### B.4 MATCHED LEVEL CONSTRUCTIONS

Levels 2–5 expand the search space while preserving the verifier, canonicalizer, and solution-mode definition. Level 2 is harder for the same model; Levels 3–5 are calibrated against a fixed Level-1 reference using larger models, with an exception for Level-4 `MathIR`.

**Level 2.** Level 2 contains 384 training, 128 development, and 128 evaluation problems, with no shared problem identities across its splits or with Level 1. The training distribution of valid-key counts matches that of Level 1. For `Graph`, `Countdown`, `Python`, and `MathIR`, the development and evaluation distributions match separate Level-1 reference sets used for dataset construction. `PantryPlan` uses its existing Level-1 splits as references; its 64-prompt development histogram is scaled by two to define the target counts for the 128-prompt Level-2 development set.

The Level-1 construction reference sets differ from the evaluation sets used to compare trained policies. Consequently, those evaluation sets have different valid-key-count distributions across levels in `Graph`, `Countdown`, and `Python`. Cross-level differences reflect both problem structure and evaluation population, and cannot isolate difficulty at a fixed distribution of mode counts. Within each level, replay and control conditions use the same prompts, seeds, fresh-sample objective, and training budget.

Table 5: **Structural differences between Levels 1 and 2.** The verifier, canonicalizer, and definition of a solution mode are unchanged; Level 2 expands the search space.

| Domain         | Level 1                              | Level 2 structural change                              |
|----------------|--------------------------------------|--------------------------------------------------------|
| Graph coloring | Three hidden vertices                | Four hidden vertices                                   |
| Countdown      | Three operands                       | Four operands, with values up to 14                    |
| Python factors | Four executed cases, all at most 96  | Four cases, including a value above 96                 |
| MathIR         | One-sided linear families            | Variable-on-both-sides families                        |
| PantryPlan     | Base dietary/composition constraints | More active dietary exclusions and tighter composition |

On every domain’s development split, fixed Qwen2.5-0.5B has `pass@8` in  $[\cdot 10, \cdot 90]$ , below its Level-1 construction reference when both levels use identical decoding settings.

**Scale-calibrated Level 3.** Level 3 is calibrated using Qwen2.5-3B, with Qwen2.5-0.5B on Level 1 as the reference. Model weights remain fixed. Construction parameters are fitted on development problems and assessed on held-out prompts. In every domain, the calibration requires absolute differences from the reference of at most  $\cdot 04$  in `pass@1` and  $\cdot 08$  in `pass@8`, measured using four independent groups of eight samples per prompt. `distinct@8` is excluded from the fitting objective.

Both models use their native chat templates, temperature 1, top- $p$  1, and a 192-token generation limit. `Graph` prompts request only a boxed final answer. The other domains receive task-specific solution guidance and use decoding constrained to legal output syntax, without filtering candidates by correctness. These criteria establish an empirical match to the reference success rates rather than a formal two-sided equivalence test. The Level-3 `Python` hint ablation removes the strategy guidance (App. I); its difficulty is therefore outside the scope of this calibration.

**Levels 4 and 5.** Levels 4 and 5 are calibrated with 7B and 14B models, respectively, against the same Qwen2.5-0.5B Level-1 reference, using pass@1 and pass@8 tolerances of .04 and .08. Level 5 meets these criteria in all five domains. Level 4 meets them in Graph, Countdown, Python, and PantryPlan, but not MathIR.

For MathIR, the reference pass@8/pass@1 ratio is 5.50, whereas the largest ratio among the evaluated 7B constructions is 4.16. A ratio of mixture means cannot exceed the largest component ratio, so mixing these constructions cannot reproduce the reference ratio exactly.

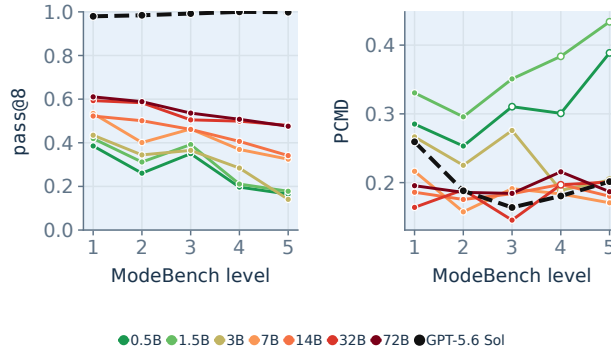
The heterogeneity gap,  $1 - (1 - \bar{P})^8 - \overline{\text{pass@8}}$ , compares observed pass@8 with its value if every prompt had the same correctness  $\bar{P}$ . The measured gaps are .169–.398 at 7B and .060 for the reference. Because the gap depends on mean correctness  $\bar{P}$ , these values are not directly comparable across correctness levels. Each Level-4 domain contains 384 training, 128 development, and 128 evaluation prompts, disjoint from other levels. Level-4 MathIR appears in Figs. 3 and 14 but is not difficulty-matched to the reference.

### B.5 MODEL PERFORMANCE ACROSS ALL FIVE LEVELS

Fig. 3 compares Qwen2.5-Instruct across five levels and domains; Fig. 16 adds three model families, for 375 model–domain–level measurements before benchmark training. Each measurement uses the same 128 held-out prompts per domain and level, with four independent groups of eight responses per prompt, totaling 4,096 responses. For pass@8 and distinct@8, we score each group, average within prompts, and then average prompts. For PCMD, we pool the four groups within each prompt (App. A.1).

Evaluation uses native chat templates, temperature 1, top- $p = 1$ , and a 192-token output limit. Graph prompts request boxed final answers; other domains use task-specific solution guidance and syntax-constrained decoding. These settings differ from the training evaluations: under the training template and its larger sample, the same untrained Level-1 Python model defines PCMD on 73 of 128 prompts, with mean .168 (Section 5.1). Training effects are therefore measured against the untrained model under the corresponding evaluation conditions.

Success does not necessarily decrease monotonically with level. pass@8 at the highest evaluated level is below its Level-1 value in 33 of 35 model–domain series. Model scale and benchmark level also affect diversity differently: the median range of PCMD across scales at a fixed level is .251, compared with .104 across levels at a fixed scale.



**Figure 13: Accuracy and solution diversity vary differently across benchmark levels.** Left: pass@8; right: PCMD, averaged with equal domain weights for each Qwen2.5-Instruct scale (colored lines) and GPT-5.6 Sol (black dashed line). This Qwen evaluation augments the 128 held-out prompts with the 384 training-split prompts where available; none were used to train these checkpoints. Filled diversity markers indicate at least 30 prompts with two or more correct responses in every domain. Hollow markers average the available estimates when at least one domain has insufficient support; the set of contributing domains can differ when an estimate is undefined.

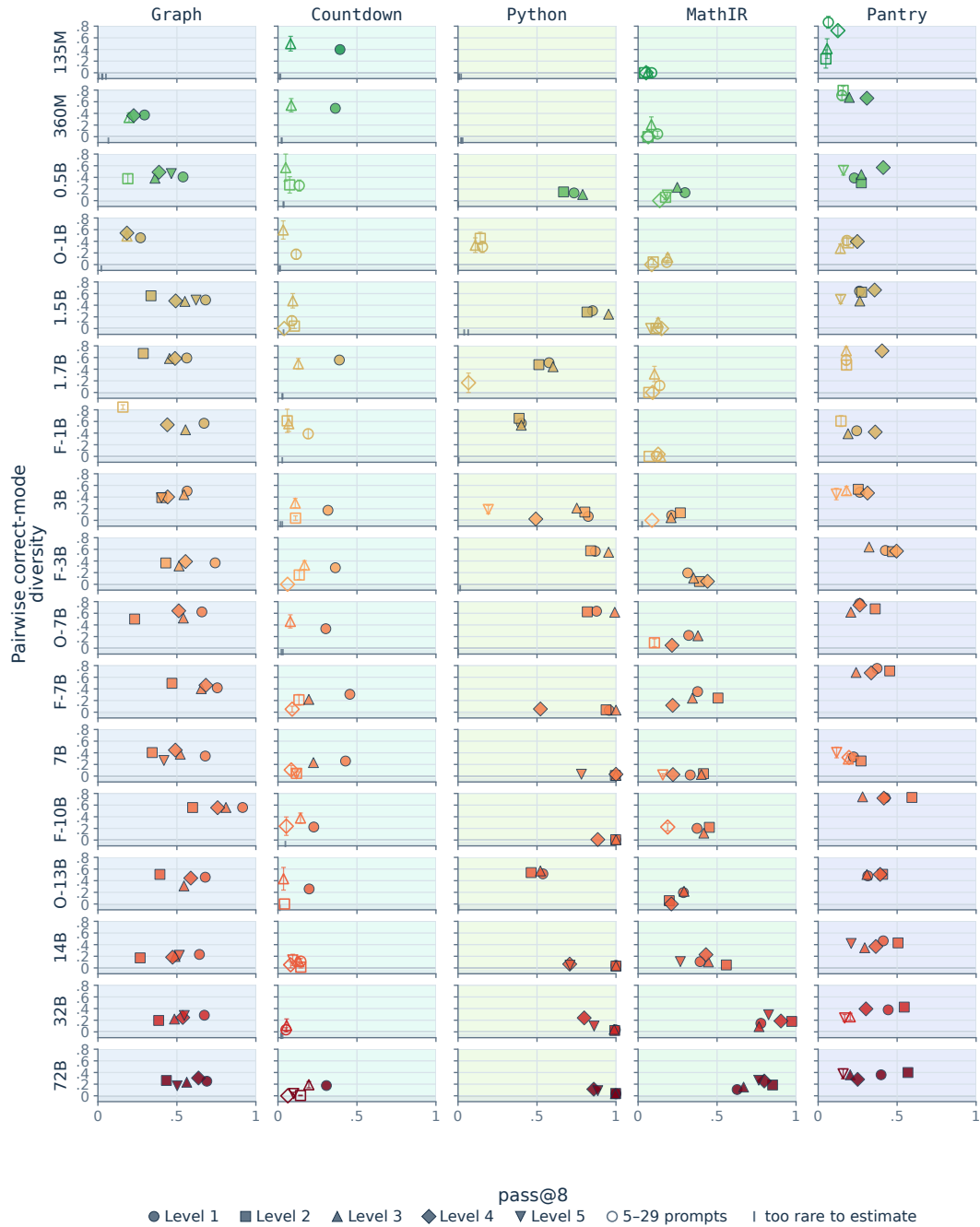


Figure 14: **High accuracy can coexist with low solution diversity.** Each row shows one checkpoint before benchmark training, with domains in columns,  $\text{pass}@8$  on the horizontal axis, and PCMD on the vertical axis. Color indicates parameter count and marker shape indicates level. Each measurement uses four groups of eight responses to 128 held-out prompts at temperature 1; PCMD pools the groups within each prompt. Filled markers indicate at least 30 prompts with two or more correct responses. Hollow markers indicate 5–29 such prompts and show one standard error. Ticks below the axis mark measurements with fewer than five eligible prompts.

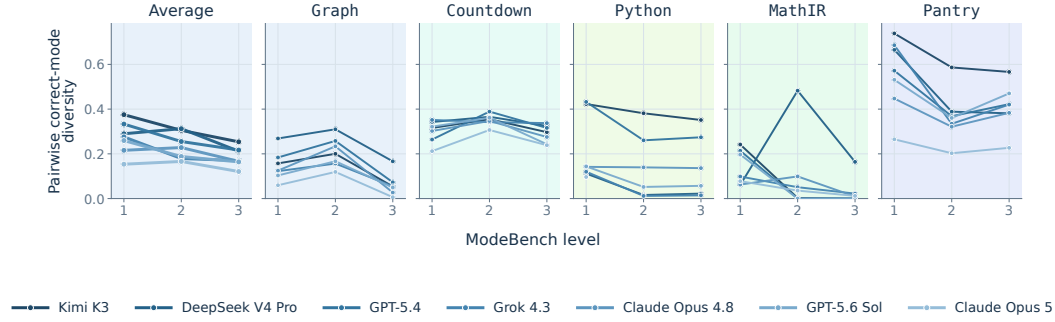


Figure 15: **Mean solution diversity is lower at Level 3 than at Level 1 for every evaluated hosted deployment.** Lines show PCMD for 7 deployments, with eight responses to each of 128 held-out prompts per domain and level. Panels show individual domains and their equally weighted mean. For each deployment, the set of domains contributing to the mean is fixed across levels. Darker lines indicate higher overall diversity. Hollow markers fall below the 30-prompt support threshold and are excluded from connecting lines. Higher levels change problem structure and add strategy guidance in MathIR, Python, and PantryPlan (App. I); these curves reflect both changes.

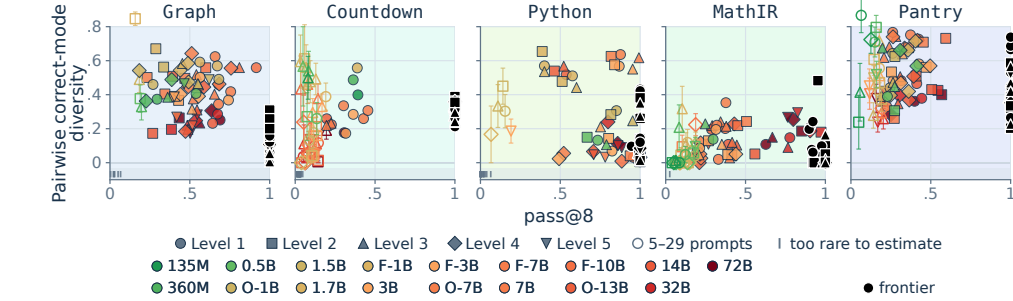


Figure 16: **Models of similar size can differ in solution diversity.** SmolLM2, Qwen2.5, Falcon3, and OLMo-2 checkpoints are compared by  $\text{pass}@8$  and PCMD across five domains and the available levels, before benchmark training. Color indicates parameter count on a logarithmic scale; marker shape indicates level. Black markers show the 7 hosted deployments. Local models use four groups of eight responses to 128 held-out prompts per domain and level; hosted deployments use eight responses per prompt. Filled markers indicate at least 30 prompts with two or more correct responses. Hollow markers indicate 5-29 and show one standard error. Ticks below the axis indicate fewer than five eligible prompts, for which no diversity estimate is reported.

## B.6 OTHER MODEL FAMILIES AT MATCHED SCALE

Fig. 16 compares SmolLM2, Qwen2.5, Falcon3, and OLMo-2 on the same 128 held-out prompts per domain and level. All use four groups of eight responses, the same verifier, and the same eligibility thresholds. Comparisons across families use the 4 shared levels; only Qwen2.5 is evaluated at Level 5. Color denotes parameter count.

Families differ in diversity even at similar parameter counts. Near 7B, Qwen2.5, Falcon3, and OLMo-2 have different PCMD values despite similar  $\text{pass}@8$ . Their ordering persists in the 10-15B range, where Falcon3 and OLMo-2 are close. Within parameter-count bands, the median range of PCMD across family means is .166, compared with .198 across scales within a family, over the 4 shared levels. The earlier .251 summary instead compares all seven Qwen scales at each level. These descriptive comparisons do not isolate which pretraining or post-training choices account for the differences between models.

## B.7 TRAINING AND EVALUATION SETTINGS

**Table 6: Paired comparisons use the same numbers of optimizer updates and fresh rollouts.** Treatment and control share the settings below. Both compute replay scores and execute the backward pass, but the replay gradient is exactly zero in the control. Their buffer contents can differ, so replay workloads, total floating-point operations, and wall-clock time are not matched. The control receives no additional training in place of replay computation (App. Q.1).

| Component       | Setting                                                                                                                                                                                                                                                                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data            | 384 training and 128 disjoint evaluation prompts per domain and level; evaluation prompts are excluded from training and replay.                                                                                                                                                               |
| Fresh rollouts  | $G = 16$ , temperature 1, top- $p = 1$ ; one prompt group per update.                                                                                                                                                                                                                          |
| Optimization    | Eight training passes (3,072 updates), with one PPO epoch per batch. $\beta_{\text{KL}} = 0$ except in the reference-KL comparison (App. P.12), which varies only this coefficient. Paired conditions share initialization, prompt order, optimizer, hardware class, and checkpoint frequency. |
| Evaluation      | At passes 0, .5, . . . , 8: four groups of eight responses per prompt at temperature 1 and top- $p = 1$ , plus a separate greedy response at temperature 0. Final comparisons use the last checkpoint.                                                                                         |
| Replay          | Capacity $M = 16$ keys per prompt; buffers are selected in round-robin order, one per update; $\lambda_{\text{replay}} = .10$ . The control has zero replay gradient; buffer contents and computational workloads may differ between conditions.                                               |
| Training seeds  | Five per model–domain combination; paired comparisons use common seeds, with sample sizes reported alongside the results.                                                                                                                                                                      |
| Learning rate   | Qwen2.5-0.5B and Falcon3-1B: constant $2 \times 10^{-7}$ , with Falcon optimizer states stored on the CPU. Qwen2.5-3B: cosine decay from a $10^{-7}$ peak to a $10^{-8}$ floor, with 10% warmup.                                                                                               |
| Optimizer       | Fused Adam in bf16 with decoupled weight decay, $\beta_1 = .9$ , $\beta_2 = .95$ , weight decay 0, gradient-norm clip 1.0, with gradient checkpointing. CPU offloading uses the corresponding CPU optimizer implementation with the same hyperparameters.                                      |
| Policy update   | Parameters are synchronized after every update, and each rollout is used once in a single PPO epoch. The symmetric clipping threshold is $\varepsilon = .2$ . The importance ratio equals 1 when the surrogate is evaluated in this update scheme, so clipping is inactive.                    |
| Response tokens | Graph and Countdown: 192; Python: 192/512/192 at 0.5B/1B/3B; MathIR: 64/128/64; PantryPlan: 8.                                                                                                                                                                                                 |

For each training seed, `pass@8` and `distinct@8` are averaged over four eight-response groups per prompt and all 128 evaluation prompts; `mean@8` is the fraction of correct responses. Responses are independent within a group. Independence across groups, required for the pooled PCMD interpretation, depends on the sampling procedure described in App. A.1. Evaluation outputs are excluded from training and replay, and do not determine checkpoint selection or stopping.

Comparisons use paired final checkpoints. Comparisons with five matched training seeds use nominal pointwise 95% Student- $t$  intervals; other main checkpoint comparisons report sample sizes and descriptive means. Diversity comparisons use the prompt-eligibility thresholds and interval procedures specified alongside their results. Intervals condition on the fixed evaluation prompts and sampled responses. The replay-weighting comparison instead uses a bootstrap over paired training seeds.

## C EXACT DOMAIN PROMPTS

Each domain’s training prompt contains one system message and one user message, without few-shot examples. Training and evaluation use the same template within each method. The examples below reproduce the first prompt from each Level-1 evaluation split in Qwen’s chat format. Falcon3 uses the same message text with its own `<|system|>` and `<|user|>` markers. Models evaluated before benchmark training use their tokenizer’s native chat template (App. B.5). App. B.4 describes the different problems and prompting conditions at Levels 2–5.

The examples preserve the prompt text. A continuation line indented by two spaces indicates a display line break; replacing that break and indentation with one space recovers the original text. In the `PantryPlan` example, a line break without this indentation is part of the prompt.

## C.1 GRAPH COLORING

384 training and 128 evaluation prompts.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Color this graph with vertices 1 through 6. The undirected edges are: (1,2), (1,3),
(3,5), (3,6), (4,6), (5,6). Assign each vertex one color from {1, 2, 3} so that no
edge has the same color on both endpoints. A partial coloring is 2??1?2, where ?
means uncolored. Keep every shown digit fixed. There are exactly 3 question marks.
Return exactly one final answer and no explanation: exactly 3 digits, each one 1,
2, or 3, for the missing positions from left to right, inside \boxed{}.<|im_end|>
<|im_start|>assistant
```

## C.2 COUNTDOWN

384 training and 128 evaluation prompts.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Using the numbers [3, 6, 9], create an arithmetic expression that equals 18. Use
each given number exactly once. You may use +, -, *, /, and parentheses. Return
exactly one final answer and no explanation: the expression inside
\boxed{}.<|im_end|>
<|im_start|>assistant
```

## C.3 PYTHON FACTORS

384 training and 128 evaluation prompts.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Write one pure Python function with the exact form lambda n: EXPR. An external
Python tool will call it once for each n in [18, 82, 91, 93]. For every call,
return an integer d satisfying 1 < d < n and n % d == 0. Different valid return
vectors are different solution modes. You may use integer literals, n, +, -, *,
//, %, comparisons, Boolean operators, and conditional expressions; calls,
imports, attributes, containers, and other names are forbidden. Return exactly the
one-line lambda inside \boxed{} and no explanation.<|im_end|>
<|im_start|>assistant
```

#### C.4 MATHIR ACTION MENU

384 training and 128 evaluation prompts.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
MathIR linear-menu-v1. Solve for x by choosing an executable action sequence. Every
selected action is applied exactly to BOTH sides of the current equation, followed
by exact simplification. Bindings: a=2, b=-9, c=-1. Initial equation: x/a + b = c.
Action menu: A means div(a); B means add(b); C means sub(b); D means sub(c); E
means sub(mul(a,b)); F means mul(a). Choose 1-4 action IDs. Illegal operations,
repeated states, and paths that do not finish with x isolated are rejected. Return
only the IDs separated by semicolons inside the answer box; for example formatting
only: \boxed{A;C}. Do not return x's numeric value.<|im_end|>
<|im_start|>assistant
```

#### C.5 PANTRYPLAN

384 training and 128 evaluation prompts.

```
<|im_start|>system
Return only six binary digits and no other text. Each digit maps to the
corresponding Pantry row in printed order. A trusted environment will choose exact
quantities on precisely the selected rows.<|im_end|>
<|im_start|>user
Create one feasible high fiber snack from this frozen pantry.
Nutrition values are per 100 g. Quantities are exact integer grams.

Pantry:
- almonds: available=100g, step=25g, minimum-if-used=50g; energy=584kcal,
  protein=21.5g, fiber=10.8g, sodium=0.0mg; tags=tree_nut
- banana: available=125g, step=25g, minimum-if-used=50g; energy=97.0kcal,
  protein=0.74g, fiber=4.62g, sodium=0.0mg; tags=none
- carrots: available=150g, step=25g, minimum-if-used=50g; energy=45.0kcal,
  protein=0.941g, fiber=3.1g, sodium=86.6mg; tags=none
- grape_tomatoes: available=100g, step=25g, minimum-if-used=50g; energy=27.0kcal,
  protein=0.83g, fiber=2.1g, sodium=6.0mg; tags=none
- navel_orange: available=150g, step=25g, minimum-if-used=50g; energy=47.0kcal,
  protein=0.91g, fiber=2.0g, sodium=9.0mg; tags=none
- sunflower_seeds: available=125g, step=25g, minimum-if-used=50g; energy=571kcal,
  protein=18.9g, fiber=7.22g, sodium=0.0mg; tags=none

Use 2 to 4 ingredients.
Forbidden tags: none
Targets:
- mass_g: minimum 125, maximum 200
- energy_kcal: minimum 239.4, maximum 432.25
- protein_g: minimum 6.423
- fiber_g: minimum 3.478
- sodium_mg: maximum 37.6

Choose the ingredient support as a six-bit mask in the exact Pantry row order. Bit 1
includes that row and bit 0 excludes it.<|im_end|>
<|im_start|>assistant
```

#### D REPLAY ALGORITHM

Verified replay combines learning from newly sampled responses with a likelihood loss on stored, verified solutions. Each prompt  $x$  has a buffer  $\mathcal{B}_x$  of at most  $M$  exemplars, with one response per canonical key. A round-robin cursor  $\sigma$  selects one prompt's buffer at each update, and all keys in that buffer contribute equally to the replay loss. Table 2 defines the notation; App. A.3 relates this objective to diversity.

### D.1 FRESH TASK ADVANTAGES

For a group of  $G$  fresh responses with verified binary rewards  $R_i$  and  $R = \sum_i R_i$ , Dr.GRPO uses  $A_i = R_i - R/G$ , while MaxRL uses

$$A_i^{\text{MaxRL}} = \begin{cases} GR_i/R - 1, & R > 0, \\ 0, & R = 0. \end{cases}$$

The centered MaxRL estimator (Tajwar et al., 2026) assigns advantage  $-1$  to failures in mixed groups and gives each success greater weight when successes are scarce. Both objectives assign zero task advantage to every response in an all-correct or all-failure group. Lemma P.3 derives MaxRL’s mean objective as a harmonic combination of pass@ $k$  terms. These advantages apply to fresh responses; stored exemplars contribute through the separate replay loss.

### D.2 OPTIMIZER UPDATE

---

**Algorithm 1 Verified replay update.** The fresh objective selects Re:Dr or Re:Max.

---

**Require:** Policy  $\pi_\theta$ ; prompt  $x$  and verifier specification  $s_x$ ; group size  $G$ ; fresh objective  $o \in \{\text{MaxRL}, \text{Dr.GRPO}\}$ ; replay weight  $\lambda_{\text{replay}}$ ; buffer capacity  $M$ ; buffers and cursor  $(\mathcal{B}, \sigma)$

**Ensure:** Updated  $(\theta', \mathcal{B}', \sigma')$

```

1: Draw the on-policy group  $\mathcal{G} = \{y_i \sim \pi_\theta(\cdot | x)\}_{i=1}^G$ 
2: for  $i = 1, \dots, G$  do
3:   Execute  $\tilde{c}_i \leftarrow V(y_i, s_x)$  and observe task reward  $R_i$ 
4:   Set  $c_i \leftarrow \tilde{c}_i$  iff  $R_i > 0 \wedge \tilde{c}_i \neq \perp$ ; otherwise set  $c_i \leftarrow \perp$ 
5: end for
6: Compute fresh task advantages  $A_i^o$  from  $\{R_j\}_{j=1}^G$ ; for  $o = \text{MaxRL}$  use the formula in App. D.1
7: After the full group is validated, deterministically retain one policy exemplar for each new  $c_i \neq \perp$  in  $\mathcal{B}_x$ , up to capacity  $M$ 
8:  $(z, \sigma') \leftarrow \text{NEXTBUFFER}(\{\mathcal{B}_u\}_u, \sigma)$  by a fixed round-robin cursor over prompts; the replayed prompt  $z$  need not be the prompt  $x$  that supplied the fresh group
9: if a replay buffer  $\mathcal{B}_z$  exists then
10:   For each exemplar  $e_c \in \mathcal{B}_z$ , compute  $\ell_c = |e_c|^{-1} \log \pi_\theta(e_c | z)$ 
11:    $L_{\text{replay}} \leftarrow -|\mathcal{B}_z|^{-1} \sum_{c \in \mathcal{B}_z} \ell_c$ 
12: else
13:    $L_{\text{replay}} \leftarrow 0$ 
14: end if
15: Set the replay-loss normalization  $w_{\text{rep}} \leftarrow (G - 1)/G^2$ 
16:  $\theta' \leftarrow \text{UPDATE}(\theta, L_{\text{PPO}}(A^o) + w_{\text{rep}} \lambda_{\text{replay}} L_{\text{replay}})$ 

```

---

### D.3 BUFFER CONSTRUCTION AND SCHEDULING

Each buffer is associated with an exact prompt-token sequence. A response is eligible for storage if it has positive task reward and a valid canonical key, including responses that reach the generation limit. The entire group is validated before the buffer is updated. For each newly observed key, we retain the lexicographically smallest normalized response-token sequence; new keys are inserted in sorted order. Selection therefore does not depend on the order of responses within the group. Each buffer holds at most  $M = 16$  exemplars, one per key, with no eviction or replacement. Keys first observed after the buffer is full receive no replay support.

At each update, a deterministic round-robin schedule selects one nonempty buffer, including buffers containing a single exemplar. The choice does not use reward magnitudes, evaluation results, a target mode count, or entropy. The policy scores every stored response in the selected buffer by teacher forcing, conditioned on its original prompt. Each exemplar contributes its mean response-token log probability, excluding prompt and padding tokens.

The replay loss is the negative mean score over the selected keys, weighted by  $\lambda_{\text{replay}}(G - 1)/G^2$  (Algorithm 1). The normalization  $(G - 1)/G^2$  matches the weight of a single success in a centered Dr.GRPO group: its advantage is  $(G - 1)/G$ , followed by averaging over  $G$  responses (Lemma P.2). This loss uses only discovered training exemplars; it requires neither the complete valid-key catalogue nor evaluation outputs.

#### D.4 CONTROL WITHOUT REPLAY GRADIENTS

Replay and control use the same validation, buffer construction, scheduling, and exemplar-scoring procedures. The control sets the replay gradient to zero while retaining the forward and backward computations. Both conditions combine the fresh-task and replay gradients in one optimizer step; the replay gradient is their only difference in the learning objective.

As the policies diverge, they can generate different responses and accumulate different buffers, which changes the selected exemplars and their lengths. The conditions match optimizer updates and fresh-rollout budgets, but not realized floating-point operations or wall-clock time. The control receives no additional task-gradient updates, fresh rollouts, or training time in place of replay computation. The comparison isolates the contribution of the replay learning signal under this design, without establishing an advantage at equal total compute (App. Q.1).

#### D.5 VERIFIER REQUIREMENTS AND APPLICABILITY

Replay requires a deterministic rule for grouping verified responses into solution modes. Its canonical key must be computable for every training response at reasonable cost and must reflect the task’s definition of a solution. In MODEBENCH, one executable validator determines both correctness and identity, so buffer construction requires no learned judge or embedding-similarity threshold.

Final-answer grading, as used for MATH, GSM8K, and AIME, does not distinguish derivations of the same answer. When a prompt has only one correct answer, an answer-based key places every successful response in the same mode. The buffer then holds one exemplar, and replay reinforces it without balancing alternative derivations. Measuring and replaying distinct derivations requires a finer definition of identity.

MathIR uses executed state trajectories as keys. Its catalogue includes verified four-state routes, but none appears among 27,450 verified draws. Across 115 solved Level-1 prompts, these draws cover 67% of the certified two-state modes and 52% of the three-state modes. Different modes rarely appear for the same prompt. Terminal PCMD averages .006, corresponding to 1.01 effective verified modes per prompt. These results provide limited evidence for diversity gains among alternative derivations.

Across the 6 comparisons, including MathIR in the five-domain mean reduces the PCMD gain by .010–.074 relative to the corresponding four-domain mean. At Qwen2.5-0.5B, the Re:Dr gain is +.294 across all five domains and +.364 across the other four. App. N.1 reports hosted-model results for the same four-domain subset.

For open-ended derivations, a learned judge could supply a novelty criterion, but that criterion would require independent validation. Judge decisions can be inconsistent or disagree with task correctness, and replay may reinforce distinctions that the judge rewards without improving solution diversity. Fixing the judge model, prompt, and decoding rule does not by itself ensure a valid equivalence relation. Where executable keys are available, they can be used to assess the judge and measure diversity separately from its training signal. We do not evaluate judge-based replay here.

#### D.6 EXEMPLAR QUALITY AND LENGTH

Canonical verification establishes that a response satisfies the task constraints. It does not validate every intermediate reasoning step: a verified response may be verbose or include an unsound explanation. Replay weights exemplars by solution key without ranking their reasoning quality.

Each key occupies one buffer entry and contributes its exemplar’s mean response-token log probability,  $\ell_c = |e_c|^{-1} \log \pi_\theta(e_c | z)$ . Longer responses therefore do not receive more buffer entries or loss weight proportional to their token count. This normalization still affects the learning target. In a categorical surrogate with one exemplar of fixed length  $L_b$  per key, the effective weights are proportional to  $L_b^{-1}$ . With full coverage, the limiting distribution favors shorter exemplars and is uniform only when lengths are equal (App. P.7). This conclusion concerns the surrogate; mean-token normalization does not imply uniform key probabilities in a language model or guarantee concise, sound explanations.

Additional length or reasoning-quality filters would change which exemplars enter or remain in the buffer, and hence change the replay target. We use no such filters beyond the stated generation limits and verification rules. Restricting length can exclude valid alternatives: MathIR

includes longer state trajectories, and the observed lambdas for rarer `Python` return vectors are longer than those for the modal vector. Evaluating a quality filter would therefore require measuring both solution quality and diversity among retained modes.

## E FIXED SEMANTIC-MAXENT COMPARATOR

Fixed Semantic-MaxEnt adds a frequency-based bonus to the fresh-response advantage. It counts previously observed verified keys but does not optimize the likelihood of stored exemplars. For response  $i$  in a group of  $G$  responses to prompt  $x$ , let  $n_x(a)$  count earlier successful responses with key  $a$ , and let  $m_{-i}(a)$  count successful responses with that key among the other members of the current group. Write  $N_x = \sum_a n_x(a)$ ,  $S_{-i} = \sum_a m_{-i}(a) \leq G - 1$ , and let  $\mathcal{K}_i$  contain all keys observed in these two sources. Adding one pseudocount per observed key and reserving one for unseen keys gives, for  $a \in \mathcal{K}_i$ ,

$$\hat{p}_i(a) = \frac{n_x(a) + m_{-i}(a) + 1}{N_x + S_{-i} + |\mathcal{K}_i| + 1}, \quad \hat{p}_i(\text{UNSEEN}) = \frac{1}{N_x + S_{-i} + |\mathcal{K}_i| + 1}. \quad (8)$$

A response whose key is outside  $\mathcal{K}_i$  receives the probability assigned to the unseen category. Invalid and unsuccessful responses do not contribute to the counts.

**Counts and advantage calculation.** Counts are specific to each exact prompt-token sequence and accumulate throughout training without decay or reset. Only responses with positive reward and a valid key are counted. Each response is scored using previous groups and the other successful responses in its current group. Counts for the full current group are added only after scoring, so a response never contributes to its own predicted probability. The added advantage for an eligible successful response is held constant during differentiation and equals

$$A_{i,t}^{\text{sem}} = \eta \frac{\min\{-\log \hat{p}_i(a_i), \kappa\} - \mathbb{E}_{\hat{p}_i}[\min\{-\log \hat{p}_i(a), \kappa\}]}{\kappa}, \quad \eta = .10, \kappa = 5. \quad (9)$$

The same  $A_{i,t}^{\text{sem}}$  applies at every token position  $t$  of response  $i$ . Invalid and unsuccessful responses receive zero bonus. The fixed scale  $\eta$  bounds the centered bonus in magnitude, while  $\kappa$  clips the surprisal. The bonus can favor an underrepresented verified mode when that mode appears in a fresh response, but it adds no direct likelihood term for a stored exemplar absent from the group. If neither previous groups nor the other successful responses provide a key, the unseen category has probability one and the bonus is zero. App. F.5 reports the comparison.

**Relation to conditional entropy.** For the true conditional distribution  $q_\theta = p_\theta(\cdot | \mathcal{C})$ ,

$$\nabla_\theta H(q_\theta) = \mathbb{E}_{a \sim q_\theta} [(-\log q_\theta(a)) \nabla_\theta \log q_\theta(a)]. \quad (10)$$

The frequency predictor generally does not provide an unbiased estimator of this gradient. To see the distinction, hold a predictor score vector  $s$  and baseline  $b$  fixed independently of the scored action, for example by conditioning on the other responses in its group. Averaging the gradients of successful responses under the unconditional policy gives

$$\mathbb{E}_{Z \sim p} [\mathbf{1}\{Z \in \mathcal{C}\} (s_Z - b) \nabla_\theta \log p_Z] = P \mathbb{E}_{c \sim q} [s_c \nabla_\theta \log q_c] + (\mathbb{E}_q[s] - b) \nabla_\theta P.$$

The predictor’s baseline need not equal the current conditional mean  $\mathbb{E}_q[s]$ . The second term can therefore change correctness as well as the allocation among correct modes. Even with the exact, unclipped score  $s_c = -\log q_c$  and baseline  $b = H(q)$ , averaging over all rollout responses yields  $P \nabla_\theta H(q)$ , scaled by the probability of success.

The distinction also matters when only one correct key exists. Its conditional entropy is zero, yet probability assigned to the unseen category can make its semantic advantage negative after it has been observed. The bonus is therefore generally not an exact conditional-entropy gradient, and it provides no guarantee of individual-mode preservation.

## Part III. Training results

### F EMPIRICAL RESULTS AND COMPARISONS

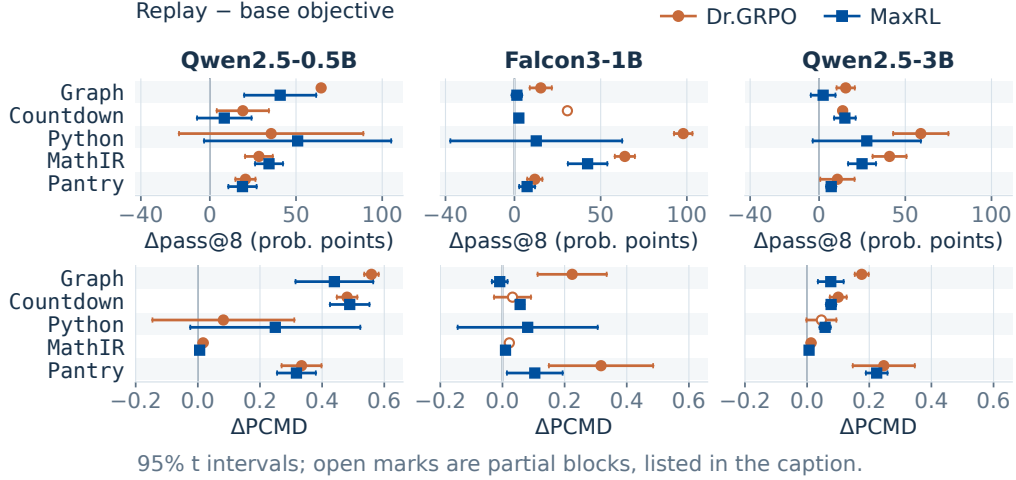


Figure 17: **Replay raises mean correctness at all three model scales.** Final-checkpoint differences: Re:Dr minus Dr.GRPO (circles) and Re:Max minus MaxRL (squares), measured in percentage points for  $\text{pass@8}$  and in PCMD units for diversity. Filled markers use five paired seeds; hollow markers use fewer. For Re:Dr, correctness uses four seeds in Falcon Countdown; diversity uses four in Falcon Countdown, one in Falcon MathIR, and two in Qwen2.5-3B Python. Diversity requires at least 20 eligible prompts per arm; blanks indicate no eligible comparison. Bars show nominal paired 95% Student- $t$  intervals, requiring five seeds for correctness and at least two for diversity.

#### F.1 TERMINAL COMPARISONS ACROSS SCALES

Paired conditions share initialization, training and evaluation prompts, eight training passes, evaluation requests, the verifier, and replay schedule. The control uses the same buffer construction and scoring procedures with zero replay gradient. Policies can develop different buffers and workloads, so total compute is not matched (App. Q.1).

Table 7: **Correctness and solution diversity after eight training passes.** Dr.GRPO and Re:Dr use five paired seeds for correctness, except Falcon Countdown ( $n = 4$ ). Diversity means average each condition’s eligible seeds, requiring at least 30 prompts with two or more correct responses per seed. Superscripts mark counts below five; dashes mark no eligible estimate. These separately eligible means need not yield the paired effects in Table 8, which requires 20 prompts per arm. Both conditions use the same red-to-green scale within each metric.

| Model        | Domain                 | Dr.GRPO |      | Re:Dr  |                   |
|--------------|------------------------|---------|------|--------|-------------------|
|              |                        | pass@8  | PCMD | pass@8 | PCMD              |
| Qwen2.5-0.5B | Graph coloring         | .323    | .001 | .969   | .560              |
|              | Countdown              | .480    | .007 | .672   | .487              |
|              | Python factors         | .172    | —    | .528   | .000 <sup>2</sup> |
|              | MathIR                 | .512    | .001 | .796   | .018              |
|              | PantryPlan             | .522    | .000 | .729   | .334              |
| Falcon3-1B   | Graph coloring         | .656    | .204 | .809   | .428              |
|              | Countdown <sup>4</sup> | .325    | .102 | .632   | .134              |
|              | Python factors         | .000    | —    | .979   | .060              |
|              | MathIR                 | .095    | —    | .736   | .022              |
|              | PantryPlan             | .633    | .000 | .751   | .317              |
| Qwen2.5-3B   | Graph coloring         | .682    | .224 | .836   | .399              |
|              | Countdown              | .609    | .012 | .746   | .112              |
|              | Python factors         | .069    | —    | .659   | .098              |
|              | MathIR                 | .278    | .001 | .686   | .014              |
|              | PantryPlan             | .638    | .105 | .744   | .351              |

Table 8: **Replay improves mean correctness and solution diversity at each tested scale.** Final-checkpoint differences between Re:Dr and Dr.GRPO, by domain and model. Superscripts indicate fewer than five paired seeds; dashes indicate no eligible comparison. Diversity effects require at least 20 eligible prompts per arm. Each Avg weights a fixed set of domains equally within seed: all five for correctness, and domains eligible for every seed for diversity. The diversity mean uses Graph and PantryPlan at Falcon3-1B, all domains except Python at Qwen2.5-3B, and all five at 0.5B. Eligible prompt subsets may differ between the two arms.

|              | $\Delta$ pass@8 |                     |        |        |        |                     | $\Delta$ PCMD |                     |                     |                     |        |        |
|--------------|-----------------|---------------------|--------|--------|--------|---------------------|---------------|---------------------|---------------------|---------------------|--------|--------|
|              | Count           |                     |        | Pantry |        |                     | Count         |                     |                     | Pantry              |        |        |
|              | Graph           | down                | Python | MathIR | Plan   | Avg                 | Graph         | down                | Python              | MathIR              | Plan   | Avg    |
| Qwen2.5-0.5B | +0.646          | +0.191              | +0.356 | +0.285 | +0.207 | +0.337              | +0.559        | +0.480              | +0.082              | +0.017              | +0.334 | +0.294 |
| Falcon3-1B   | +0.153          | +0.307 <sup>4</sup> | +0.979 | +0.640 | +0.118 | +0.442 <sup>4</sup> | +0.224        | +0.032 <sup>4</sup> | —                   | +0.022 <sup>1</sup> | +0.317 | +0.271 |
| Qwen2.5-3B   | +0.154          | +0.137              | +0.590 | +0.409 | +0.106 | +0.279              | +0.176        | +0.100              | +0.046 <sup>2</sup> | +0.013              | +0.247 | +0.134 |

**Dr.GRPO retention contrasts.** Tables 7 and 8 report means for each condition and paired replay effects, respectively. Adding replay to Dr.GRPO increases mean pass@8 and distinct@8 in all 14 comparisons with five paired seeds; distinct@8 increases in all 74 individual seed pairs. Because distinct@8 also depends on correctness, we use PCMD to measure diversity conditional on a correct response.

For Qwen2.5-0.5B, Countdown pass@8 rises from .480 to .672, with a PCMD gain of +.480 (95% interval [+0.447, +0.513]). Python correctness rises from .172 to .528, with a diversity gain of +0.082 ([−.146, +0.310]). MathIR correctness rises from .512 to .796, with a diversity gain of +0.017 ([+0.003, +0.031]). These are nominal Student-*t* intervals over five paired training seeds, without multiple-comparison adjustment. Countdown’s final PCMD values are .007 for Dr.GRPO and .487 for Re:Dr. Its keys distinguish computation routes, but these held-out evaluations do not track the survival of particular routes discovered during training.

Fig. 18 shows the results by domain. Two-method effects use shared seeds; factorial interactions use seeds shared by all four methods. Diversity comparisons additionally require enough verified responses to estimate the conditional distribution.

Diversity effects in Fig. 17 and Table 8 use paired seeds for which both arms have at least 20 eligible evaluation prompts. The separate-arm means in Table 7 and Fig. 18 instead require 30 eligible prompts per seed in each arm. Correctness, absolute diversity, and paired diversity effects can therefore use different seed sets; eligible prompt subsets can also differ between arms. Diversity-effect intervals use Student-*t* with  $n - 1$  degrees of freedom for  $n \geq 2$  paired seeds; single-seed effects have no interval. Correctness and mode-count intervals require five paired seeds. All intervals are nominal 95% estimates, without adjustment for multiple comparisons.

For Qwen2.5-0.5B, Re:Max increases PCMD by +.439, +.489, and +.317 on Graph, Countdown, and PantryPlan, respectively; each five-seed interval excludes zero. Falcon Graph has a diversity effect of −.009 ([−.034, +0.016]) and a correctness effect of +0.014 ([−.016, +0.044]), both inconclusive. Thus a positive cross-domain average does not imply improvement in every domain.

For Qwen2.5-3B, the five-seed PCMD effects are +.076, +.077, +.058, +.006, and +.224 for Graph, Countdown, Python, MathIR, and PantryPlan, respectively. All diversity intervals exclude zero, although the MathIR effect is small. Correctness intervals exclude zero for Countdown, MathIR, and PantryPlan, but include zero for Graph and Python. Across scales, Re:Max exceeds MaxRL in mean PCMD in 14 of 15 measurable comparisons; Re:Dr exceeds Dr.GRPO in 14 of 14. These counts summarize the signs of the estimated effects, not their statistical significance.

**Cross-domain factorial summaries.** Weighting all five domains equally within each paired seed, Re:Max improves pass@8 over MaxRL by +0.307 at Qwen2.5-0.5B (95% interval [+0.208, +0.406]), +0.133 at Falcon3-1B ([+0.043, +0.223]), and +0.154 at Qwen2.5-3B ([+0.086, +0.222]). Each comparison uses five seeds. The corresponding Re:Dr effects are +0.337, +0.442, and +0.279. The Falcon Dr.GRPO comparison uses four common seeds and has no interval; the other two use five.

At Qwen2.5-3B, MaxRL and Re:Max have mean correctness .547 and .701, and mean distinct@8 .744 and 1.107, respectively. Their paired mode-count difference is +0.363 ([+0.285, +0.441]). The cross-domain PCMD means in Table 8 use a fixed set of domains eligible for every contributing seed.

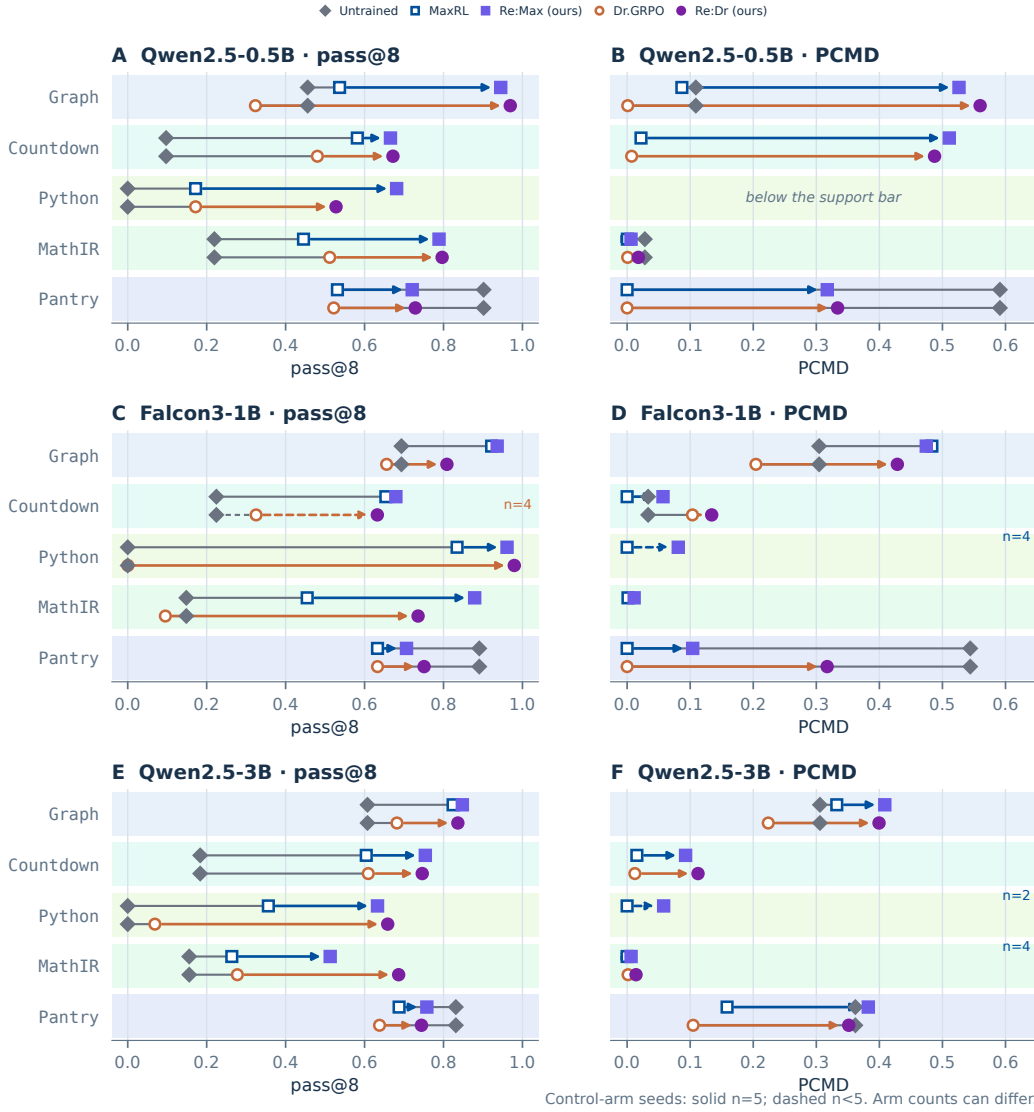


Figure 18: **Replay gains vary by domain and model scale.** Final pass@8 and PCMD at three model scales. Hollow and filled markers denote training without and with replay; circles denote Dr.GRPO, squares MaxRL, and gray diamonds the model before training. Correctness uses five paired seeds except Falcon Countdown Dr.GRPO ( $n = 4$ ). Diversity averages each arm’s eligible seeds separately, requiring at least 30 eligible prompts per seed. Lines are solid for five control seeds and dashed for fewer. Seed counts may differ between arms; estimates below the threshold are omitted. Table 8 reports paired diversity effects using a 20-prompt threshold. Error bars are omitted.

## F.2 LEVEL 1 AT QWEN2.5-7B

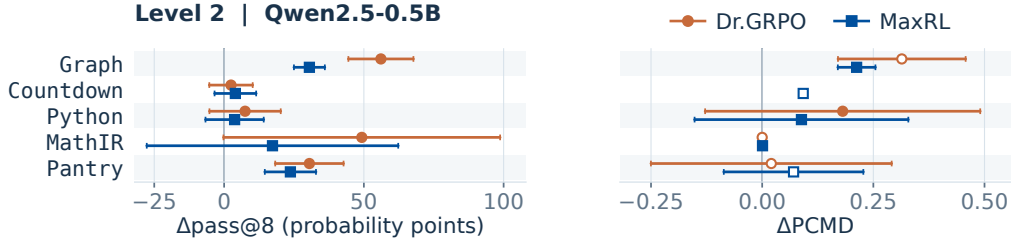
We also compare MaxRL and Re:Max at Qwen2.5-7B on Level 1 in 3 domains. The 6 training runs form one matched treatment–control pair per domain, each trained for 3072 updates with the same prompts, horizon, verifier, evaluation procedure, and replay schedule as the smaller-scale comparisons. With one matched seed per domain, these effects are descriptive and have no confidence intervals.

On MathIR, MaxRL’s observed pass@8 reaches zero at update 960 and remains zero at subsequent evaluations. The final evaluation contains no verified response, so PCMD cannot be estimated. Table 9 reports it as undefined rather than zero. Re:Max, evaluated on the same prompts and seed, reaches .744 pass@8 and .017 PCMD. The correctness result remains measurable even though a paired diversity effect is unavailable.

Where both conditions have measurable diversity, Re:Max increases PCMD by +.298 on Graph and +.137 on Countdown. These point estimates exceed the corresponding 3B means, but one seed does not establish a trend with model scale. Mean  $\text{pass@8}$  across all 3 domains is .342 for MaxRL and .789 for Re:Max. Mean PCMD is .028 and .245, respectively, over the 2 domains where both conditions define it. Thus the correctness mean includes MathIR, while the diversity mean does not. Table 9: **Level-1 correctness and solution diversity at Qwen2.5-7B**. Results from 6 runs with 1 matched seed per domain after 3072 updates. On MathIR, MaxRL produces no verified response: its correctness is zero, while its PCMD and the paired diversity effect are undefined (–). Means use 3 domains for correctness and the 2 jointly eligible domains for diversity. No confidence intervals are reported per domain for these single-seed comparisons.

| Domain    | MaxRL  |      | Re:Max |      | Re:Max – MaxRL |       |
|-----------|--------|------|--------|------|----------------|-------|
|           | pass@8 | PCMD | pass@8 | PCMD | pass@8         | PCMD  |
| Countdown | .631   | .001 | .805   | .138 | +.174          | +.137 |
| Graph     | .396   | .055 | .818   | .353 | +.422          | +.298 |
| MathIR    | .000   | –    | .744   | .017 | +.744          | –     |
| Mean      | .342   | .028 | .789   | .245 | +.447          | +.217 |

### F.3 RESULTS ON HARDER BENCHMARK LEVELS



Replay – base; 95% t intervals. Open marks are partial blocks; the panels do not share a cohort.

Figure 19: **Replay improves Level-2 Graph accuracy and solution diversity.** Final Re:Dr minus Dr:GRPO (circles) and Re:Max minus MaxRL (squares) effects for Qwen2.5-0.5B, in percentage points of  $\text{pass@8}$  and PCMD units. Correctness uses five paired seeds per domain. Diversity requires at least 20 eligible prompts per arm. Re:Dr uses three seeds for Graph and PantryPlan, four for MathIR, and none for Countdown; Re:Max uses one for Countdown and four for PantryPlan. Other comparisons use five. Hollow markers indicate fewer than five pairs; blanks indicate no eligible pair. Bars show nominal paired 95% Student- $t$  intervals for comparisons with at least two seeds.

Fig. 10 compares replay effects across Levels 1–3 using fixed domain sets. Correctness gives equal weight to Graph, Countdown, Python, and PantryPlan within each of five common training seeds. Diversity first averages eligible seeds within each arm and domain, then weights Graph, MathIR, and PantryPlan equally. Its contributing seeds can differ across arms and levels, so differences between these diversity means are descriptive and have no intervals. At Level 2, only PantryPlan has sufficient verified responses before training to estimate diversity; the initial three-domain diversity mean is therefore undefined.

Each Level-2 method has five final training seeds per domain. The diversity comparisons in Fig. 19 additionally require eligible prompts in both arms. Graph improves under both objectives: Re:Max increases PCMD by +.212 ([+.170, +.255], five seeds), and Re:Dr by +.314 ([+.170, +.458], three seeds). The corresponding Python effects are +.088 and +.181, with intervals including zero; MathIR effects are near zero. PantryPlan correctness improves by +.305 under Re:Dr and +.237 under Re:Max, each measured over five seeds.

Each Level-2 MathIR prompt admits one shortest derivation and four longer alternatives. Concentration on the shortest route therefore yields near-zero diversity despite the existence of other valid modes. The levels also use different held-out problems and can differ in mode-count distributions and prompt guidance (App. B.4). Cross-level comparisons consequently do not isolate the causal effect of difficulty.

For each common seed and endpoint  $Y$ , the replay interaction is

$$I = (Y^{\text{Re:Max}} - Y^{\text{MaxRL}}) - (Y^{\text{Re:Dr}} - Y^{\text{Dr.GRPO}}).$$

A negative interaction means that replay adds less under MaxRL than under Dr.GRPO; it does not imply that replay harms MaxRL. On Level-2 Graph, MaxRL alone improves pass@8 over Dr.GRPO by +.233 ([+.140, +.325]), while the replay interaction is -.256 ([-.329, -.183]). Replay improves both objectives, with a smaller additional benefit under the stronger fresh objective.

Table 10: **A stronger fresh objective need not increase the additional benefit of replay.** Contrasts use five common seeds per domain across all four methods. The interaction is (Re : Max - MaxRL) - (Re : Dr - Dr.GRPO). Here  $P_8 = \text{pass@8}$ ,  $D_8 = \text{distinct@8}$ , and  $B_8 = D_8 - P_8$ . Entries report paired means and nominal 95% Student- $t$  intervals. Both mode-count metrics depend on correctness. The conditional-diversity analysis in Fig. 19 applies separate eligibility requirements, so its seed populations can differ from those in this table.

| Domain     | $n$ | Contrast                          | $\Delta P_8$          | $\Delta D_8$          | $\Delta B_8$          |
|------------|-----|-----------------------------------|-----------------------|-----------------------|-----------------------|
| Graph      | 5   | MaxRL - Dr.GRPO                   | +.233 [+ .140, +.325] | +.322 [+ .162, +.482] | +.089 [+ .019, +.159] |
| Graph      | 5   | Re:Max - Re:Dr                    | -.023 [-.062, +.016]  | -.029 [-.198, +.140]  | -.006 [-.142, +.130]  |
| Graph      | 5   | Replay $\times$ MaxRL interaction | -.256 [-.329, -.183]  | -.351 [-.446, -.255]  | -.095 [-.186, -.003]  |
| Countdown  | 5   | MaxRL - Dr.GRPO                   | -.014 [-.073, +.045]  | -.014 [-.073, +.045]  | +.000 [+ .000, +.000] |
| Countdown  | 5   | Re:Max - Re:Dr                    | +.002 [-.008, +.012]  | -.002 [-.047, +.044]  | -.004 [-.042, +.035]  |
| Countdown  | 5   | Replay $\times$ MaxRL interaction | +.016 [-.047, +.078]  | +.012 [-.068, +.092]  | -.004 [-.042, +.035]  |
| Python     | 5   | MaxRL - Dr.GRPO                   | +.000 [+ .000, +.000] | +.000 [+ .000, +.000] | +.000 [+ .000, +.000] |
| Python     | 5   | Re:Max - Re:Dr                    | -.037 [-.232, +.157]  | -.188 [-.988, +.613]  | -.150 [-1.014, +.713] |
| Python     | 5   | Replay $\times$ MaxRL interaction | -.037 [-.232, +.157]  | -.188 [-.988, +.613]  | -.150 [-1.014, +.713] |
| MathIR     | 5   | MaxRL - Dr.GRPO                   | +.283 [-.193, +.759]  | +.283 [-.193, +.759]  | +.000 [+ .000, +.000] |
| MathIR     | 5   | Re:Max - Re:Dr                    | -.036 [-.158, +.085]  | -.035 [-.158, +.088]  | +.002 [-.003, +.006]  |
| MathIR     | 5   | Replay $\times$ MaxRL interaction | -.320 [-.864, +.225]  | -.318 [-.860, +.224]  | +.002 [-.003, +.006]  |
| PantryPlan | 5   | MaxRL - Dr.GRPO                   | +.019 [-.067, +.105]  | +.019 [-.081, +.119]  | +.000 [-.021, +.021]  |
| PantryPlan | 5   | Re:Max - Re:Dr                    | -.049 [-.113, +.014]  | -.061 [-.249, +.128]  | -.012 [-.145, +.122]  |
| PantryPlan | 5   | Replay $\times$ MaxRL interaction | -.068 [-.168, +.032]  | -.080 [-.279, +.119]  | -.012 [-.134, +.110]  |

**Level-2 Graph results.** Before training, Qwen2.5-0.5B Graph pass@8 is .438 on the Level-1 development split and .148 on Level 2. After eight Level-2 training passes, Dr.GRPO and MaxRL reach .202 and .434, compared with .763 and .739 for Re:Dr and Re:Max. The paired correctness gains are +.561 and +.305, with 95% Student- $t$  intervals [+ .445, +.677] and [+ .250, +.360] over five seeds each. The lowest replay accuracy (.711) exceeds the highest accuracy without replay (.477). Replay adds 1.019 and .668 distinct verified modes for the two objectives, respectively, yielding distinct@8 of 1.243 and 1.214.

For the three eligible Dr.GRPO seed pairs, PCMD rises from .039 to .353; for the five MaxRL pairs, it rises from .112 to .325. Expressed on the effective-mode scale, these changes are 1.04 to 1.55 and 1.13 to 1.48, corresponding to ratios of 1.49 and 1.32. Relative increases in PCMD are unstable near a zero baseline, so we report absolute changes instead. The joint improvements in correctness and diversity do not establish that greater diversity causes better correctness.

#### F.4 ACCURACY AND VERIFIED MODES THROUGHOUT TRAINING

Figs. 20–23 show correctness and diversity throughout training, using four groups of eight responses to each of 128 held-out prompts. Lines show seed means; bands show the range across seeds, rather than confidence intervals. A seed contributes to conditional diversity only when at least five prompts have two or more correct responses. The contributing seeds and prompts can therefore change across methods and checkpoints.

#### F.5 SUPPORTING COMPARATORS

GRPO normalizes rewards within each group; UCPO redistributes the advantage among fresh on-policy responses; sparse RLEP-Dr reuses verified successes without uniform weighting by canonical key. We evaluate GRPO in 75 training runs across three model scales. UCPO and RLEP-Dr cover the two smaller scales, with 50 and 47 runs, respectively. RLEP-Dr uses two seeds for Falcon Python and five elsewhere. Fixed Semantic-MaxEnt adds a frequency-based bonus to fresh responses (App. E). Its four-method factorial covers ten model-domain combinations at the two smaller scales, with five common seeds except Falcon Countdown ( $n = 4$ ). The 3B comparison uses one seed to evaluate adding the bonus to replay (App. G).

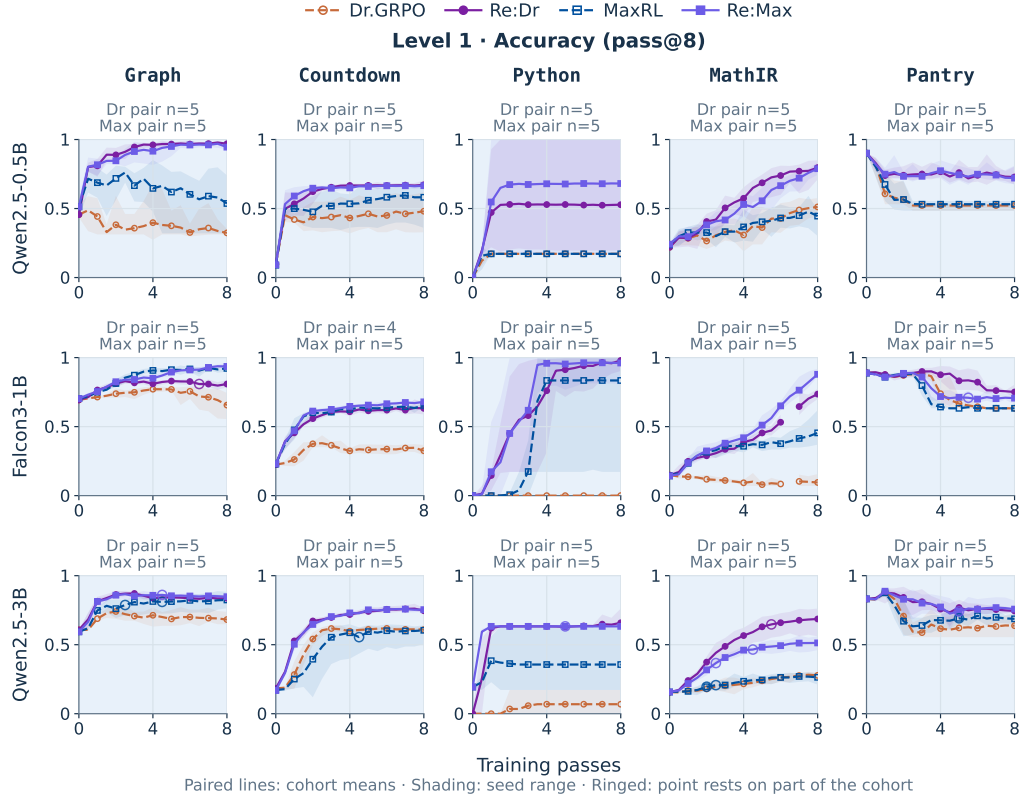


Figure 20: **Replay improves final accuracy across model scales.** pass@8 over eight training passes for Dr.GRPO, Re:Dr, MaxRL, and Re:Max. Rows show Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B; columns show domains. Panel labels give the numbers of paired seeds at the final checkpoint. Lines show seed means and bands show seed ranges, rather than confidence intervals. Ringed points use fewer seeds than the final comparison.

For GRPO, UCPO, and RLEP-Dr, the diversity contrasts in this subsection average per-prompt differences over prompts with at least two correct responses in both arms. Each seed must have at least 30 such jointly eligible prompts. Table 1 instead uses separate 20-prompt thresholds for each arm. On Qwen2.5-0.5B Graph, one UCPO seed and four RLEP-Dr seeds meet the joint criterion; no Python comparison meets it at any scale.

For Qwen2.5-0.5B, UCPO improves Countdown pass@8 by  $+0.133$  ( $[+0.005, +0.262]$ ), while its PCMD effect is inconclusive:  $+0.002$  ( $[-0.018, +0.023]$ ). Its MathIR effects are also inconclusive. RLEP-Dr improves MathIR correctness by  $+0.126$  ( $[+0.030, +0.222]$ ), with zero measured diversity change; its Countdown and Falcon PantryPlan effects are inconclusive. Across GRPO, UCPO, and RLEP-Dr, seven of 34 five-seed correctness intervals lie above zero. None of the 19 five-seed diversity intervals lies above zero; one lies below zero, for Falcon Countdown GRPO:  $-0.105$  ( $[-0.147, -0.063]$ ). At Qwen2.5-3B, all five GRPO correctness intervals and the three available diversity intervals include zero. These are nominal paired 95% Student- $t$  intervals over training seeds.

The uniform versus frequency-weighted replay comparison uses five paired seeds in each of nine model-domain combinations: all five Qwen2.5-0.5B domains, plus Graph and PantryPlan at both larger scales. Changing only the within-buffer weights increases extra verified modes by  $+0.318$  per prompt in the Qwen2.5-0.5B average ( $[+0.272, +0.358]$ , paired-seed bootstrap). The correctness effect is inconclusive at  $+0.010$  ( $[-0.119, +0.139]$ ). This result supports a benefit from uniform weighting beyond reusing successful responses in proportion to their frequency, although extra-mode counts still depend on correctness (App. F.7).

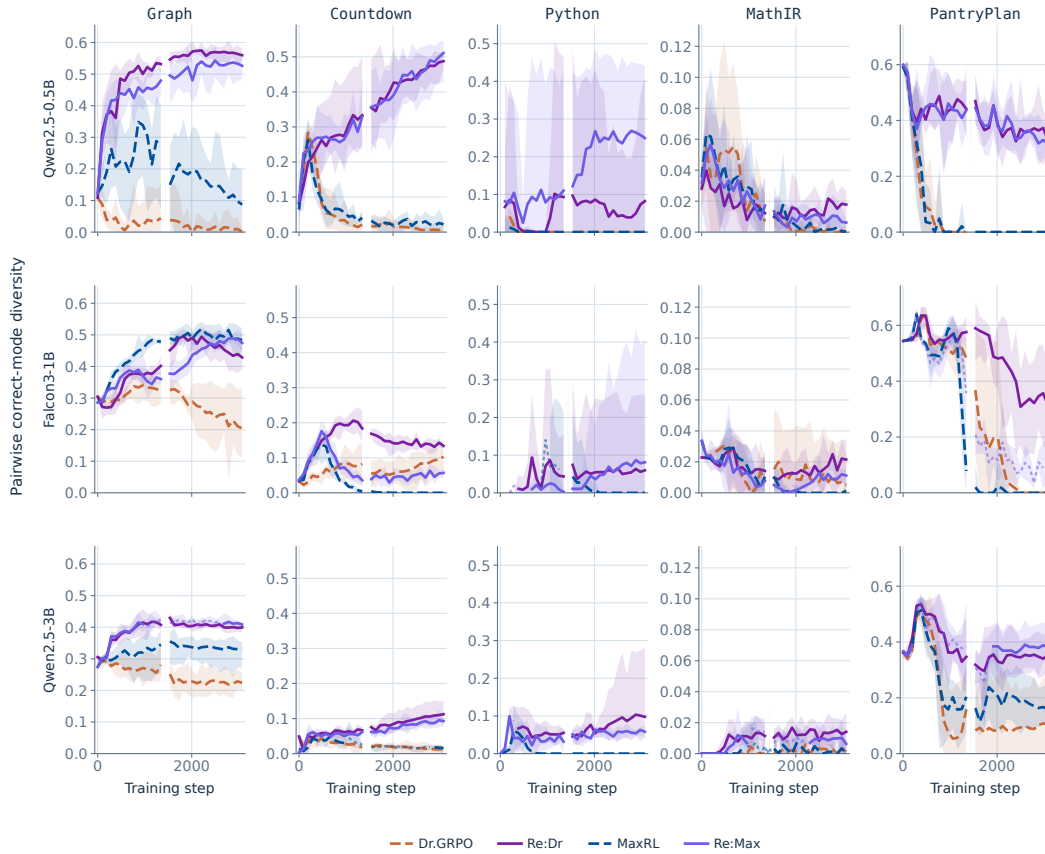


Figure 21: **Replay limits the loss of `PantryPlan` solution diversity at every scale.** PCMD throughout training, with model rows and method colours as in Fig. 20. Both fresh objectives reach zero terminal `PantryPlan` diversity at 0.5B and 1B, whereas both replay variants retain multiple solution modes. Lines average one to five eligible seeds per checkpoint; bands show seed ranges and are absent for a single seed. A seed contributes when at least five prompts yield at least two correct responses. Fine dotted segments use fewer seeds, so changes can reflect sample composition. Gaps indicate undefined estimates. Vertical scales are shared within each domain.

RLEP-Dr’s offline response pool uses temperature .7 and  $\text{top-}p = .95$ ; evaluation uses temperature 1 and  $\text{top-}p = 1$ . Its offline data collection also differs from the online construction of prompt-specific verified replay buffers.

**Comparator settings.** Each comparator shares the non-objective settings of its paired control (Table 6). UCPO uses the published default  $\omega = .2$ , and RLEP-Dr uses the published combination of 16 fresh and 2 replay responses. GAPO has no free coefficient; its reward is affinely mapped to the control’s unit range because Dr.GRPO does not normalize by the group standard deviation. SetPO uses  $\lambda = 1.0$ , giving a diversity term approximately 5.3% of the task advantage; its paper provides no coefficient for the 0.5B verifier-reward setting. Semantic-MaxEnt uses  $\eta = .10$  and  $\kappa = 5$ . Replay uses  $\lambda_{\text{replay}} = .10$  and capacity  $M = 16$ , equal to the rollout group size. These settings are not tuned to MODEBENCH outcomes, and no method receives a learning-rate sweep. The results therefore compare the methods at the stated settings; hyperparameter tuning could improve correctness or diversity.

**Interpreting zero diversity.** In Table 1, an em-dash denotes an undefined paired difference. An exact .000 instead denotes a measured zero, with identical values for both methods in every eligible seed. The `Python` PCMD column contains measured zeros because, for each eligible prompt, both methods’ verified responses share a single execution key. High correctness can coexist with this concentration: four of the five SetPO `Python` seeds reach `pass@8` 1.000 and terminal PCMD .000. Every held-out prompt has at least one correct response, but the verified responses for each prompt use a single mode. Per-response correctness also increases in all 15 control comparisons (App. H.5).

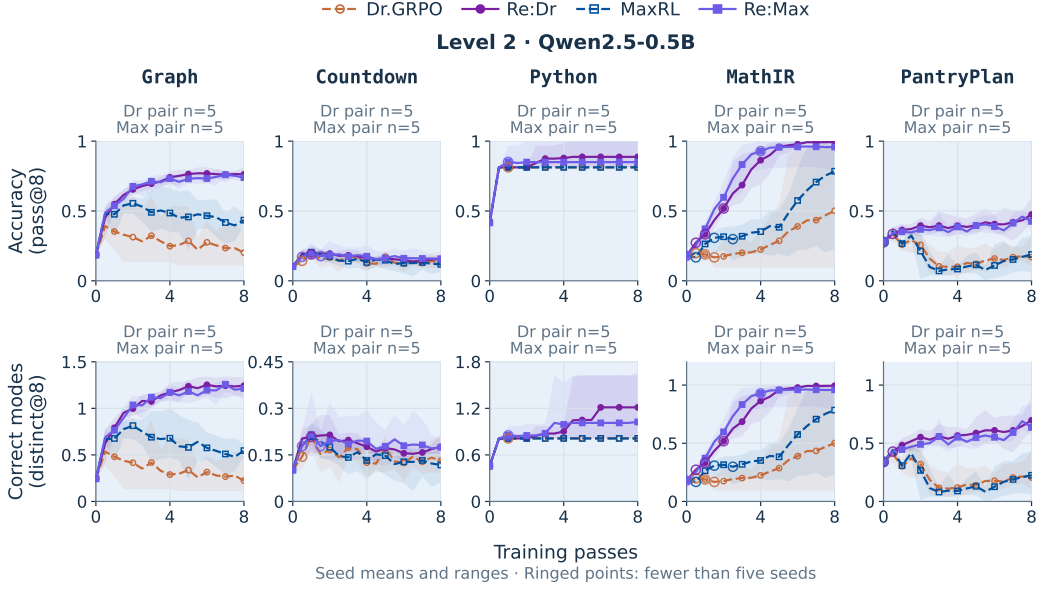


Figure 22: **Replay improves final accuracy and verified-mode counts at Level 2.** Qwen2.5-0.5B trajectories for Dr.GRPO, Re:Dr, MaxRL, and Re:Max across five domains. Rows show `pass@8` and `distinct@8`; the latter depends on both correctness and solution diversity. Each objective pair has five matched seeds at the terminal checkpoint. Lines show seed means and bands show seed ranges. Ringed points use fewer than five seeds; Fig. 23 reports success-conditional diversity.

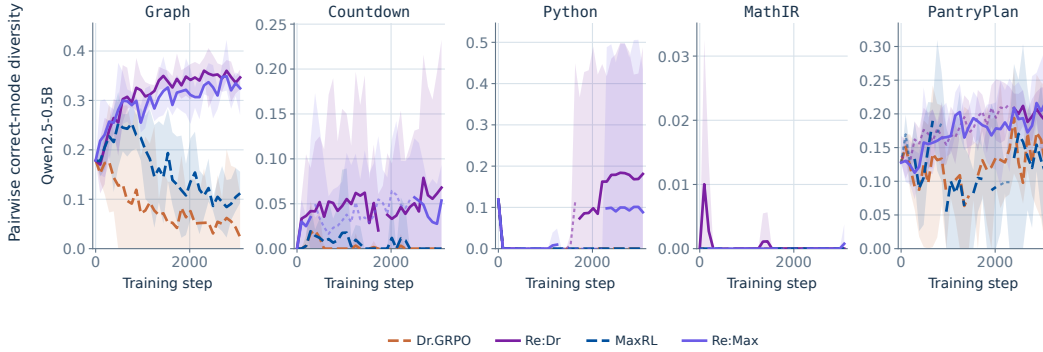


Figure 23: **Replay preserves greater solution diversity on Level-2 Graph.** PCMD during Qwen2.5-0.5B training for the four primary methods. At the terminal checkpoint, Re:Dr and Re:Max reach .346 and .325, compared with .023 for Dr.GRPO and .112 for MaxRL. MathIR remains near zero under all four methods. Lines average eligible seeds and bands show seed ranges. Each seed requires at least five prompts with at least two correct responses. Fine dotted segments indicate fewer contributing seeds, and gaps indicate that the estimate is undefined.

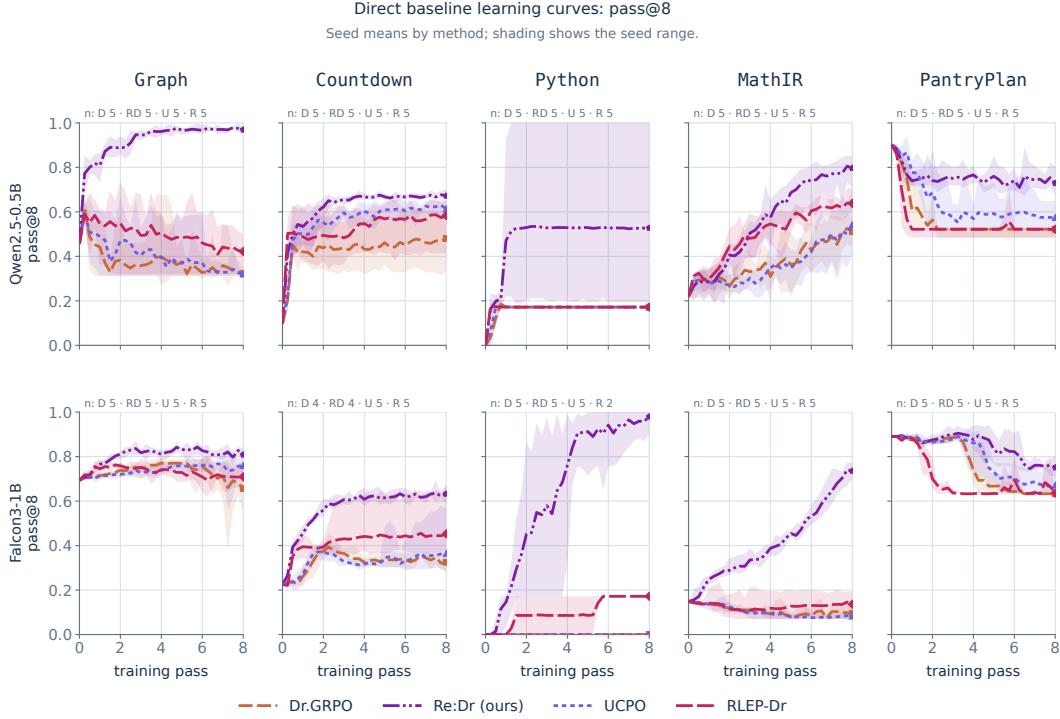


Figure 24: **Re:Dr has the highest terminal mean accuracy in every displayed comparison.** pass@8 during Level-1 training for Dr.GRPO, Re:Dr, UCPO, and sparse RLEP-Dr. Rows show Qwen2.5-0.5B and Falcon3-1B; columns show the five domains. Lines show seed means and bands show seed ranges; gaps indicate unavailable estimates. Dr.GRPO and Re:Dr use paired seeds. Comparisons use five seeds except Falcon Countdown Dr.GRPO/Re:Dr ( $n = 4$ ) and Falcon Python RLEP-Dr ( $n = 2$ ), as indicated in the panels.

## F.6 DIVERSITY-PRESERVING COMPARATORS

GAPO (Anschel et al., 2025) and SetPO (Li et al., 2026a) target output diversity within the fresh rollout group. Both are evaluated on Level-1 Qwen2.5-0.5B across five domains and five seeds. GAPO assigns verifier-positive rollouts the reward  $[1 - (f_i - 1/L) + 1]/2$ , with zero for failures, where  $f_i$  is the key’s frequency among the group’s verified rollouts and  $L$  is the prompt’s certified number of valid modes. SetPO augments each rollout’s advantage by its marginal contribution to kernelized set diversity, measured by removing that rollout from the group, with coefficient  $\lambda = 1$ . It computes similarities using a fixed all-MiniLM-L6-v2 sentence encoder, mean pooling, and cosine similarity clamped to  $[0, 1]$ . Encoding the responses adds computation.

GAPO increases PCMD on Countdown (+.340,  $[+.112, +.567]$ ) and PantryPlan (+.653,  $[+.610, +.696]$ ), while its Python effect is near zero with an interval spanning zero. Python has more valid modes than the sixteen-rollout group can represent on 95% of training prompts, with up to 3,600 modes. This limits within-group coverage but does not prevent a policy from assigning mass across the full support. MathIR effects are small across methods; the largest is Re:Dr’s +.017 ( $[+.003, +.031]$ ). SetPO increases PCMD on PantryPlan (+.266,  $[+.177, +.356]$ ); its other four domain intervals include zero. Five-domain mean effects are +.225 for GAPO, +.066 for SetPO, and +.320 for Re:Max, although GAPO’s PantryPlan effect is larger than Re:Max’s. The effective number of modes,  $N_{\text{eff}}^{(2)}$ , on PantryPlan is 2.88 for GAPO and 1.36 for SetPO, compared with 1.00 for their shared control. SetPO also raises Python pass@8 from .338 to .834, a relative increase of 147%. Relative changes in PCMD are undefined for PantryPlan and Python because the control has PCMD = .000. GAPO adjusts rewards using frequencies within the fresh rollout group, whereas replay balances the likelihood term across verified modes accumulated in the buffer.

GAPO and SetPO share a Dr.GRPO control with matching hardware and nonoverlapping evaluation draws. Its terminal PCMD differs from the Dr.GRPO control used in the main re-

play comparison by a mean of  $+0.003$  over twenty paired comparisons, ranging from  $-0.014$  to  $+0.034$ . These observed differences are small relative to the five-domain effects above, but do not establish equivalence between the controls.

#### F.7 UNIFORM VERSUS FRESH-FREQUENCY REPLAY

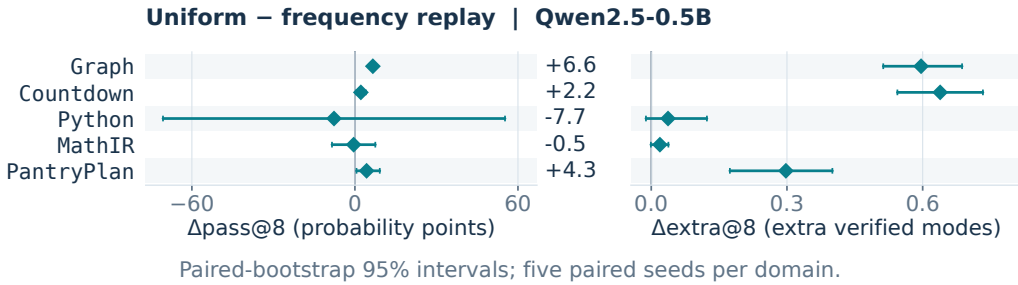
The weighting ablation uses the same buffer construction and update rule, changing only the within-buffer target: uniform weights over retained keys versus weights proportional to verifier-positive fresh-rollout counts. The resulting policies can produce different buffer contents during training. Positive contrasts below favor uniform weighting. We report  $B_8 = \text{distinct@8} - \text{pass@8}$ , extra verified modes per prompt, and  $P_8 = \text{pass@8}$ . The five-domain Qwen2.5-0.5B aggregate averages domain effects within each matched seed, giving equal domain weight.

Paired-seed intervals enumerate all  $5^5 = 3,125$  ordered bootstrap resamples of the five paired training seeds and use linearly interpolated 2.5th and 97.5th percentiles. Each resampled seed carries its complete domain vector for the five-domain aggregate. These nominal intervals quantify seed variability within the fixed benchmark; prompts and domains are not resampled, and intervals are not adjusted across comparisons.

The Qwen2.5-0.5B five-domain effect is  $+0.318$  extra modes per prompt ( $[+0.272, +0.358]$ ), with positive within-seed domain averages in all five seeds. Graph, Countdown, and PantryPlan drive this result; Python and MathIR remain inconclusive. The  $P_8$  effect is  $+0.010$  ( $[-0.119, +0.139]$ ), establishing neither a gain nor equivalence. Python’s  $P_8$  effects range from  $-0.781$  to  $+0.828$  across seeds despite small extra-mode effects in four seeds.

Falcon3-1B Graph also favors uniform weighting on extra modes ( $+0.233$ ,  $[+0.182, +0.320]$ ). Its correctness measures differ:  $\Delta P_8 = +0.045$  ( $[-0.0004, +0.0910]$ ) is inconclusive, while the effect on mean per-response correctness is negative at  $-0.0152$  ( $[-0.0214, -0.0090]$ ). Falcon3-1B PantryPlan is inconclusive on both outcomes:  $\Delta B_8 = -0.146$  ( $[-0.564, +0.414]$ ) and  $\Delta P_8 = -0.021$  ( $[-0.063, +0.026]$ ). Qwen2.5-3B Graph has five pairs and an extra-mode effect of  $+0.079$  ( $[+0.011, +0.135]$ ); its  $P_8$  interval includes zero. Qwen2.5-3B PantryPlan also has five pairs, with  $\Delta B_8 = +0.207$  ( $[+0.038, +0.377]$ ) and inconclusive  $\Delta P_8$  ( $+0.011$ ,  $[-0.020, +0.043]$ ).

Rehearsing self-generated verified successes is related to rejection-sampling fine-tuning (Dong et al., 2023); here the likelihood term is length-normalized and combined with the on-policy objective. This ablation isolates the effect of uniform versus fresh-frequency weighting within that replay procedure. Uniform weighting increases extra verified modes in several domains. The inconclusive  $P_8$  effects do not establish that weighting leaves accuracy unchanged or that replay’s accuracy and diversity effects have separate causes.



**Figure 25: Uniform weighting preserves more verified alternatives.** Uniform minus frequency-weighted replay at Qwen2.5-0.5B, Level 1, with five matched seeds. Extra modes are  $B_8 = \text{distinct@8} - \text{pass@8}$ . Bars show nominal 95% paired-seed bootstrap intervals. The mean gain is .318 extra modes, with an inconclusive correctness effect. Extra modes remain coupled to correctness; they do not measure success-conditional PCMD.

## G FIXED SEMANTIC-MAXENT COMPARISONS

A four-method factorial crosses replay with a fixed Semantic-MaxEnt bonus, comparing Dr.GRPO and Re:Dr with and without the bonus across five domains at Qwen2.5-0.5B and Falcon3-1B. All three contrasts within a model–domain combination use the same training seeds: five in nine combinations and four for Falcon Countdown. Table 11 reports terminal paired means after eight training passes, with nominal pointwise 95% Student- $t$  intervals using  $n - 1$  degrees of freedom. These intervals describe training-seed variability on the fixed evaluation prompts and are unadjusted across comparisons.

At Qwen2.5-3B, one training seed compares Re:Dr with and without the bonus across all five domains. No uncertainty interval is available from a single seed. The absence of a semantic-only comparison also prevents estimating the interaction between replay and the bonus. These results characterize the evaluated coefficients rather than an optimal entropy strength. Extra-mode counts depend on correctness as well as diversity and do not measure success-conditional solution diversity.

**Table 11: Fixed Semantic-MaxEnt increases verified-mode counts on Qwen2.5-0.5B PantryPlan with and without replay.** Without replay is semantic-only minus Dr.GRPO; on replay is semantic-plus-replay minus Re:Dr; interaction subtracts the former effect from the latter. Extra modes equal `distinct@8` minus `pass@8`. Entries give paired means and nominal 95% Student- $t$  intervals; the single-seed 3B effects have no intervals.

| Model        | Domain     | $n$ | Contrast       |        | $\Delta$ pass@8 | $\Delta$ distinct@8 | $\Delta$ extra modes |
|--------------|------------|-----|----------------|--------|-----------------|---------------------|----------------------|
| Qwen2.5-0.5B | Graph      | 5   | Without replay | 0.002  | [−0.028, 0.031] | 0.004               | [−0.034, 0.041]      |
|              |            | 5   | On replay      | −0.002 | [−0.027, 0.023] | −0.022              | [−0.179, 0.135]      |
|              |            | 5   | Interaction    | −0.004 | [−0.035, 0.028] | −0.026              | [−0.193, 0.141]      |
|              | Countdown  | 5   | Without replay | 0.141  | [−0.047, 0.330] | 0.184               | [0.011, 0.358]       |
|              |            | 5   | On replay      | −0.003 | [−0.033, 0.028] | 0.115               | [0.022, 0.209]       |
|              |            | 5   | Interaction    | −0.144 | [−0.326, 0.037] | −0.069              | [−0.233, 0.095]      |
|              | Python     | 5   | Without replay | 0.000  | [0.000, 0.000]  | 0.000               | [0.000, 0.000]       |
|              |            | 5   | On replay      | −0.002 | [−0.717, 0.714] | 0.016               | [−0.747, 0.780]      |
|              |            | 5   | Interaction    | −0.002 | [−0.717, 0.714] | 0.016               | [−0.747, 0.780]      |
|              | MathIR     | 5   | Without replay | −0.121 | [−0.349, 0.106] | −0.121              | [−0.347, 0.105]      |
|              |            | 5   | On replay      | 0.031  | [−0.035, 0.097] | 0.035               | [−0.044, 0.114]      |
|              |            | 5   | Interaction    | 0.153  | [−0.018, 0.324] | 0.156               | [−0.050, 0.362]      |
|              | PantryPlan | 5   | Without replay | 0.201  | [0.119, 0.283]  | 1.120               | [0.888, 1.352]       |
|              |            | 5   | On replay      | 0.141  | [0.044, 0.238]  | 1.095               | [0.368, 1.823]       |
|              |            | 5   | Interaction    | −0.061 | [−0.161, 0.040] | −0.025              | [−0.826, 0.777]      |
| Falcon3-1B   | Graph      | 5   | Without replay | 0.036  | [−0.192, 0.264] | 0.205               | [−0.483, 0.893]      |
|              |            | 5   | On replay      | 0.001  | [−0.040, 0.043] | −0.006              | [−0.113, 0.101]      |
|              |            | 5   | Interaction    | −0.035 | [−0.237, 0.168] | −0.211              | [−0.886, 0.464]      |
|              | Countdown  | 4   | Without replay | 0.004  | [−0.092, 0.100] | −0.042              | [−0.258, 0.173]      |
|              |            | 4   | On replay      | 0.036  | [−0.028, 0.101] | 0.047               | [−0.082, 0.177]      |
|              |            | 4   | Interaction    | 0.032  | [−0.098, 0.161] | 0.090               | [−0.188, 0.367]      |
|              | Python     | 5   | Without replay | 0.002  | [−0.003, 0.008] | 0.002               | [−0.003, 0.008]      |
|              |            | 5   | On replay      | −0.133 | [−0.308, 0.043] | −0.091              | [−0.503, 0.321]      |
|              |            | 5   | Interaction    | −0.135 | [−0.310, 0.040] | −0.093              | [−0.507, 0.321]      |
|              | MathIR     | 5   | Without replay | −0.002 | [−0.063, 0.060] | −0.002              | [−0.063, 0.060]      |
|              |            | 5   | On replay      | 0.086  | [0.004, 0.168]  | 0.108               | [0.003, 0.214]       |
|              |            | 5   | Interaction    | 0.087  | [0.029, 0.146]  | 0.110               | [0.043, 0.176]       |
|              | PantryPlan | 5   | Without replay | 0.011  | [−0.019, 0.041] | 0.091               | [−0.161, 0.342]      |
|              |            | 5   | On replay      | 0.025  | [−0.044, 0.094] | 0.382               | [−0.702, 1.465]      |
|              |            | 5   | Interaction    | 0.014  | [−0.052, 0.081] | 0.291               | [−0.646, 1.228]      |
| Qwen2.5-3B   | Graph      | 1   | On replay      |        | −0.016          | −0.010              | 0.006                |
|              | Countdown  | 1   | On replay      |        | 0.039           | 0.146               | 0.107                |
|              | Python     | 1   | On replay      |        | −0.590          | −1.215              | −0.625               |
|              | MathIR     | 1   | On replay      |        | 0.033           | 0.010               | −0.023               |
|              | PantryPlan | 1   | On replay      |        | 0.035           | 0.393               | 0.357                |

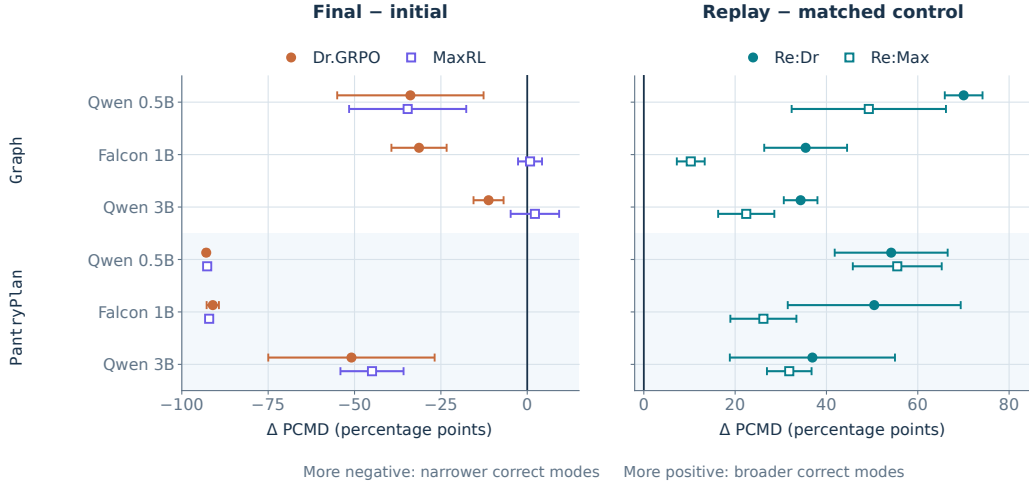


Figure 26: **Replay increases correct-mode diversity at a 64-response budget.** Left: Dr.GRPO reduces PCMD in all six model–domain comparisons; MaxRL’s change is inconclusive for Falcon3-1B and Qwen2.5-3B Graph. Right: replay increases PCMD in all twelve comparisons with matched trained controls. Points show means over five paired seeds; bars give nominal 95% Student- $t$  intervals. Each contrast uses its own jointly eligible Level-1 prompts. Horizontal scales differ between panels.

## H CONCENTRATION AND REPLAY MECHANISMS

### H.1 CONCENTRATION WITH 64 RESPONSES PER PROMPT

We evaluate initial and terminal checkpoints on the 128 held-out Level-1 Graph and PantryPlan prompts, drawing 64 responses per prompt from new sampling streams. This evaluation is separate from the 32-response comparison in Fig. 8. It covers Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B, with five training seeds for Dr.GRPO, Re:Dr, MaxRL, and Re:Max. For each model, the five initial evaluations use independent sampling from the same weights. Each initial evaluation is paired with one training seed and shared across the four methods.

**Matched comparisons.** We estimate PCMD from correct-output pairs and average promptwise differences over prompts with at least two correct responses in both conditions, then average the five seed effects equally. Thus positive changes indicate more diverse correct modes. Matching prompts within each comparison differs from subtracting means over separately eligible prompts. The eligible population ranges from 37 to 127 of the 128 prompts across comparisons and seeds. Effects from different comparisons therefore cannot be subtracted to reconstruct a trajectory on a common prompt population. Tables 12 and 13 give all 36 contrasts and their eligible-prompt ranges. Intervals are nominal 95% paired Student- $t$  intervals with four degrees of freedom. They describe variation over the five training-seed and initial-sampling pairs, conditional on the fixed prompts and initial weights. They are unadjusted for multiple comparisons and do not quantify uncertainty over unseen tasks.

**Concentration and mitigation.** Dr.GRPO reduces PCMD in all six model–domain comparisons and all 30 seed effects. Replay increases PCMD relative to its matched trained control under both objectives in all twelve comparisons and all 60 seed effects. All eighteen nominal intervals exclude zero (Fig. 26). These directions also hold when pooling correct pairs, using each policy’s own eligible prompts, or restricting to the intersection of eligible prompts across seeds. The intersection can be small: for Qwen2.5-0.5B Graph, only five prompts are jointly eligible across all seeds in comparisons involving Dr.GRPO.

MaxRL alone does not consistently reduce diversity: its Falcon3-1B and Qwen2.5-3B Graph intervals include zero. An improvement over a trained control also need not restore initial diversity. Both replay variants remain below initial PCMD in PantryPlan at every scale, whereas their Graph comparisons exceed the initial value. These results describe diversity within each comparison’s eligible prompt population; they do not establish recovery of the same individual modes.

Table 12: 64-response evaluation: trained policy minus initialization. Changes in PCMD are in probability units; positive values indicate more diverse correct modes. Brackets give nominal 95% paired Student- $t$  intervals ( $n = 5$ , four degrees of freedom). The last column gives the range of jointly eligible prompts across seeds, out of 128. Each row uses its own matched prompt population.

| Domain              | Contrast          | $\Delta\text{PCMD}$ [95% interval] | Joint prompts |
|---------------------|-------------------|------------------------------------|---------------|
| <b>Qwen2.5-0.5B</b> |                   |                                    |               |
| Graph               | Dr.GRPO – initial | −0.338 [−0.550, −0.126]            | 37–45         |
|                     | Re:Dr – initial   | +0.217 [+0.190, +0.243]            | 97–103        |
|                     | MaxRL – initial   | −0.346 [−0.516, −0.177]            | 37–86         |
| PantryPlan          | Re:Max – initial  | +0.155 [+0.107, +0.203]            | 97–101        |
|                     | Dr.GRPO – initial | −0.929 [−0.936, −0.923]            | 62–68         |
|                     | Re:Dr – initial   | −0.357 [−0.445, −0.269]            | 109–122       |
|                     | MaxRL – initial   | −0.927 [−0.932, −0.921]            | 62–71         |
|                     | Re:Max – initial  | −0.355 [−0.438, −0.272]            | 108–124       |
| <b>Falcon3-1B</b>   |                   |                                    |               |
| Graph               | Dr.GRPO – initial | −0.313 [−0.393, −0.233]            | 75–99         |
|                     | Re:Dr – initial   | +0.043 [+0.006, +0.081]            | 100–107       |
|                     | MaxRL – initial   | +0.008 [−0.027, +0.043]            | 104–109       |
| PantryPlan          | Re:Max – initial  | +0.119 [+0.096, +0.141]            | 106–109       |
|                     | Dr.GRPO – initial | −0.910 [−0.928, −0.893]            | 81–86         |
|                     | Re:Dr – initial   | −0.378 [−0.507, −0.250]            | 104–110       |
|                     | MaxRL – initial   | −0.921 [−0.925, −0.916]            | 81            |
|                     | Re:Max – initial  | −0.519 [−0.575, −0.462]            | 106–116       |
| <b>Qwen2.5-3B</b>   |                   |                                    |               |
| Graph               | Dr.GRPO – initial | −0.112 [−0.156, −0.069]            | 78–86         |
|                     | Re:Dr – initial   | +0.226 [+0.212, +0.240]            | 88–96         |
|                     | MaxRL – initial   | +0.022 [−0.048, +0.092]            | 81–93         |
| PantryPlan          | Re:Max – initial  | +0.254 [+0.234, +0.274]            | 90–96         |
|                     | Dr.GRPO – initial | −0.509 [−0.750, −0.268]            | 69–99         |
|                     | Re:Dr – initial   | −0.115 [−0.157, −0.073]            | 101–110       |
|                     | MaxRL – initial   | −0.449 [−0.541, −0.358]            | 83–98         |
|                     | Re:Max – initial  | −0.116 [−0.208, −0.024]            | 105–110       |

**Correctness, coverage, and sampling budget.** All 24 trained-versus-initial comparisons increase per-response correctness averaged over all evaluation prompts, in every seed. Nevertheless, every PantryPlan method-scale comparison has lower  $\text{pass@64}$  than initialization in every seed. Averaging over all evaluation prompts, replay improves mean  $\text{pass@8}$ ,  $\text{distinct@8}$ ,  $\text{pass@64}$ , and  $\text{distinct@64}$  relative to its trained control in all twelve comparisons; per-response correctness improves in seven of twelve. On jointly eligible prompts, however, replay lowers mean correctness in eleven of twelve comparisons. The concentration effects therefore do not hold correctness fixed.

The eight-response metrics average over subsets of eight responses from each 64-response pool (rarefaction). For Qwen2.5-3B Graph, Dr.GRPO increases per-response correctness by 14.49 percentage points and  $\text{distinct@8}$  by 0.112 modes per prompt, while reducing PCMD by 11.20 points and  $\text{distinct@64}$  by 0.220 modes. The change in verified-mode count reverses sign between eight and 64 responses in every seed. A gain at one sampling budget thus need not persist at a larger budget. These evaluations measure the distribution of correct responses and the modes observed in finite samples; they do not establish exact mode extinction, recovery of the full support, or a causal effect of diversity on accuracy.

Table 13: 64-response evaluation: replay minus its matched trained control. Changes in PCMD are in probability units; positive values indicate more diverse correct modes. Brackets give nominal 95% paired Student- $t$  intervals ( $n = 5$ , four degrees of freedom). The last column gives the range of jointly eligible prompts across seeds, out of 128. Each row uses its own matched prompt population.

| Domain              | Contrast        | $\Delta\text{PCMD}$ [95% interval] | Joint prompts |
|---------------------|-----------------|------------------------------------|---------------|
| <b>Qwen2.5-0.5B</b> |                 |                                    |               |
| Graph               | Re:Dr – Dr.GRPO | +0.700 [+0.659, +0.742]            | 40–46         |
|                     | Re:Max – MaxRL  | +0.493 [+0.324, +0.662]            | 41–107        |
| PantryPlan          | Re:Dr – Dr.GRPO | +0.542 [+0.418, +0.665]            | 62–68         |
|                     | Re:Max – MaxRL  | +0.555 [+0.458, +0.652]            | 61–71         |
| <b>Falcon3-1B</b>   |                 |                                    |               |
| Graph               | Re:Dr – Dr.GRPO | +0.354 [+0.264, +0.445]            | 73–101        |
|                     | Re:Max – MaxRL  | +0.103 [+0.072, +0.133]            | 122–127       |
| PantryPlan          | Re:Dr – Dr.GRPO | +0.505 [+0.315, +0.694]            | 80–84         |
|                     | Re:Max – MaxRL  | +0.262 [+0.190, +0.334]            | 81            |
| <b>Qwen2.5-3B</b>   |                 |                                    |               |
| Graph               | Re:Dr – Dr.GRPO | +0.343 [+0.307, +0.380]            | 94–107        |
|                     | Re:Max – MaxRL  | +0.224 [+0.163, +0.286]            | 105–123       |
| PantryPlan          | Re:Dr – Dr.GRPO | +0.369 [+0.188, +0.550]            | 72–101        |
|                     | Re:Max – MaxRL  | +0.319 [+0.270, +0.367]            | 89–99         |

## H.2 CONDITIONAL CONCENTRATION AND SAMPLING DEPENDENCE

Conditional concentration measures repetition of solution keys among correct outputs. For a fixed prompt with correctness probability  $P > 0$ , write  $q_c = \Pr[V(Y, s_x) = c \mid V(Y, s_x) \neq \perp]$  and define

$$C(q) = \sum_c q_c^2, \quad \bar{D}_2 = 2P - P^2 C(q). \quad (11)$$

Here  $C$  is the probability that two independent correct outputs share a key, and  $\bar{D}_2$  is the expected number of distinct verified keys in two outputs. The identity follows from  $\bar{D}_2 = \sum_c [1 - (1 - Pq_c)^2]$ . At  $P = 0$ ,  $C$  is undefined and  $\bar{D}_2 = 0$ . Expected distinct keys,  $\bar{D}_K$ , differ from extra verified modes,  $B = \text{distinct@}K - \text{pass@}K$ . Sampled diversity can change through either correctness or conditional concentration. Lower  $C$  alone does not imply a more uniform distribution in the majorization ordering or larger total support. Collisions also do not determine whether any particular unobserved key has zero probability.

**Connection to the conditional theory.** Under the assumptions of Theorem P.4, the time change in its proof gives  $dq_c/d\tau = q_c(q_c - C)$ . Consequently,

$$\frac{dC}{d\tau} = 2 \left( \sum_c q_c^3 - C^2 \right) = 2 \text{Var}_{c \sim q}(q_c) \geq 0. \quad (12)$$

The variance vanishes exactly when  $q$  is uniform on its positive support. Concentration therefore increases strictly along this ideal categorical flow whenever the positive coordinates are unequal. The conclusion is specific to this flow: it does not establish a monotonic increase during neural-network training or a monotonic decrease under finite-buffer replay.

**Estimator and comparison population.** For a prompt with  $R \geq 2$  correct samples, the collision estimator is  $\hat{C} = 1 - \widehat{\text{PCMD}}$ , using the pair statistic in App. A.1. It is unbiased for promptwise  $C$  under independent sampling from a fixed policy. Estimates are undefined at  $R < 2$ ; joint selection of eligible prompts across conditions imposes additional assumptions, as detailed in App. A.1.

For paired training seed  $s$ , let  $E_s$  contain the prompts with defined estimates in both conditions. When  $E_s$  is nonempty, the seed contrast is  $|E_s|^{-1} \sum_{x \in E_s} (\hat{C}_{B,x} - \hat{C}_{A,x})$ . Defined seed estimates receive equal weight. Initial-to-final contrasts compare each method’s own initial evaluation with step 3,072; replay contrasts pair terminal evaluations. Nominal Student- $t$  intervals require five defined seeds; smaller samples are descriptive. Intervals describe variation among seed means, without simultaneous coverage or uncertainty over unseen prompts. Tables report defined seeds and eligible prompts.

The analysis of hosted models instead pools correct pairs across prompts. Under independent sampling at a common budget, its ratio of expected colliding to correct pair counts is  $\sum_x P_x^2 C(q_x) / \sum_x P_x^2$ ; the finite-sample ratio is not generally unbiased for this target. Training comparisons weight eligible prompts equally. Changes in correctness can alter both eligibility and pooled pair weights, so these aggregates estimate different quantities.

**Sampling dependence and sensitivity analyses.** The following comparisons use four groups of eight responses per prompt, separate from the 64-response evaluation in App. H.1. The groups have overlapping sampling streams: under the vLLM V0 seed convention, they contain eleven distinct nominal stream identifiers. The concentration estimator retains one representative per identifier, chosen independently of correctness and key identity. Treating these identifiers as distinct streams is an assumption of the analysis. Equal identifiers can produce different outputs, and different identifiers do not guarantee independent draws.

Two sensitivity comparisons assign disjoint sets of five and six stream identifiers to the two conditions. Split 0 assigns the lower five to A and the upper six to B; split 1 reverses them. Eligibility is determined separately for each split. Streams also repeat across prompts, so the splits do not establish independence across prompts or conditions. Comparisons remain descriptive where the estimator’s independent-sampling or joint-selection assumptions do not hold.

The reported `mean@8`, `pass@8`, and `distinct@8` average over the four intact groups. Overlapping streams make these group estimates dependent, but do not change the expectation of their average.

**Concentration before and after training.** At Level 1, all twelve `Graph/PantryPlan` `Dr.GRPO/GRPO` comparisons across the three scales show increased concentration, with nominal intervals above zero. Per-sample correctness also increases on their jointly eligible prompts. `Graph`’s increases range from .098 to .274 and `PantryPlan`’s from .447 to .886. Both disjoint splits agree in direction in eleven of these twelve comparisons; `Falcon3-1B Graph` with `GRPO` is the exception. The direction of the estimated effect can therefore depend on the sampling split. These results do not establish exact mode disappearance or increased concentration in every RLVR run.

**Effect of replay on concentration.** At Level 1, `Re:Dr` has lower concentration than `Dr.GRPO` in all six `Graph` and `PantryPlan` comparisons. Changes range from  $-.756$  to  $-.273$  in `Graph` and  $-.433$  to  $-.300$  in `PantryPlan`; all six nominal intervals lie below zero, and both disjoint splits agree in direction. These comparisons also have lower per-sample correctness under replay on their jointly eligible prompts. The collision contrast therefore describes a change in the conditional correct-key distribution on that population; it neither fixes correctness nor implies improvement in every accuracy measure. The `pass@8` and sampled-diversity metrics over the full evaluation set measure different outcomes.

Fewer prompts are eligible in the other domains. Level-1 `Dr.GRPO/GRPO Python` initial evaluations yield no jointly eligible prompts. Some `Countdown` initial-to-final comparisons contain only one or two eligible prompts per seed. Several `MathIR` conditions already have observed collision probability one, leaving little room for a measured increase. A diversity increase alone therefore does not imply reduced conditional concentration.

For `Qwen2.5-3B MaxRL` at Level 1, replay lowers concentration in `Countdown`, `Python`, and `PantryPlan`, with nominal intervals below zero; `MathIR`’s contrast is zero. `Graph`’s eleven-stream contrast is negative, but its two disjoint splits disagree in direction, as they do for `Falcon3-1B Graph` with `MaxRL`. At Level 2, both `Graph` replay contrasts are negative and both splits agree. `PantryPlan` has few jointly eligible prompts: `Re:Dr` has three defined seeds with 2–16 eligible prompts and mean  $+.495$ ; `Re:Max` has four defined seeds with 1–19 eligible prompts and mean  $+.439$ . Both splits also give positive means, with fewer defined seeds. Neither contrast has a five-seed interval. Diversity gains over the full evaluation set can therefore coexist with greater conditional concentration on these small jointly eligible subsets.

**Table conventions.** Positive  $\Delta\hat{C}$  indicates greater concentration. Eligible prompts have  $R \geq 2$  correct representatives in both conditions, with eligibility determined separately for each contrast and split. The range  $|E_s|$  counts eligible prompts out of 128 across available seed pairs, including zero;  $n$  counts seeds with a defined estimate, out of five. Within each seed, eligible

prompts receive equal weight; defined seeds then receive equal weight. The nominal 95% intervals are unadjusted, Student- $t$  intervals, are not truncated to the parameter range, and appear only at  $n = 5$ . Dashes indicate that a contrast or interval cannot be estimated under these criteria. Replay comparisons give Re:Dr minus Dr:GRPO or Re:Max minus MaxRL at step 3,072, whereas initial-to-final comparisons measure the change within each method.

| Domain     | Objective | $n$ | $\Delta\hat{C}$ | nominal 95% interval | $ E_s $ | split 0 ( $n$ ) | split 1 ( $n$ ) |
|------------|-----------|-----|-----------------|----------------------|---------|-----------------|-----------------|
| Graph      | Dr:GRPO   | 5   | +0.242          | [+0.110, +0.373]     | 17–40   | +0.257 (5)      | +0.284 (5)      |
| Graph      | GRPO      | 5   | +0.221          | [+0.097, +0.344]     | 15–28   | +0.268 (5)      | +0.287 (5)      |
| Graph      | MaxRL     | 5   | +0.145          | [-0.033, +0.324]     | 18–40   | +0.115 (5)      | +0.174 (5)      |
| Graph      | Re:Dr     | 5   | -0.507          | [-0.558, -0.456]     | 45–49   | -0.519 (5)      | -0.465 (5)      |
| Graph      | Re:Max    | 5   | -0.399          | [-0.458, -0.339]     | 40–51   | -0.436 (5)      | -0.348 (5)      |
| Countdown  | Dr:GRPO   | 5   | +1.000          | [+1.000, +1.000]     | 1–2     | – (0)           | +1.000 (5)      |
| Countdown  | GRPO      | 5   | +0.964          | [+0.863, +1.065]     | 1–2     | – (0)           | +0.920 (5)      |
| Countdown  | MaxRL     | 4   | +0.977          | –                    | 0–2     | – (0)           | +1.000 (4)      |
| Countdown  | Re:Dr     | 5   | +0.467          | [+0.117, +0.816]     | 1–3     | – (0)           | +0.713 (4)      |
| Countdown  | Re:Max    | 5   | +0.420          | [+0.359, +0.480]     | 1–2     | – (0)           | +0.493 (5)      |
| Python     | Dr:GRPO   | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| Python     | GRPO      | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| Python     | MaxRL     | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| Python     | Re:Dr     | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| Python     | Re:Max    | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| MathIR     | Dr:GRPO   | 5   | +0.000          | [+0.000, +0.000]     | 5–8     | +0.000 (5)      | +0.000 (1)      |
| MathIR     | GRPO      | 5   | +0.040          | [-0.071, +0.151]     | 4–10    | +0.000 (5)      | +0.250 (2)      |
| MathIR     | MaxRL     | 5   | +0.022          | [-0.039, +0.084]     | 5–9     | +0.000 (5)      | +0.067 (5)      |
| MathIR     | Re:Dr     | 5   | +0.059          | [-0.046, +0.163]     | 8–11    | +0.000 (5)      | +0.367 (2)      |
| MathIR     | Re:Max    | 5   | +0.063          | [-0.009, +0.134]     | 9–11    | +0.000 (5)      | +0.250 (5)      |
| PantryPlan | Dr:GRPO   | 5   | +0.833          | [+0.819, +0.847]     | 54–64   | +1.000 (5)      | +1.000 (5)      |
| PantryPlan | GRPO      | 5   | +0.849          | [+0.845, +0.853]     | 64–65   | +1.000 (5)      | +1.000 (5)      |
| PantryPlan | MaxRL     | 5   | +0.859          | [+0.836, +0.881]     | 54–65   | +1.000 (5)      | +1.000 (5)      |
| PantryPlan | Re:Dr     | 5   | +0.413          | [+0.345, +0.481]     | 78–89   | +0.543 (5)      | +0.538 (5)      |
| PantryPlan | Re:Max    | 5   | +0.446          | [+0.324, +0.567]     | 80–88   | +0.593 (5)      | +0.550 (5)      |

Table 14: **Initial-to-final conditional concentration: Qwen2.5-0.5B, Level 1.** Each row compares the same method at initialization and step 3,072; positive  $\Delta\hat{C}$  means greater concentration among correct outputs.  $n$  counts seeds with defined estimates;  $|E_s|$  gives the range of eligible prompt counts. Splits 0 and 1 reverse the assignment of disjoint stream identifiers, with eligibility determined separately. Nominal 95% Student- $t$  intervals require five defined seed estimates.

| Domain     | Objective | $n$ | $\Delta\hat{C}$ | nominal 95% interval | $ E_s $ | split 0 ( $n$ ) | split 1 ( $n$ ) |
|------------|-----------|-----|-----------------|----------------------|---------|-----------------|-----------------|
| Graph      | Dr:GRPO   | 5   | +0.233          | [+0.124, +0.343]     | 41–54   | +0.059 (5)      | +0.428 (5)      |
| Graph      | GRPO      | 5   | +0.098          | [+0.026, +0.170]     | 48–56   | -0.186 (5)      | +0.273 (5)      |
| Graph      | MaxRL     | 3   | -0.113          | –                    | 48–52   | -0.083 (3)      | +0.039 (3)      |
| Graph      | Re:Dr     | 5   | -0.007          | [-0.065, +0.050]     | 37–45   | +0.048 (5)      | +0.045 (5)      |
| Graph      | Re:Max    | 2   | -0.062          | –                    | 51–52   | -0.300 (2)      | -0.015 (2)      |
| Countdown  | Dr:GRPO   | 5   | +0.022          | [-0.038, +0.082]     | 6–8     | +0.000 (5)      | – (0)           |
| Countdown  | GRPO      | 5   | +0.180          | [+0.128, +0.231]     | 6–11    | +0.126 (5)      | – (0)           |
| Countdown  | MaxRL     | 2   | +0.182          | –                    | 11      | +0.143 (2)      | – (0)           |
| Countdown  | Re:Dr     | 4   | +0.035          | –                    | 10–11   | +0.158 (4)      | – (0)           |
| Countdown  | Re:Max    | 1   | +0.135          | –                    | 11      | +0.143 (1)      | – (0)           |
| Python     | Dr:GRPO   | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| Python     | GRPO      | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| Python     | MaxRL     | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| Python     | Re:Dr     | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| Python     | Re:Max    | 0   | –               | –                    | –       | – (0)           | – (0)           |
| MathIR     | Dr:GRPO   | 5   | +0.105          | [+0.088, +0.122]     | 6       | – (0)           | +0.098 (5)      |
| MathIR     | GRPO      | 5   | +0.054          | [-0.099, +0.206]     | 3–7     | – (0)           | +0.233 (5)      |
| MathIR     | MaxRL     | 1   | +0.278          | –                    | 6       | +1.000 (1)      | +0.133 (1)      |
| MathIR     | Re:Dr     | 5   | +0.086          | [+0.026, +0.146]     | 6–7     | – (0)           | +0.111 (5)      |
| MathIR     | Re:Max    | 2   | +0.167          | –                    | 5–6     | +1.000 (2)      | +0.000 (2)      |
| PantryPlan | Dr:GRPO   | 5   | +0.886          | [+0.886, +0.886]     | 76      | +1.000 (5)      | +1.000 (5)      |
| PantryPlan | GRPO      | 5   | +0.863          | [+0.798, +0.928]     | 76      | +0.945 (5)      | +1.000 (5)      |
| PantryPlan | MaxRL     | 1   | +0.886          | –                    | 76      | +1.000 (1)      | +1.000 (1)      |
| PantryPlan | Re:Dr     | 4   | +0.437          | –                    | 75–83   | +0.451 (4)      | +0.618 (4)      |
| PantryPlan | Re:Max    | 2   | +0.746          | –                    | 76–78   | +0.740 (2)      | +0.978 (2)      |

Table 15: **Initial-to-final conditional concentration: Falcon3-1B, Level 1.** Each row compares the same method at initialization and step 3,072; positive  $\Delta\hat{C}$  means greater concentration among correct outputs.  $n$  counts seeds with defined estimates;  $|E_s|$  gives the range of eligible prompt counts. Splits 0 and 1 reverse the assignment of disjoint stream identifiers, with eligibility determined separately. Nominal 95% Student- $t$  intervals require five defined seed estimates.

| Domain     | Objective | $n$ | $\Delta\hat{C}$ | nominal 95% interval | $ E_s $ | split 0 ( $n$ ) | split 1 ( $n$ ) |
|------------|-----------|-----|-----------------|----------------------|---------|-----------------|-----------------|
| Graph      | Dr.GRPO   | 5   | +0.274          | [+0.232, +0.316]     | 44–47   | +0.030 (5)      | +0.339 (5)      |
| Graph      | GRPO      | 5   | +0.274          | [+0.207, +0.341]     | 46–49   | +0.046 (5)      | +0.364 (5)      |
| Graph      | MaxRL     | 5   | +0.072          | [-0.002, +0.145]     | 45–55   | -0.208 (5)      | +0.215 (5)      |
| Graph      | Re:Dr     | 5   | -0.038          | [-0.060, -0.016]     | 48–53   | -0.227 (5)      | +0.121 (5)      |
| Graph      | Re:Max    | 5   | -0.145          | [-0.198, -0.093]     | 46–52   | -0.220 (5)      | +0.165 (5)      |
| Countdown  | Dr.GRPO   | 5   | +0.160          | [+0.109, +0.211]     | 11–12   | +0.200 (5)      | -0.016 (5)      |
| Countdown  | GRPO      | 5   | +0.182          | [+0.182, +0.182]     | 11      | +0.200 (5)      | +0.000 (5)      |
| Countdown  | MaxRL     | 5   | +0.056          | [-0.046, +0.159]     | 9–11    | +0.050 (5)      | +0.000 (5)      |
| Countdown  | Re:Dr     | 5   | +0.020          | [-0.060, +0.099]     | 11–15   | +0.169 (5)      | -0.071 (5)      |
| Countdown  | Re:Max    | 5   | -0.100          | [-0.145, -0.055]     | 10–12   | +0.000 (5)      | -0.050 (5)      |
| Python     | Dr.GRPO   | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| Python     | GRPO      | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| Python     | MaxRL     | 5   | +0.000          | [+0.000, +0.000]     | 2–9     | +0.000 (2)      | +0.000 (5)      |
| Python     | Re:Dr     | 0   | –               | –                    | 0       | – (0)           | – (0)           |
| Python     | Re:Max    | 5   | -0.023          | [-0.050, +0.003]     | 9       | +0.000 (5)      | -0.040 (5)      |
| MathIR     | Dr.GRPO   | 5   | +0.000          | [+0.000, +0.000]     | 11–14   | +0.000 (5)      | +0.000 (5)      |
| MathIR     | GRPO      | 5   | +0.000          | [+0.000, +0.000]     | 9–11    | +0.000 (5)      | +0.000 (5)      |
| MathIR     | MaxRL     | 5   | +0.000          | [+0.000, +0.000]     | 7–14    | +0.000 (5)      | +0.000 (5)      |
| MathIR     | Re:Dr     | 4   | -0.012          | –                    | 14      | +0.000 (4)      | +0.000 (4)      |
| MathIR     | Re:Max    | 5   | +0.000          | [+0.000, +0.000]     | 12–15   | +0.000 (5)      | +0.000 (5)      |
| PantryPlan | Dr.GRPO   | 5   | +0.447          | [+0.266, +0.627]     | 49–67   | +0.478 (5)      | +0.245 (5)      |
| PantryPlan | GRPO      | 5   | +0.469          | [+0.411, +0.526]     | 49–62   | +0.514 (5)      | +0.271 (5)      |
| PantryPlan | MaxRL     | 5   | +0.431          | [+0.297, +0.565]     | 59–69   | +0.394 (5)      | +0.246 (5)      |
| PantryPlan | Re:Dr     | 5   | +0.184          | [+0.081, +0.288]     | 64–71   | +0.087 (5)      | +0.037 (5)      |
| PantryPlan | Re:Max    | 5   | +0.142          | [+0.018, +0.266]     | 68–75   | +0.046 (5)      | -0.015 (5)      |

Table 16: **Initial-to-final conditional concentration: Qwen2.5-3B, Level 1.** Each row compares the same method at initialization and step 3,072; positive  $\Delta\hat{C}$  means greater concentration among correct outputs.  $n$  counts seeds with defined estimates;  $|E_s|$  gives the range of eligible prompt counts. Splits 0 and 1 reverse the assignment of disjoint stream identifiers, with eligibility determined separately. Nominal 95% Student- $t$  intervals require five defined seed estimates.

| Domain     | Objective | $n$ | $\Delta\hat{C}$ | nominal 95% interval | $ E_s $ | split 0 ( $n$ ) | split 1 ( $n$ ) |
|------------|-----------|-----|-----------------|----------------------|---------|-----------------|-----------------|
| Graph      | Dr.GRPO   | 5   | +0.625          | [+0.295, +0.955]     | 2–10    | +0.633 (2)      | +0.860 (5)      |
| Graph      | MaxRL     | 5   | +0.452          | [+0.387, +0.518]     | 6–10    | +0.767 (4)      | +0.484 (5)      |
| Graph      | Re:Dr     | 5   | -0.204          | [-0.282, -0.126]     | 9–11    | +0.033 (5)      | -0.372 (5)      |
| Graph      | Re:Max    | 5   | -0.128          | [-0.262, +0.006]     | 9–11    | +0.333 (5)      | -0.267 (5)      |
| Countdown  | Dr.GRPO   | 1   | +0.000          | –                    | 0–1     | – (0)           | – (0)           |
| Countdown  | MaxRL     | 1   | +0.000          | –                    | 0–1     | – (0)           | – (0)           |
| Countdown  | Re:Dr     | 5   | +0.000          | [+0.000, +0.000]     | 1       | – (0)           | – (0)           |
| Countdown  | Re:Max    | 3   | +0.000          | –                    | 0–1     | – (0)           | – (0)           |
| Python     | Dr.GRPO   | 5   | +0.302          | [+0.302, +0.302]     | 32      | +0.000 (5)      | +0.378 (5)      |
| Python     | MaxRL     | 5   | +0.302          | [+0.302, +0.302]     | 32      | +0.000 (5)      | +0.378 (5)      |
| Python     | Re:Dr     | 5   | +0.094          | [-0.259, +0.448]     | 32      | -0.224 (5)      | +0.146 (5)      |
| Python     | Re:Max    | 5   | +0.199          | [-0.088, +0.485]     | 32      | -0.070 (5)      | +0.258 (5)      |
| MathIR     | Dr.GRPO   | 3   | +0.000          | –                    | 0–5     | – (0)           | +0.000 (3)      |
| MathIR     | MaxRL     | 4   | +0.000          | –                    | 0–6     | – (0)           | +0.000 (4)      |
| MathIR     | Re:Dr     | 5   | +0.000          | [+0.000, +0.000]     | 6       | – (0)           | +0.000 (5)      |
| MathIR     | Re:Max    | 4   | +0.000          | –                    | 6       | – (0)           | +0.000 (4)      |
| PantryPlan | Dr.GRPO   | 3   | -0.578          | –                    | 0–3     | -0.500 (1)      | -1.000 (1)      |
| PantryPlan | MaxRL     | 3   | -0.238          | –                    | 1–5     | -1.000 (1)      | -1.000 (1)      |
| PantryPlan | Re:Dr     | 2   | -0.049          | –                    | 15–16   | -0.075 (2)      | -0.050 (2)      |
| PantryPlan | Re:Max    | 3   | +0.020          | –                    | 18–20   | +0.092 (3)      | -0.174 (3)      |

Table 17: **Initial-to-final conditional concentration: Qwen2.5-0.5B, Level 2.** Each row compares the same method at initialization and step 3,072; positive  $\Delta\hat{C}$  means greater concentration among correct outputs.  $n$  counts seeds with defined estimates;  $|E_s|$  gives the range of eligible prompt counts. Splits 0 and 1 reverse the assignment of disjoint stream identifiers, with eligibility determined separately. Nominal 95% Student- $t$  intervals require five defined seed estimates.

| Domain     | Control | $n$ | $\Delta\hat{C}$ | nominal 95% interval | $ E_s $ | split 0 ( $n$ ) | split 1 ( $n$ ) |
|------------|---------|-----|-----------------|----------------------|---------|-----------------|-----------------|
| Graph      | Dr.GRPO | 5   | -0.756          | [-0.831, -0.680]     | 37-40   | -0.853 (5)      | -0.807 (5)      |
| Graph      | MaxRL   | 5   | -0.587          | [-0.759, -0.414]     | 41-73   | -0.625 (5)      | -0.605 (5)      |
| Countdown  | Dr.GRPO | 5   | -0.529          | [-0.605, -0.452]     | 42-78   | -0.529 (5)      | -0.535 (5)      |
| Countdown  | MaxRL   | 5   | -0.547          | [-0.635, -0.459]     | 45-84   | -0.578 (5)      | -0.530 (5)      |
| Python     | Dr.GRPO | 5   | -0.090          | [-0.341, +0.160]     | 22      | -0.115 (5)      | -0.062 (5)      |
| Python     | MaxRL   | 5   | -0.266          | [-0.577, +0.044]     | 22      | -0.298 (5)      | -0.272 (5)      |
| MathIR     | Dr.GRPO | 5   | -0.008          | [-0.017, +0.001]     | 44-68   | -0.006 (5)      | -0.008 (5)      |
| MathIR     | MaxRL   | 5   | -0.005          | [-0.013, +0.003]     | 32-56   | -0.007 (5)      | -0.002 (5)      |
| PantryPlan | Dr.GRPO | 5   | -0.433          | [-0.518, -0.347]     | 54-66   | -0.442 (5)      | -0.494 (5)      |
| PantryPlan | MaxRL   | 5   | -0.417          | [-0.547, -0.288]     | 53-71   | -0.363 (5)      | -0.508 (5)      |

Table 18: **Replay and conditional concentration: Qwen2.5-0.5B, Level 1.** Each row gives Re:Dr minus Dr.GRPO or Re:Max minus MaxRL at step 3,072; negative  $\Delta\hat{C}$  means lower concentration among correct outputs.  $n$  counts seeds with defined estimates;  $|E_s|$  gives the range of eligible prompt counts. Splits 0 and 1 reverse the assignment of disjoint stream identifiers, with eligibility determined separately. Nominal 95% Student- $t$  intervals require five defined seed estimates.

| Domain     | Control | $n$ | $\Delta\hat{C}$ | nominal 95% interval | $ E_s $ | split 0 ( $n$ ) | split 1 ( $n$ ) |
|------------|---------|-----|-----------------|----------------------|---------|-----------------|-----------------|
| Graph      | Dr.GRPO | 5   | -0.309          | [-0.523, -0.094]     | 31-64   | -0.129 (5)      | -0.452 (5)      |
| Graph      | MaxRL   | 5   | -0.021          | [-0.040, -0.002]     | 95-100  | +0.100 (5)      | -0.013 (5)      |
| Countdown  | Dr.GRPO | 4   | -0.168          | —                    | 33-42   | -0.139 (4)      | -0.233 (4)      |
| Countdown  | MaxRL   | 5   | -0.063          | [-0.079, -0.047]     | 78-81   | -0.062 (5)      | -0.063 (5)      |
| Python     | Dr.GRPO | 0   | —               | —                    | 0       | — (0)           | — (0)           |
| Python     | MaxRL   | 5   | -0.093          | [-0.350, +0.165]     | 22-128  | -0.039 (5)      | -0.090 (5)      |
| MathIR     | Dr.GRPO | 5   | +0.004          | [-0.007, +0.015]     | 9-14    | +0.000 (5)      | +0.000 (5)      |
| MathIR     | MaxRL   | 5   | -0.001          | [-0.010, +0.009]     | 40-72   | +0.000 (5)      | +0.000 (5)      |
| PantryPlan | Dr.GRPO | 5   | -0.404          | [-0.600, -0.209]     | 74-81   | -0.435 (5)      | -0.352 (5)      |
| PantryPlan | MaxRL   | 5   | -0.129          | [-0.223, -0.035]     | 81      | -0.203 (5)      | -0.043 (5)      |

Table 19: **Replay and conditional concentration: Falcon3-1B, Level 1.** Each row gives Re:Dr minus Dr.GRPO or Re:Max minus MaxRL at step 3,072; negative  $\Delta\hat{C}$  means lower concentration among correct outputs.  $n$  counts seeds with defined estimates;  $|E_s|$  gives the range of eligible prompt counts. Splits 0 and 1 reverse the assignment of disjoint stream identifiers, with eligibility determined separately. Nominal 95% Student- $t$  intervals require five defined seed estimates.

| Domain     | Control | $n$ | $\Delta\hat{C}$ | nominal 95% interval | $ E_s $ | split 0 ( $n$ ) | split 1 ( $n$ ) |
|------------|---------|-----|-----------------|----------------------|---------|-----------------|-----------------|
| Graph      | Dr.GRPO | 5   | -0.273          | [-0.299, -0.247]     | 54-71   | -0.363 (5)      | -0.151 (5)      |
| Graph      | MaxRL   | 5   | -0.162          | [-0.261, -0.062]     | 68-83   | -0.169 (5)      | +0.059 (5)      |
| Countdown  | Dr.GRPO | 5   | -0.137          | [-0.183, -0.090]     | 70-75   | -0.153 (5)      | -0.139 (5)      |
| Countdown  | MaxRL   | 5   | -0.105          | [-0.126, -0.084]     | 56-76   | -0.113 (5)      | -0.095 (5)      |
| Python     | Dr.GRPO | 2   | -0.190          | —                    | 0-22    | -0.221 (2)      | -0.141 (2)      |
| Python     | MaxRL   | 5   | -0.153          | [-0.265, -0.041]     | 22-81   | -0.203 (5)      | -0.083 (5)      |
| MathIR     | Dr.GRPO | 5   | -0.010          | [-0.022, +0.003]     | 23-34   | -0.017 (5)      | +0.003 (5)      |
| MathIR     | MaxRL   | 5   | +0.000          | [+0.000, +0.000]     | 21-27   | +0.000 (5)      | +0.000 (5)      |
| PantryPlan | Dr.GRPO | 5   | -0.300          | [-0.420, -0.179]     | 71-83   | -0.337 (5)      | -0.305 (5)      |
| PantryPlan | MaxRL   | 5   | -0.292          | [-0.319, -0.265]     | 81-89   | -0.355 (5)      | -0.237 (5)      |

Table 20: **Replay and conditional concentration: Qwen2.5-3B, Level 1.** Each row gives Re:Dr minus Dr.GRPO or Re:Max minus MaxRL at step 3,072; negative  $\Delta\hat{C}$  means lower concentration among correct outputs.  $n$  counts seeds with defined estimates;  $|E_s|$  gives the range of eligible prompt counts. Splits 0 and 1 reverse the assignment of disjoint stream identifiers, with eligibility determined separately. Nominal 95% Student- $t$  intervals require five defined seed estimates.

| Domain     | Control | $n$ | $\Delta\hat{C}$ | nominal 95% interval | $ E_s $ | split 0 ( $n$ ) | split 1 ( $n$ ) |
|------------|---------|-----|-----------------|----------------------|---------|-----------------|-----------------|
| Graph      | Dr.GRPO | 5   | -0.577          | [-0.736, -0.418]     | 9-25    | -0.599 (5)      | -0.544 (5)      |
| Graph      | MaxRL   | 5   | -0.405          | [-0.472, -0.339]     | 28-42   | -0.262 (5)      | -0.416 (5)      |
| Countdown  | Dr.GRPO | 4   | -0.230          | —                    | 0-9     | -0.281 (4)      | -0.292 (4)      |
| Countdown  | MaxRL   | 4   | -0.073          | —                    | 0-10    | +0.000 (3)      | -0.118 (4)      |
| Python     | Dr.GRPO | 5   | -0.209          | [-0.563, +0.146]     | 104     | -0.219 (5)      | -0.234 (5)      |
| Python     | MaxRL   | 5   | -0.106          | [-0.397, +0.184]     | 104     | -0.076 (5)      | -0.121 (5)      |
| MathIR     | Dr.GRPO | 5   | +0.000          | [+0.000, +0.000]     | 12-122  | +0.000 (5)      | +0.000 (5)      |
| MathIR     | MaxRL   | 5   | -0.001          | [-0.004, +0.002]     | 29-125  | +0.000 (5)      | -0.001 (5)      |
| PantryPlan | Dr.GRPO | 3   | +0.495          | —                    | 0-16    | +0.656 (1)      | +0.554 (2)      |
| PantryPlan | MaxRL   | 4   | +0.439          | —                    | 0-19    | +0.267 (3)      | +0.208 (2)      |

Table 21: **Replay and conditional concentration: Qwen2.5-0.5B, Level 2.** Each row gives Re:Dr minus Dr.GRPO or Re:Max minus MaxRL at step 3,072; negative  $\Delta\hat{C}$  means lower concentration among correct outputs.  $n$  counts seeds with defined estimates;  $|E_s|$  gives the range of eligible prompt counts. Splits 0 and 1 reverse the assignment of disjoint stream identifiers, with eligibility determined separately. Nominal 95% Student- $t$  intervals require five defined seed estimates.

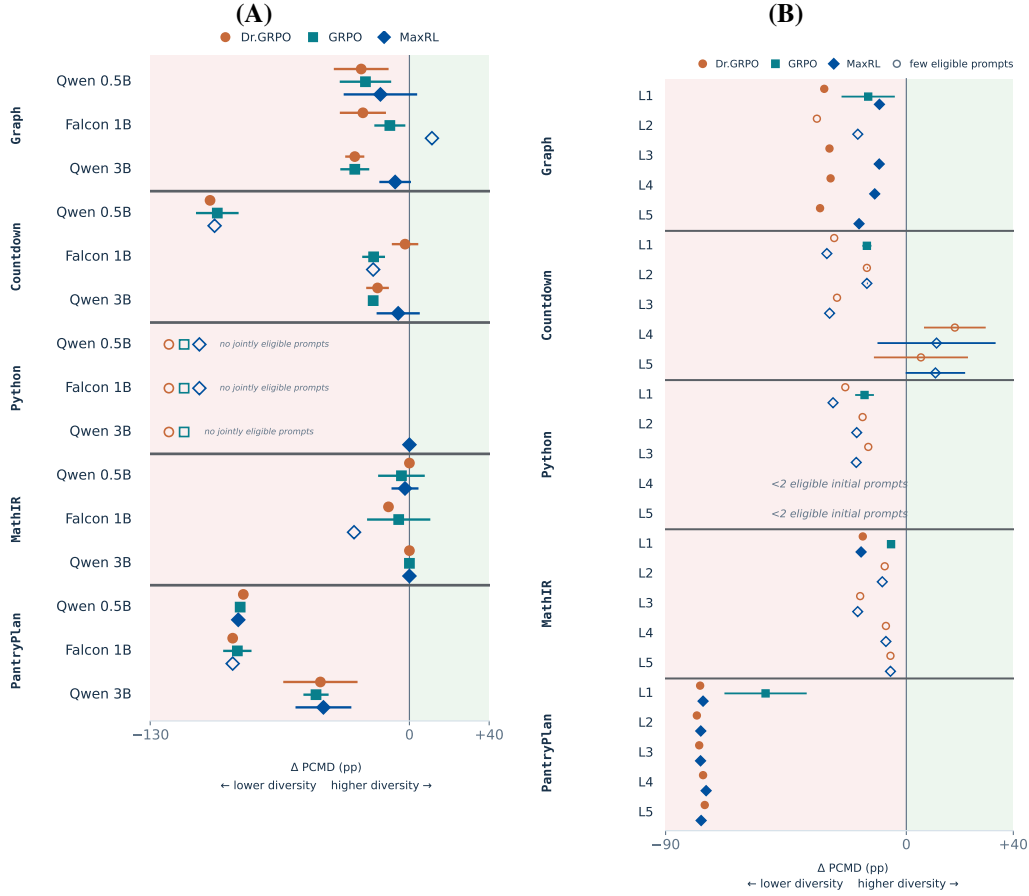


Figure 27: **Graph and PantryPlan lose conditional solution diversity across scales and task levels.** Changes in PCMD, in percentage points. (A) Initial-to-final changes at Level 1 across three model scales and three objectives. Each seed averages over prompts with at least two correct responses at both checkpoints. Filled marks and bars show five-seed means and nominal paired 95% Student-*t* intervals. Open marks show means from fewer than five seeds, without intervals; annotated hollow symbols identify methods with no jointly eligible prompts. Dr.GRPO and GRPO reduce diversity in Graph and PantryPlan at every scale. (B) Level-1-trained policies evaluated on Levels 1–5, with Levels 2–5 measuring transfer. Available evaluations are averaged within each scale, then scales receive equal weight. Evaluations with different context limits contribute separately within a scale but do not constitute additional training seeds. Each scale uses one initial checkpoint evaluation. Bars appear only when one scale contributes and give nominal 95% Student-*t* intervals over its training seeds. Open marks indicate that at least one contributing scale has fewer than 30 eligible initial prompts or fewer than 30 eligible trained prompt observations in total. Missing initial references have fewer than two eligible prompts. GRPO appears only at Level 1. Graph and PantryPlan means decrease at every level; Countdown increases at Levels 4–5.

### H.3 DIVERSITY BEFORE AND AFTER RLVR-ONLY TRAINING

We evaluate Dr.GRPO and GRPO before training and after eight passes across three models and five domains. The 150 runs each use four  $K = 8$  response groups per checkpoint. Averaged equally across the five domains, extra verified modes decrease by 98.2%–99.4% at Qwen2.5-0.5B and 41.5%–71.6% at larger scales. The `pass@8` mean falls in 10 of the 30 domain–method–scale comparisons, while `macro distinct@8` ends above its initial value at Qwen2.5-3B. Fewer extra verified modes can therefore coexist with a greater probability of sampling at least one correct response. Finite samples do not establish that unobserved solution keys have zero probability.

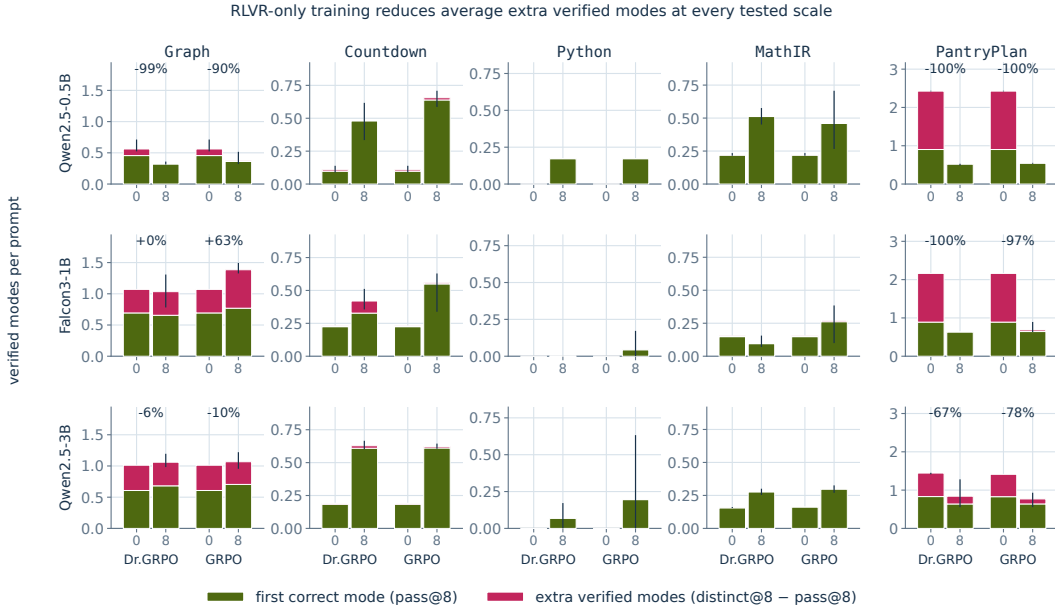


Figure 28: **RLVR-only training reduces average extra verified modes at every tested scale.** Bars decompose  $\text{distinct@8}$  into  $\text{pass@8}$  and extra verified modes before training and after eight passes. Panels show Dr.GRPO and GRPO across three models and five domains: 150 runs, with five seeds per model–domain–method combination and four eight-response groups per checkpoint. Whiskers show seed ranges; each method uses its own initial evaluation. Percentage changes in extra modes are shown for Graph and PantryPlan, whose initial means exceed .05. The other three domains start at .011 or less.

The domain average of PCMD includes only domains measurable at both endpoints: Graph, MathIR, and PantryPlan at 0.5B; Graph, Countdown, and PantryPlan at 1B; and Graph and PantryPlan at 3B. It declines by 98.5% and 99.7% at Qwen2.5-0.5B under GRPO and Dr.GRPO, respectively, and by 50.8%–65.1% at the larger scales. Terminal macro PCMD rounds to .001 for Dr.GRPO at Qwen2.5-0.5B. The decline thus also appears in diversity conditional on success. Because eligibility depends on the number of correct samples, these descriptive averages do not hold the prompt population or correctness fixed. App. H.2 reports contrasts on jointly eligible prompts. The decline is not universal: Falcon3-1B Countdown has increasing PCMD under Dr.GRPO, from an initial value of .033.

Although the initial weights are shared, each method uses its own evaluation samples. Differences between initial estimates therefore reflect sampling variability.

Graph and PantryPlan have substantial initial diversity at every scale, whereas the other domains start with at most .011 extra modes per prompt. To avoid unstable ratios near zero, retained-diversity percentages are reported only when the initial extra-mode count exceeds .05. The final  $\text{distinct@8}$  is below its initial value in six of the 15 Dr.GRPO model–domain combinations. For Re:Dr, this occurs in two available comparisons, both on PantryPlan; Falcon3-1B Countdown uses four seeds. Conditional diversity comparisons also depend on which prompts yield enough correct responses. App. H.2 reports paired contrasts on jointly eligible prompts, including seed counts and uncertainty. These held-out evaluations do not track individual modes stored during training.

Uniform sampling over the enumerable action representations gives a diversity reference for Graph and PantryPlan (App. A.1). No corresponding reference is specified for the other domains. On PantryPlan, uniform sampling of bit masks yields higher expected  $\text{distinct@8}$  than the final sampled value of every model under both Dr.GRPO and Re:Dr.

#### H.4 VANISHING TASK GRADIENTS AND REPEATED OUTPUTS

A Dr.GRPO group with identical rewards has exactly zero fresh task gradient. Only mixed groups, containing both correct and incorrect responses, provide a nonzero task advantage. Optimizer

Table 22: **Uniform actions and untrained checkpoints provide absolute diversity references.** *Uniform* gives expected `distinct@8` under uniform sampling of the action representation: 27 color strings for `Graph` and 64 bit masks for `PantryPlan`. *Initial* gives the Dr.GRPO pass-0 reference; the final two columns give sampled means after eight training passes.

| Model        | Domain     | Uniform | Initial | Dr.GRPO | Re:Dr |
|--------------|------------|---------|---------|---------|-------|
| Qwen2.5-0.5B | Graph      | 1.633   | .561    | .325    | 2.438 |
|              | PantryPlan | 2.153   | 2.426   | .522    | 1.539 |
| Falcon3-1B   | Graph      | 1.633   | 1.070   | 1.034   | 1.717 |
|              | PantryPlan | 2.153   | 2.162   | .633    | 1.576 |
| Qwen2.5-3B   | Graph      | 1.633   | 1.012   | 1.060   | 1.637 |
|              | PantryPlan | 2.153   | 1.445   | .841    | 1.514 |

momentum can still change the policy when the current task gradient is zero. The table below reports the fraction of mixed groups over 3,072 Qwen2.5-0.5B updates, averaged across five seeds.

| Domain     | Dr.GRPO | Re:Dr |
|------------|---------|-------|
| Graph      | .144    | .954  |
| Countdown  | .140    | .456  |
| Python     | .015    | .185  |
| MathIR     | .364    | .453  |
| PantryPlan | .123    | .789  |

Thus 98.5% of `Python` control updates carry zero fresh task gradient. Replay contributes a likelihood gradient from verified responses and can also change the policy so that subsequent fresh groups contain both correct and incorrect responses.

All five Qwen2.5-0.5B `Python` control seeds exhibit observed output collapse. At pass 0, eight-response groups average 7.99 distinct completion strings per prompt. By steps 288–480, all 32 temperature-1 responses for each prompt are identical within each seed and match its greedy completion. This holds on all 128 prompts and persists at every subsequent evaluation through step 3,072. The 32 responses span four groups with overlapping sampling streams (App. H.2); they are not 32 independent draws. Within eight-response groups, the terminal mean is 1.00 distinct completion per prompt for the control and 1.97 for replay. The repeated completion can be incorrect. These finite observations do not rule out low-probability alternatives.

This sparse binary-reward setting allows fresh task gradients to vanish. Sampling strategies that favor mixed groups, larger groups, a reference-policy KL penalty, token-entropy regularization, or broader decoding could change this behavior. The present comparison does not isolate their effects.

## H.5 CONTROL BEHAVIOR AT FIXED LEARNING RATES

The controls use a constant learning rate of  $2 \times 10^{-7}$  at Qwen2.5-0.5B and Falcon3-1B and a peak rate of  $10^{-7}$  with cosine decay at Qwen2.5-3B (Table 6). Learning rates are not swept. Table 23 reports changes in accuracy and the frequency of informative response groups under these schedules. The results characterize the tested settings without identifying an optimal schedule for either method.

**Accuracy changes during training.** Using an absolute-change threshold of .005, 8 of the 15 control comparisons improve in `pass@8` and 6 decline. The mean change is +0.039. The largest increase is +0.426 (Countdown at Qwen2.5-3B); the largest decrease is −0.380 (PantryPlan at Qwen2.5-0.5B). Per-response correctness rises in all 15 comparisons, including 7 without a `pass@8` increase above that threshold. In PantryPlan at Falcon3-1B, per-response correctness rises from 0.318 to 0.633, while `pass@8` falls from 0.891 to 0.633. This combination is consistent with prior observations of increased single-response accuracy alongside reduced sampling coverage (Yue et al., 2025).

The gap between `pass@8` and `mean@8` is at least .02 in 12 initial model–domain comparisons and below .02 in 7 terminal comparisons. This gap reflects variation in correctness within response groups; it does not identify repeated strings or conditional diversity among correct solution keys. The completion-string analysis in App. H.4 directly measures repetition for the Qwen2.5-0.5B `Python` controls.

Table 23: **Per-response accuracy increases in every control comparison, while pass@8 can fall.** Accuracy compares initial and last available evaluations. Columns show unpaired five-run means for each model–domain combination. mean@8 measures per-response correctness. Mixed gives the mean fraction of rollout groups containing both correct and incorrect responses. An unmixed group has zero fresh task advantage, although optimizer momentum can still change the policy. Training schedules match update counts and fresh-rollout budgets; total compute is discussed in App. Q.2.

| Model        | Domain     | control pass@8 |          |          | control mean@8 |          | mixed groups | Re:Dr pass@8 |
|--------------|------------|----------------|----------|----------|----------------|----------|--------------|--------------|
|              |            | initial        | terminal | $\Delta$ | initial        | terminal |              |              |
| Qwen2.5-0.5B | Graph      | .456           | .323     | −.132    | .178           | .323     | .144         | .969         |
|              | Countdown  | .097           | .480     | +.383    | .014           | .465     | .140         | .672         |
|              | Python     | .000           | .172     | +.172    | .000           | .172     | .015         | .528         |
|              | MathIR     | .220           | .512     | +.292    | .037           | .464     | .364         | .796         |
|              | PantryPlan | .902           | .522     | −.380    | .344           | .522     | .123         | .729         |
| Falcon3-1B   | Graph      | .693           | .656     | −.037    | .178           | .300     | .777         | .809         |
|              | Countdown  | .225           | .328     | +.103    | .037           | .268     | .400         | .636         |
|              | Python     | .000           | .000     | +.000    | .000           | .000     | .009         | .979         |
|              | MathIR     | .148           | .095     | −.053    | .023           | .068     | .158         | .736         |
|              | PantryPlan | .891           | .633     | −.258    | .318           | .633     | .523         | .751         |
| Qwen2.5-3B   | Graph      | .607           | .682     | +.075    | .215           | .360     | .706         | .836         |
|              | Countdown  | .184           | .609     | +.426    | .049           | .569     | .281         | .746         |
|              | Python     | .000           | .069     | +.069    | .000           | .069     | .011         | .659         |
|              | MathIR     | .156           | .278     | +.122    | .061           | .168     | .152         | .686         |
|              | PantryPlan | .831           | .638     | −.193    | .267           | .558     | .482         | .744         |

**Sparse fresh reward information.** Initial pass@8 is zero on Python at all three scales. In Python at Falcon3-1B, only a fraction 0.009 of updates contain mixed rewards, leaving zero fresh task advantage in 99.1% of updates. A learning-rate change cannot make an identically zero current task gradient nonzero, but it can change the effect of informative updates, optimizer momentum, and subsequent sampling. The Falcon3-1B Python control’s terminal pass@8 rounds to .000, while Re:Dr ends at .979. This is an accuracy contrast in a sparse-success setting; conditional mode diversity requires at least two correct responses per prompt and is undefined when that sampling condition is unmet.

**Scope of the comparison.** Replay and control share the learning rate, schedule, initialization, prompt stream, optimizer, update count, and fresh-rollout budget. The comparison measures the effect of replay under these settings. Per-response correctness increases in the controls, but the absence of a learning-rate sweep limits conclusions about their optimal performance. Other learning rates could change accuracy, diversity, and the relative benefit of replay; the present results do not determine those changes.

The frequency of mixed groups helps characterize these comparisons. When a control rarely produces mixed groups (App. H.4), replay provides a likelihood gradient on updates with no fresh task advantage. We therefore also summarize replay effects in subsets with increasing control accuracy. Across all 15 model–domain combinations, the mean Re:Dr minus Dr.GRPO effect is +.352 in pass@8 and +.189 in PCMD. Among the 9 combinations with any positive initial-to-final change in control pass@8, the effects are +.379 and +.118. Among the 6 that also have nonzero initial pass@8 and at least 10% mixed rollout groups, they are +.247 and +.136. Every comparison in these subsets has positive effects on both metrics; the smallest effects in the six-comparison subset are +.137 in pass@8 and +.013 in PCMD. Its mean diversity effect is about two thirds of the mean across all comparisons. Table 23 gives the control measurements that define these subsets.

## H.6 LIKELIHOOD CHANGES WITH A FIXED REPLAY BUFFER

We examine the likelihoods of stored responses during Qwen2.5-0.5B Re:Dr training on Graph, using five seeds. The buffer contents and counts of fresh verified responses are held fixed from immediately before the fresh rollout at step 384. Uniform replay continues through step 3,072 without adding modes. For each stored exemplar, identified by prompt and canonical mode, we measure mean token log probability  $s$  and sequence log probability  $\ell$  whenever it is replayed. Changes compare its first and final replay visits after the buffer is fixed; these steps can differ across exemplars.

The 3,406 exemplars span 1,575 seed-prompt pairs, with 8–9 replay visits per exemplar. The pooled median change is  $+0.038$  nat/token (95% interval  $[+0.025, +0.048]$ ), while the 10th percentile is  $-0.256$  and 2.61% of exemplars lose more than  $.5$  nat/token ( $[1.54, 3.77]\%$ ). The largest observed intermediate decrease is  $-1.328$  nat/token. Sequence log probability has median change  $+0.154$  nat/sequence ( $[+0.104, +0.198]$ ), with 21.99% of exemplars losing more than  $.5$  nat/sequence. Lengths range from 4 to 100 tokens, so the token and sequence thresholds measure different changes.

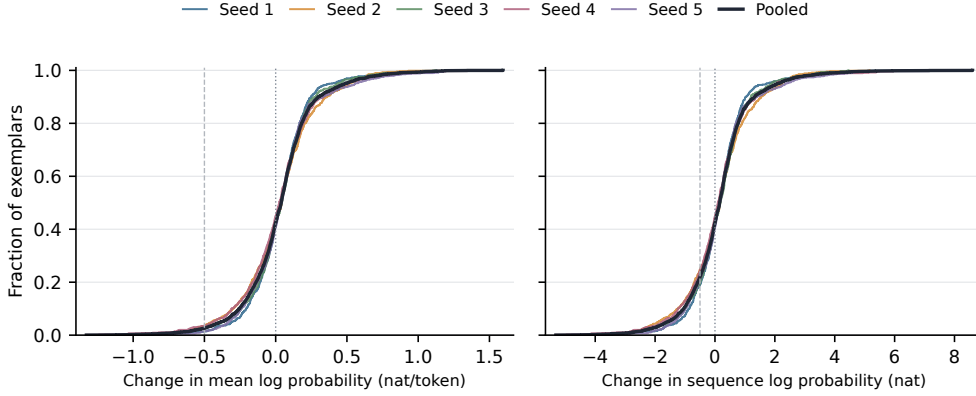


Figure 29: **Median exemplar scores rise, while some exemplars decline.** Empirical cumulative distributions of first-to-final score changes for 3,406 fixed-buffer exemplars from five Qwen2.5-0.5B Re:Dr Graph seeds. Panels show mean token and sequence log probability. Colored curves represent individual seeds; the pooled curve weights exemplars equally. Dotted and dashed vertical lines mark zero and  $-0.5$ , respectively. Positive changes indicate higher likelihood for the stored token sequence. Sequence likelihood does not measure the total probability of generating a canonical solution mode.

Pooled summaries weight exemplars equally. Descriptive 95% percentile intervals use 10,000 hierarchical bootstrap replicates, resampling prompts with replacement within each training seed and then resampling the five seeds with replacement.

Scores are measured at replay visits, so changes between visits are unobserved. Some stored exemplars lose likelihood despite continued replay. A matched fixed-buffer control without replay would be needed to isolate the effect of replay. These sequence-level observations also do not test the mode-probability floor in Theorem P.22.

## Part IV. What diversity represents

### I MATCHED PROMPT-HINT ABLATION

#### I.1 REMOVING STRATEGY HINTS FROM UNCHANGED PROBLEMS

We compare original and neutral prompts on `Python`, `MathIR`, and `PantryPlan` at Levels 2 and 3. Each of the six domain-level combinations uses 32 problems selected independently of model outputs from the 128-problem evaluation split. We sample eight responses per problem under each wording. The original wording uses the standard prompt. The neutral wording removes the `Python` nested-conditional example and suggestions to test small divisors or dispatch on listed values, the `MathIR` ordered algebraic-isolation suggestion, and the `PantryPlan` preference for ingredients high in energy and protein but low in sodium such as seeds or oats. `MathIR` changes “those operations” to “the operations” to preserve the instruction to return a menu ID. The problem instance, answer specification, executable verifier, and remaining format requirements are unchanged.

The paired effects are neutral minus original on empirical  $P_8 = \text{pass@8}$  and  $D_8 = \text{distinct@8}$ , with  $B_8 = D_8 - P_8$  as extra verified modes per problem. Incorrect, empty, and truncated outputs remain in their eight-draw groups; only verifier-accepted outputs contribute a verified key. Results use strict verification. A sensitivity analysis with normalized formatting preserves every strict success and canonical key and applies the same rules to both arms.

Intervals use 20,000 whole-problem bootstrap resamples, paired across arms and stratified by domain and level. Domain-level combinations receive equal weight in aggregate estimates. The intervals are pointwise 95% percentile intervals without multiplicity adjustment. With 32 problems per domain and level, an interval crossing zero does not establish equivalence. A zero-width interval reflects zero variation in the observed bootstrap resamples, including identical paired  $P_8$  outcomes, and does not establish population equivalence. Correct-pair collision and its certified-support uniform reference use the same correct-pair weights. Eight draws per prompt do not identify total unseen support.

We evaluate the initial Qwen2.5-0.5B-Instruct checkpoint and Level-2-trained Dr.GRPO and Re:Dr checkpoints. `Python` and `MathIR` use five matched training seeds; `PantryPlan` uses two. Level 3 tests transfer from Level 2 training. All checkpoints evaluate the same selected problems under the benchmark syntax constraints and a 192-token generation budget. Five-seed intervals resample whole training seeds and paired problems; `PantryPlan` intervals condition on its two checkpoints, and initial-model intervals condition on one instruction-tuned checkpoint. These effects concern the specified models, problems, and interface.

PCMD is reported for five of the 18 comparisons in Table 24. Each requires at least thirty checkpoint-problem groups with at least two verified draws under each wording. The same 32 problems are evaluated at each checkpoint, so these group counts include repeated problems across training seeds. Within a comparison, PCMD weights eligible groups equally; checkpoints with more eligible problems therefore receive more weight. The population depends on success under both wordings and need not represent all evaluated problems. Three eligible comparisons are `MathIR`: all observed verified pairs within each eligible group share a canonical key, giving PCMD = .000 under both wordings. This observation does not limit the number of valid keys. In the two Re:Dr comparisons on `Python`, hint removal changes PCMD from .231 to .173 at Level 2 and from .256 to .323 at Level 3, on 66 and 57 eligible checkpoint-problem groups, respectively. The two point estimates have opposite signs; without intervals, they do not establish a shared direction or the reliability of either concentration effect. At Level 3, conditional diversity uses four training seeds, while  $P_8$  and  $D_8$  use all five; the fifth seed has too few jointly verified responses. A dash denotes an unavailable conditional estimate, not a zero effect.

Formatting normalization adds no verified responses for any Qwen2.5-0.5B checkpoint in this panel; all tabulated estimates and intervals therefore agree under the two grading rules.

Table 24: **Removing Python guidance reduces correctness and observed breadth.** Qwen2.5-0.5B checkpoints under strict verification, with original  $\rightarrow$  neutral wording.  $P_8 = \text{pass@8}$ ,  $D_8 = \text{distinct@8}$ , and  $B_8 = D_8 - P_8$  use eight draws on each of 32 problems per domain and level. Effects are neutral minus original.  $P_8$ ,  $D_8$ , and  $B_8$  average the displayed number  $s$  of checkpoints. Brackets are pointwise 95% bootstrap intervals; paired problems and training seeds are resampled for five-seed groups; PantryPlan intervals condition on its two checkpoints, and initial-model intervals on one checkpoint. PCMD estimates the probability that two correct responses have different canonical keys, averaging checkpoint–problem groups with at least two verified draws in each arm. Its paired population depends on correctness, and repeated problems at different checkpoints count separately. Only 5 of 18 comparisons meet the minimum of 30 such groups; dashes denote the others. These PCMD point estimates have no reported uncertainty intervals.

| Model               | Domain/level  | $P_8$ : O $\rightarrow$ N | $D_8$ : O $\rightarrow$ N | $\Delta P_8$ [95%]      | $\Delta D_8$ [95%]   | $\Delta B_8$ | PCMD: O $\rightarrow$ N | $\Delta$ PCMD |
|---------------------|---------------|---------------------------|---------------------------|-------------------------|----------------------|--------------|-------------------------|---------------|
| Initial ( $s = 1$ ) | Python L2     | 0.656 $\rightarrow$ 0.125 | 0.69 $\rightarrow$ 0.12   | -0.531 [-0.719, -0.344] | -0.56 [-0.75, -0.38] | -0.03        | -                       | -             |
| Initial ( $s = 1$ ) | MathIR L2     | 0.250 $\rightarrow$ 0.281 | 0.25 $\rightarrow$ 0.28   | +0.031 [+0.000, +0.094] | +0.03 [+0.00, +0.09] | +0.00        | -                       | -             |
| Initial ( $s = 1$ ) | PantryPlan L2 | 0.156 $\rightarrow$ 0.188 | 0.19 $\rightarrow$ 0.22   | +0.031 [+0.000, +0.094] | +0.03 [+0.00, +0.09] | +0.00        | -                       | -             |
| Initial ( $s = 1$ ) | Python L3     | 0.719 $\rightarrow$ 0.031 | 0.81 $\rightarrow$ 0.03   | -0.688 [-0.844, -0.531] | -0.78 [-0.97, -0.59] | -0.09        | -                       | -             |
| Initial ( $s = 1$ ) | MathIR L3     | 0.219 $\rightarrow$ 0.219 | 0.22 $\rightarrow$ 0.22   | +0.000 [-0.125, +0.125] | +0.00 [-0.12, +0.12] | +0.00        | -                       | -             |
| Initial ( $s = 1$ ) | PantryPlan L3 | 0.281 $\rightarrow$ 0.312 | 0.66 $\rightarrow$ 0.69   | +0.031 [+0.000, +0.094] | +0.03 [-0.06, +0.12] | +0.00        | -                       | -             |
| Dr.GRPO ( $s = 5$ ) | MathIR L2     | 0.531 $\rightarrow$ 0.506 | 0.53 $\rightarrow$ 0.51   | -0.025 [-0.100, +0.031] | -0.03 [-0.10, +0.03] | +0.00        | .000 $\rightarrow$ .000 | +0.000        |
| Dr.GRPO ( $s = 5$ ) | MathIR L3     | 0.194 $\rightarrow$ 0.156 | 0.19 $\rightarrow$ 0.16   | -0.037 [-0.113, +0.019] | -0.04 [-0.11, +0.02] | +0.00        | -                       | -             |
| Dr.GRPO ( $s = 2$ ) | PantryPlan L2 | 0.219 $\rightarrow$ 0.203 | 0.28 $\rightarrow$ 0.25   | -0.016 [-0.156, +0.125] | -0.03 [-0.19, +0.12] | -0.02        | -                       | -             |
| Dr.GRPO ( $s = 2$ ) | PantryPlan L3 | 0.172 $\rightarrow$ 0.172 | 0.27 $\rightarrow$ 0.25   | +0.000 [-0.078, +0.078] | -0.02 [-0.16, +0.11] | -0.02        | -                       | -             |
| Dr.GRPO ( $s = 5$ ) | Python L2     | 0.812 $\rightarrow$ 0.000 | 0.81 $\rightarrow$ 0.00   | -0.812 [-0.938, -0.656] | -0.81 [-0.94, -0.66] | +0.00        | -                       | -             |
| Dr.GRPO ( $s = 5$ ) | Python L3     | 1.000 $\rightarrow$ 0.000 | 1.00 $\rightarrow$ 0.00   | -1.000 [-1.000, -1.000] | -1.00 [-1.00, -1.00] | +0.00        | -                       | -             |
| Re:Dr ( $s = 5$ )   | MathIR L2     | 0.994 $\rightarrow$ 0.975 | 0.99 $\rightarrow$ 0.97   | -0.019 [-0.075, +0.000] | -0.02 [-0.07, +0.00] | +0.00        | .000 $\rightarrow$ .000 | +0.000        |
| Re:Dr ( $s = 5$ )   | MathIR L3     | 0.312 $\rightarrow$ 0.275 | 0.31 $\rightarrow$ 0.28   | -0.037 [-0.138, +0.044] | -0.04 [-0.14, +0.04] | +0.00        | .000 $\rightarrow$ .000 | +0.000        |
| Re:Dr ( $s = 2$ )   | PantryPlan L2 | 0.516 $\rightarrow$ 0.500 | 0.75 $\rightarrow$ 0.73   | -0.016 [-0.109, +0.078] | -0.02 [-0.20, +0.17] | +0.00        | -                       | -             |
| Re:Dr ( $s = 2$ )   | PantryPlan L3 | 0.234 $\rightarrow$ 0.281 | 0.39 $\rightarrow$ 0.50   | +0.047 [+0.000, +0.109] | +0.11 [+0.05, +0.19] | +0.06        | -                       | -             |
| Re:Dr ( $s = 5$ )   | Python L2     | 0.887 $\rightarrow$ 0.556 | 1.24 $\rightarrow$ 0.68   | -0.331 [-0.525, -0.150] | -0.57 [-0.85, -0.32] | -0.24        | .231 $\rightarrow$ .173 | -.058         |
| Re:Dr ( $s = 5$ )   | Python L3     | 1.000 $\rightarrow$ 0.494 | 1.42 $\rightarrow$ 0.66   | -0.506 [-0.825, -0.194] | -0.76 [-1.07, -0.38] | -0.26        | .256 $\rightarrow$ .323 | +.067         |

**Two-seed PantryPlan effects.** Intervals condition on the two checkpoints and do not quantify training-seed variability. The effects for each seed are shown below; each pair gives  $(\Delta P_8, \Delta D_8)$ .

| Method  | Level | First seed      | Second seed     |
|---------|-------|-----------------|-----------------|
| Dr.GRPO | 2     | (+0.000, +0.03) | (-0.031, -0.09) |
| Dr.GRPO | 3     | (+0.031, +0.06) | (-0.031, -0.09) |
| Re:Dr   | 2     | (-0.031, -0.06) | (+0.000, +0.03) |
| Re:Dr   | 3     | (+0.031, +0.12) | (+0.062, +0.09) |

**Scope of the wording intervention.** The Python intervention removes an executable nested-conditional example along with divisor-testing and dispatch guidance, and it shortens the prompt. Its effect concerns this combined change; it does not isolate a preference cue from example or length effects. Lower  $\text{distinct@8}$  together with lower  $\text{pass@8}$  can reflect reduced correctness rather than increased concentration among correct outputs. The PCMD columns in Table 24 measure conditional diversity on jointly eligible checkpoint–problem groups. Eligibility itself depends on correctness, so these comparisons do not hold the full evaluation population or its accuracy fixed.

**Observed prompt dependence.** Python shows a large loss of correctness after guidance removal. Across the five Dr.GRPO seeds, mean  $P_8$  falls from .8125 to zero at Level 2 and from 1.0 to zero at Level 3;  $D_8$  falls identically and  $B_8 = 0$  in both arms. With no correct neutral Dr.GRPO outputs, its concentration among correct outputs is undefined. Re:Dr retains neutral  $P_8 = .55625$  and  $.49375$  and  $D_8 = .675$  and  $.65625$  at Levels 2 and 3, respectively, while losing extra verified modes:  $\Delta B_8 = -.2375$  and  $-.25625$ . The initial checkpoint also loses Python correctness (Table 24). These observations identify substantial dependence on the combined Python wording intervention; a raw mode-count decline cannot be interpreted solely as a redistribution among correct answers.

**Python responses after hint removal.** Across 5,120 Dr.GRPO Python responses from five seeds, two levels, and both wordings, all 2,560 outputs generated with the original wording yield the same extracted lambda expression, whose body matches the conditional example in the prompt; 2,320 of these responses pass verification:

```
lambda n: 2 if n % 2 == 0 else 3 if n % 3 == 0 else 5
```

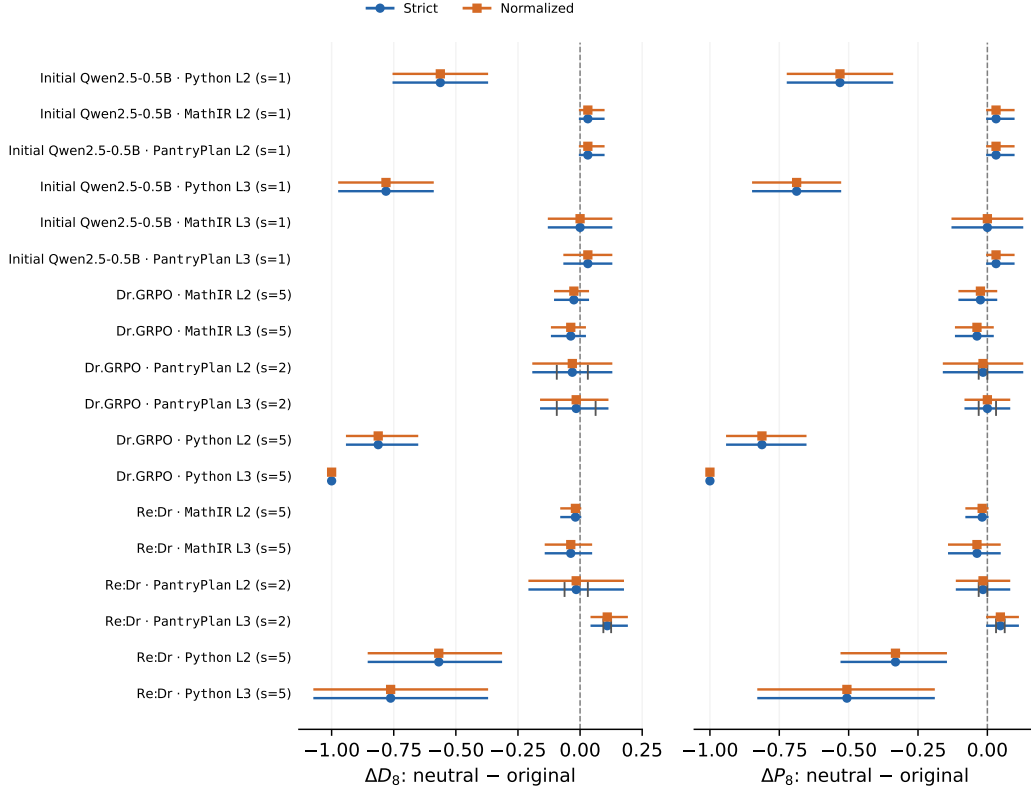


Figure 30: **Python guidance removal produces the largest wording effects in this panel.** Neutral-minus-original  $D_8$  (left) and  $P_8$  (right) for the initial Qwen2.5-0.5B checkpoint and Level-2-trained Dr.GRPO and Re:Dr on Python, MathIR, and PantryPlan at Levels 2 and 3, with 32 paired problems and eight draws per arm. Blue circles use strict verification; orange squares use formatting normalization. Horizontal bars are pointwise 95% bootstrap intervals; the vertical dashed line marks zero. Five-seed groups resample training seeds and paired problems; PantryPlan and initial-model intervals condition on two checkpoints and one checkpoint, respectively. Gray marks show the individual effects for the two PantryPlan training seeds.

Of the 2,560 neutral-wording responses, 1,816 (70.94%) are invalid Python expressions, 59 violate other restricted-language rules, 376 return non-integers, 276 fail the proper-divisor condition, and 33 (1.29%) have no extracted candidate after reaching the token limit. Syntax errors are the largest failure category. The categories follow the first rejection during extraction, parsing, and execution. They describe output failures without identifying the training mechanism responsible.

**Concentration after hint removal.** Every trained MathIR checkpoint has  $B_8 = 0$  under both wordings at both levels. Within each problem’s eight-draw group, every observed pair of correct outputs shares a canonical key at every trained checkpoint and level: correct-pair collision is 1.0, versus the uniform reference of .2 over five certified modes. Table 24 reports PCMD .000 under both wordings in every eligible MathIR comparison. Removing this hint therefore leaves observed correct outputs concentrated. At Level 2, Re:Dr retains mean  $P_8 = .975$  without the order hint, compared with .99375 originally. Across its five checkpoints, the neutral arm has 4,179 colliding pairs out of 4,179 correct pairs, from 154 eligible checkpoint–problem groups; the original arm has 4,382 out of 4,382 pairs from 157 such groups. Both arms evaluate the same 32 problems at each checkpoint. Thus observed concentration persists after removing the order hint. The comparison does not measure unseen modes, and the remaining interface constraints still apply. PantryPlan effects are mixed. Re:Dr at Level 3 gains .109375 in  $D_8$ , comprising .046875 in  $P_8$  and .0625 in  $B_8$ , but this comparison has only two training seeds. App. K evaluates hosted models under both wordings with 64 draws on sixteen problems per domain–level combination. Its problem population and sampling budget differ from the eight-draw evaluation here.

## J DECODING TEMPERATURE, NUCLEUS, AND BUDGET

**Temperature at fixed checkpoints.** Qwen2.5-0.5B Dr.GRPO and Re:Dr checkpoints after eight training passes are evaluated at  $T \in \{.5, .7, 1.0, 1.3, 1.6, 2.0\}$ , with  $\text{top-}p = 1$  and four eight-response groups per prompt. Five domains, two methods, and five seeds give 300 checkpoint-setting evaluations (Fig. 31A). The prompt template, evaluation split, response limit, and sampling seeds are fixed across temperatures. Correct keys are pooled across response groups for each prompt; the groups have overlapping sampling streams (App. H.2). Estimates therefore describe these sampled pools. A seed contributes to the PCMD mean when at least 30 prompts yield at least two correct responses. Eligible prompts and seeds can differ across temperatures and methods.

Across the six temperatures, the largest control PCMD is .021 in Graph, .016 in Countdown, .002 in MathIR, and .004 in PantryPlan. Re:Dr at  $T = 1$  has corresponding means .562, .485, .020, and .330. These four domains retain five eligible control seeds at every temperature. Python controls have no seed reaching the 30-prompt threshold, so their conditional diversity is unidentified under this reporting rule. The sweep therefore shows low measured control diversity in the four eligible domains, without establishing that unobserved modes have zero probability or that every decoding choice would behave the same way.

**Hardware sensitivity.** All evaluations in the temperature sweep use NVIDIA RTX A5000 GPUs. A paired comparison of A100 and A5000 evaluations at  $T = 1$  gives mean absolute PCMD difference .0032 across twelve comparisons, with maximum .0131. These differences describe the tested hardware comparisons; they do not bound hardware sensitivity at other checkpoints or settings.

**Larger groups and nucleus truncation.** A separate set of Qwen2.5-0.5B checkpoints trained for twelve passes is evaluated at eleven settings (Fig. 31B): the six temperatures above; two groups of  $K = 32$  at  $T \in \{1.0, 1.3, 1.6\}$ ; and four groups of  $K = 8$  with  $\text{top-}p = .95$  at  $T \in \{1.0, 1.6\}$ . The  $K = 32$  settings quadruple the group size and double total responses per prompt from 32 to 64. Five domains, two methods, and five seeds give 550 checkpoint-setting evaluations. The replay variant combines separate mass and within-buffer balancing terms with novelty and semantic-entropy shaping. Its objective and training duration therefore differ from the eight-pass Re:Dr comparison.

Control PCMD is zero throughout the grid in Graph. Its range is .006 to .031 in Countdown, .000 to .003 in MathIR, and .037 to .069 in PantryPlan. All eleven settings retain five eligible control seeds in these domains; Python controls remain below the reporting threshold. The replay markers provide a descriptive reference for this separate variant, without identifying the effect of the Re:Dr objective.

Three comparisons hold temperature and nucleus sampling fixed while increasing both group size and total response budget:  $K = 8$  and  $K = 32$  at  $T \in \{1.0, 1.3, 1.6\}$ ,  $\text{top-}p = 1$ . Across these matched settings, control PCMD changes by at most .010 in absolute value, and no  $K = 32$  control mean exceeds .054. Graph stays at .000; Countdown moves from .014–.027 to .013–.023; MathIR from .000–.001 to .001–.003; and PantryPlan from .050–.063 to .045–.053. Each mean uses five eligible seeds. The larger budget therefore does not move these controls out of their low-PCMD range. This compares sampled pools at fixed checkpoints; it does not establish that unobserved modes carry zero probability.

**Scope of the decoding comparison.** Both grids evaluate fixed terminal policies at one model scale. They do not measure the effects of changing rollout temperature, optimizer settings, or training duration. Correctness and conditional eligibility can change with decoding settings; a missing PCMD estimate indicates insufficient eligible prompts, rather than disappearance of all correct solution modes.

## K SAMPLING-BUDGET ABLATION

### K.1 SAMPLING DESIGN AND ESTIMANDS

Large-budget  $\text{pass@k}$  evaluation can reveal differences hidden at smaller sampling budgets (Yue et al., 2025). We select sixteen problems independently of model outputs for each of Python, MathIR, and PantryPlan at Levels 2 and 3, using the prompt-hint ablation popu-

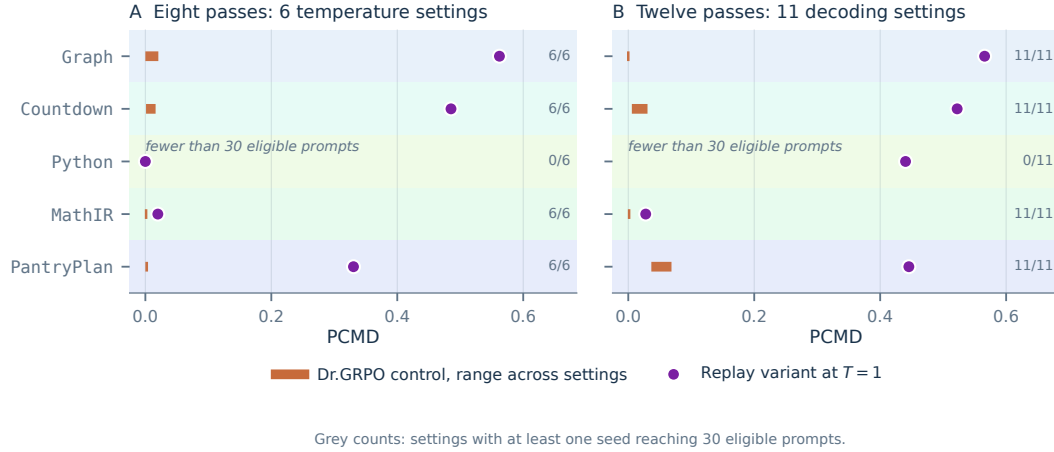


Figure 31: **Control diversity remains low across the tested decoding settings.** Qwen2.5-0.5B terminal PCMD across five domains. (A) Eight-pass Dr.GRPO/Re:Dr checkpoints at six temperatures ( $T = .5-2.0$ ): 300 checkpoint–setting evaluations. (B) Separate twelve-pass checkpoints at eleven temperature, group-size, and nucleus settings: 550 evaluations, using a different replay objective. Bars span control means across settings; they are not confidence intervals. Markers show the corresponding replay variant at  $T = 1$ , top- $p = 1$ , and  $K = 8$ . Means use seeds with at least 30 prompts having at least two correct responses, so control and replay populations need not match. Control bars use five seeds outside Python; Python controls have none, while its replay markers use two seeds in A and three in B. Counts give eligible settings. Ranges narrower than .004 are displayed at width .004 for visibility.

lation. Each problem has original and neutral wordings, with 64 responses per wording from each checkpoint or deployment. Eight-draw estimates use subsets of these same response pools. The initial Qwen2.5-0.5B-Instruct checkpoint has no training-seed replication. Dr.GRPO and Re:Dr use five matched seeds in Python and MathIR and two in PantryPlan. These checkpoints are trained at Level 2, so Level 3 measures transfer.

For a 64-draw pool, let  $c$  be the number of correct outputs and  $n_j$  the count of verified canonical key  $j$ . At  $k \in \{1, 2, 4, 8, 16, 32, 64\}$ , we average over all size- $k$  subsets without replacement:

$$P_k = 1 - \frac{\binom{64-c}{k}}{\binom{64}{k}}, \quad D_k = \sum_j \left[ 1 - \frac{\binom{64-n_j}{k}}{\binom{64}{k}} \right], \quad B_k = D_k - P_k.$$

Here  $P_k$  is the probability of at least one correct response,  $D_k$  is the expected number of distinct verified modes, and  $B_k$  counts modes beyond the first success. Infeasible numerator combinations are zero. Incorrect, empty, refused, and truncated responses remain in the pool; only verifier-accepted responses contribute keys. The curve points are correlated because they share a response pool and describe finite-sample discovery. The gains  $G_P = P_{64} - P_8$ ,  $G_D = D_{64} - D_8$ , and  $G_B = B_{64} - B_8$  distinguish improvements in first-success probability from the discovery of additional modes; they do not estimate complete unseen support.

We report strict-verifier results and a formatting-normalization sensitivity analysis that preserves every strict success and canonical key. Intervals use 20,000 whole-problem bootstrap replicates, paired across wordings and stratified by domain and level. Five-seed intervals also resample matched training seeds with shared problem indices; PantryPlan intervals condition on its two checkpoints. Each domain–level comparison contains sixteen problems, and intervals are unadjusted for multiple comparisons. Zero-width intervals indicate no observed resampled variation and do not establish equivalence.

For breadth at a fixed number of correct draws, we average over subsets of  $m$  correct responses:

$$R_m = \sum_j \left[ 1 - \frac{\binom{c-n_j}{m}}{\binom{c}{m}} \right].$$

Paired comparisons require  $c \geq m$  under both wordings. They therefore condition on observed success, without equating accuracy or estimating diversity for excluded problems. Eligibility varies with  $m$ , method, and grading. Trained-checkpoint means weight seeds equally

and are undefined if any seed lacks eligible prompts. Conditional intervals retain only bootstrap replicates in which the corresponding mean is defined.

The 64-draw summaries use two prompt weightings. Pooled collision  $C$  divides the total number of equal-key correct pairs by the total number of correct pairs within each checkpoint or deployment; a prompt therefore receives weight proportional to  $\binom{c}{2}$ . Promptwise PCMD averages  $1 - \sum_j \binom{n_j}{2} / \binom{c}{2}$  equally over prompts with  $c \geq 2$ . Each wording’s absolute estimate uses its own eligible prompts, whereas the neutral-minus-original contrast  $\Delta_{\text{PCMD}}$  uses only jointly eligible prompts. Trained-checkpoint estimates then receive equal weight across seeds.

Support references use certified lower bounds  $L \leq M$ : two valid modes in `Python`, five in `MathIR`, and the problem-specific certificate count in `PantryPlan`. Thus  $1/L$  is an upper reference for collision under a uniform distribution over full support, and  $L[1 - (1 - 1/L)^m]$  is a lower reference for uniform expected breadth. These references describe uniform sampling; they do not bound model breadth, and observed distinct counts may exceed  $L$ .

In the tables, O/N denotes original/neutral wording, and  $\Delta G_D$  is the neutral-minus-original discovery gain.  $U$  denotes the uniform reference under the corresponding weighting.  $E_O/E_N$  count separately eligible checkpoint–problem pairs with  $c \geq 2$ ;  $Q_O/Q_N$  count correct response pairs. The counts sum across checkpoints, whereas the estimates weight checkpoints equally. Separate eligibility counts do not give the size of the jointly eligible population used for a contrast.

Table 25: **Discovery gains from eight to 64 draws.** Strict verification; O/N denotes original/neutral wording. Each row summarizes sixteen problems for one deployment, domain, and level. Intervals resample paired problems and condition on the deployment; they do not include variation across model training seeds.

| Model       | Domain/level  | $G_P$ : O/N | $G_D$ : O/N | $G_B$ : O/N | $\Delta G_D$ [95%]      |
|-------------|---------------|-------------|-------------|-------------|-------------------------|
| GPT-5.4     | Python L2     | 0.014/0.000 | 2.017/3.051 | 2.003/3.051 | +1.033 [−0.044, +2.371] |
| GPT-5.6 Sol | Python L2     | 0.000/0.405 | 0.000/0.866 | 0.000/0.462 | +0.866 [+0.545, +1.231] |
| Grok 4.3    | Python L2     | 0.000/0.000 | 0.667/1.013 | 0.667/1.013 | +0.346 [−0.583, +1.128] |
| GPT-5.4     | Python L3     | 0.025/0.000 | 2.100/2.340 | 2.076/2.340 | +0.240 [−0.379, +0.930] |
| GPT-5.6 Sol | Python L3     | 0.000/0.586 | 0.000/0.759 | 0.000/0.174 | +0.759 [+0.581, +0.952] |
| Grok 4.3    | Python L3     | 0.000/0.000 | 0.424/1.576 | 0.424/1.576 | +1.152 [+0.640, +1.689] |
| GPT-5.4     | MathIR L2     | 0.043/0.042 | 0.043/0.909 | 0.000/0.868 | +0.867 [+0.587, +1.111] |
| GPT-5.6 Sol | MathIR L2     | 0.027/0.004 | 0.027/0.686 | 0.000/0.682 | +0.659 [+0.416, +0.888] |
| Grok 4.3    | MathIR L2     | 0.000/0.001 | 0.223/0.322 | 0.223/0.321 | +0.099 [−0.226, +0.438] |
| GPT-5.4     | MathIR L3     | 0.000/0.000 | 0.055/0.663 | 0.055/0.663 | +0.609 [+0.342, +0.888] |
| GPT-5.6 Sol | MathIR L3     | 0.000/0.000 | 0.000/0.515 | 0.000/0.515 | +0.515 [+0.318, +0.714] |
| Grok 4.3    | MathIR L3     | 0.000/0.000 | 0.167/0.805 | 0.167/0.805 | +0.638 [+0.348, +0.935] |
| GPT-5.4     | PantryPlan L2 | 0.001/0.015 | 1.158/2.583 | 1.156/2.568 | +1.426 [+0.423, +2.641] |
| GPT-5.6 Sol | PantryPlan L2 | 0.086/0.101 | 0.917/1.789 | 0.831/1.688 | +0.872 [+0.245, +1.575] |
| Grok 4.3    | PantryPlan L2 | 0.000/0.000 | 1.022/3.377 | 1.022/3.377 | +2.354 [+1.592, +3.123] |
| GPT-5.4     | PantryPlan L3 | 0.000/0.002 | 1.666/3.219 | 1.666/3.218 | +1.553 [+0.776, +2.367] |
| GPT-5.6 Sol | PantryPlan L3 | 0.032/0.129 | 1.625/1.842 | 1.592/1.713 | +0.217 [−0.489, +0.921] |
| Grok 4.3    | PantryPlan L3 | 0.000/0.000 | 1.810/4.466 | 1.810/4.466 | +2.656 [+1.698, +3.640] |

## K.2 HOSTED-MODEL DISCOVERY: WORDING, FORMATTING, AND ADDITIONAL DRAWS

The hosted evaluation contains 36,864 responses from GPT-5.4, GPT-5.6 Sol, and Grok 4.3. Each deployment receives sixteen problems from each of `Python`, `MathIR`, and `PantryPlan` at Levels 2 and 3, with two wordings and 64 draws per wording. All three deployments use medium reasoning effort and an 8,192-token output cap. These settings differ from the reasoning-disabled main-table evaluation. The `Python` comparisons use the same mathematical problems under both wordings. The hosted models receive no task-specific training in this evaluation.

**Neutral wording broadens the GPT models’ correct `MathIR` solutions.** Both GPT deployments yield at most one verified mode per Level-2 problem in 64 original-wording draws. Among problems with at least eight correct responses under both wordings, neutral wording increases mean breadth at eight correct draws by .794 ([.623, .990]) for GPT-5.4 and .872 ([.606, 1.123]) for GPT-5.6 Sol. The Level-3 effects are .204 ([.106, .320]) and .860 ([.563, 1.141]). These comparisons include fourteen and fifteen jointly eligible problems at Level 2, respectively, and sixteen for each GPT deployment at Level 3. Grok 4.3’s corresponding intervals include zero at both levels. The pointwise

Table 26: **Discovery gains from eight to 64 draws.** Formatting normalization; O/N denotes original/neutral wording. Each row summarizes sixteen problems for one deployment, domain, and level. Intervals resample paired problems and condition on the deployment; they do not include variation across model training seeds.

| Model       | Domain/level  | $G_P$ : O/N | $G_D$ : O/N | $G_B$ : O/N | $\Delta G_D$ [95%]      |
|-------------|---------------|-------------|-------------|-------------|-------------------------|
| GPT-5.4     | Python L2     | 0.000/0.000 | 2.156/3.216 | 2.156/3.216 | +1.060 [-0.071, +2.386] |
| GPT-5.6 Sol | Python L2     | 0.008/0.010 | 0.113/0.794 | 0.105/0.784 | +0.681 [+0.351, +1.066] |
| Grok 4.3    | Python L2     | 0.000/0.000 | 0.667/1.013 | 0.667/1.013 | +0.346 [-0.583, +1.128] |
| GPT-5.4     | Python L3     | 0.000/0.000 | 2.431/2.604 | 2.431/2.604 | +0.173 [-0.657, +1.075] |
| GPT-5.6 Sol | Python L3     | 0.011/0.007 | 0.120/0.490 | 0.109/0.483 | +0.370 [+0.112, +0.674] |
| Grok 4.3    | Python L3     | 0.000/0.000 | 0.424/1.576 | 0.424/1.576 | +1.152 [+0.640, +1.689] |
| GPT-5.4     | MathIR L2     | 0.043/0.042 | 0.043/0.909 | 0.000/0.868 | +0.867 [+0.587, +1.111] |
| GPT-5.6 Sol | MathIR L2     | 0.027/0.004 | 0.027/0.686 | 0.000/0.682 | +0.659 [+0.416, +0.888] |
| Grok 4.3    | MathIR L2     | 0.000/0.001 | 0.223/0.322 | 0.223/0.321 | +0.099 [-0.226, +0.438] |
| GPT-5.4     | MathIR L3     | 0.000/0.000 | 0.055/0.663 | 0.055/0.663 | +0.609 [+0.342, +0.888] |
| GPT-5.6 Sol | MathIR L3     | 0.000/0.000 | 0.000/0.515 | 0.000/0.515 | +0.515 [+0.318, +0.714] |
| Grok 4.3    | MathIR L3     | 0.000/0.000 | 0.167/0.805 | 0.167/0.805 | +0.638 [+0.348, +0.935] |
| GPT-5.4     | PantryPlan L2 | 0.001/0.015 | 1.158/2.583 | 1.156/2.568 | +1.426 [+0.423, +2.641] |
| GPT-5.6 Sol | PantryPlan L2 | 0.004/0.004 | 0.954/1.834 | 0.950/1.830 | +0.880 [+0.237, +1.547] |
| Grok 4.3    | PantryPlan L2 | 0.000/0.000 | 1.022/3.377 | 1.022/3.377 | +2.354 [+1.592, +3.123] |
| GPT-5.4     | PantryPlan L3 | 0.000/0.002 | 1.666/3.219 | 1.666/3.218 | +1.553 [+0.776, +2.367] |
| GPT-5.6 Sol | PantryPlan L3 | 0.001/0.020 | 1.586/2.462 | 1.585/2.443 | +0.877 [+0.272, +1.436] |
| Grok 4.3    | PantryPlan L3 | 0.000/0.000 | 1.810/4.466 | 1.810/4.466 | +2.656 [+1.698, +3.640] |

Table 27: **Pair weighting changes the summary of correct-solution diversity.** Strict 64-draw estimates for hosted deployments. Collision  $C$  weights prompts by their number of correct pairs; PCMD weights eligible prompts equally. Absolute estimates for each wording use their own eligible prompts, whereas  $\Delta_{PCMD}$  uses only jointly eligible prompts. Thus  $\Delta_{PCMD}$  need not equal the difference between the displayed absolute estimates, and  $E_O/E_N$  are separate eligibility counts. Each deployment is fixed; intervals resample paired problems.

| Model       | Domain/level  | $C_O$ | $C_N$ | $PCMD_O$ | $PCMD_N$ | $U_O^C$ | $U_O^{PCMD}$ | $E_O/E_N$ | $Q_O/Q_N$   | $\Delta_{PCMD}$ [95%]   |
|-------------|---------------|-------|-------|----------|----------|---------|--------------|-----------|-------------|-------------------------|
| GPT-5.4     | Python L2     | 0.790 | 0.665 | 0.257    | 0.341    | 0.500   | 0.500        | 16/16     | 10374/21720 | +0.084 [-0.006, +0.186] |
| GPT-5.6 Sol | Python L2     | 1.000 | 0.974 | 0.000    | 0.184    | 0.500   | 0.500        | 3/13      | 4467/6074   | +0.021 [+0.000, +0.032] |
| Grok 4.3    | Python L2     | 0.963 | 0.938 | 0.037    | 0.062    | 0.500   | 0.500        | 16/16     | 32256/32256 | +0.025 [+0.002, +0.048] |
| GPT-5.4     | Python L3     | 0.732 | 0.620 | 0.334    | 0.384    | 0.500   | 0.500        | 16/16     | 6747/18043  | +0.050 [-0.026, +0.128] |
| GPT-5.6 Sol | Python L3     | 1.000 | 0.966 | 0.000    | 0.130    | 0.500   | 0.500        | 1/10      | 1953/2097   | +0.031 [+0.031, +0.031] |
| Grok 4.3    | Python L3     | 0.981 | 0.893 | 0.019    | 0.107    | 0.500   | 0.500        | 16/16     | 32193/32256 | +0.087 [+0.041, +0.154] |
| GPT-5.4     | MathIR L2     | 1.000 | 0.756 | 0.000    | 0.228    | 0.200   | 0.800        | 16/15     | 28440/28227 | +0.228 [+0.161, +0.302] |
| GPT-5.6 Sol | MathIR L2     | 1.000 | 0.678 | 0.000    | 0.302    | 0.200   | 0.800        | 16/15     | 28645/28377 | +0.302 [+0.200, +0.402] |
| Grok 4.3    | MathIR L2     | 0.968 | 0.985 | 0.033    | 0.014    | 0.200   | 0.800        | 16/16     | 31600/29677 | -0.020 [-0.076, +0.017] |
| GPT-5.4     | MathIR L3     | 0.998 | 0.944 | 0.002    | 0.056    | 0.200   | 0.800        | 16/16     | 32256/32256 | +0.054 [+0.027, +0.087] |
| GPT-5.6 Sol | MathIR L3     | 1.000 | 0.676 | 0.000    | 0.324    | 0.200   | 0.800        | 16/16     | 32256/32256 | +0.324 [+0.208, +0.432] |
| Grok 4.3    | MathIR L3     | 0.966 | 0.949 | 0.033    | 0.051    | 0.200   | 0.800        | 16/16     | 30520/32193 | +0.019 [-0.046, +0.064] |
| GPT-5.4     | PantryPlan L2 | 0.680 | 0.499 | 0.252    | 0.483    | 0.069   | 0.934        | 16/16     | 20097/17436 | +0.231 [+0.069, +0.395] |
| GPT-5.6 Sol | PantryPlan L2 | 0.674 | 0.624 | 0.275    | 0.474    | 0.076   | 0.934        | 16/15     | 12941/5001  | +0.180 [+0.058, +0.327] |
| Grok 4.3    | PantryPlan L2 | 0.684 | 0.412 | 0.315    | 0.589    | 0.066   | 0.934        | 16/16     | 32005/32193 | +0.274 [+0.143, +0.411] |
| GPT-5.4     | PantryPlan L3 | 0.546 | 0.392 | 0.431    | 0.610    | 0.065   | 0.937        | 16/16     | 22332/21975 | +0.179 [+0.103, +0.257] |
| GPT-5.6 Sol | PantryPlan L3 | 0.566 | 0.652 | 0.422    | 0.441    | 0.061   | 0.937        | 16/16     | 15229/4946  | +0.019 [-0.136, +0.172] |
| Grok 4.3    | PantryPlan L3 | 0.673 | 0.294 | 0.330    | 0.703    | 0.063   | 0.937        | 16/16     | 31830/31881 | +0.373 [+0.224, +0.523] |

intervals resample the sixteen paired problems within each domain and level, conditioning on the deployment. They do not include variation across model training seeds. These breadth contrasts describe jointly eligible problems without equating the wordings' accuracy.

**Additional draws reveal more PantryPlan modes after first success.** Under neutral wording, GPT-5.4 gains 2.583 and 3.219 verified modes between eight and 64 draws at Levels 2 and 3; Grok 4.3 gains 3.377 and 4.466. Grok 4.3's empirical pass probability is already 1.0 at eight draws at both levels, so these gains consist entirely of additional modes. At Level 3 its distinct count rises from 3.972 to 8.438. On jointly eligible problems at eight correct draws, neutral minus original breadth is 1.781 ([1.226, 2.338]) for Grok 4.3 and .934 ([.670, 1.184]) for GPT-5.4; GPT-5.6 Sol's Level-3 contrast is inconclusive. These counts measure finite-sample discovery rather than exhaustive solution coverage.

**Formatting limits GPT-5.6 Sol's strict Python scores.** GPT-5.6 Sol's original-wording strict pass probability is approximately .1875 at Level 2 and .0625 at Level 3 from eight through 64 draws. Formatting normalization gives  $P_{64} = 1.0$  under both wordings at both levels, with all four  $P_8$  values above .989. The low strict scores therefore include failures to express valid solutions in the required

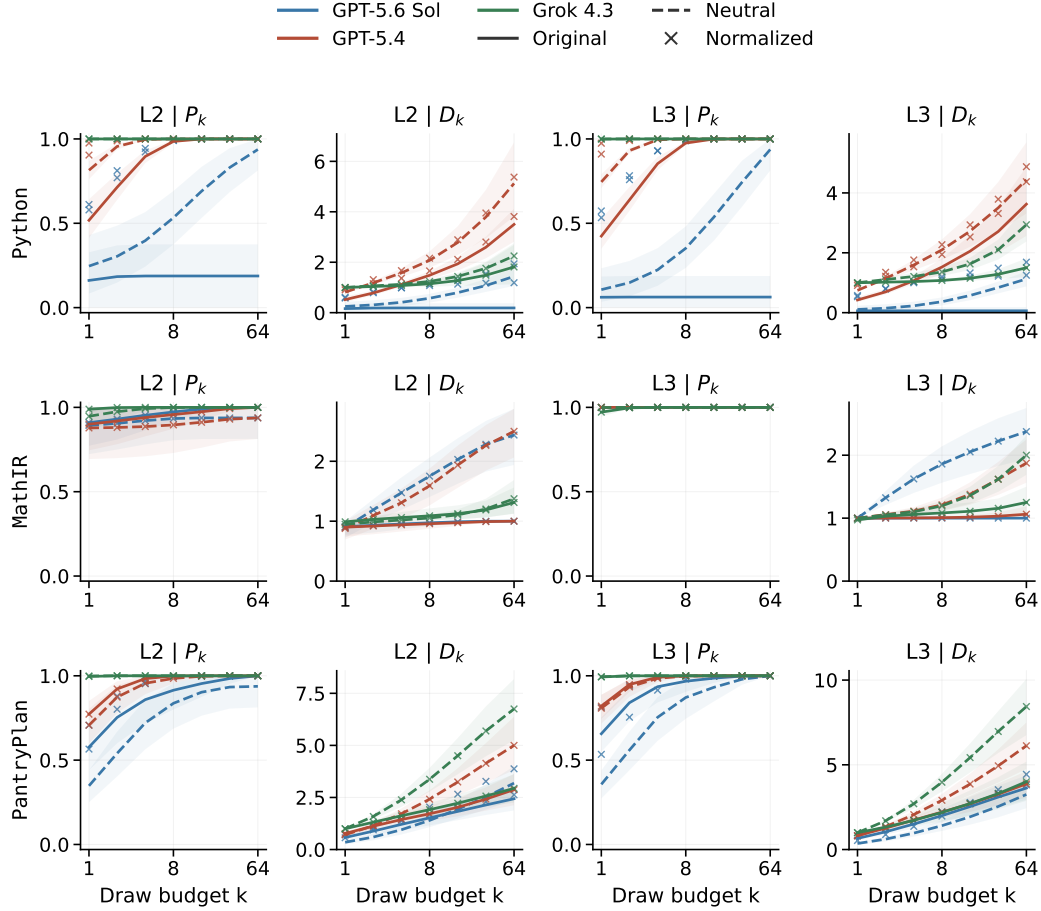


Figure 32: **Additional draws reveal solution modes beyond early success.** Hosted deployments; rows show Python, MathIR, and PantryPlan, with paired  $P_k$  and  $D_k$  columns for Levels 2 and 3. Curves show pass probability  $P_k$  and distinct verified modes  $D_k$  against draw budget  $k$ , obtained by averaging subsets of sixteen 64-draw problem pools per domain and level. Colours identify deployments; solid/dashed lines denote original/neutral wording. Lines use strict verification, crosses formatting normalization, and shading pointwise 95% bootstrap intervals. Intervals resample paired problems within each domain and level and condition on the deployment.

format. After normalization, the neutral-minus-original difference in discovery gains from eight to 64 draws is .681 ([.351, 1.066]) at Level 2 and .370 ([.112, .674]) at Level 3. Normalization changes response formatting before applying the same verifier and canonical-key extraction rules.

### K.3 ALL FIVE DOMAINS AND ALL THREE LEVELS AT 512 DRAWS

We evaluate GPT-5.6 Sol on thirty-two problems from each of five domains at Levels 1, 2, and 3, using the original prompt wording. Problems are selected independently of model outputs. Each problem receives 512 responses: 480 problems and 245,760 responses in total. Requests use medium reasoning, an 8,192-token output limit, and unspecified temperature and top- $p$ , with model snapshot gpt-5.6-sol-2026-07-09. Invalid, refused, and truncated answers remain in the sampling denominator.

Figure 5 uses formatting-normalized grading; Table 30 also gives strict-verifier results on the same response pools. In draw-index order, the first 64 responses solve 474 of 480 problems after normalization and 418 under strict grading. At 512 responses, these counts are 477 and 436, respectively.

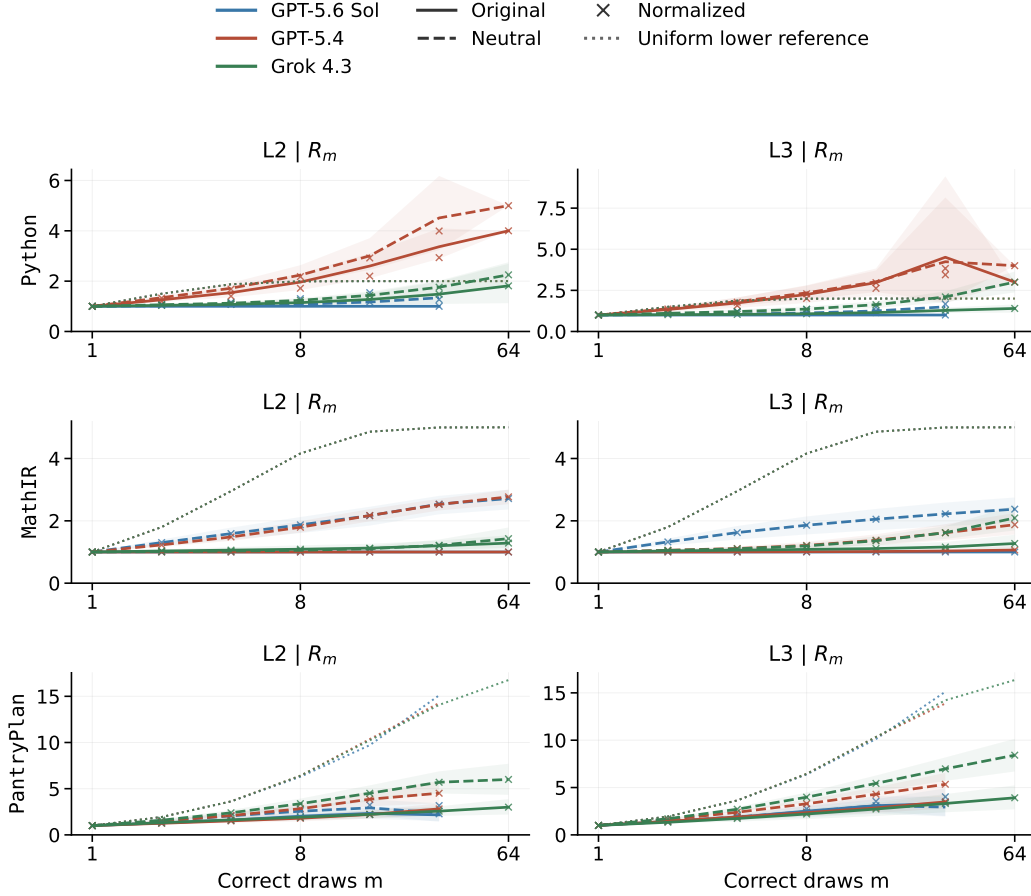


Figure 33: **Neutral wording broadens GPT solutions in MathIR at fixed correct-draw budgets.** Hosted deployments; rows show Python, MathIR, and PantryPlan, and columns Levels 2 and 3. Curves show  $R_m$ , the expected distinct verified modes in  $m$  correct draws, on prompts with at least  $m$  correct outputs under both wordings. Eligibility varies with  $m$ , method, and grading; gaps mark undefined means. Dotted lines give uniform lower references on prompts eligible under strict grading in both wordings, with the same prompt and checkpoint weighting as the strict curves. Colours identify deployments; solid/dashed lines denote original/neutral wording. Lines use strict verification, crosses formatting normalization, and shading pointwise 95% bootstrap intervals. Intervals resample paired problems within each domain and level and condition on the deployment.

For each pool of size  $N = 512$ , with  $c$  correct responses and verified-mode multiplicities  $n_j$ , the expected success indicator and number of distinct verified modes in a uniformly sampled  $k$ -subset are

$$P_N(k) = 1 - \frac{\binom{N-c}{k}}{\binom{N}{k}}, \quad D_N(k) = \sum_j \left[ 1 - \frac{\binom{N-n_j}{k}}{\binom{N}{k}} \right],$$

respectively. We average all thirty-two problems equally within each domain and level, including problems with no correct response. Every curve uses the complete 512-response pools at budgets  $k \in \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512\}$ . The rarefied estimate at  $k = 64$  need not equal the result from the first 64 responses in draw-index order. The final-doubling contrast is  $D_{512}(512) - D_{512}(256)$ . Pointwise 95% intervals use 20,000 whole-problem bootstrap replicates within domain and level; the same resampled problems form both sides of each contrast. Intervals do not provide simultaneous coverage across budgets or cells.

Graph support exhausts all fixed-color-compatible assignments. Countdown references enumerate binary expression trees; the validator also accepts unary negation and retains it in canonical keys, so these counts are lower bounds. The Python, MathIR, and PantryPlan references are also certified lower bounds. Discovered modes can exceed a lower bound without exhausting the available modes.

Table 28: **Discovery gains from eight to 64 draws.** Strict verification; O/N denotes original/neutral wording. Qwen2.5-0.5B-Instruct checkpoint means use five training seeds in Python and MathIR and two in PantryPlan;  $s = 1$  denotes the single initial checkpoint. Five-seed intervals resample seeds and paired prompts; PantryPlan intervals condition on its two checkpoints. Formatting normalization gives the same values in every local cell.

| Model                         | Domain/level  | $G_P$ : O/N | $G_D$ : O/N | $G_B$ : O/N | $\Delta G_D$ [95%]      |
|-------------------------------|---------------|-------------|-------------|-------------|-------------------------|
| Dr.GRPO ( $s = 5$ )           | Python L2     | 0.000/0.000 | 0.000/0.000 | 0.000/0.000 | +0.000 [+0.000, +0.000] |
| Initial Qwen 0.5B ( $s = 1$ ) | Python L2     | 0.147/0.105 | 0.303/0.165 | 0.156/0.060 | -0.138 [-0.217, -0.061] |
| Re:Dr ( $s = 5$ )             | Python L2     | 0.000/0.231 | 0.058/0.463 | 0.058/0.232 | +0.405 [+0.185, +0.620] |
| Dr.GRPO ( $s = 5$ )           | Python L3     | 0.000/0.033 | 0.000/0.033 | 0.000/0.000 | +0.033 [+0.000, +0.120] |
| Initial Qwen 0.5B ( $s = 1$ ) | Python L3     | 0.218/0.096 | 0.593/0.096 | 0.376/0.000 | -0.497 [-0.766, -0.270] |
| Re:Dr ( $s = 5$ )             | Python L3     | 0.000/0.239 | 0.153/0.531 | 0.153/0.292 | +0.379 [+0.111, +0.678] |
| Dr.GRPO ( $s = 5$ )           | MathIR L2     | 0.021/0.017 | 0.021/0.017 | 0.000/0.000 | -0.004 [-0.068, +0.045] |
| Initial Qwen 0.5B ( $s = 1$ ) | MathIR L2     | 0.344/0.235 | 0.344/0.235 | 0.000/0.000 | -0.109 [-0.383, +0.164] |
| Re:Dr ( $s = 5$ )             | MathIR L2     | 0.010/0.011 | 0.010/0.011 | 0.000/0.000 | +0.001 [-0.036, +0.050] |
| Dr.GRPO ( $s = 5$ )           | MathIR L3     | 0.023/0.018 | 0.026/0.018 | 0.004/0.000 | -0.008 [-0.065, +0.038] |
| Initial Qwen 0.5B ( $s = 1$ ) | MathIR L3     | 0.254/0.246 | 0.359/0.360 | 0.105/0.114 | +0.001 [-0.123, +0.095] |
| Re:Dr ( $s = 5$ )             | MathIR L3     | 0.064/0.056 | 0.064/0.056 | 0.000/0.000 | -0.008 [-0.099, +0.059] |
| Dr.GRPO ( $s = 2$ )           | PantryPlan L2 | 0.332/0.323 | 0.718/0.736 | 0.386/0.414 | +0.018 [-0.422, +0.433] |
| Initial Qwen 0.5B ( $s = 1$ ) | PantryPlan L2 | 0.440/0.419 | 0.857/0.955 | 0.416/0.537 | +0.099 [-0.201, +0.392] |
| Re:Dr ( $s = 2$ )             | PantryPlan L2 | 0.141/0.147 | 0.415/0.429 | 0.274/0.282 | +0.014 [-0.123, +0.152] |
| Dr.GRPO ( $s = 2$ )           | PantryPlan L3 | 0.268/0.258 | 0.949/1.025 | 0.681/0.766 | +0.076 [-0.237, +0.465] |
| Initial Qwen 0.5B ( $s = 1$ ) | PantryPlan L3 | 0.259/0.384 | 0.927/1.078 | 0.668/0.694 | +0.151 [-0.109, +0.399] |
| Re:Dr ( $s = 2$ )             | PantryPlan L3 | 0.204/0.149 | 0.568/0.461 | 0.364/0.312 | -0.107 [-0.321, +0.098] |

Table 29: **Pair weighting changes the summary of correct-solution diversity.** Strict 64-draw estimates for Qwen2.5-0.5B-Instruct checkpoints. Collision  $C$  weights prompts by their number of correct pairs; PCMD weights eligible prompts equally. Absolute estimates for each wording use their own eligible prompts, whereas  $\Delta_{\text{PCMD}}$  uses only jointly eligible prompts. Thus  $\Delta_{\text{PCMD}}$  need not equal the difference between the displayed absolute estimates, and  $E_O/E_N$  are separate eligibility counts. Checkpoint estimates receive equal weight; a seed with no eligible prompt makes the mean undefined.

| Model             | Domain/level  | $C_O$ | $C_N$ | PCMD <sub>O</sub> | PCMD <sub>N</sub> | $U_O^C$ | $U_O^{\text{PCMD}}$ | $E_O/E_N$ | $Q_O/Q_N$     | $\Delta_{\text{PCMD}}$ [95%] |
|-------------------|---------------|-------|-------|-------------------|-------------------|---------|---------------------|-----------|---------------|------------------------------|
| Dr.GRPO           | Python L2     | 1.000 | --    | 0.000             | --                | 0.500   | 0.500               | 65/0      | 130977/0      | --                           |
| Initial Qwen 0.5B | Python L2     | 0.791 | 0.875 | 0.097             | 0.167             | 0.500   | 0.500               | 13/3      | 1270/24       | -0.252 [-0.395, +0.019]      |
| Re:Dr             | Python L2     | 0.809 | 0.800 | 0.191             | 0.156             | 0.500   | 0.500               | 71/53     | 140722/18289  | -0.037 [-0.139, +0.037]      |
| Dr.GRPO           | Python L3     | 1.000 | --    | 0.000             | --                | 0.500   | 0.500               | 80/0      | 161280/0      | --                           |
| Initial Qwen 0.5B | Python L3     | 0.875 | 1.000 | 0.101             | 0.000             | 0.500   | 0.500               | 16/1      | 1010/3        | -0.513 [-0.513, -0.513]      |
| Re:Dr             | Python L3     | 0.805 | 0.797 | 0.195             | 0.227             | 0.500   | 0.500               | 80/52     | 160275/12893  | +0.030 [-0.023, +0.117]      |
| Dr.GRPO           | MathIR L2     | 1.000 | 1.000 | 0.000             | 0.000             | 0.200   | 0.800               | 43/42     | 80766/74119   | +0.000 [+0.000, +0.000]      |
| Initial Qwen 0.5B | MathIR L2     | 1.000 | 1.000 | 0.000             | 0.000             | 0.200   | 0.800               | 3/3       | 64/64         | +0.000 [+0.000, +0.000]      |
| Re:Dr             | MathIR L2     | 1.000 | 1.000 | 0.000             | 0.000             | 0.200   | 0.800               | 79/79     | 155016/151485 | +0.000 [+0.000, +0.000]      |
| Dr.GRPO           | MathIR L3     | 0.980 | 1.000 | 0.038             | 0.000             | 0.200   | 0.800               | 16/18     | 24457/25393   | -0.038 [-0.202, +0.000]      |
| Initial Qwen 0.5B | MathIR L3     | 0.883 | 0.912 | 0.128             | 0.162             | 0.200   | 0.800               | 7/6       | 282/181       | +0.013 [-0.014, +0.052]      |
| Re:Dr             | MathIR L3     | 1.000 | 1.000 | 0.000             | 0.000             | 0.200   | 0.800               | 34/34     | 46891/50301   | +0.000 [+0.000, +0.000]      |
| Dr.GRPO           | PantryPlan L2 | 0.374 | 0.276 | 0.733             | 0.780             | 0.047   | 0.953               | 9/10      | 101/193       | +0.220 [-0.030, +0.470]      |
| Initial Qwen 0.5B | PantryPlan L2 | 0.849 | 0.787 | 0.587             | 0.623             | 0.052   | 0.945               | 7/10      | 212/178       | -0.078 [-0.300, +0.064]      |
| Re:Dr             | PantryPlan L2 | 0.600 | 0.613 | 0.343             | 0.218             | 0.051   | 0.942               | 16/18     | 10144/8510    | -0.093 [-0.220, +0.007]      |
| Dr.GRPO           | PantryPlan L3 | 0.203 | 0.228 | 0.808             | 0.680             | 0.047   | 0.950               | 10/12     | 134/165       | -0.156 [-0.359, +0.081]      |
| Initial Qwen 0.5B | PantryPlan L3 | 0.375 | 0.313 | 0.307             | 0.341             | 0.049   | 0.945               | 7/6       | 531/705       | -0.017 [-0.109, +0.091]      |
| Re:Dr             | PantryPlan L3 | 0.636 | 0.666 | 0.337             | 0.198             | 0.051   | 0.941               | 10/11     | 4006/3946     | -0.112 [-0.319, +0.008]      |

Figure 5 divides the mean distinct-mode count by the mean enumerated mode count for the same thirty-two problems. This is a ratio of means, rather than the mean fraction recovered per problem; the denominator is held fixed when scaling interval endpoints. The independently certified references in Table 30 can be smaller than these enumerated counts. In Python, the table certifies at least two modes per problem, while the mean enumerated counts used in the figure are 222, 335.75, and 354.75 at Levels 1, 2, and 3.

These curves describe random subsets of the observed response pools. Interpreting them as predictions for fresh sampling at each budget also requires stable sampling behavior across requests; a common model snapshot and decoding settings do not ensure this. Small or zero gains at the largest budgets do not rule out rare unseen modes.

#### K.4 OBSERVED SAMPLING-BUDGET EFFECTS

The 64-draw comparison in App. K uses 36,864 hosted and 110,592 local responses, 147,456 in total. Each of the six hosted model-wording cohorts contributes 6,144 responses. The local panel contains 25 checkpoints: each trained checkpoint is evaluated in its own domain at Levels 2 and 3, and the

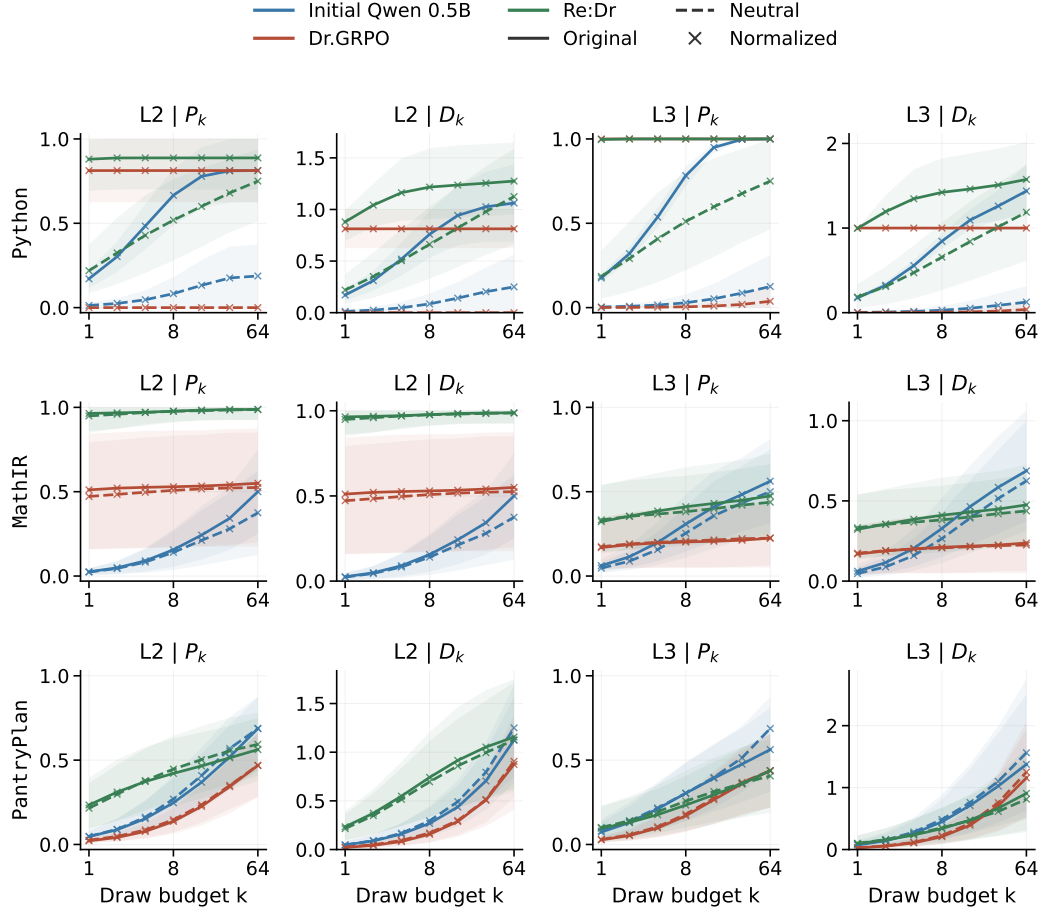


Figure 34: **Additional draws reveal few new MathIR modes after training.** Qwen2.5-0.5B-Instruct checkpoints; rows show Python, MathIR, and PantryPlan, with paired  $P_k$  and  $D_k$  columns for Levels 2 and 3. Curves show pass probability  $P_k$  and distinct verified modes  $D_k$  against draw budget  $k$ , obtained by averaging subsets of sixteen 64-draw problem pools per domain and level. Colours identify methods; solid/dashed lines denote original/neutral wording. Lines use strict verification, crosses formatting normalization, and shading pointwise 95% bootstrap intervals. Means weight checkpoints equally: five training seeds in Python and MathIR, two in PantryPlan, and one initial checkpoint. Five-seed intervals resample seeds and paired problems; PantryPlan and initial intervals condition on the observed checkpoints.

initial checkpoint covers all six cells. Every response remains in its sampling denominator. Formatting normalization adds no verified local responses, so the two grading rules give identical local estimates. These sixteen-problem cells are separate from the thirty-two-problem, 512-draw study in App. K.3.

**Prompt weighting changes the diversity summary.** Tables 27 and 29 compare two summaries of the full 64-response pools. Within each checkpoint or deployment, collision  $C$  pools correct pairs across prompts, weighting each prompt in proportion to  $\binom{c}{2}$ . Promptwise PCMD instead gives each prompt with at least two verified responses equal weight. Trained checkpoint scores then receive equal weight across seeds. Across 70 defined model-cell-wording entries, the median absolute difference between promptwise PCMD and  $1 - C$  is .003, but nineteen entries differ by at least .05. For the initial checkpoint under original wording, Level-2 Pantry has  $1 - C = .151$  and promptwise PCMD of .587; Level 3 has .625 and .307, respectively. Both entries use seven of sixteen problems. The direction of the difference depends on which prompts supply the most correct pairs.

Neither weighting holds overall accuracy or the eligible population fixed. For example, initial-checkpoint Level-3 Python has sixteen eligible problems under original wording and one under neutral wording. Absolute columns use each wording’s own eligible prompts, while

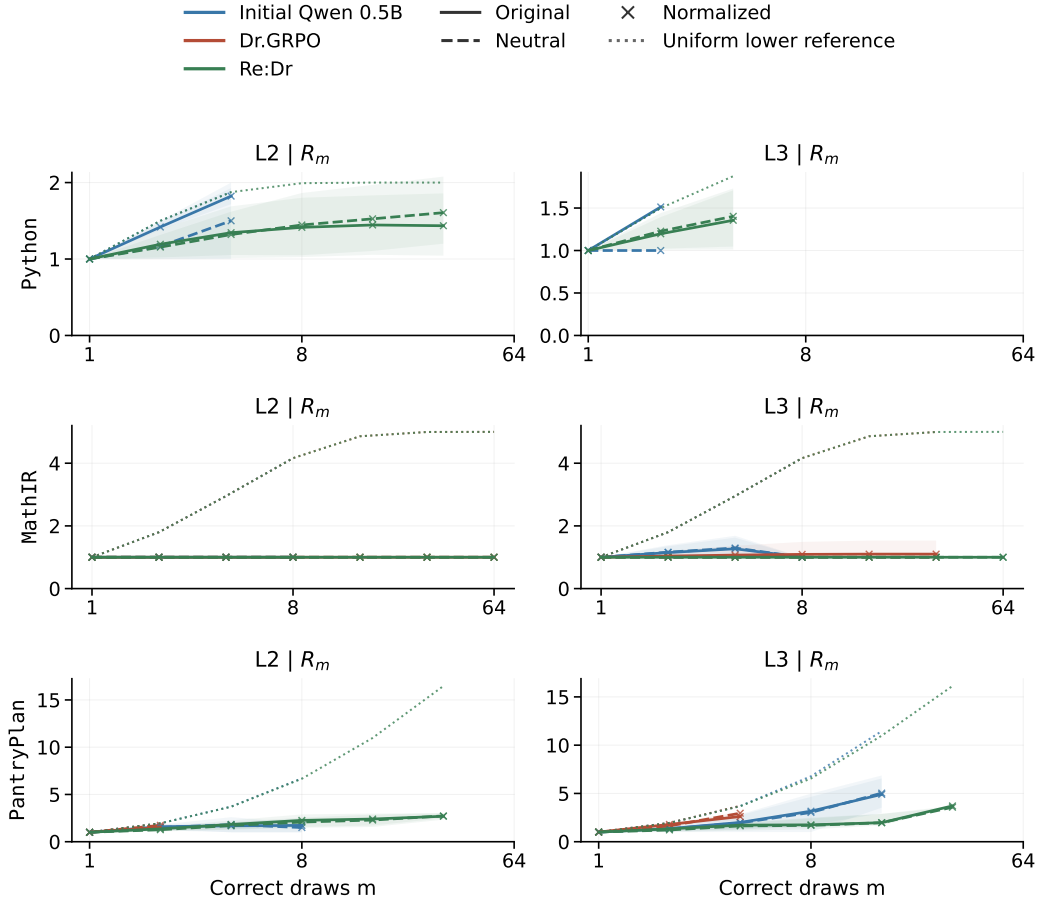


Figure 35: **MathIR correct solutions remain concentrated after training.** Qwen2.5-0.5B-Instruct checkpoints; rows show Python, MathIR, and PantryPlan, and columns Levels 2 and 3. Curves show  $R_m$ , the expected distinct verified modes in  $m$  correct draws, on prompts with at least  $m$  correct outputs under both wordings. Eligibility varies with  $m$ , method, and grading; gaps mark undefined means. Dotted lines give uniform lower references on prompts eligible under strict grading in both wordings, with the same prompt and checkpoint weighting as the strict curves. Colours identify methods; solid/dashed lines denote original/neutral wording. Lines use strict verification, crosses formatting normalization, and shading pointwise 95% bootstrap intervals. Means weight checkpoints equally: five training seeds in Python and MathIR, two in PantryPlan, and one initial checkpoint. Five-seed intervals resample seeds and paired problems; PantryPlan and initial intervals condition on the observed checkpoints.

paired differences use jointly eligible prompts. Equal prompt weights describe conditional diversity within these selected populations (App. A.1).

**MathIR: concentration largely persists through 64 draws.** At Level 2, both trained methods have  $B_k = 0$  at every measured budget under both wordings. Re:Dr has original  $P_8 = .9774$  and  $P_{64} = .9875$ ; neutral wording gives .9767 and .9875. Its  $D_k$  equals  $P_k$  throughout, so the small late gain consists entirely of first successes. At exactly eight correct draws, both wordings have  $R_8 = 1$  on the same 76 eligible checkpoint–problem groups, versus the certified-support uniform lower reference  $5[1 - (4/5)^8] = 4.1611$ . These groups reuse the same 16 problems across five checkpoints. At Level 3, all neutral trained groups and all original Re:Dr groups also contain at most one observed correct key. There is one exception in original DrGRPO: one seed–problem group contains two keys with counts 28 and 9, giving mean  $B_{64} = .0125$  and  $B_{64} - B_8 = .0035$ . Its equal-seed correct-pair collision is .9803; the other trained MathIR cells have collision 1.0, against the uniform upper reference .2. Thus the larger budget reveals a rare additional mode without supporting an exact single-mode claim for every trained condition.

| Domain     | Level | Known support (range) | $D/P$ at 512 |             | $\Delta D$ , 256 to 512 [95% CI] |                      |
|------------|-------|-----------------------|--------------|-------------|----------------------------------|----------------------|
|            |       |                       | Normalized   | Strict      | Normalized                       | Strict               |
| Graph      | 1     | 6.97 (4–12)           | 2.531/100.0  | 2.531/100.0 | 0.222 [0.142, 0.305]             | 0.222 [0.142, 0.305] |
| Graph      | 2     | 6.03 (4–12)           | 2.625/100.0  | 2.625/100.0 | 0.268 [0.170, 0.373]             | 0.268 [0.170, 0.373] |
| Graph      | 3     | 6.66 (4–12)           | 1.719/100.0  | 1.719/100.0 | 0.137 [0.059, 0.228]             | 0.137 [0.059, 0.228] |
| Countdown  | 1     | $\geq 4.53$ (2–6)     | 3.469/100.0  | 3.375/100.0 | 0.148 [0.070, 0.243]             | 0.164 [0.083, 0.259] |
| Countdown  | 2     | $\geq 4.40$ (3–7)     | 3.156/100.0  | 3.094/100.0 | 0.218 [0.123, 0.320]             | 0.235 [0.136, 0.344] |
| Countdown  | 3     | $\geq 4.25$ (2–8)     | 2.812/100.0  | 2.562/100.0 | 0.181 [0.090, 0.289]             | 0.191 [0.097, 0.297] |
| Python     | 1     | $\geq 2.00$ (2)       | 3.750/100.0  | 2.875/100.0 | 0.592 [0.438, 0.750]             | 0.492 [0.334, 0.656] |
| Python     | 2     | $\geq 2.00$ (2)       | 1.438/100.0  | 0.562/43.8  | 0.073 [0.020, 0.135]             | 0.148 [0.078, 0.227] |
| Python     | 3     | $\geq 2.00$ (2)       | 1.344/100.0  | 0.281/28.1  | 0.013 [0.000, 0.032]             | 0.086 [0.031, 0.156] |
| MathIR     | 1     | $\geq 5.00$ (5)       | 1.906/93.8   | 1.906/93.8  | 0.071 [0.012, 0.153]             | 0.071 [0.012, 0.153] |
| MathIR     | 2     | $\geq 5.00$ (5)       | 1.000/100.0  | 1.000/100.0 | 0.002 [0.000, 0.006]             | 0.002 [0.000, 0.006] |
| MathIR     | 3     | $\geq 5.00$ (5)       | 0.969/96.9   | 0.969/96.9  | 0.000 [0.000, 0.000]             | 0.000 [0.000, 0.000] |
| PantryPlan | 1     | $\geq 16.90$ (8–39)   | 6.906/100.0  | 6.906/100.0 | 0.622 [0.458, 0.799]             | 0.622 [0.458, 0.799] |
| PantryPlan | 2     | $\geq 16.50$ (8–39)   | 4.875/100.0  | 4.625/100.0 | 0.702 [0.476, 0.951]             | 0.702 [0.472, 0.962] |
| PantryPlan | 3     | $\geq 16.87$ (8–40)   | 5.562/100.0  | 5.250/100.0 | 0.719 [0.527, 0.914]             | 0.678 [0.489, 0.875] |

**Table 30: Late mode discovery is largest in PantryPlan and Level-1 Python.** GPT-5.6 Sol with original wording and 512 responses for each of thirty-two problems per domain and level.  $D/P$  gives mean distinct verified modes and the percentage of problems solved at least once. Support gives each cell’s mean and range: exact for Graph and a certified lower bound elsewhere. Strict and normalized grading use identical pools. The paired contrast  $\Delta D = D_{512}(512) - D_{512}(256)$  measures the gain between random subsets of the complete pool, not chronological new modes. Brackets give pointwise 95% whole-problem bootstrap intervals.

**Python: prompt dependence and late discovery coexist.** Original DrGRPO has unchanged  $P_k = D_k = .8125$  at Level 2 and 1.0 at Level 3 from eight through 64 draws. Neutral DrGRPO has zero correct responses out of 5,120 at Level 2 and three out of 5,120 at Level 3. Each of those three successes occurs alone in its checkpoint–problem group, so Level 3 mean  $P_{64} = D_{64} = .0375$  while correct-pair collision remains undefined. Re:Dr behaves differently. Under original wording its pass probability is already unchanged from eight to 64, but  $D_k$  gains .0582 at Level 2 and .1529 at Level 3. Under neutral wording,  $P_k$  rises from .5186 to .7500 and from .5109 to .7500, respectively;  $D_k$  rises from .6618 to 1.1250 and from .6561 to 1.1875. These final mode counts remain below the original 1.2750 and 1.5750. The corresponding additional-mode gains are .2318 [.0574, .4565] and .2923 [.0659, .5445], with the seed-and-prompt intervals. Larger budgets therefore recover both first successes and additional verified alternatives in the neutral replay condition. These unconditional gains do not identify a change in the conditional distribution of correct solutions by themselves.

**Pantry: eight draws understate observed discovery.** Across the trained Pantry cells and both wordings,  $D_{64} - D_8$  ranges from .4151 to 1.0247, with additional-mode gains ranging from .2739 to .7664. The initial checkpoint also gains .8565–1.0778 modes. In the two observed training seeds, Level 3 point estimates change method ordering: original DrGRPO/Re:Dr have  $D_8 = .2075/.3381$  but  $D_{64} = 1.1563/.9063$ ; neutral wording gives .2253/.3515 at eight and 1.2500/.8125 at 64. This is a descriptive ranking of these checkpoints, not evidence of a population-level method advantage. Pantry has only two training seeds, and fixed-correct-draw comparisons at larger  $m$  often have few or no jointly eligible problems. These comparisons condition on observed success, and an equal-seed mean is undefined when either seed has no eligible problem.

Several trained conditions remain concentrated at 64 draws, while others continue discovering solution modes. Even a flat finite-pool curve leaves open the possibility of rare unseen modes. Here the  $k = 8$  points are rarefaction estimates from sixteen 64-response pools per cell; App. I uses a separate eight-draw evaluation on thirty-two problems per cell. The trained checkpoints use Level-2 training, so their Level-3 results measure transfer. Hosted results concern fixed deployments.

## L INFERENCE: ADAPTATION, MODEL MIXTURES, AND COARSER KEYS

We evaluate adaptation to changed requirements, complementarity between models, and sensitivity to the definition of a solution mode. Model weights remain fixed during inference. Pointwise 95% percentile intervals use 20,000 paired problem-bootstrap replicates. Each resampled problem carries all its perturbations, model conditions, and training seeds together; hosted comparisons resample

within each domain-level cell. These intervals condition on the observed response pools and fixed training seeds. They do not estimate variability across new response pools or training seeds, and the 21 model pairs are dependent comparisons without multiplicity adjustment.

### L.1 PANTRY PORTFOLIOS UNDER CHANGED REQUIREMENTS

We evaluate the initial Qwen2.5-0.5B checkpoint and Level-2 Dr.GRPO and Re:Dr checkpoints from two matched training seeds on 32 held-out problems. All 192 single-ingredient outages come from task specifications independently of generated answers; exhaustive search of the original quantity grid finds 181 feasible revised tasks. Verified feasibility witnesses are withheld from the model. The five feasible additional dietary restrictions are reported separately because only five problem clusters qualify.

Each portfolio contains eight plans. We test these plans unchanged against the revised requirements, then allow at most eight sequential recovery calls if none survives, stopping at the first valid plan. Unresolved portfolios consume all eight calls. The common recovery interface shows the revised task, initial portfolio, previous recovery outputs and deterministic verifier feedback. Cumulative success includes portfolios that needed no recovery. We average perturbations within each source problem, then weight problems equally.

The three sampling strategies use the same checkpoints, verifier, and 192-token response cap. Ordinary sampling uses temperature 1.0. Tuned temperature is chosen from  $\{0.7, 1.0, 1.3\}$  by outage survival on 16 separate development problems (384 responses per checkpoint). Diversity prompting makes eight sequential calls, requesting a new support while retaining the complete output history, including failures. These strategies match initial response counts; their context lengths and total token costs differ.

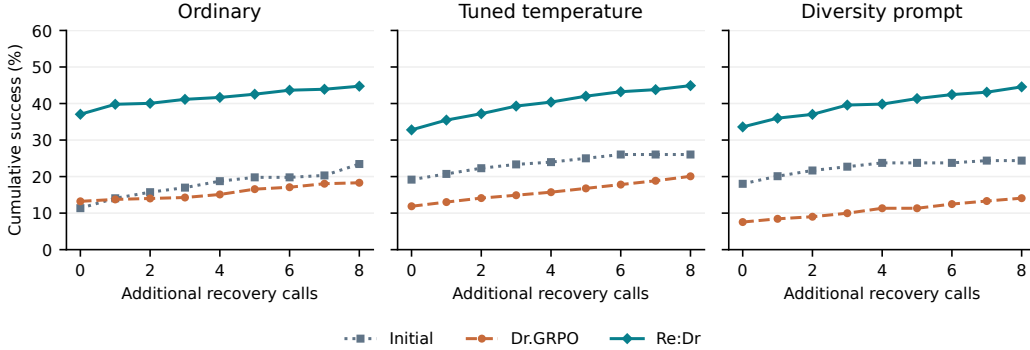


Figure 36: **Re:Dr improves portfolio survival and success within eight recovery calls.** Panels compare ordinary sampling, tuned temperature, and diversity prompting on the same 181 feasible ingredient outages from 32 `Pantry` problems. The horizontal axis is the additional recovery-call budget; the vertical axis includes success without recovery at zero calls. Curves average outages within problems, then problems equally. Solid diamonds show Re:Dr, dashed circles Dr.GRPO, and dotted squares the initial checkpoint; trained curves average two matched seeds. Pointwise 95% intervals for paired differences appear in Table 32.

With ordinary sampling, Re:Dr improves survival over Dr.GRPO by 23.83 percentage points [12.16, 35.89], reduces capped recovery burden by 2.08 calls [1.14, 3.03], and improves success within eight extra calls by 26.43 points [14.69, 38.18]. Mean input and output token totals fall by 2,539.27 [1, 618.01, 3, 468.86] and 427.40 [386.83, 466.30], respectively. However, original correctness also increases by 1.234 correct responses out of eight [.578, 1.984]. Restricting each seed comparison to equal observed correct counts leaves a 1.05-point survival gain [0.00, 3.53]: within each problem we average its eligible seed comparisons, then average the 19 problems with at least one eligible comparison. This descriptive subset changes both problem and seed composition; equal observed counts do not establish equal underlying accuracy. The adaptation gains therefore do not isolate diversity as their causal mechanism.

Table 31: **Re:Dr portfolios adapt better to ingredient outages across all three sampling strategies.** Eight initial plans face 181 feasible outages from 32 problems or five additional dietary restrictions from five problems. Survival is success without recovery; By 8 includes success within eight recovery calls (both percentages). Calls count unresolved cases as eight. Token totals charge the initial portfolio once per revised task. All entries average perturbations within problems, then problems equally; dietary estimates use only five problems.

| Change  | Checkpoint | Strategy          | Survival | By 8  | Calls | Input tokens | Output tokens |
|---------|------------|-------------------|----------|-------|-------|--------------|---------------|
| Outages | Initial    | Ordinary          | 11.35    | 23.44 | 6.63  | 11301.75     | 376.49        |
|         | Initial    | Tuned temperature | 19.17    | 26.04 | 6.13  | 10864.88     | 370.17        |
|         | Initial    | Diversity prompt  | 18.02    | 24.38 | 6.22  | 11794.81     | 352.87        |
|         | Dr.GRPO 43 | Ordinary          | 9.79     | 14.01 | 7.09  | 12439.22     | 823.70        |
|         | Dr.GRPO 43 | Tuned temperature | 13.44    | 16.77 | 6.79  | 12017.57     | 775.44        |
|         | Dr.GRPO 43 | Diversity prompt  | 5.21     | 8.39  | 7.44  | 14103.23     | 848.09        |
|         | Dr.GRPO 46 | Ordinary          | 16.67    | 22.60 | 6.47  | 11265.39     | 637.59        |
|         | Dr.GRPO 46 | Tuned temperature | 10.31    | 23.39 | 6.75  | 11531.40     | 644.94        |
|         | Dr.GRPO 46 | Diversity prompt  | 9.90     | 19.79 | 6.90  | 13124.10     | 731.00        |
|         | Re:Dr 43   | Ordinary          | 39.38    | 47.97 | 4.47  | 9078.92      | 297.98        |
|         | Re:Dr 43   | Tuned temperature | 28.23    | 40.94 | 5.19  | 9718.01      | 310.50        |
|         | Re:Dr 43   | Diversity prompt  | 36.93    | 47.19 | 4.68  | 10465.22     | 344.02        |
|         | Re:Dr 46   | Ordinary          | 34.74    | 41.51 | 4.94  | 9547.16      | 308.52        |
|         | Re:Dr 46   | Tuned temperature | 37.34    | 48.85 | 4.53  | 9148.77      | 300.05        |
|         | Re:Dr 46   | Diversity prompt  | 30.26    | 41.93 | 5.06  | 10584.02     | 313.50        |
| Dietary | Initial    | Ordinary          | 0.00     | 20.00 | 6.80  | 11392.80     | 371.20        |
|         | Initial    | Tuned temperature | 20.00    | 40.00 | 6.40  | 11175.20     | 387.20        |
|         | Initial    | Diversity prompt  | 0.00     | 0.00  | 8.00  | 13630.20     | 405.00        |
|         | Dr.GRPO 43 | Ordinary          | 0.00     | 0.00  | 8.00  | 13310.20     | 849.20        |
|         | Dr.GRPO 43 | Tuned temperature | 0.00     | 0.00  | 8.00  | 13394.60     | 856.00        |
|         | Dr.GRPO 43 | Diversity prompt  | 0.00     | 0.00  | 8.00  | 15045.40     | 939.60        |
|         | Dr.GRPO 46 | Ordinary          | 0.00     | 20.00 | 7.40  | 12063.00     | 650.40        |
|         | Dr.GRPO 46 | Tuned temperature | 0.00     | 0.00  | 8.00  | 12544.20     | 615.40        |
|         | Dr.GRPO 46 | Diversity prompt  | 0.00     | 0.00  | 8.00  | 14132.40     | 725.60        |
|         | Re:Dr 43   | Ordinary          | 20.00    | 60.00 | 5.20  | 9732.60      | 326.00        |
|         | Re:Dr 43   | Tuned temperature | 40.00    | 40.00 | 4.80  | 9402.80      | 307.60        |
|         | Re:Dr 43   | Diversity prompt  | 40.00    | 40.00 | 4.80  | 10693.60     | 355.60        |
|         | Re:Dr 46   | Ordinary          | 40.00    | 80.00 | 3.20  | 7853.20      | 271.20        |
|         | Re:Dr 46   | Tuned temperature | 40.00    | 80.00 | 3.60  | 8127.80      | 265.20        |
|         | Re:Dr 46   | Diversity prompt  | 20.00    | 40.00 | 5.00  | 10738.20     | 341.00        |

Table 32: **Re:Dr raises outage survival and reduces recovery calls.** Entries are Re:Dr minus Dr.GRPO, averaged over two matched seeds on 32 paired problems. Survival and success by eight calls are percentage-point differences; calls include unresolved cases. Brackets are pointwise 95% paired problem-bootstrap intervals.

| Strategy          | $\Delta$ survival    | $\Delta$ by 8 calls  | $\Delta$ calls       |
|-------------------|----------------------|----------------------|----------------------|
| Ordinary          | 23.83 [12.16, 35.89] | 26.43 [14.69, 38.18] | -2.08 [-3.03, -1.14] |
| Tuned temperature | 20.91 [9.19, 32.89]  | 24.82 [14.04, 35.52] | -1.91 [-2.78, -1.05] |
| Diversity prompt  | 26.04 [16.93, 35.65] | 30.47 [21.51, 39.27] | -2.30 [-3.04, -1.57] |

Per-perturbation token totals represent deploying one portfolio and recovering from one change in requirements. Input counts include reused prefixes and do not measure the compute saved by prefix caching. Table 33 counts each response once across calibration, the two controlled portfolio strategies, and recovery for all three strategies. It excludes ordinary initial-portfolio generation. Training costs are outside both inference cost summaries.

Table 33: **Inference costs include temperature selection and recovery.** Each checkpoint selects  $T$  using 384 responses on 16 development problems. Inference totals include this calibration, initial portfolios for tuned-temperature and diversity prompting, and recovery under all three strategies on 32 evaluation problems. Ordinary initial-portfolio generation is excluded. Each response is counted once; input counts include reused prefixes. Seconds measure generation time, exclude training, and are not hardware-normalized compute. Labels (a) and (b) identify the two paired training replicates.

| Checkpoint  | Scope           | $T$ | Responses | Input tokens | Output tokens | Seconds |
|-------------|-----------------|-----|-----------|--------------|---------------|---------|
| Dr.GRPO (a) | Calibration     | 1.3 | 384       | 229392       | 19871         | 136.3   |
| Dr.GRPO (a) | Inference total | 1.3 | 4874      | 4870884      | 272484        | 3302.0  |
| Dr.GRPO (b) | Calibration     | 1.3 | 384       | 229392       | 15544         | 126.5   |
| Dr.GRPO (b) | Inference total | 1.3 | 4634      | 4365524      | 220458        | 2792.9  |
| Re:Dr (a)   | Calibration     | 0.7 | 384       | 229392       | 8552          | 111.1   |
| Re:Dr (a)   | Inference total | 0.7 | 3559      | 3160284      | 92227         | 1527.7  |
| Re:Dr (b)   | Calibration     | 1.0 | 384       | 229392       | 8585          | 112.5   |
| Re:Dr (b)   | Inference total | 1.0 | 3592      | 3169337      | 88300         | 1510.2  |
| Initial     | Calibration     | 1.3 | 384       | 229392       | 9521          | 123.5   |
| Initial     | Inference total | 1.3 | 4416      | 4035069      | 116619        | 1957.4  |

## L.2 COMPLEMENTARITY ACROSS HOSTED MODELS

Seven hosted models each produce eight responses to the same 1,920 prompts (five domains, three levels, 128 prompts per cell), totaling 107,520 responses. Grading is strict, with the original Opus 5 Python interface. These conditions differ from the normalized hosted comparison with the revised Opus 5 interface. For each of the 21 model pairs, a mixed portfolio draws four responses without replacement from each model’s eight-response pool. We compute its expected distinct-mode count exactly over all  $\binom{8}{4}^2 = 4,900$  subset pairs, then compare with each constituent’s complete eight-response portfolio and their average. These are expectations within the finite response pools, conditional on the outputs generated.

Every pair increases expected distinct solution modes over the average constituent by .115–.266. Ten pairs improve over both constituents in point estimates; nine have positive pointwise intervals against both. On prompts with at least four correct responses from each model, we also compare a mixture of two correct draws per model with four correct draws from either constituent. All pairwise gains over the average constituent remain positive (.073–.177 modes). This fixes the number of correct draws used to measure diversity; it conditions on a different eligible subset for each pair and does not match accuracy or the number of attempts needed to obtain them.

Cross-model collision is .624–.740 versus .704–.819 for mean within-model collision, averaging prompts with  $\geq 2$  correct draws per model equally per pair. Absence of observed overlap does not establish disjoint supports. Average-constituent gains persist with coarser keys and formatting-normalized grading. Equal response counts leave provider tokens, compute, costs, and reasoning interfaces unmatched; these finite-pool comparisons do not identify causal training mechanisms.

## L.3 SENSITIVITY TO TASK-RELEVANT EQUIVALENCES

Graph groups global color renamings that fix every anchored color; vertex identities remain fixed. Python groups opposite members  $d$  and  $n/d$  of the same unordered proper factor pair, retaining input-vector order. Countdown flattens associative additions and multiplications and sorts their children, preserving operands, multiplicity, and subtraction/division boundaries. MathIR keeps its simplified state trajectory. Pantry keeps ingredient supports, whose differences can matter under outages. The same domain-specific map applies to each model’s correct responses. Correctness is unchanged; deterministic merging cannot increase the observed distinct-mode count or reduce correct-pair collision.

Uniform mass over fine keys generally induces unequal mass across coarse groups. For Graph and Python, exhaustive enumeration determines both the number of coarse groups and their sizes, allowing separate collision references for uniform fine-key mass and uniform coarse-group mass. Countdown’s full verifier support is unknown, so neither uniform reference has a numerical value.

The original-wording Python Level-3 gain falls from .419 to zero; Level 2 falls from .431 to .075. Neutral-wording coarse gains remain .556 at Level 2 and .500 at Level 3. Thus the original Python effect can reflect opposite members of the same factor pair, and the conclusion depends on wording and the meaning of output identity. Neither divisor vectors nor factor-pair vectors identify different algorithms.

## M PORTFOLIO SURVIVAL UNDER WITHDRAWN OPTIONS

A portfolio can remain useful when a solved task loses an option. We evaluate this survival across all five domains in 474 terminal checkpoint evaluations, each with 128 prompts and four groups of eight responses per prompt. The groups have overlapping sampling streams (App. H.2). Replay’s largest gains in expected distinct modes at four verified draws occur in Graph, Countdown, Python and Pantry. Across 38 matched domain–scale–level–objective comparisons, PCMD gains correlate with survival gains at four verified draws (Pearson  $r = .57$ ).

**One withdrawn option.** Each task loses one option: a colour at one vertex (Graph), an arithmetic operation (Countdown), a returned divisor value (Python), a menu action (MathIR), or an ingredient (Pantry; see App. L.1). Options are enumerated from the task specification. The solution-mode enumeration matches the certified counts for all 1,280 prompt rows. A surviving

Table 34: **Every model pair increases expected distinct modes over its average constituent.** A 4+4 mixture samples without replacement from the two eight-response pools; A and B name the constituents in order. Panel A compares with their average: fine and coarse keys use all 1,920 prompts, while the four-correct-draw comparison uses  $n$  prompts with at least four correct responses per model (2+2 mixed versus four per constituent). Panel B compares fine-key distinct modes with each constituent and pass@8 with their average. Entries are distinct-mode differences except pass@8, in percentage points. Brackets are pointwise 95% paired problem-bootstrap intervals, stratified by domain and level; comparisons share models and prompts.

| A                             |                 |                  |        |                  |                 |                         |              |     |  |
|-------------------------------|-----------------|------------------|--------|------------------|-----------------|-------------------------|--------------|-----|--|
| Pair                          |                 | Fine             |        | Coarse           |                 | Four correct draws      |              | $n$ |  |
| DeepSeek V4 Pro + Kimi K3     | 0.148           | [0.136, 0.160]   | 0.142  | [0.131, 0.154]   | 0.107           | [0.098, 0.115]          | 1858         |     |  |
| DeepSeek V4 Pro + Opus 4.8    | 0.198           | [0.185, 0.211]   | 0.183  | [0.171, 0.197]   | 0.153           | [0.142, 0.164]          | 1847         |     |  |
| DeepSeek V4 Pro + Opus 5      | 0.262           | [0.249, 0.275]   | 0.252  | [0.240, 0.265]   | 0.177           | [0.165, 0.190]          | 1510         |     |  |
| DeepSeek V4 Pro + GPT-5.4     | 0.133           | [0.122, 0.143]   | 0.127  | [0.116, 0.138]   | 0.098           | [0.090, 0.107]          | 1695         |     |  |
| DeepSeek V4 Pro + GPT-5.6 Sol | 0.235           | [0.223, 0.247]   | 0.224  | [0.212, 0.237]   | 0.154           | [0.142, 0.166]          | 1396         |     |  |
| DeepSeek V4 Pro + Grok 4.3    | 0.123           | [0.112, 0.134]   | 0.117  | [0.106, 0.127]   | 0.091           | [0.083, 0.099]          | 1859         |     |  |
| Kimi K3 + Opus 4.8            | 0.187           | [0.174, 0.200]   | 0.172  | [0.159, 0.185]   | 0.138           | [0.127, 0.148]          | 1885         |     |  |
| Kimi K3 + Opus 5              | 0.266           | [0.253, 0.279]   | 0.255  | [0.242, 0.268]   | 0.162           | [0.150, 0.175]          | 1543         |     |  |
| Kimi K3 + GPT-5.4             | 0.127           | [0.115, 0.139]   | 0.112  | [0.101, 0.124]   | 0.083           | [0.075, 0.092]          | 1720         |     |  |
| Kimi K3 + GPT-5.6 Sol         | 0.202           | [0.190, 0.214]   | 0.195  | [0.183, 0.207]   | 0.093           | [0.083, 0.104]          | 1417         |     |  |
| Kimi K3 + Grok 4.3            | 0.115           | [0.104, 0.126]   | 0.110  | [0.099, 0.121]   | 0.073           | [0.066, 0.080]          | 1900         |     |  |
| Opus 4.8 + Opus 5             | 0.187           | [0.176, 0.199]   | 0.182  | [0.171, 0.193]   | 0.098           | [0.087, 0.109]          | 1534         |     |  |
| Opus 4.8 + GPT-5.4            | 0.201           | [0.186, 0.215]   | 0.185  | [0.172, 0.199]   | 0.156           | [0.144, 0.168]          | 1713         |     |  |
| Opus 4.8 + GPT-5.6 Sol        | 0.247           | [0.234, 0.261]   | 0.237  | [0.223, 0.250]   | 0.151           | [0.138, 0.164]          | 1409         |     |  |
| Opus 4.8 + Grok 4.3           | 0.180           | [0.167, 0.194]   | 0.171  | [0.158, 0.184]   | 0.141           | [0.130, 0.153]          | 1894         |     |  |
| Opus 5 + GPT-5.4              | 0.228           | [0.215, 0.241]   | 0.217  | [0.204, 0.230]   | 0.158           | [0.145, 0.171]          | 1493         |     |  |
| Opus 5 + GPT-5.6 Sol          | 0.169           | [0.157, 0.182]   | 0.161  | [0.149, 0.174]   | 0.143           | [0.130, 0.156]          | 1370         |     |  |
| Opus 5 + Grok 4.3             | 0.231           | [0.219, 0.244]   | 0.225  | [0.213, 0.237]   | 0.145           | [0.133, 0.157]          | 1547         |     |  |
| GPT-5.4 + GPT-5.6 Sol         | 0.156           | [0.145, 0.167]   | 0.147  | [0.136, 0.158]   | 0.094           | [0.084, 0.104]          | 1400         |     |  |
| GPT-5.4 + Grok 4.3            | 0.130           | [0.119, 0.142]   | 0.121  | [0.111, 0.132]   | 0.095           | [0.086, 0.104]          | 1722         |     |  |
| GPT-5.6 Sol + Grok 4.3        | 0.211           | [0.199, 0.224]   | 0.204  | [0.192, 0.216]   | 0.118           | [0.107, 0.130]          | 1418         |     |  |
| B                             |                 |                  |        |                  |                 |                         |              |     |  |
| Pair                          | $\Delta D$ vs A |                  |        | $\Delta D$ vs B  |                 | $\Delta$ pass@8 vs mean |              |     |  |
| DeepSeek V4 Pro + Kimi K3     | 0.279           | [0.254, 0.304]   |        | 0.017            | [-0.008, 0.043] | 0.22                    | [0.08, 0.39] |     |  |
| DeepSeek V4 Pro + Opus 4.8    | 0.089           | [0.064, 0.114]   |        | 0.307            | [0.282, 0.332]  | 0.31                    | [0.15, 0.49] |     |  |
| DeepSeek V4 Pro + Opus 5      | -0.061          | [-0.082, -0.039] |        | 0.585            | [0.560, 0.610]  | 7.48                    | [7.19, 7.78] |     |  |
| DeepSeek V4 Pro + GPT-5.4     | 0.097           | [0.074, 0.121]   |        | 0.168            | [0.145, 0.192]  | 0.64                    | [0.41, 0.90] |     |  |
| DeepSeek V4 Pro + GPT-5.6 Sol | -0.003          | [-0.025, 0.019]  |        | 0.473            | [0.447, 0.500]  | 7.16                    | [6.71, 7.62] |     |  |
| DeepSeek V4 Pro + Grok 4.3    | 0.059           | [0.037, 0.082]   |        | 0.186            | [0.163, 0.210]  | 0.34                    | [0.17, 0.52] |     |  |
| Kimi K3 + Opus 4.8            | -0.053          | [-0.080, -0.027] |        | 0.427            | [0.401, 0.452]  | 0.19                    | [0.07, 0.32] |     |  |
| Kimi K3 + Opus 5              | -0.188          | [-0.212, -0.165] |        | 0.719            | [0.692, 0.747]  | 7.34                    | [7.08, 7.61] |     |  |
| Kimi K3 + GPT-5.4             | -0.040          | [-0.064, -0.015] |        | 0.293            | [0.267, 0.320]  | 0.73                    | [0.48, 1.00] |     |  |
| Kimi K3 + GPT-5.6 Sol         | -0.167          | [-0.191, -0.144] |        | 0.571            | [0.543, 0.600]  | 7.38                    | [6.92, 7.85] |     |  |
| Kimi K3 + Grok 4.3            | -0.080          | [-0.104, -0.056] |        | 0.309            | [0.284, 0.335]  | 0.12                    | [0.03, 0.23] |     |  |
| Opus 4.8 + Opus 5             | -0.027          | [-0.045, -0.009] |        | 0.401            | [0.379, 0.423]  | 7.42                    | [7.14, 7.71] |     |  |
| Opus 4.8 + GPT-5.4            | 0.274           | [0.248, 0.300]   |        | 0.127            | [0.101, 0.153]  | 0.93                    | [0.65, 1.24] |     |  |
| Opus 4.8 + GPT-5.6 Sol        | 0.118           | [0.096, 0.141]   |        | 0.377            | [0.350, 0.403]  | 7.51                    | [7.03, 7.99] |     |  |
| Opus 4.8 + Grok 4.3           | 0.226           | [0.202, 0.250]   |        | 0.135            | [0.110, 0.160]  | 0.11                    | [0.02, 0.23] |     |  |
| Opus 5 + GPT-5.4              | 0.515           | [0.489, 0.541]   | -0.060 | [-0.082, -0.037] | 6.30            | [5.90, 6.71]            |              |     |  |
| Opus 5 + GPT-5.6 Sol          | 0.254           | [0.232, 0.277]   |        | 0.084            | [0.063, 0.106]  | 2.11                    | [1.66, 2.57] |     |  |
| Opus 5 + Grok 4.3             | 0.490           | [0.467, 0.514]   | -0.028 | [-0.048, -0.007] | 7.41            | [7.13, 7.69]            |              |     |  |
| GPT-5.4 + GPT-5.6 Sol         | -0.046          | [-0.067, -0.025] |        | 0.359            | [0.335, 0.383]  | 5.61                    | [5.15, 6.08] |     |  |
| GPT-5.4 + Grok 4.3            | 0.102           | [0.078, 0.127]   |        | 0.159            | [0.135, 0.182]  | 0.93                    | [0.65, 1.23] |     |  |
| GPT-5.6 Sol + Grok 4.3        | 0.386           | [0.360, 0.412]   |        | 0.037            | [0.015, 0.058]  | 7.55                    | [7.08, 8.04] |     |  |

Table 35: **Coarser keys reduce observed mode counts and increase collision.** Strict grading uses eight responses per hosted model on 1,920 prompts.  $D@8$  averages all prompts; collision averages prompts with at least two correct responses, with eligibility varying by model. Brackets are pointwise 95% paired problem-bootstrap intervals, stratified by domain and level.

| Model           | Fine $D@8$ |                | Coarse $D@8$ |                | Fine collision |                | Coarse collision |                |
|-----------------|------------|----------------|--------------|----------------|----------------|----------------|------------------|----------------|
| DeepSeek V4 Pro | 1.826      | [1.792, 1.859] | 1.762        | [1.729, 1.795] | 0.720          | [0.710, 0.731] | 0.739            | [0.729, 0.750] |
| Kimi K3         | 2.087      | [2.047, 2.129] | 1.952        | [1.913, 1.992] | 0.688          | [0.678, 0.699] | 0.723            | [0.712, 0.733] |
| Opus 4.8        | 1.608      | [1.575, 1.641] | 1.556        | [1.525, 1.588] | 0.795          | [0.785, 0.805] | 0.811            | [0.801, 0.821] |
| Opus 5          | 1.180      | [1.155, 1.206] | 1.152        | [1.128, 1.177] | 0.856          | [0.845, 0.866] | 0.867            | [0.857, 0.877] |
| GPT-5.4         | 1.755      | [1.718, 1.792] | 1.679        | [1.644, 1.715] | 0.729          | [0.717, 0.741] | 0.750            | [0.739, 0.762] |
| GPT-5.6 Sol     | 1.349      | [1.317, 1.382] | 1.312        | [1.281, 1.345] | 0.778          | [0.765, 0.790] | 0.794            | [0.782, 0.806] |
| Grok 4.3        | 1.698      | [1.665, 1.732] | 1.645        | [1.613, 1.679] | 0.791          | [0.782, 0.800] | 0.807            | [0.798, 0.816] |

Table 36: **Python diversity gains depend on the factor-pair equivalence and prompt wording.** Entries are Re:Dr minus Dr.GRPO in distinct modes from eight responses, under strict grading. Level-2 trained Qwen2.5-0.5B endpoints are evaluated on 32 paired problems at each level and averaged over the displayed fixed seeds. Brackets are pointwise 95% paired problem-bootstrap intervals, conditional on those checkpoints. MathIR and Pantry use identical fine and coarse keys.

| Domain | Level | Wording  | Seeds | Fine $\Delta D@8$ |                 | Coarse $\Delta D@8$ |                 |
|--------|-------|----------|-------|-------------------|-----------------|---------------------|-----------------|
| Python | 2     | original | 5     | 0.431             | [0.406, 0.456]  | 0.075               | [0.025, 0.138]  |
| MathIR | 2     | original | 5     | 0.463             | [0.387, 0.531]  | 0.463               | [0.387, 0.531]  |
| Pantry | 2     | original | 2     | 0.469             | [0.203, 0.750]  | 0.469               | [0.203, 0.750]  |
| Python | 3     | original | 5     | 0.419             | [0.400, 0.444]  | 0.000               | [0.000, 0.000]  |
| MathIR | 3     | original | 5     | 0.119             | [0.019, 0.225]  | 0.119               | [0.019, 0.225]  |
| Pantry | 3     | original | 2     | 0.125             | [-0.109, 0.375] | 0.125               | [-0.109, 0.375] |
| Python | 2     | neutral  | 5     | 0.675             | [0.519, 0.838]  | 0.556               | [0.438, 0.675]  |
| MathIR | 2     | neutral  | 5     | 0.469             | [0.375, 0.562]  | 0.469               | [0.375, 0.562]  |
| Pantry | 2     | neutral  | 2     | 0.484             | [0.234, 0.766]  | 0.484               | [0.234, 0.766]  |
| Python | 3     | neutral  | 5     | 0.656             | [0.537, 0.775]  | 0.500               | [0.419, 0.575]  |
| MathIR | 3     | neutral  | 5     | 0.119             | [0.025, 0.213]  | 0.119               | [0.025, 0.213]  |
| Pantry | 3     | neutral  | 2     | 0.250             | [-0.031, 0.562] | 0.250               | [-0.031, 0.562] |

certified mode establishes feasibility; absence from this enumeration does not establish infeasibility when the certificate is a lower bound on the verifier’s full support. Of 9,864 options, 9,118 retain at least one certified mode. Among these, 8,308 are binding: they also remove at least one certified mode. We evaluate both the binding set and the full set with a feasibility certificate.

**What counts as surviving.** Each portfolio contains the verified draws among eight responses. Survival means that at least one retained solution mode remains valid. Graph, Countdown and Python determine this directly from the outcome key, including verified keys outside the enumeration. Countdown contributes 2 such keys, observed 50 times among the cohort’s 846,398 verified draws: the verifier accepts unary negation, which the expression enumeration omits. For MathIR, re-enumerating the reduced menu tests whether the same executed state path remains realizable, possibly through a different action sequence. Pantry checks the certified witness plan for each ingredient-support key. All observed MathIR and Pantry keys belong to their respective enumerations.

**Survival conditional on correct draws.** Raw survival depends on both correctness and the distribution of valid modes. At budgets  $m = 1, 2, 4$ , we compute the exact expected survival of a uniformly selected  $m$ -element subset of a portfolio’s verified draws, as in App. L.2. Replay-control contrasts use only prompt-group pairs with at least  $m$  verified draws in both arms; the eligible population can change with  $m$ . This fixes the number of correct draws, rather than policy accuracy. Differences can still reflect individual modes’ resistance to a withdrawal; an increasing gain with  $m$  alone does not isolate diversity. Expected distinct modes at the same budget provide a separate measure of portfolio breadth.

Raw survival improves in 39 of 40 domain-scale-level-objective comparisons. At four verified draws, the point estimate remains positive in 28 of 39 eligible comparisons. These conditional comparisons use a smaller population than raw survival. Re:Dr’s expected distinct-mode gain is largest in Graph, Countdown, Python and Pantry, with pooled survival gains reaching 19.6 percentage points. In MathIR, the corresponding gains are only .004 distinct modes and .14 survival points; the latter interval rounds to [.00, .30]. These small positive estimates agree descriptively with the small PCMD gain in Table 8.

Graph’s one-draw contrast is zero under this withdrawal distribution. Each valid colouring loses one colour choice per variable vertex, so every single mode survives the same fraction of binding with-

Table 37: **Binding withdrawals remove certified modes while retaining a feasibility certificate.** Each row covers 128 prompts. Support is the mean certified mode count per prompt; Options counts single withdrawals. Binding counts options that retain at least one certified mode and remove at least one. Kept is the mean retained fraction of certified modes across binding options. Pairs counts binding pairs of withdrawals, and Kept<sub>2</sub> gives the corresponding retained fraction.

| Domain    | Level | Prompts | Support | Options | Binding | Kept | Pairs | Kept <sub>2</sub> |
|-----------|-------|---------|---------|---------|---------|------|-------|-------------------|
| Graph     | 1     | 128     | 6.3     | 1152    | 818     | 0.57 | 1957  | 0.34              |
| Graph     | 2     | 128     | 6.1     | 1536    | 1068    | 0.57 | 3457  | 0.35              |
| Countdown | 1     | 128     | 4.5     | 512     | 147     | 0.48 | 21    | 0.44              |
| Countdown | 2     | 128     | 4.4     | 512     | 185     | 0.49 | 40    | 0.37              |
| Python    | 1     | 128     | 229.4   | 1433    | 1433    | 0.68 | 7716  | 0.50              |
| Python    | 2     | 128     | 251.7   | 1647    | 1647    | 0.71 | 10381 | 0.55              |
| MathIR    | 1     | 128     | 5.0     | 768     | 448     | 0.40 | 384   | 0.23              |
| MathIR    | 2     | 128     | 5.0     | 768     | 512     | 0.45 | 512   | 0.25              |
| Pantry    | 1     | 128     | 18.2    | 768     | 651     | 0.44 | 1231  | 0.18              |
| Pantry    | 2     | 128     | 18.2    | 768     | 653     | 0.45 | 1228  | 0.19              |

Table 38: **Replay improves survival in Graph, Python and Pantry at four verified draws.** Each replay arm is compared with its fresh objective. Survival differences are percentage points;  $\Delta D_4$  is the difference in expected distinct modes at four verified draws. Raw uses all eight responses; Budget columns use the indicated number of verified draws. Fixed-correct-draw results pool jointly eligible prompt-group pairs across matched seeds, scales and levels; raw results use all paired groups. Binding options are used except in the all column, which includes every option with a feasibility certificate. Brackets are pointwise 95% intervals from 20,000 paired bootstrap replicates. Each resampled cluster is a prompt within one level and scale, with all its groups and training seeds kept together; copies at different scales are separate clusters. Intervals condition on the observed training seeds.

| Domain    | Arm    | Raw   | Budget 1             | Budget 2            | Budget 4             | Budget 4 (all) | $\Delta D_4$ |
|-----------|--------|-------|----------------------|---------------------|----------------------|----------------|--------------|
| Graph     | Re:Dr  | 34.78 | 0.00 [0.00, 0.00]    | 10.51 [8.76, 12.31] | 19.65 [15.61, 23.82] | 15.66          | 1.290        |
| Graph     | Re:Max | 18.98 | 0.00 [-0.00, 0.00]   | 6.62 [5.23, 7.98]   | 10.99 [8.25, 13.81]  | 8.31           | 0.646        |
| Countdown | Re:Dr  | 5.32  | 3.51 [1.88, 5.30]    | 6.52 [4.27, 8.94]   | 9.58 [6.81, 12.63]   | 2.53           | 0.626        |
| Countdown | Re:Max | 1.95  | -2.40 [-4.62, -0.21] | -1.25 [-3.70, 1.14] | 0.10 [-2.62, 2.78]   | -0.05          | 0.490        |
| Python    | Re:Dr  | 38.99 | -0.22 [-1.25, 0.91]  | 4.09 [3.17, 5.08]   | 7.67 [6.82, 8.58]    | 7.67           | 0.317        |
| Python    | Re:Max | 22.84 | 1.53 [0.87, 2.24]    | 3.85 [3.04, 4.72]   | 5.68 [4.80, 6.64]    | 5.68           | 0.198        |
| MathIR    | Re:Dr  | 33.61 | 0.01 [0.00, 0.05]    | 0.12 [0.03, 0.23]   | 0.14 [0.00, 0.30]    | 0.08           | 0.004        |
| MathIR    | Re:Max | 21.64 | 0.00 [-0.04, 0.05]   | 0.03 [-0.03, 0.10]  | 0.09 [-0.01, 0.21]   | 0.06           | 0.003        |
| Pantry    | Re:Dr  | 16.55 | -1.11 [-1.65, -0.54] | 4.36 [3.82, 4.87]   | 9.78 [8.90, 10.65]   | 9.20           | 0.810        |
| Pantry    | Re:Max | 13.04 | -0.64 [-1.04, -0.21] | 3.12 [2.72, 3.53]   | 6.61 [5.83, 7.39]    | 6.20           | 0.547        |

drawals. Including nonbinding options reduces the contrast because those options retain every certified mode. For example, 63% of Countdown Level-1 options with a feasibility certificate are nonbinding. The all-options column and the binding column therefore describe different withdrawal populations.

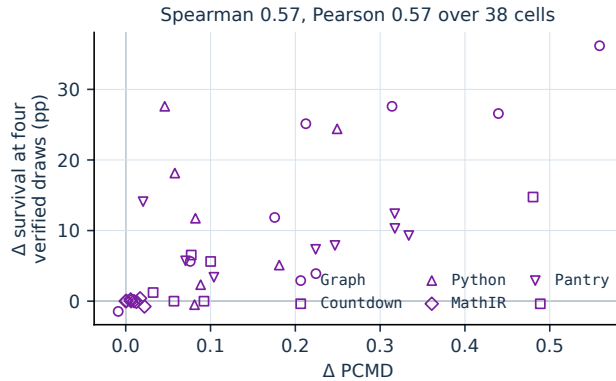


Figure 37: **Diversity gains correlate with survival gains after a withdrawal.** Each mark is one replay-fresh-objective contrast in a domain, scale and level. The horizontal axis is the terminal PCMD difference; the vertical axis is the percentage-point survival difference at four verified draws, averaged equally over eligible seed-specific contrasts. Survival uses jointly eligible prompt-group pairs within each seed. Marker shape identifies the domain; zero lines mark no change. The 38 comparisons have Pearson and Spearman correlations of .57; points show estimates without intervals. The association is descriptive, and the two metrics need not use the same eligible population.

## M.1 TWO WITHDRAWALS AT ONCE

We evaluate pairs drawn from the binding single withdrawals. Of 29,213 pairs, 26,927 retain at least one certified mode and are included in the survival comparison. For Graph, Countdown, Python and Pantry, a certified mode survives the pair exactly when it survives both individual withdrawals. MathIR can differ: removing two actions can eliminate every realization of a state path even when either action alone can be removed. Re-enumeration gives equality with the single-withdrawal intersection for all 1,344 MathIR pairs in this evaluation.

**Table 39: Two withdrawals retain survival gains in Graph, Python and Pantry.** Entries are replay–fresh-objective differences in percentage points at two or four verified draws. One and Two use binding single options and binding pairs with a feasibility certificate, respectively; their eligible prompt populations can differ. Estimates pool jointly eligible groups. Brackets are pointwise 95% paired prompt-bootstrap intervals with the clustering in Table 38.

| Domain    | Arm    | One @2 | One @4               | Two @2 | Two @4               |
|-----------|--------|--------|----------------------|--------|----------------------|
| Graph     | Re:Dr  | 10.51  | 19.65 [15.61, 23.82] | 10.02  | 23.65 [19.14, 28.25] |
| Graph     | Re:Max | 6.62   | 10.99 [8.25, 13.81]  | 5.93   | 12.51 [9.37, 15.69]  |
| Countdown | Re:Dr  | 6.52   | 9.58 [6.81, 12.63]   | 3.29   | 6.11 [0.87, 12.22]   |
| Countdown | Re:Max | -1.25  | 0.10 [-2.62, 2.78]   | -5.53  | -4.87 [-11.80, 0.76] |
| Python    | Re:Dr  | 4.09   | 7.67 [6.82, 8.58]    | 5.37   | 10.10 [8.77, 11.55]  |
| Python    | Re:Max | 3.85   | 5.68 [4.80, 6.64]    | 5.64   | 8.16 [6.79, 9.64]    |
| MathIR    | Re:Dr  | 0.12   | 0.14 [0.00, 0.30]    | 0.18   | 0.20 [0.01, 0.45]    |
| MathIR    | Re:Max | 0.03   | 0.09 [-0.01, 0.21]   | 0.04   | 0.12 [-0.01, 0.29]   |
| Pantry    | Re:Dr  | 4.36   | 9.78 [8.90, 10.65]   | 2.85   | 7.23 [6.51, 7.95]    |
| Pantry    | Re:Max | 3.12   | 6.61 [5.83, 7.39]    | 2.03   | 4.94 [4.35, 5.57]    |

Two withdrawals retain less certified support on average (Table 37). Re:Dr’s pooled survival gain at four verified draws rises in Graph, Python and MathIR, and falls in Countdown and Pantry. Countdown Level 1 starts with only 4.5 certified modes per prompt; Pantry Level 1 retains only 18% of its certified modes under a binding pair. These support differences accompany the effect-size differences but do not identify their cause. Across domain–scale–level–objective comparisons, 25 of 39 point estimates are positive at four verified draws, compared with 28 for single withdrawals.

## M.2 DECODING SETTINGS AS A SUBSTITUTE

We evaluate binding Level-1 withdrawals on 128 prompts per domain using the separate twelve-pass Qwen2.5-0.5B cohort in App. J. Its replay variant combines separate mass and within-buffer balancing terms with novelty and semantic-entropy shaping; it differs from Re:Dr in both objective and training duration. Five seeds per arm are evaluated at 11 settings: six temperatures from .5 to 2.0, three settings with larger groups, and two with top- $p = .95$ . Larger groups increase  $K$  from 8 to 32, but use two groups rather than four: total responses per prompt double from 32 to 64. The reference is  $T = 1$ , top- $p = 1$ , and four groups of eight. For each domain, the control setting with the highest four-verified-draw survival is selected on these evaluation prompts; replay uses the reference setting.

**Table 40: Replay survival exceeds the selected control setting in all five domains.** Survival at four verified draws is expressed as percentages. Control, Control grid and Replay are equal-seed means on each setting’s own eligible groups; the grid column gives the control range and Best its maximizing temperature.  $\Delta$  instead pools jointly eligible groups for reference replay minus selected control, so it need not equal the difference of the displayed means. Brackets are pointwise 95% intervals from 20,000 paired prompt-bootstrap replicates, keeping all groups and five seeds of a prompt together. The selected setting stays fixed in the bootstrap; intervals condition on that selection and the observed seeds.

| Domain    | Control | Control grid | Best      | Replay | $\Delta$ vs best     |
|-----------|---------|--------------|-----------|--------|----------------------|
| Graph     | 58.61   | 58.59–58.85  | $T_2$     | 95.91  | 36.97 [35.53, 38.42] |
| Countdown | 15.49   | 12.57–16.16  | $T_{1.6}$ | 40.09  | 28.87 [18.98, 38.87] |
| Python    | 70.54   | 70.36–70.54  | $T_{0.5}$ | 77.56  | 8.33 [7.15, 9.69]    |
| MathIR    | 72.23   | 72.18–72.55  | $T_2$     | 73.60  | 1.37 [0.61, 2.30]    |
| Pantry    | 50.65   | 50.02–51.79  | $T_2$     | 64.09  | 12.73 [10.22, 15.31] |

Control survival spans at most 3.6 percentage points across the grid. The paired replay gains range from 1.4–37.0 points, with all five pointwise intervals above zero. Every selected control setting uses eight-response groups and top- $p = 1$ . Thus the tested decoding changes do not close the survival gap in this cohort. The finite samples do not establish that alternative modes have zero probability, and selecting a setting on evaluation outcomes does not measure its performance on new prompts.

### M.3 RECOVERY AFTER THE CHANGE

Recovery is evaluated at the Level-1 Qwen2.5-0.5B terminal Dr.GRPO and Re:Dr checkpoints from two matched training seeds in each of five domains. Each domain has 32 test and 16 temperature-calibration prompts in a disjoint split. Prompts without a binding withdrawal that retains a certified mode are excluded, leaving 28 test and 15 calibration prompts in Countdown and all prompts elsewhere. Up to four binding options per test prompt give 533 distinct withdrawals, shared by both arms and seeds.

Each initial portfolio contains eight responses. Ordinary sampling uses its terminal evaluation group at  $T = 1$ . The temperature strategy generates a group at the temperature in  $\{.7, 1.0, 1.3\}$  maximizing mean survival on the calibration prompts. The diversity strategy generates eight responses sequentially, including the preceding responses in each prompt. All strategies receive the same recovery procedure: if no initial mode survives, generate one response per call at  $T = 1$ , up to eight calls, with the revised task, initial portfolio and previous failed attempts in context. Stop at the first verified response whose mode survives the withdrawal.

Survival uses the mode predicates defined above. In MathIR, a retained or newly generated state path can count as surviving even if realizing it under the reduced menu requires a different action sequence. Pantry responses are six-bit ingredient-support masks projected onto verified quantities by the environment (App. C.5); survival is evaluated on the resulting support key. These outcomes measure retained mode feasibility, not necessarily reuse of the original response text unchanged.

**Table 41: Replay improves ordinary and temperature-tuned recovery in every domain.** Saved is the percentage of withdrawals with a surviving initial mode; By 8 includes those resolved within eight additional responses. Calls is the mean additional response count, with zero for initial survival and eight for unresolved cases. Each row pools the same withdrawals over two matched seeds. Fresh denotes Dr.GRPO and Replay denotes Re:Dr; the three strategies differ in initial portfolio generation and share the recovery procedure.

| Domain    | Strategy          | Fresh saved | Fresh by 8 | Fresh calls | Replay saved | Replay by 8 | Replay calls |
|-----------|-------------------|-------------|------------|-------------|--------------|-------------|--------------|
| Graph     | Ordinary          | 15.62       | 22.66      | 6.38        | 83.20        | 91.02       | 1.01         |
| Graph     | Tuned temperature | 15.62       | 20.31      | 6.55        | 77.34        | 89.06       | 1.36         |
| Graph     | Diversity prompt  | 29.69       | 30.47      | 5.58        | 62.50        | 74.61       | 2.44         |
| Countdown | Ordinary          | 2.70        | 2.70       | 7.78        | 9.46         | 10.81       | 7.23         |
| Countdown | Tuned temperature | 2.70        | 2.70       | 7.78        | 9.46         | 13.51       | 7.05         |
| Countdown | Diversity prompt  | 2.70        | 2.70       | 7.78        | 5.41         | 9.46        | 7.41         |
| Python    | Ordinary          | 14.06       | 14.06      | 6.88        | 69.92        | 77.73       | 2.21         |
| Python    | Tuned temperature | 14.06       | 14.06      | 6.88        | 71.88        | 75.78       | 2.12         |
| Python    | Diversity prompt  | 14.06       | 14.06      | 6.88        | 62.50        | 67.97       | 2.79         |
| MathIR    | Ordinary          | 38.39       | 44.20      | 4.60        | 58.93        | 65.18       | 2.92         |
| MathIR    | Tuned temperature | 37.05       | 45.09      | 4.61        | 63.84        | 67.86       | 2.67         |
| MathIR    | Diversity prompt  | 41.07       | 45.09      | 4.58        | 47.32        | 53.57       | 3.85         |
| Pantry    | Ordinary          | 31.25       | 31.64      | 5.48        | 47.66        | 52.34       | 4.04         |
| Pantry    | Tuned temperature | 31.25       | 44.53      | 4.81        | 53.12        | 62.11       | 3.38         |
| Pantry    | Diversity prompt  | 56.64       | 58.20      | 3.39        | 33.98        | 45.70       | 4.71         |

**Table 42: Recovery benefits depend on the initial portfolio strategy.** Entries are paired Re:Dr–Dr.GRPO differences on the same withdrawals under each strategy. Saved and By 8 are percentage points; Calls is additional responses. Brackets are pointwise 95% intervals from 20,000 paired prompt-bootstrap replicates. Each prompt’s withdrawals and both training seeds move together; intervals condition on those two seeds.

| Domain    | Strategy          | $\Delta$ saved          | $\Delta$ by 8          | $\Delta$ calls       |
|-----------|-------------------|-------------------------|------------------------|----------------------|
| Graph     | Ordinary          | 67.58 [57.81, 76.56]    | 68.36 [59.38, 76.95]   | −5.37 [−6.05, −4.65] |
| Graph     | Tuned temperature | 61.72 [51.95, 71.09]    | 68.75 [61.33, 75.78]   | −5.20 [−5.84, −4.52] |
| Graph     | Diversity prompt  | 32.81 [21.09, 44.14]    | 44.14 [34.77, 53.12]   | −3.14 [−3.95, −2.30] |
| Countdown | Ordinary          | 6.76 [1.35, 13.89]      | 8.11 [1.43, 16.18]     | −0.55 [−1.12, −0.11] |
| Countdown | Tuned temperature | 6.76 [1.35, 13.89]      | 10.81 [3.33, 19.32]    | −0.73 [−1.35, −0.21] |
| Countdown | Diversity prompt  | 2.70 [0.00, 6.58]       | 6.76 [1.32, 13.64]     | −0.38 [−0.76, −0.06] |
| Python    | Ordinary          | 55.86 [44.92, 65.23]    | 63.67 [51.56, 74.22]   | −4.67 [−5.45, −3.76] |
| Python    | Tuned temperature | 57.81 [46.88, 67.19]    | 61.72 [50.39, 71.48]   | −4.76 [−5.54, −3.87] |
| Python    | Diversity prompt  | 48.44 [37.11, 59.38]    | 53.91 [41.80, 65.62]   | −4.09 [−4.99, −3.14] |
| MathIR    | Ordinary          | 20.54 [11.16, 30.09]    | 20.98 [12.71, 29.82]   | −1.68 [−2.37, −1.02] |
| MathIR    | Tuned temperature | 26.79 [18.26, 35.65]    | 22.77 [13.89, 32.27]   | −1.94 [−2.66, −1.28] |
| MathIR    | Diversity prompt  | 6.25 [−4.09, 16.67]     | 8.48 [−0.46, 18.26]    | −0.73 [−1.50, 0.03]  |
| Pantry    | Ordinary          | 16.41 [8.20, 25.39]     | 20.70 [11.72, 30.47]   | −1.44 [−2.18, −0.76] |
| Pantry    | Tuned temperature | 21.88 [13.28, 31.25]    | 17.58 [10.94, 24.61]   | −1.43 [−2.00, −0.91] |
| Pantry    | Diversity prompt  | −22.66 [−31.25, −13.67] | −12.50 [−21.88, −2.73] | 1.32 [0.55, 2.05]    |

With ordinary or temperature-tuned initial portfolios, replay resolves more withdrawals and uses fewer recovery responses in every domain. All ten resolution contrasts and all ten response-count contrasts

have pointwise intervals excluding zero. Ordinary sampling saves 0.55–5.37 responses per withdrawal. Summed over all three strategies and both seeds, replay uses 9,872 recovery responses and the control uses 18,335. These totals cover recovery only; they exclude initial portfolios and temperature calibration. Response counts do not match token lengths, context lengths or training compute.

The diversity prompt increases the control’s initial survival in `Graph` and `Pantry`, relative to ordinary sampling, with intervals above zero. For replay, initial-survival point estimates decrease in every domain. The intervention changes both the request for alternatives and the accumulated context, so their effects are not separated. With both arms using the diversity prompt, `Pantry` favours the control by 22.66 points of initial survival and 12.50 points of resolution by eight calls. Replay’s initial-survival difference is positive in the other four domains, ranging from 2.7 to 48.4 points, although the `Countdown` and `MathIR` intervals include zero. `Pantry`’s six-bit response format permits enumerating support masks, but this comparison does not establish that enumerability explains the reversal.

**Interpretation and limits.** In `Graph`, `Countdown`, `Python` and `Pantry`, `Re:Dr`’s broader four-draw portfolios accompany larger survival gains than in `MathIR`. The positive association with `PCMD` is descriptive. Fixing the number of verified draws conditions on jointly eligible groups; it does not fix policy accuracy, and eligibility can change with the draw budget. Feasibility is established by a surviving certified mode, which may belong to a lower bound on the verifier’s full support. The interventions cover single and paired exclusions, rather than arbitrary task revisions. Recovery uses domain-specific mode predicates and yields a strategy-dependent reversal in `Pantry`. Finally, the intervals quantify prompt-sampling uncertainty within the observed training cohorts; they do not estimate variability over new training seeds.

## Part V. Fixed deployments

### N INFERENCE-ONLY CONCENTRATION IN A HOSTED MODEL

GPT-5.6 Sol’s verified responses concentrate on a small subset of solution modes under the MODEBENCH interface. This inference-only evaluation measures the output distribution of one deployment, independently of the paper’s matched training interventions.

**Evaluation setup.** The deployment is `gpt-5.6-sol`, with model snapshot `gpt-5.6-sol-2026-07-09`. Each of five domains contributes 128 held-out prompts at each of three levels. Eight stateless requests per prompt yield 15,360 responses. Requests use medium reasoning, an 8,192-token output limit, no tools and no conversation state. Temperature is 1.0 and top- $p$  is .98, the service defaults. The hosted natural-language interface does not enforce the local training evaluation’s token-level grammar constraints.

Level 1 uses the native evaluation splits; Levels 2 and 3 use the matched constructions described in App. B.4. Level 3 extends the test population beyond the two levels of the training intervention. All cells contain multi-answer problems. The Level-2/Level-3 support histograms match those of a separate Level-1 reference population, which differs from the evaluated Level-1 split in `Graph`, `Countdown` and `Python`. Prompts differ across levels and are not paired by row index. Calibration against small-model references does not guarantee increasing difficulty for this hosted model.

Each prompt has eight returned responses. Stateless requests do not establish independent provider randomness. Hidden reasoning content is not exposed.

Table 43: **Correct responses frequently share a solution mode at all three levels.** GPT-5.6 Sol with medium reasoning, under strict grading. Each level averages five domains equally, with 128 prompts and eight responses per prompt in each domain. M and P are mean@8 and pass@8 (%); D is mean distinct@8. C is correct-pair collision (%), weighted by correct pairs within each domain. Brackets are pointwise 95% intervals from whole-prompt resampling.

| Level | M    | P    | D    | C (95% interval)  |
|-------|------|------|------|-------------------|
| 1     | 83.0 | 95.8 | 1.71 | 74.0 [71.9, 76.1] |
| 2     | 62.9 | 77.5 | 1.16 | 82.3 [80.3, 84.4] |
| 3     | 68.7 | 80.3 | 1.18 | 84.5 [82.8, 86.2] |

**Metrics and uncertainty.** Per-response correctness is mean@8; pass@8 is the fraction of prompts with at least one verified response. Raw distinct@8 counts verified keys across eight draws, including zero for unsolved prompts. For prompt  $x$ , let  $c_x$  be its number of correct draws and  $n_{xc}$  its count for correct key  $c$ . The within-cell correct-pair collision estimate is

$$\hat{C} = \frac{\sum_x \sum_c \binom{n_{xc}}{2}}{\sum_x \binom{c_x}{2}}.$$

It describes concentration conditional on successful draws, with prompts weighted by their number of correct pairs. Level summaries average the five domain estimates equally. When the certified correct-key count  $m_x$  is available, a uniform-correct-key reference replaces the numerator by  $\sum_x \binom{c_x}{2} / m_x$ . This differs from a uniform-action baseline and is a descriptive reference, not a normative sampling requirement. Countdown’s verifier accepts modes beyond its enumerated binary-expression library, including unary negation, so its total support is unknown and this reference is omitted. Intervals use 2,000 whole-prompt bootstrap replicates, resampling within each domain and level and keeping all eight draws together. They are pointwise percentile intervals for the empirical prompt population; generations are not resampled separately.

**Formatting sensitivity.** Strict grading applies the executable validators directly. The normalized comparison applies a fixed set of typography rules to the same responses, then uses those validators without repairing calculations or program logic. The rules convert a leading LaTeX lambda to Python `lambda`, unescape percent signs and underscores, remove LaTeX spacing commands, simplify sized parentheses and translate balanced LaTeX fractions into parenthesized division. Numeric/parenthesis

Table 44: **High Graph correctness coexists with concentrated solution modes.** GPT-5.6 Sol with medium reasoning and strict executable grading. Each cell contains 128 prompts and 1,024 responses. M and P are mean@8 and pass@8 (%); D is mean distinct@8, including unsolved prompts. C is correct-pair collision (%), weighted by correct pairs, with pointwise 95% whole-prompt bootstrap intervals. U is the uniform-correct-key collision reference (%); a dash denotes unknown total support.

| Level | Domain     | M     | P     | D    | C (95% interval)     | U    |
|-------|------------|-------|-------|------|----------------------|------|
| 1     | Graph      | 99.2  | 100.0 | 1.30 | 89.7 [86.3, 92.8]    | 18.3 |
| 1     | Countdown  | 86.1  | 98.4  | 1.77 | 71.2 [66.1, 75.9]    | –    |
| 1     | Python     | 40.4  | 90.6  | 1.11 | 81.8 [76.9, 87.2]    | 0.8  |
| 1     | MathIR     | 89.3  | 89.8  | 1.38 | 80.3 [75.4, 85.0]    | 20.0 |
| 1     | PantryPlan | 100.0 | 100.0 | 3.00 | 46.9 [42.2, 51.5]    | 6.5  |
| 2     | Graph      | 99.3  | 100.0 | 1.48 | 83.3 [79.6, 87.0]    | 18.9 |
| 2     | Countdown  | 65.4  | 86.7  | 1.57 | 64.7 [59.0, 70.6]    | –    |
| 2     | Python     | 8.5   | 11.7  | 0.12 | 100.0 [100.0, 100.0] | 0.7  |
| 2     | MathIR     | 86.2  | 93.0  | 0.93 | 100.0 [100.0, 100.0] | 20.0 |
| 2     | PantryPlan | 55.3  | 96.1  | 1.72 | 63.6 [56.8, 70.3]    | 6.9  |
| 3     | Graph      | 100.0 | 100.0 | 1.16 | 95.1 [92.4, 97.3]    | 19.0 |
| 3     | Countdown  | 71.2  | 89.8  | 1.38 | 76.4 [71.1, 81.8]    | –    |
| 3     | Python     | 9.4   | 14.1  | 0.14 | 100.0 [100.0, 100.0] | 0.9  |
| 3     | MathIR     | 97.7  | 97.7  | 0.98 | 100.0 [100.0, 100.0] | 20.0 |
| 3     | PantryPlan | 65.3  | 100.0 | 2.22 | 51.0 [45.3, 57.3]    | 6.8  |

adjacency becomes explicit multiplication only when the containing group has no division sign. Normalization leaves Graph, MathIR and Level-1 PantryPlan unchanged. Strict successes retain their original keys. Normalization verifies 2,580 additional outputs; the added correct responses change both the eligible correct pairs and their weights, so collision can increase or decrease. This comparison measures sensitivity to these formatting conventions on the evaluated responses.

Table 45: **Formatting normalization improves correctness while substantial collision remains.** The same 128 prompts and 1,024 responses per cell as Table 44, with typography normalization before executable verification. M, P, D, C and U use the same definitions; C weights correct pairs and includes pointwise 95% whole-prompt bootstrap intervals. Additional verified responses change the eligible pairs and their weights.

| Level | Domain     | M     | P     | D    | C (95% interval)     | U    |
|-------|------------|-------|-------|------|----------------------|------|
| 1     | Graph      | 99.2  | 100.0 | 1.30 | 89.7 [86.3, 92.8]    | 18.3 |
| 1     | Countdown  | 99.9  | 100.0 | 1.93 | 67.9 [63.3, 72.4]    | –    |
| 1     | Python     | 92.4  | 100.0 | 1.41 | 85.6 [82.3, 88.8]    | 1.4  |
| 1     | MathIR     | 89.3  | 89.8  | 1.38 | 80.3 [75.4, 85.0]    | 20.0 |
| 1     | PantryPlan | 100.0 | 100.0 | 3.00 | 46.9 [42.2, 51.5]    | 6.5  |
| 2     | Graph      | 99.3  | 100.0 | 1.48 | 83.3 [79.6, 87.0]    | 18.9 |
| 2     | Countdown  | 98.5  | 100.0 | 1.99 | 63.9 [59.5, 68.4]    | –    |
| 2     | Python     | 60.7  | 99.2  | 1.12 | 94.8 [91.9, 97.2]    | 1.1  |
| 2     | MathIR     | 86.2  | 93.0  | 0.93 | 100.0 [100.0, 100.0] | 20.0 |
| 2     | PantryPlan | 71.3  | 100.0 | 1.96 | 64.0 [58.1, 69.6]    | 6.8  |
| 3     | Graph      | 100.0 | 100.0 | 1.16 | 95.1 [92.4, 97.3]    | 19.0 |
| 3     | Countdown  | 99.0  | 100.0 | 1.70 | 75.7 [71.2, 80.1]    | –    |
| 3     | Python     | 56.5  | 98.4  | 1.09 | 94.3 [90.8, 97.2]    | 1.0  |
| 3     | MathIR     | 97.7  | 97.7  | 0.98 | 100.0 [100.0, 100.0] | 20.0 |
| 3     | PantryPlan | 75.2  | 100.0 | 2.35 | 53.0 [47.3, 58.9]    | 6.7  |

**Concentration across levels.** The strict five-domain collision average is 74.0%, 82.3% and 84.5% at Levels 1–3; the normalized comparison gives 74.1%, 81.2% and 83.6%. This aggregate pattern is not monotonic within every domain. Graph collision is 89.7%, 83.3%, then 95.1%; normalized PantryPlan collision is 46.9%, 64.0%, then 53.0%. Normalized overall correctness also rebounds from 83.2% at Level 2 to 85.7% at Level 3. Thus a higher level label is not equivalent to a measured increase in difficulty for this model, and raw distinct-mode counts also change with correctness and support.

Level-3 Graph separates concentration from failed responses: all 1,024 outputs verify, eight draws produce 1.16 distinct keys on average, and 3,407 of 3,584 correct pairs share a key (95.1%). The uniform-correct-key reference is 19.0%; the excess is 76.1 percentage points (95% interval [73.5, 78.3]). The domain tables show the corresponding correctness, diversity and collision estimates for every domain and level.

**Interface and interpretation.** Level-2/Level-3 MathIR instructions prescribe moving the right-side variable term left, removing the constant and then dividing. Python instructions suggest a nested small-divisor conditional chain, and PantryPlan instructions prefer particular ingredient families. These preferences can concentrate the very keys being measured. In addition, Level-1 PantryPlan requests a support mask whose quantities are supplied by a trusted projection; Levels 2–3 ask the model to provide quantities. MathIR verifies symbolic action compliance, which can reject a numerically valid shortcut that omits a required symbolic step. Neither these failures nor the cross-level differences isolate numerical reasoning ability. Canonical keys describe executed outputs, not hidden reasoning strategies. Observed keys provide a lower bound on accessible support: eight draws cannot establish that an unseen mode has zero probability. The results describe concentration under the specified prompts and deployment configuration; they identify neither a training cause nor a causal difficulty or model-size effect.

#### N.1 HOSTED CONCENTRATION ON THE SUCCESS-CONDITIONAL AXIS

Table 46 reports promptwise PCMD under strict grading and the original task wording. Within each cell it averages equally over prompts with at least two verified responses; a cell is reportable when at least thirty prompts qualify. The collision tables above instead weight prompts by their number of correct pairs. For GPT-5.6 Sol under strict grading,  $1 - \hat{C}$  and promptwise PCMD summarize the same responses with different weights (App. A.1). Fig. 40 shows the corresponding strict, original-wording cell estimates in its diversity row. Its accuracy row uses formatting normalization and revised Opus 5 Python wording, so the two rows have different grading and, for that deployment and domain, different prompt conditions.

For GPT-5.6 Sol, strict MathIR PCMD is .000 at Levels 2 and 3: none of the 110 and 125 eligible prompts, respectively, contains two different verified keys. Level-3 Graph has PCMD .049, although its prompts admit between four and twelve valid colorings. PantryPlan has higher diversity, with PCMD .531, .386 and .455 across the three levels. Python at Levels 2 and 3 has only 14 and 15 eligible prompts out of 128, below the thirty-prompt reporting threshold. These missing entries reflect strict formatting failures, including LaTeX-escaped Python; this deployment has no provider refusals in those cells. Opus 5’s corresponding missing entries have a different cause: 1,021 and 1,024 of its 1,024 responses are refusals.

**The Level-4 probe.** The GPT-5.6 Sol Level-4 points in Fig. 3 average to .181 across 5 domains, using the pair-pooled  $1 - \hat{C}$  statistic and formatting normalization. This evaluation uses the direct OpenAI service, whereas Levels 1–3 use Azure. The deployments may differ, so their contrast does not isolate a level effect. Python is particularly sensitive to grading: strict grading verifies two of 1,024 responses, compared with 675 after normalization. Normalization changes the correct-response population on which diversity is measured. This difference measures sensitivity to output formatting; it does not establish correctness for the remaining responses.

**Concentration across deployments.** The seven strict-grading macros range from .142 for Claude Opus 5 to .320 for DeepSeek V4 Pro. They use the thirteen domain–level cells that are reportable for every deployment, excluding Python at Levels 2 and 3. The transformed summaries  $N_{\text{eff}}^{(2)} = 1/(1 - \text{macro PCMD})$  range from 1.17 to 1.47. These are transforms of the mean diversity, not averages of per-cell effective mode counts or bounds on unseen modes. At the largest observed macro, the corresponding mean probability of repeating a correct mode is about .68 under these prompt and cell weights.

MathIR defines modes by canonical derivation paths; the other domains use alternative output structures. Removing the three MathIR cells raises every deployment’s macro. Across the ten remaining common cells, the largest transformed summary is 1.58, for Kimi K3. Python contributes only Level 1 to this comparison, so this is an equal-cell mean, not an equal-domain mean

over four domains. The low diversity summaries therefore persist without `MathIR`, although their magnitudes depend on the mode definitions and averaging population.

Averaging each deployment over its own reportable cells instead changes the ranking: DeepSeek V4 Pro has .280 across its fifteen cells and falls below Kimi K3. Domain rankings also differ: GPT-5.4 and Kimi K3 have higher `Python` diversity, while DeepSeek V4 Pro has higher `Graph` diversity. Every deployment’s per-level mean is lower at Level 3 than at Level 1, although the intermediate level need not be monotone. Fig. 15 shows the normalized cohort one deployment per row.

Table 46: **Seven hosted deployments have low diversity among correct solutions.** Strict grading, original wording, 128 prompts per domain–level cell and eight stateless requests per prompt. Within-cell PCMD weights eligible prompts equally. Macro PCMD averages the same thirteen cells for every deployment, omitting `Python` L2/L3;  $N_{\text{eff}}^{(2)}$  transforms that macro. L1–L3 columns average each deployment’s own reportable cells: five domains per level, except four at L2/L3 for GPT-5.6 Sol and Opus 5. Thus the macro is not the mean of those three columns. A cell requires thirty prompts with at least two verified responses. Entries are point estimates.

| Deployment      | Macro PCMD | $N_{\text{eff}}^{(2)}$ | Macro PCMD by level |      |      |
|-----------------|------------|------------------------|---------------------|------|------|
|                 |            |                        | L1                  | L2   | L3   |
| DeepSeek V4 Pro | .320       | 1.47                   | .289                | .319 | .232 |
| Kimi K3         | .302       | 1.43                   | .374                | .307 | .253 |
| GPT-5.4         | .264       | 1.36                   | .338                | .247 | .224 |
| Grok 4.3        | .239       | 1.31                   | .275                | .179 | .171 |
| GPT-5.6 Sol     | .226       | 1.29                   | .256                | .229 | .183 |
| Claude Opus 4.8 | .215       | 1.27                   | .217                | .229 | .168 |
| Claude Opus 5   | .142       | 1.17                   | .140                | .164 | .121 |

**Level comparisons also change prompt guidance.** The Level-2 and Level-3 system messages add strategy guidance to `MathIR`, `Python` and `PantryPlan`: an algebraic isolation order, a nested small-divisor chain and a preference over ingredient families. Level 1 lacks this guidance. In the normalized original-wording cohort, pair-pooled  $1 - \hat{C}$  falls from Level 1 to Level 3 by .115 on average across the 20 measured deployment–domain pairs in these domains, compared with .048 across the 14 pairs in `Graph` and `Countdown`. Averaging the two unguided domains within each model, 7 of 7 deployments decline. The guided and unguided domains also differ in their tasks and solution spaces, so this comparison does not estimate the effect of guidance.

The prompt ablations test that effect within specified conditions. Removing `MathIR` guidance from the local 0.5B checkpoints leaves observed PCMD at .000 under both wordings (App. I). The separate 64-draw hosted study finds higher Level-2 `MathIR` diversity under neutral wording for both GPT deployments at fixed correct-draw budgets, with intervals excluding zero; Grok’s intervals include zero (App. K.2). These comparisons condition on jointly eligible prompts and do not match overall accuracy. The different responses to wording concern different models and evaluation populations. Cross-level narrowing describes concentration under the specified prompts; it does not isolate task difficulty from prompting or deployment differences.

## O COMPARISON ACROSS HOSTED DEPLOYMENTS

Figure 40 compares accuracy and success-conditional solution-mode diversity across hosted deployments. Its accuracy row uses a common formatting normalizer and the revised Opus 5 `Python` instructions at each level; hollow marks show that deployment’s original `Python` condition. Other accuracy cells use the original instructions. The diversity row uses strict grades and original instructions for every deployment, with the prompt-level estimand and eligibility criteria of App. N.1. Accuracy includes every response, including refusals and verification failures. Deployment settings differ, so the figure describes the stated configurations and does not rank models under matched prompts and compute.

The tables and `Graph` figure use the original instructions for every deployment. The `Python` comparison reports both instruction conditions under strict and formatting-normalized grad-

ing. Level 3 extends the test tasks; performance across levels need not follow a common difficulty ordering for all deployments.

**Evaluation population.** The deployments are GPT-5.6 Sol, Claude Opus 5, GPT-5.4, Grok 4.3, Kimi K3, Claude Opus 4.8, DeepSeek V4 Pro. The original-instruction cohort contains 107,520 responses: 15,360 per deployment, with the same 128 held-out prompts in each of five domains and three levels, and eight stateless responses per prompt. Requests use no model-side tools or conversation history, and the evaluation includes no task training or replay intervention.

**Provider configurations.** All deployments request medium reasoning effort and an 8,192-token native output limit. Claude Opus 5 and Claude Opus 4.8 use Anthropic Messages with adaptive thinking; the other deployments use hosted API routes.

The requested reasoning setting and output limit do not match compute across providers: effort labels and token accounting have provider-specific meanings. Temperature and nucleus sampling use provider defaults, whose exact values are unknown when the provider does not return them.

Strict grading checks the original executable answer; formatting-normalized grading applies the same typography corrections across deployments before the executable checks. The accuracy row of the main display uses normalized grades and the revised Opus 5 `Python` instruction at all three levels; its other accuracy cells use the original instructions. The diversity row uses strict grades and original instructions for every deployment. The tables below report original-instruction results under both grading rules. Prompt preferences and the Level-1 `PantryPlan` interface difference limit comparisons across levels. `Countdown` has no certified finite-total-support uniform reference. This inference-only comparison does not identify a training cause, a model-size effect, or a difficulty effect. Collision intervals are pointwise estimates for individual cells, not intervals for paired deployment differences.

The 15,360 scored DeepSeek V4 Pro responses include verification failures and truncations. Usage totals omit requests without reported usage and therefore give a lower bound on the cost of all attempted requests.

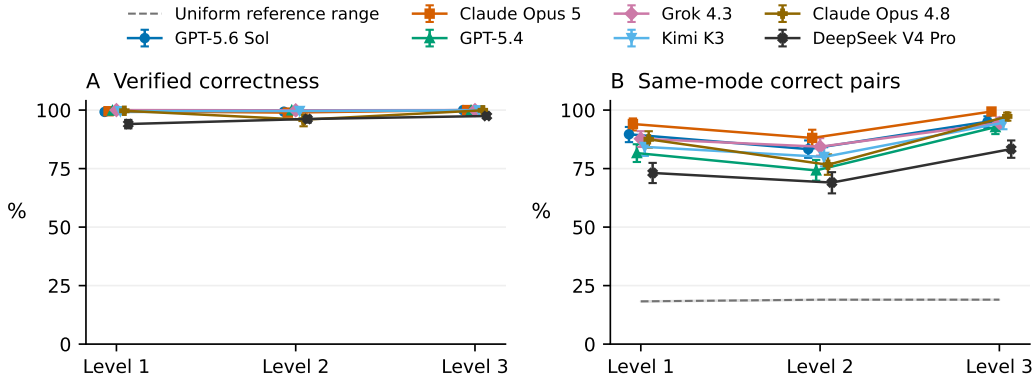


Figure 38: **Hosted Graph answers are usually correct but concentrated relative to uniform sampling.** Panel A shows strict per-response accuracy; panel B shows the fraction of correct pairs sharing a solution mode, pooling pairs across prompts. Colors and markers identify seven deployments, each with the same 128 prompts per level and eight stateless responses per prompt. Error bars are pointwise 95% intervals from resampling whole prompts. The gray band spans the deployments’ uniform-correct-mode references, weighted by their observed correct-pair counts; the dashed line is their mean. Level labels do not imply equal or monotonically increasing difficulty across deployments.

#### O.1 PYTHON PROMPT SENSITIVITY

Opus 5 receives the same 384 `Python` tasks under the original instructions and a direct arithmetic-expression instruction without a system message. Each condition contains eight responses from separate requests per task across three levels. Adaptive thinking, medium effort, an 8,192-token

output limit, allowed operators, public inputs, and graders are the same in both conditions. The revised instruction asks for a one-line boxed Python lambda returning any proper divisor for each listed input, without a preferred divisor. Task wording and system-message presence both change; provider state and collection time can also differ. The comparison describes these two configurations and does not isolate which change accounts for their difference. All responses contribute to accuracy and mode counts, including provider refusals.

Table 47: **Opus 5 rarely refuses Python tasks with direct instructions and no system message.** Each row contains 128 tasks with eight responses each (1,024 responses). R counts provider-declared refusals; A is per-response accuracy (%); D is mean `distinct@8`; C is correct-pair collision (%), pooling pairs within each level. Both conditions use the same executable grader and formatting normalizer. Original Level-2/3 collision is undefined because no task has two correct responses.

| Condition             | L | R    | A    | Strict |      | C     | Normalized |      | C |
|-----------------------|---|------|------|--------|------|-------|------------|------|---|
|                       |   |      |      | D      | D    |       | A          | D    |   |
| Original              | 1 | 409  | 38.1 | 0.91   | 93.0 | 60.0  | 1.12       | 90.4 |   |
|                       | 2 | 1021 | 0.3  | 0.02   | –    | 0.3   | 0.02       | –    |   |
|                       | 3 | 1024 | 0.0  | 0.00   | –    | 0.0   | 0.00       | –    |   |
| Plain user, no system | 1 | 0    | 74.9 | 1.42   | 80.2 | 100.0 | 1.49       | 79.9 |   |
|                       | 2 | 0    | 71.8 | 1.31   | 84.7 | 100.0 | 1.38       | 85.7 |   |
|                       | 3 | 3    | 76.2 | 1.38   | 82.7 | 99.7  | 1.41       | 84.1 |   |

The revised condition has no token-limit truncations. Strict grading accepts 2,282 responses; formatting normalization accepts 787 additional responses by converting escaped percent signs (`\%`) to Python modulo (`%`). Every non-refused answer therefore passes the executable checks after normalization. Normalized collision is 79.9/85.7/84.1%, versus conditional uniform references of 1.43/1.15/1.15%. All eight responses per task contribute to these results, including the three refusals.

Accuracy differences use a paired bootstrap of whole eight-response task groups across conditions (20,000 replicates). Normalized accuracy gains at Levels 1/2/3 are 40.0 [34.7, 45.5], 99.7 [99.3, 100.0], and 99.7 [99.2, 100.0] percentage points (pointwise 95% intervals). Tasks are paired; output draws are not paired across conditions. Collision weights tasks by their numbers of correct pairs, which differ between conditions. Its raw difference therefore compares different correct-pair populations and does not estimate a prompt effect at matched accuracy or on a common set of pairs.

**First-valid retry selection.** A retry condition replaces each of the revised-wording cohort’s three refused responses with the first verified-valid answer to an additional request. All three replacements pass formatting-normalized verification on the first retry and reproduce a solution mode already observed for that task. Selecting 3,072 valid responses therefore requires 3,075 requests. The selected set’s 100% accuracy is a consequence of selection, not unconditional model accuracy. Its `distinct@8` is unchanged at 1.492, 1.375, 1.414 at Levels 1–3. Level-3 correct-pair collision changes from 84.0909% to 84.1518%; the other levels are unchanged. This diagnostic uses variable request effort. The main figure uses the fixed eight-response condition, including all three refusals, without these replacements.

## O.2 REQUESTED-TEMPERATURE SENSITIVITY

Grok 4.3 and Kimi K3 each receive requested temperatures  $T = 1.0$  and  $T = 1.5$  on the same 120 tasks: eight prompts in each of five domains and three levels. Eight responses from separate requests per prompt give 960 responses per deployment–temperature condition and 3,840 in total. Within each deployment, the two conditions share all non-temperature request settings and run concurrently. Accuracy and mode counts include all responses, including verification failures and truncated outputs.

The providers accept both requested temperature settings, but their responses do not report effective sampling temperatures. DeepSeek is outside this comparison because its thinking-mode interface ignores the temperature setting. The results concern these requested settings on this 120-task subset and do not establish a general temperature response for all hosted deployments.

Each level summary averages five domains equally; All averages the 15 domain–level cells equally. Collision pools correct pairs within each cell, so tasks with more correct responses receive more weight. Eligible tasks and pair weights can differ between temperatures; the contrast does not hold accuracy or the correct-pair population fixed. An undefined cell makes the corresponding average

undefined. Pointwise 95% intervals use 20,000 paired whole-prompt bootstrap resamples, stratified by domain and level. Each resampled prompt includes all eight responses. Prompts are paired across conditions; output draws are not. Intervals have no multiplicity adjustment.

**Observed sensitivity.** For Grok, the observed diversity change is +0.050 [-0.058, +0.158] modes, and the collision change is -0.78 [-3.86, +2.12] percentage points. Both intervals include zero, so neither shows a clear diversity gain. The per-response accuracy change is -2.29 [-3.85, -0.94] percentage points. This does not establish invariance to temperature in general.

Kimi returns 696/960 token-limited outputs at  $T = 1.5$ , compared with 0 at  $T = 1.0$ . Strict accuracy falls from 96.56% to 21.25%, while raw diversity falls from 1.933 to 0.983 modes. The large increase in truncation accompanies losses in accuracy and distinct modes. These results do not isolate correct-output concentration at matched accuracy. The five-domain collision mean is undefined at  $T = 1.5$  because some cells have no eligible correct pairs. Neither model has provider-declared refusals in this experiment.

† marks a defined contrast whose interval is omitted because some bootstrap resamples have undefined cells. Five-domain averages require all five cell estimates; conditioning on only the estimable resamples would change the uncertainty calculation.

**Table 48: Higher requested temperature yields no clear Grok diversity gain and much lower Kimi accuracy.** Strict executable grades compare  $T = 1.5$  with  $T = 1.0$ . Each condition contains eight responses on each of 120 tasks. Level summaries average five domains equally; All averages 15 domain-level cells. A is per-response accuracy (%); D is mean distinct@8; C is correct-pair collision (%), pooling pairs within each cell. Differences subtract  $T = 1.0$  from  $T = 1.5$ : mode counts for D and percentage points for A/C. Brackets give pointwise 95% intervals from paired whole-prompt bootstrap resampling. † marks an omitted interval when some resamples have undefined cells; – denotes an undefined estimate or omitted interval.

| Deployment | Level | Metric | $T = 1.0$ | $T = 1.5$ | Difference [95% interval] |
|------------|-------|--------|-----------|-----------|---------------------------|
| Grok 4.3   | 1     | A      | 98.12     | 95.62     | -2.50 [-4.06, -0.94]      |
|            |       | D      | 1.900     | 2.000     | +0.100 [-0.100, +0.300]   |
|            |       | C      | 72.89     | 73.16     | +0.27 [-6.31, +6.37]      |
|            | 2     | A      | 99.06     | 99.38     | +0.31 [-0.94, +1.56]      |
|            |       | D      | 1.425     | 1.425     | +0.000 [-0.175, +0.200]   |
|            |       | C      | 86.61     | 85.20     | -1.41 [-6.26, +3.30]      |
|            | 3     | A      | 98.75     | 94.06     | -4.69 [-9.06, -0.94]      |
|            |       | D      | 1.500     | 1.550     | +0.050 [-0.125, +0.225]   |
|            |       | C      | 84.04     | 82.85     | -1.20 [-5.48, +2.79]      |
|            | All   | A      | 98.65     | 96.35     | -2.29 [-3.85, -0.94]      |
|            |       | D      | 1.608     | 1.658     | +0.050 [-0.058, +0.158]   |
|            |       | C      | 81.18     | 80.40     | -0.78 [-3.86, +2.12]      |
| Kimi K3    | 1     | A      | 92.50     | 22.50     | -70.00 [-75.62, -63.75]   |
|            |       | D      | 2.200     | 1.075     | -1.125 [-1.475, -0.775]   |
|            |       | C      | 66.23     | 53.12     | -13.11 [-]†               |
|            | 2     | A      | 98.12     | 20.31     | -77.81 [-82.50, -72.81]   |
|            |       | D      | 1.850     | 0.900     | -0.950 [-1.300, -0.625]   |
|            |       | C      | 72.18     | 66.55     | -5.63 [-]†                |
|            | 3     | A      | 99.06     | 20.94     | -78.12 [-83.75, -72.81]   |
|            |       | D      | 1.750     | 0.975     | -0.775 [-1.025, -0.525]   |
|            |       | C      | 78.33     | –         | –                         |
|            | All   | A      | 96.56     | 21.25     | -75.31 [-78.44, -72.19]   |
|            |       | D      | 1.933     | 0.983     | -0.950 [-1.133, -0.767]   |
|            |       | C      | 72.25     | –         | –                         |

Countdown has no finite-total-support uniform reference; uniform-reference comparisons therefore use the other four domains. Cross-level differences also reflect different task populations and do not establish a common ordering of model difficulty. These inference-only results do not identify a training or parameter-scale cause of concentration.

Table 49: **Formatting normalization leaves the temperature sensitivity largely unchanged.** Only rows differing from strict grading appear; all other rows are identical. Each condition contains eight responses on each of 120 tasks. Level summaries average five domains equally; All averages 15 domain-level cells. A is per-response accuracy (%); D is mean `distinct@8`; C is correct-pair collision (%), pooling pairs within each cell. Differences subtract  $T = 1.0$  from  $T = 1.5$ : mode counts for D and percentage points for A/C. Brackets give pointwise 95% intervals from paired whole-prompt bootstrap resampling. † marks an omitted interval when some resamples have undefined cells; – denotes an undefined estimate or omitted interval.

| Deployment | Level | Metric | $T = 1.0$ | $T = 1.5$ | Difference [95% interval] |
|------------|-------|--------|-----------|-----------|---------------------------|
| Kimi K3    | 1     | A      | 92.50     | 22.81     | -69.69 [-75.31, -63.44]   |
|            | 1     | C      | 66.23     | 53.61     | -12.62 [-]†               |
|            | 3     | A      | 99.38     | 20.94     | -78.44 [-84.06, -73.12]   |
|            | 3     | C      | 78.18     | –         | –                         |
|            | All   | A      | 96.67     | 21.35     | -75.31 [-78.44, -72.19]   |
|            | All   | C      | 72.20     | –         | –                         |

### O.3 GPT-5.6 SOL: TEMPERATURE, PASS@8 AND VERIFIED MODES

Figure 39 plots empirical `pass@8` against `distinct@8` for  $T \in \{0.0, 0.5, 1.0, 1.5, 2.0\}$ , with reasoning `none`. Each condition uses the same 480 held-out prompts (32 per domain-level cell), with eight stateless responses per prompt: 3,840 responses per temperature and 19,200 in total. A prompt contributes one to `pass@8` if at least one of its eight answers is verified correct, and contributes its number of distinct correct canonical keys to `distinct@8`. Both metrics average over all prompts, including those with no correct answer. The empirical `pass@8` uses the observed eight-response groups;  $1 - (1 - p)^8$  applied to pooled accuracy would ignore differences among prompts. Raw `distinct@8` depends on both success and variation among correct solution modes, so its changes need not track `pass@8`.

Prompts are selected independently of model outputs within each domain-level cell. The evaluation comprises 120-prompt and 360-prompt subsets collected at different times. Temperatures are interleaved for the 360-prompt subset; zero and nonzero temperatures are collected separately for the 120-prompt subset. Service changes can therefore confound temperature contrasts, and prompt composition can affect comparisons between subsets.

Requests use the same prompt wording, verifiers and sampling controls except temperature, with an 8,192-token output cap. Returned metadata identifies `gpt-5.6-sol-2026-07-09`, reasoning `none`, `top_p = 0.98` and the requested temperature in every condition. The exposed settings do not establish unchanged internal service behavior or independent provider randomness. A zero-temperature request does not establish deterministic outputs. The deployment rejects non-default temperatures with medium reasoning and rejects temperature 2.5; this sweep therefore does not test temperature robustness under medium reasoning.

Strict grading applies the executable validators directly. Normalized grading uses the typography rules in App. N before the same validators, preserving strict successes and their mode keys.

Each level averages five domains equally; All averages all 15 cells. Pointwise 95 percent intervals use 20,000 paired whole-prompt bootstrap resamples stratified by domain and level, keeping all eight responses together and matching prompts across temperatures. Individual model responses are not paired across conditions. There is no multiplicity adjustment. A zero-width empirical interval reflects constant observed groups, not certain success on unseen prompts.

**Coverage and sampled solution modes.** At  $T = 0.0$  and  $T = 2.0$ , normalized `pass@8` is 65.62% and 79.38%, respectively, while `distinct@8` is 0.673 and 1.146. Across these 5 measured temperatures, the highest observed aggregate `pass@8` occurs at  $T = 1.5$  and the highest `distinct@8` occurs at  $T = 2.0$ . These five temperatures do not establish a population optimum or a continuous Pareto boundary, and level-specific curves differ.

Per-response accuracy changes from 62.11% to 55.94% between the endpoints. That diagnostic is distinct from `pass@8`: success within eight draws and the reliability of an individual draw need not change together as temperature varies.

Table 50: **Nonzero temperatures improve eight-draw coverage in this sweep.** P is solved prompts (% pass@8); D is distinct@8; A is per-response accuracy (%). G is the grading: S strict, N formatting-normalized. Each temperature uses the same 480 prompts (32 per domain-level cell), eight responses each, and reasoning none. Level rows average five domains equally; All averages all 15 cells. Brackets give pointwise 95% whole-prompt bootstrap intervals.

| $T$ | Level | G | P [95% interval]  | D [95% interval]     | A [95% interval]  |
|-----|-------|---|-------------------|----------------------|-------------------|
| 0.0 | 1     | S | 55.0 [48.8, 61.3] | 0.569 [0.500, 0.637] | 53.9 [47.7, 60.1] |
|     | 2     |   | 45.0 [39.4, 50.6] | 0.456 [0.400, 0.512] | 39.5 [34.4, 44.7] |
|     | 3     |   | 49.4 [42.5, 56.2] | 0.519 [0.444, 0.594] | 44.5 [38.3, 50.9] |
|     | All   |   | 49.8 [46.2, 53.3] | 0.515 [0.475, 0.554] | 46.0 [42.6, 49.4] |
| 0.5 | 1     | S | 70.6 [64.4, 76.9] | 0.956 [0.850, 1.069] | 57.5 [51.9, 63.0] |
|     | 2     |   | 50.0 [44.4, 55.6] | 0.525 [0.463, 0.594] | 38.6 [33.9, 43.4] |
|     | 3     |   | 54.4 [48.1, 60.6] | 0.581 [0.506, 0.662] | 43.0 [37.1, 48.9] |
|     | All   |   | 58.3 [54.8, 61.9] | 0.688 [0.640, 0.738] | 46.4 [43.3, 49.5] |
| 1.0 | 1     | S | 75.6 [69.4, 81.2] | 1.181 [1.050, 1.312] | 57.3 [52.0, 62.7] |
|     | 2     |   | 51.9 [45.6, 58.1] | 0.581 [0.506, 0.662] | 37.6 [33.2, 42.0] |
|     | 3     |   | 60.6 [54.4, 66.9] | 0.713 [0.625, 0.806] | 42.7 [37.3, 48.1] |
|     | All   |   | 62.7 [59.2, 66.2] | 0.825 [0.767, 0.885] | 45.9 [42.9, 48.8] |
| 1.5 | 1     | S | 80.6 [75.0, 86.2] | 1.413 [1.269, 1.556] | 56.6 [51.7, 61.4] |
|     | 2     |   | 58.1 [51.9, 64.4] | 0.650 [0.569, 0.738] | 38.4 [34.3, 42.7] |
|     | 3     |   | 65.6 [59.4, 71.2] | 0.838 [0.731, 0.956] | 43.0 [38.0, 48.2] |
|     | All   |   | 68.1 [64.8, 71.5] | 0.967 [0.900, 1.035] | 46.0 [43.3, 48.8] |
| 2.0 | 1     | S | 83.8 [78.1, 88.8] | 1.519 [1.375, 1.663] | 59.7 [54.8, 64.5] |
|     | 2     |   | 54.4 [48.1, 60.6] | 0.600 [0.525, 0.675] | 35.3 [31.4, 39.3] |
|     | 3     |   | 61.9 [56.2, 67.5] | 0.819 [0.719, 0.925] | 40.5 [35.8, 45.5] |
|     | All   |   | 66.7 [63.3, 70.0] | 0.979 [0.915, 1.044] | 45.2 [42.6, 47.8] |
| 0.0 | 1     | N | 70.6 [65.0, 76.2] | 0.725 [0.662, 0.788] | 70.5 [64.5, 76.2] |
|     | 2     |   | 60.6 [55.0, 66.2] | 0.613 [0.556, 0.669] | 54.5 [49.0, 59.9] |
|     | 3     |   | 65.6 [59.4, 71.9] | 0.681 [0.606, 0.756] | 61.4 [55.2, 67.6] |
|     | All   |   | 65.6 [62.3, 69.0] | 0.673 [0.635, 0.710] | 62.1 [58.8, 65.5] |
| 0.5 | 1     | N | 76.9 [71.2, 82.5] | 1.038 [0.931, 1.150] | 71.7 [66.4, 77.0] |
|     | 2     |   | 68.1 [62.5, 73.8] | 0.706 [0.644, 0.769] | 53.7 [48.8, 58.6] |
|     | 3     |   | 71.9 [65.6, 77.5] | 0.775 [0.700, 0.850] | 58.1 [52.3, 63.9] |
|     | All   |   | 72.3 [69.2, 75.4] | 0.840 [0.792, 0.890] | 61.2 [58.2, 64.3] |
| 1.0 | 1     | N | 79.4 [73.8, 84.4] | 1.238 [1.106, 1.369] | 70.4 [65.2, 75.5] |
|     | 2     |   | 69.4 [63.7, 75.0] | 0.756 [0.681, 0.831] | 51.6 [47.1, 56.2] |
|     | 3     |   | 76.9 [71.2, 82.5] | 0.894 [0.806, 0.988] | 55.2 [49.8, 60.7] |
|     | All   |   | 75.2 [72.1, 78.3] | 0.963 [0.904, 1.021] | 59.1 [56.2, 62.0] |
| 1.5 | 1     | N | 82.5 [77.5, 87.5] | 1.494 [1.344, 1.644] | 68.5 [63.7, 73.3] |
|     | 2     |   | 75.0 [69.4, 80.6] | 0.844 [0.762, 0.931] | 49.8 [45.5, 54.3] |
|     | 3     |   | 81.9 [76.2, 87.5] | 1.038 [0.931, 1.156] | 54.7 [49.6, 59.8] |
|     | All   |   | 79.8 [76.7, 82.9] | 1.125 [1.056, 1.196] | 57.7 [54.9, 60.5] |
| 2.0 | 1     | N | 85.0 [80.0, 90.0] | 1.606 [1.456, 1.762] | 69.6 [64.8, 74.3] |
|     | 2     |   | 72.5 [66.9, 78.1] | 0.800 [0.731, 0.869] | 45.7 [41.7, 49.8] |
|     | 3     |   | 80.6 [75.0, 86.2] | 1.031 [0.931, 1.137] | 52.5 [47.7, 57.5] |
|     | All   |   | 79.4 [76.2, 82.5] | 1.146 [1.079, 1.212] | 55.9 [53.3, 58.6] |

The paired  $T = 2.0$  minus  $T = 0.0$  contrast is 13.8 [10.8, 16.7] pass@8 percentage points and 0.473 [0.417, 0.529] distinct correct modes, with pointwise 95% intervals.

**Response outcomes.** At every temperature ( $T = 0.0$ ,  $T = 0.5$ ,  $T = 1.0$ ,  $T = 1.5$  and  $T = 2.0$ ), 0 of 3840 outputs are token-limited and 0 are provider-declared refusals. Unsuccessful answers remain in the eight-response groups.

**Sensitivity across prompt subsets.** We compare the same temperature contrast within each prompt subset. The subsets differ in prompts and collection periods, so differences between them do not isolate a causal effect of time.

Across all 480 prompts, the paired  $T = 2$  minus  $T = 1.5$  contrast is -0.4 [-2.5, 1.7] pass@8 percentage points and 0.021 [-0.029, 0.071] distinct modes (95% intervals). 120-prompt subset: the paired  $T = 2$  minus  $T = 1.5$  contrast is -1.7 [-5.0, 0.8] pass@8 percentage points and -0.008 [-0.083,

0.067] distinct modes (95% intervals). 360-prompt subset: the paired  $T = 2$  minus  $T = 1.5$  contrast is 0.0 [-2.8, 2.8]  $\text{pass@8}$  percentage points and 0.031 [-0.033, 0.094] distinct modes (95% intervals).

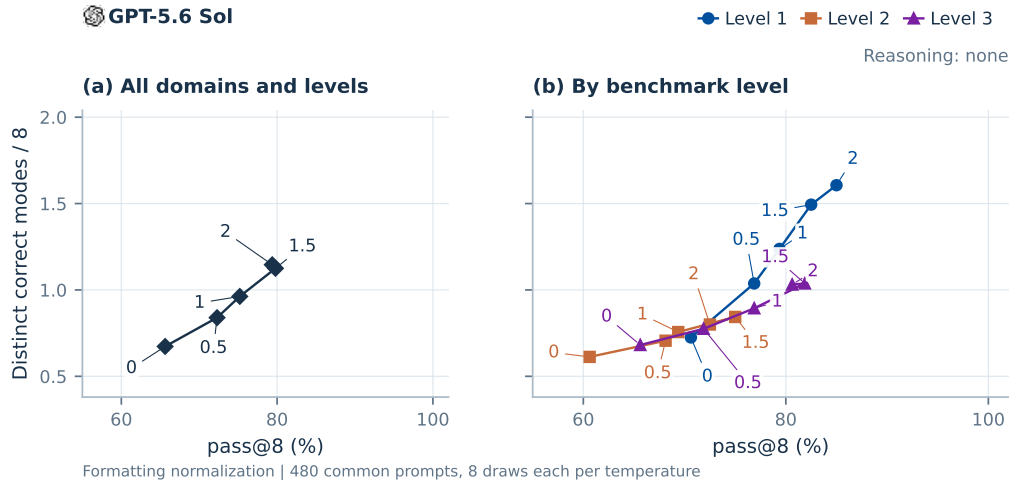


Figure 39: **Nonzero temperatures yield greater observed eight-draw coverage.** GPT-5.6 Sol with reasoning none, typography normalization, and the same 480 prompts at each of five temperatures (32 per domain-level cell, eight responses each). Left: an equal mean over all 15 domain-level cells; right: equal means over five domains at each level, identified by color and marker. The horizontal axis is empirical  $\text{pass@8}$ ; the vertical axis averages distinct correct solution modes per eight responses, including unsolved prompts as zero. Labels give requested temperatures; segments connect measured point estimates in temperature order. Table 50 gives pointwise 95% whole-prompt bootstrap intervals. Distinct-mode counts depend on both success and variation among correct answers, and need not move with  $\text{pass@8}$ .

#### O.4 NATIVE PROVIDER OUTCOMES AND UNDEFINED CONCENTRATION

Provider metadata distinguish refusal and filtering from executable correctness. A refusal does not establish that the underlying model cannot solve the task. Classification uses native stop reasons, refusal fields and filter metadata; it does not detect refusals expressed only in ordinary answer text. An empty visible answer is a failed draw, including when the response contains reasoning output. Empty-answer and refusal counts therefore need not agree.

Correct-pair collision requires at least two correct draws on one prompt. It is undefined otherwise, as is a five-domain mean containing any undefined domain. Dashes denote these undefined values. Accuracy and distinct-mode counts include all responses, with zero correct modes on unsuccessful prompts.

Native completion states over the 15,360 original-instruction responses per deployment are:

- GPT-5.6 Sol — completed: 15,360
- Claude Opus 5 — end\_turn: 12,838, refusal: 2,522
- GPT-5.4 — completed: 15,355, incomplete: 5
- Grok 4.3 — stop: 15,360
- Kimi K3 — length: 15, stop: 15,345
- Claude Opus 4.8 — end\_turn: 15,355, max\_tokens: 5
- DeepSeek V4 Pro — length: 632, stop: 14,728

Claude Opus 5 returns provider-declared refusals for 409, 1,021, 1,024 of the 1,024 Python requests at Levels 1, 2 and 3, respectively. Across those three levels, the provider labels the refusals cyber: 2,454. These labels describe service behavior on factor-selection tasks; they do not measure concentration among correct Python outputs.

Table 51: **Provider-declared refusals occur only for Opus 5 in this cohort.** Each original-instruction cell contains 128 prompts with eight responses each (1,024 responses). R counts native refusal signals, F native content filters, and E responses without visible answer text, including reasoning-only outputs. Counters can overlap and are separate from executable grades. Only nonzero cells appear; the other 83 of 105 cells have  $R = F = E = 0$ , including all cells for GPT-5.6 Sol, Grok 4.3, Kimi K3. Zero R does not exclude refusals expressed only in answer text.

| Deployment      | L | Domain     | R    | F | E    |
|-----------------|---|------------|------|---|------|
| Claude Opus 5   | 1 | Python     | 409  | 0 | 409  |
|                 | 2 | Python     | 1021 | 0 | 1021 |
|                 | 2 | MathIR     | 35   | 0 | 35   |
|                 | 3 | Python     | 1024 | 0 | 1024 |
| GPT-5.4         | 3 | MathIR     | 33   | 0 | 32   |
|                 | 1 | PantryPlan | 0    | 0 | 1    |
|                 | 2 | Countdown  | 0    | 0 | 1    |
|                 | 3 | Countdown  | 0    | 0 | 1    |
| Claude Opus 4.8 | 3 | PantryPlan | 0    | 0 | 2    |
|                 | 3 | Countdown  | 0    | 0 | 3    |
| DeepSeek V4 Pro | 3 | PantryPlan | 0    | 0 | 1    |
|                 | 1 | Graph      | 0    | 0 | 59   |
|                 | 1 | MathIR     | 0    | 0 | 4    |
|                 | 1 | PantryPlan | 0    | 0 | 13   |
|                 | 2 | Graph      | 0    | 0 | 38   |
|                 | 2 | Countdown  | 0    | 0 | 1    |
|                 | 2 | MathIR     | 0    | 0 | 59   |
|                 | 2 | PantryPlan | 0    | 0 | 80   |
|                 | 3 | Graph      | 0    | 0 | 24   |
|                 | 3 | Countdown  | 0    | 0 | 7    |
|                 | 3 | MathIR     | 0    | 0 | 100  |
|                 | 3 | PantryPlan | 0    | 0 | 246  |

Table 53: **Verified hosted outputs concentrate on a few observed solution modes.** Each original-instruction cell contains 128 prompts with eight responses each. M/P are mean@8/pass@8 (%); D averages distinct correct solution modes per prompt, including zeros. C is the fraction of correct pairs sharing a mode (%), pooling pairs across prompts; it differs from one minus prompt-averaged PCMD. Intervals are pointwise 95% percentiles from 2,000 whole-prompt bootstrap resamples within each cell. U is the uniform-correct-mode reference (%), using certified support counts and the same pair weights. A dash in C denotes no eligible pairs; U is also undefined when total support is unknown (Countdown). Grades use the original executable answers.

| Deployment    | Level | Domain     | M     | P     | D    | C (95% interval)     | U    |
|---------------|-------|------------|-------|-------|------|----------------------|------|
| GPT-5.6 Sol   | 1     | Graph      | 99.2  | 100.0 | 1.30 | 89.7 [86.3, 92.8]    | 18.3 |
|               | 1     | Countdown  | 86.1  | 98.4  | 1.77 | 71.2 [66.1, 75.9]    | –    |
|               | 1     | Python     | 40.4  | 90.6  | 1.11 | 81.8 [76.9, 87.2]    | 0.8  |
|               | 1     | MathIR     | 89.3  | 89.8  | 1.38 | 80.3 [75.4, 85.0]    | 20.0 |
|               | 1     | PantryPlan | 100.0 | 100.0 | 3.00 | 46.9 [42.2, 51.5]    | 6.5  |
|               | 2     | Graph      | 99.3  | 100.0 | 1.48 | 83.3 [79.6, 87.0]    | 18.9 |
|               | 2     | Countdown  | 65.4  | 86.7  | 1.57 | 64.7 [59.0, 70.6]    | –    |
|               | 2     | Python     | 8.5   | 11.7  | 0.12 | 100.0 [100.0, 100.0] | 0.7  |
|               | 2     | MathIR     | 86.2  | 93.0  | 0.93 | 100.0 [100.0, 100.0] | 20.0 |
|               | 2     | PantryPlan | 55.3  | 96.1  | 1.72 | 63.6 [56.8, 70.3]    | 6.9  |
|               | 3     | Graph      | 100.0 | 100.0 | 1.16 | 95.1 [92.4, 97.3]    | 19.0 |
|               | 3     | Countdown  | 71.2  | 89.8  | 1.38 | 76.4 [71.1, 81.8]    | –    |
|               | 3     | Python     | 9.4   | 14.1  | 0.14 | 100.0 [100.0, 100.0] | 0.9  |
|               | 3     | MathIR     | 97.7  | 97.7  | 0.98 | 100.0 [100.0, 100.0] | 20.0 |
|               | 3     | PantryPlan | 65.3  | 100.0 | 2.22 | 51.0 [45.3, 57.3]    | 6.8  |
| Claude Opus 5 | 1     | Graph      | 99.5  | 100.0 | 1.15 | 94.0 [91.3, 96.4]    | 18.3 |
|               | 1     | Countdown  | 100.0 | 100.0 | 1.59 | 78.8 [74.6, 83.0]    | –    |
|               | 1     | Python     | 38.1  | 81.2  | 0.91 | 93.0 [88.7, 96.6]    | 0.7  |
|               | 1     | MathIR     | 90.5  | 91.4  | 1.09 | 92.2 [88.8, 95.4]    | 20.0 |
|               | 1     | PantryPlan | 98.4  | 100.0 | 1.86 | 73.5 [68.5, 78.2]    | 6.5  |
|               | 2     | Graph      | 98.9  | 100.0 | 1.32 | 88.1 [84.5, 91.6]    | 18.9 |
|               | 2     | Countdown  | 100.0 | 100.0 | 1.81 | 69.3 [65.2, 73.6]    | –    |
|               | 2     | Python     | 0.3   | 2.3   | 0.02 | –                    | –    |
|               | 2     | MathIR     | 89.1  | 98.4  | 1.06 | 96.5 [93.9, 98.6]    | 20.0 |
|               | 2     | PantryPlan | 94.5  | 100.0 | 1.56 | 80.7 [76.0, 85.3]    | 6.4  |
|               | 3     | Graph      | 100.0 | 100.0 | 1.02 | 99.4 [98.3, 100.0]   | 19.0 |
|               | 3     | Countdown  | 99.9  | 100.0 | 1.66 | 76.2 [71.7, 80.7]    | –    |
|               | 3     | Python     | 0.0   | 0.0   | 0.00 | –                    | –    |
|               | 3     | MathIR     | 95.3  | 99.2  | 1.02 | 98.8 [97.5, 99.8]    | 20.0 |
|               | 3     | PantryPlan | 93.6  | 100.0 | 1.63 | 78.8 [74.2, 83.4]    | 6.5  |
| GPT-5.4       | 1     | Graph      | 99.8  | 100.0 | 1.57 | 81.6 [77.8, 85.4]    | 18.3 |

*Strict executable grades, continued.*

| Deployment      | Level | Domain     | M     | P     | D    | C (95% interval)     | U    |
|-----------------|-------|------------|-------|-------|------|----------------------|------|
| Grok 4.3        | 1     | Countdown  | 91.8  | 100.0 | 1.73 | 76.6 [71.2, 81.3]    | –    |
|                 | 1     | Python     | 87.6  | 100.0 | 2.52 | 56.4 [51.3, 61.5]    | 1.4  |
|                 | 1     | MathIR     | 89.3  | 90.6  | 1.45 | 78.6 [74.0, 83.4]    | 20.0 |
|                 | 1     | PantryPlan | 99.9  | 100.0 | 3.16 | 42.8 [38.4, 47.4]    | 6.5  |
|                 | 2     | Graph      | 99.9  | 100.0 | 1.74 | 74.2 [69.8, 78.7]    | 19.0 |
|                 | 2     | Countdown  | 91.1  | 100.0 | 1.98 | 63.0 [58.6, 67.5]    | –    |
|                 | 2     | Python     | 47.9  | 96.1  | 1.47 | 72.9 [64.6, 80.2]    | 1.0  |
|                 | 2     | MathIR     | 87.0  | 92.2  | 0.92 | 100.0 [100.0, 100.0] | 20.0 |
|                 | 2     | PantryPlan | 77.0  | 100.0 | 1.93 | 63.2 [58.4, 68.6]    | 6.8  |
|                 | 3     | Graph      | 99.8  | 100.0 | 1.23 | 92.7 [89.7, 95.4]    | 19.0 |
|                 | 3     | Countdown  | 82.2  | 93.8  | 1.72 | 69.1 [64.4, 73.5]    | –    |
|                 | 3     | Python     | 50.1  | 97.7  | 1.58 | 74.4 [68.6, 79.8]    | 1.1  |
|                 | 3     | MathIR     | 97.7  | 97.7  | 0.98 | 100.0 [100.0, 100.0] | 20.0 |
|                 | 3     | PantryPlan | 86.5  | 100.0 | 2.34 | 57.8 [52.8, 63.4]    | 6.6  |
|                 | 1     | Graph      | 100.0 | 100.0 | 1.40 | 87.8 [84.2, 91.0]    | 18.3 |
|                 | 1     | Countdown  | 99.9  | 100.0 | 2.05 | 64.8 [60.1, 69.5]    | –    |
|                 | 1     | Python     | 99.8  | 100.0 | 1.41 | 88.0 [84.8, 91.0]    | 1.4  |
|                 | 1     | MathIR     | 92.4  | 97.7  | 1.24 | 90.1 [86.9, 93.1]    | 20.0 |
|                 | 1     | PantryPlan | 99.4  | 100.0 | 3.77 | 31.4 [27.8, 35.1]    | 6.5  |
|                 | 2     | Graph      | 99.9  | 100.0 | 1.52 | 84.3 [80.5, 88.0]    | 19.0 |
|                 | 2     | Countdown  | 99.8  | 100.0 | 1.99 | 65.8 [61.8, 70.2]    | –    |
|                 | 2     | Python     | 99.8  | 100.0 | 1.05 | 98.8 [97.8, 99.6]    | 1.1  |
|                 | 2     | MathIR     | 97.5  | 100.0 | 1.12 | 94.9 [92.4, 97.2]    | 20.0 |
|                 | 2     | PantryPlan | 99.5  | 100.0 | 2.16 | 66.6 [61.2, 71.4]    | 6.5  |
|                 | 3     | Graph      | 100.0 | 100.0 | 1.17 | 94.5 [92.0, 96.8]    | 19.0 |
|                 | 3     | Countdown  | 99.9  | 100.0 | 1.98 | 66.3 [61.7, 71.0]    | –    |
|                 | 3     | Python     | 99.9  | 100.0 | 1.04 | 98.6 [97.2, 99.6]    | 1.2  |
|                 | 3     | MathIR     | 95.0  | 100.0 | 1.05 | 97.8 [95.9, 99.3]    | 20.0 |
|                 | 3     | PantryPlan | 99.2  | 100.0 | 2.52 | 58.0 [53.1, 63.5]    | 6.5  |
| Kimi K3         | 1     | Graph      | 99.5  | 100.0 | 1.48 | 84.3 [80.4, 87.8]    | 18.3 |
|                 | 1     | Countdown  | 99.1  | 100.0 | 1.91 | 68.4 [63.8, 73.3]    | –    |
|                 | 1     | Python     | 97.9  | 100.0 | 2.52 | 57.7 [53.5, 61.9]    | 1.4  |
|                 | 1     | MathIR     | 90.2  | 92.2  | 1.47 | 75.8 [70.7, 80.7]    | 20.0 |
|                 | 1     | PantryPlan | 97.5  | 100.0 | 4.18 | 26.2 [23.1, 29.2]    | 6.5  |
|                 | 2     | Graph      | 99.6  | 100.0 | 1.62 | 80.0 [76.0, 84.1]    | 19.0 |
|                 | 2     | Countdown  | 98.4  | 100.0 | 1.98 | 64.4 [59.9, 68.7]    | –    |
|                 | 2     | Python     | 98.2  | 100.0 | 2.44 | 61.9 [57.6, 66.2]    | 1.1  |
|                 | 2     | MathIR     | 94.3  | 100.0 | 1.02 | 99.8 [99.3, 100.0]   | 20.0 |
|                 | 2     | PantryPlan | 93.3  | 100.0 | 3.20 | 41.3 [36.6, 45.9]    | 6.6  |
|                 | 3     | Graph      | 99.9  | 100.0 | 1.19 | 94.6 [91.8, 96.9]    | 19.0 |
|                 | 3     | Countdown  | 97.5  | 100.0 | 1.84 | 70.6 [66.1, 74.8]    | –    |
|                 | 3     | Python     | 98.0  | 100.0 | 2.34 | 64.7 [60.1, 69.0]    | 1.2  |
|                 | 3     | MathIR     | 98.1  | 100.0 | 1.00 | 100.0 [100.0, 100.0] | 20.0 |
|                 | 3     | PantryPlan | 95.6  | 100.0 | 3.13 | 43.4 [38.6, 48.3]    | 6.5  |
| Claude Opus 4.8 | 1     | Graph      | 99.7  | 100.0 | 1.34 | 87.5 [83.9, 91.0]    | 18.3 |
|                 | 1     | Countdown  | 95.6  | 99.2  | 1.78 | 69.8 [64.9, 74.5]    | –    |
|                 | 1     | Python     | 98.0  | 100.0 | 1.40 | 85.7 [81.7, 89.4]    | 1.4  |
|                 | 1     | MathIR     | 92.3  | 94.5  | 1.10 | 93.7 [91.1, 96.4]    | 20.0 |
|                 | 1     | PantryPlan | 96.6  | 100.0 | 2.59 | 55.3 [50.3, 60.1]    | 6.5  |
|                 | 2     | Graph      | 95.9  | 100.0 | 1.66 | 76.6 [72.3, 81.1]    | 18.8 |
|                 | 2     | Countdown  | 99.9  | 100.0 | 1.99 | 65.3 [60.9, 69.9]    | –    |
|                 | 2     | Python     | 100.0 | 100.0 | 1.34 | 86.0 [82.3, 89.6]    | 1.2  |
|                 | 2     | MathIR     | 97.0  | 100.0 | 1.24 | 90.1 [86.7, 93.4]    | 20.0 |
|                 | 2     | PantryPlan | 99.5  | 100.0 | 2.11 | 67.9 [62.8, 73.1]    | 6.5  |
|                 | 3     | Graph      | 99.9  | 100.0 | 1.08 | 97.3 [95.5, 98.9]    | 19.0 |
|                 | 3     | Countdown  | 97.9  | 99.2  | 1.80 | 72.5 [68.2, 76.6]    | –    |
|                 | 3     | Python     | 98.9  | 100.0 | 1.32 | 86.4 [82.5, 90.1]    | 1.2  |
|                 | 3     | MathIR     | 99.1  | 100.0 | 1.03 | 99.1 [98.1, 99.8]    | 20.0 |
|                 | 3     | PantryPlan | 98.5  | 100.0 | 2.33 | 61.6 [56.7, 66.6]    | 6.5  |
| DeepSeek V4 Pro | 1     | Graph      | 94.0  | 100.0 | 1.80 | 73.1 [68.8, 77.4]    | 18.1 |
|                 | 1     | Countdown  | 100.0 | 100.0 | 2.02 | 65.8 [61.4, 70.3]    | –    |
|                 | 1     | Python     | 99.3  | 100.0 | 1.41 | 88.8 [86.1, 91.4]    | 1.4  |
|                 | 1     | MathIR     | 90.1  | 92.2  | 1.11 | 93.9 [91.2, 96.2]    | 20.0 |
|                 | 1     | PantryPlan | 98.5  | 100.0 | 3.73 | 33.5 [29.8, 37.2]    | 6.5  |
|                 | 2     | Graph      | 96.2  | 100.0 | 1.97 | 69.0 [64.4, 73.5]    | 19.1 |
|                 | 2     | Countdown  | 99.9  | 100.0 | 2.04 | 63.4 [59.2, 67.7]    | –    |
|                 | 2     | Python     | 99.6  | 100.0 | 1.05 | 98.4 [97.1, 99.4]    | 1.2  |
|                 | 2     | MathIR     | 85.4  | 95.3  | 1.82 | 51.7 [49.8, 53.9]    | 20.0 |
|                 | 2     | PantryPlan | 87.1  | 100.0 | 2.34 | 61.1 [56.0, 66.2]    | 6.5  |
|                 | 3     | Graph      | 97.5  | 100.0 | 1.52 | 83.4 [79.6, 87.0]    | 18.9 |
|                 | 3     | Countdown  | 99.2  | 100.0 | 1.98 | 67.5 [63.1, 72.2]    | –    |
|                 | 3     | Python     | 99.6  | 100.0 | 1.08 | 97.8 [96.3, 99.0]    | 1.2  |
|                 | 3     | MathIR     | 88.5  | 100.0 | 1.44 | 83.6 [79.2, 87.6]    | 20.0 |
|                 | 3     | PantryPlan | 70.3  | 98.4  | 2.08 | 61.9 [55.9, 68.2]    | 6.5  |

Table 52: **Hosted deployments average fewer than three distinct correct modes per prompt.** Each row averages five domains equally under strict grading and original instructions, with 128 prompts per domain and eight responses per prompt. M/P are  $\text{mean}@8/\text{pass}@8$  (%); D averages distinct correct solution modes, including zeros. C is correct-pair collision (%), pooling pairs within each domain before averaging domains. Pointwise 95% intervals use 2,000 whole-prompt bootstrap resamples within domains. A dash denotes an undefined domain collision and hence an undefined five-domain mean. Provider settings differ, so these are descriptive deployment comparisons.

| Deployment      | Level | M    | P     | D    | C (95% interval)  |
|-----------------|-------|------|-------|------|-------------------|
| GPT-5.6 Sol     | 1     | 83.0 | 95.8  | 1.71 | 74.0 [71.9, 76.1] |
| GPT-5.6 Sol     | 2     | 62.9 | 77.5  | 1.16 | 82.3 [80.3, 84.4] |
| GPT-5.6 Sol     | 3     | 68.7 | 80.3  | 1.18 | 84.5 [82.8, 86.2] |
| Claude Opus 5   | 1     | 85.3 | 94.5  | 1.32 | 86.3 [84.5, 88.0] |
| Claude Opus 5   | 2     | 76.6 | 80.2  | 1.16 | —                 |
| Claude Opus 5   | 3     | 77.8 | 79.8  | 1.07 | —                 |
| GPT-5.4         | 1     | 93.7 | 98.1  | 2.09 | 67.2 [65.1, 69.3] |
| GPT-5.4         | 2     | 80.6 | 97.7  | 1.61 | 74.7 [72.4, 76.8] |
| GPT-5.4         | 3     | 83.3 | 97.8  | 1.57 | 78.8 [76.9, 80.7] |
| Grok 4.3        | 1     | 98.3 | 99.5  | 1.97 | 72.4 [70.7, 74.0] |
| Grok 4.3        | 2     | 99.3 | 100.0 | 1.57 | 82.1 [80.6, 83.7] |
| Grok 4.3        | 3     | 98.8 | 100.0 | 1.55 | 83.0 [81.5, 84.6] |
| Kimi K3         | 1     | 96.9 | 98.4  | 2.31 | 62.5 [60.6, 64.4] |
| Kimi K3         | 2     | 96.8 | 100.0 | 2.05 | 69.4 [67.6, 71.2] |
| Kimi K3         | 3     | 97.8 | 100.0 | 1.90 | 74.6 [73.1, 76.2] |
| Claude Opus 4.8 | 1     | 96.4 | 98.8  | 1.64 | 78.4 [76.6, 80.3] |
| Claude Opus 4.8 | 2     | 98.5 | 100.0 | 1.67 | 77.2 [75.4, 79.1] |
| Claude Opus 4.8 | 3     | 98.9 | 99.8  | 1.51 | 83.4 [81.8, 84.9] |
| DeepSeek V4 Pro | 1     | 96.4 | 98.4  | 2.01 | 71.0 [69.4, 72.7] |
| DeepSeek V4 Pro | 2     | 93.7 | 99.1  | 1.84 | 68.7 [67.0, 70.3] |
| DeepSeek V4 Pro | 3     | 91.0 | 99.7  | 1.62 | 78.8 [76.9, 80.9] |

Table 54: **Formatting normalization increases accuracy without consistently reducing collision.** Each original-instruction cell contains 128 prompts with eight responses each. M/P are  $\text{mean}@8/\text{pass}@8$  (%); D averages distinct correct solution modes per prompt, including zeros. C is the fraction of correct pairs sharing a mode (%), pooling pairs across prompts; it differs from one minus prompt-averaged PCMD. Intervals are pointwise 95% percentiles from 2,000 whole-prompt bootstrap resamples within each cell. U is the uniform-correct-mode reference (%), using certified support counts and the same pair weights. A dash in C denotes no eligible pairs; U is also undefined when total support is unknown (Countdown). Typography corrections precede the same executable verifier; mathematical values and program logic are not repaired. Only cells with at least one changed displayed entry appear; omitted cells match at the printed precision. Every displayed entry is unchanged for Grok 4.3.

| Deployment    | Level | Domain     | M     | P     | D    | C (95% interval)  | U   |
|---------------|-------|------------|-------|-------|------|-------------------|-----|
| GPT-5.6 Sol   | 1     | Countdown  | 99.9  | 100.0 | 1.93 | 67.9 [63.3, 72.4] | —   |
|               | 1     | Python     | 92.4  | 100.0 | 1.41 | 85.6 [82.3, 88.8] | 1.4 |
|               | 2     | Countdown  | 98.5  | 100.0 | 1.99 | 63.9 [59.5, 68.4] | —   |
|               | 2     | Python     | 60.7  | 99.2  | 1.12 | 94.8 [91.9, 97.2] | 1.1 |
|               | 2     | PantryPlan | 71.3  | 100.0 | 1.96 | 64.0 [58.1, 69.6] | 6.8 |
|               | 3     | Countdown  | 99.0  | 100.0 | 1.70 | 75.7 [71.2, 80.1] | —   |
| Claude Opus 5 | 3     | Python     | 56.5  | 98.4  | 1.09 | 94.3 [90.8, 97.2] | 1.0 |
|               | 3     | PantryPlan | 75.2  | 100.0 | 2.35 | 53.0 [47.3, 58.9] | 6.7 |
|               | 1     | Python     | 60.0  | 94.5  | 1.12 | 90.4 [86.2, 94.3] | 0.9 |
|               | 2     | PantryPlan | 99.9  | 100.0 | 1.61 | 79.7 [75.0, 84.3] | 6.5 |
|               | 3     | Countdown  | 100.0 | 100.0 | 1.66 | 76.1 [71.7, 80.7] | —   |
|               | 3     | PantryPlan | 99.4  | 100.0 | 1.70 | 77.3 [72.6, 82.0] | 6.5 |
| GPT-5.4       | 1     | Countdown  | 99.7  | 100.0 | 1.80 | 73.6 [68.5, 78.3] | —   |
|               | 1     | Python     | 98.7  | 100.0 | 2.57 | 56.8 [51.9, 61.6] | 1.4 |
|               | 2     | Countdown  | 98.4  | 100.0 | 2.12 | 61.1 [57.0, 65.1] | —   |
|               | 2     | Python     | 91.3  | 100.0 | 1.87 | 74.0 [69.4, 78.1] | 1.1 |
|               | 3     | Countdown  | 99.0  | 100.0 | 1.96 | 68.4 [64.2, 72.7] | —   |
|               | 3     | Python     | 91.6  | 100.0 | 1.91 | 72.6 [68.3, 77.1] | 1.2 |
| Kimi K3       | 1     | Countdown  | 99.4  | 100.0 | 1.91 | 68.4 [63.9, 73.3] | —   |
|               | 1     | Python     | 98.2  | 100.0 | 2.53 | 57.7 [53.5, 61.9] | 1.4 |
|               | 2     | Countdown  | 98.7  | 100.0 | 1.98 | 64.5 [60.1, 68.9] | —   |
|               | 2     | Python     | 98.3  | 100.0 | 2.44 | 61.9 [57.6, 66.2] | 1.1 |
|               | 2     | PantryPlan | 93.4  | 100.0 | 3.20 | 41.4 [36.7, 46.0] | 6.6 |

Formatting-normalized grades, continued.

| Deployment      | Level | Domain     | M    | P     | D    | C (95% interval)  | U   |
|-----------------|-------|------------|------|-------|------|-------------------|-----|
| Claude Opus 4.8 | 3     | Countdown  | 98.1 | 100.0 | 1.84 | 70.3 [65.8, 74.5] | –   |
|                 | 3     | Python     | 98.5 | 100.0 | 2.34 | 64.9 [60.3, 69.2] | 1.2 |
|                 | 1     | Python     | 98.6 | 100.0 | 1.40 | 85.8 [81.8, 89.5] | 1.5 |
|                 | 3     | PantryPlan | 98.7 | 100.0 | 2.33 | 61.7 [56.7, 66.6] | 6.5 |
| DeepSeek V4 Pro | 1     | Python     | 99.4 | 100.0 | 1.41 | 88.8 [86.1, 91.4] | 1.4 |
|                 | 2     | Python     | 99.7 | 100.0 | 1.05 | 98.4 [97.1, 99.4] | 1.2 |
|                 | 3     | Python     | 99.7 | 100.0 | 1.08 | 97.8 [96.3, 99.0] | 1.2 |

| Model           | Level 1           |                   |      | Level 2           |                   |                   | Level 3           |                   |                   |
|-----------------|-------------------|-------------------|------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                 | pass@8 (%)        | # modes           | PCMD | pass@8 (%)        | # modes           | PCMD              | pass@8 (%)        | # modes           | PCMD              |
| GPT-5.6 Sol     | 98.0              | 1.80              | .256 | 98.4              | 1.50              | .229 <sup>†</sup> | 99.2              | 1.45              | .183 <sup>†</sup> |
| Claude Opus 5   | 98.3 <sup>‡</sup> | 1.43 <sup>‡</sup> | .140 | 99.7 <sup>‡</sup> | 1.44 <sup>‡</sup> | .164 <sup>‡</sup> | 99.8 <sup>‡</sup> | 1.36 <sup>‡</sup> | .121 <sup>‡</sup> |
| GPT-5.4         | 98.1              | 2.11              | .338 | 98.4              | 1.72              | .247              | 99.5              | 1.68              | .224              |
| Grok 4.3        | 99.5              | 1.97              | .275 | 100.0             | 1.57              | .179              | 100.0             | 1.55              | .171              |
| Kimi K3         | 98.4              | 2.31              | .374 | 100.0             | 2.05              | .307              | 100.0             | 1.90              | .253              |
| Claude Opus 4.8 | 98.8              | 1.64              | .217 | 100.0             | 1.67              | .229              | 99.8              | 1.51              | .168              |
| DeepSeek V4 Pro | 98.4              | 2.01              | .289 | 99.1              | 1.84              | .319              | 99.7              | 1.62              | .232              |

Table 55: **High prompt success coexists with few verified solution modes.** Each deployment requests medium reasoning effort, with 128 prompts per domain and level and eight responses per prompt. `pass@8` and `# modes` use formatting-normalized grading and average the five domains equally: `pass@8` is empirical prompt success; `# modes` is mean `distinct@8`, including zero for prompts without a verified response. <sup>‡</sup> marks the three Claude Opus 5 cells using revised `Python` wording without a system message (App. O.1). `PCMD` instead uses strict grading and the original wording throughout, averaging per-prompt diversity over prompts with at least two verified draws and then equally over reportable domains. These are the per-level columns of Table 46, not its common-cell macro. <sup>†</sup> marks four cells omitting `Python` because fewer than thirty prompts qualify: GPT-5.6 Sol has strict-formatting failures; Opus 5 has provider refusals. All entries are point estimates. Table 56 gives normalized success and mode counts on the four domains with common wording.

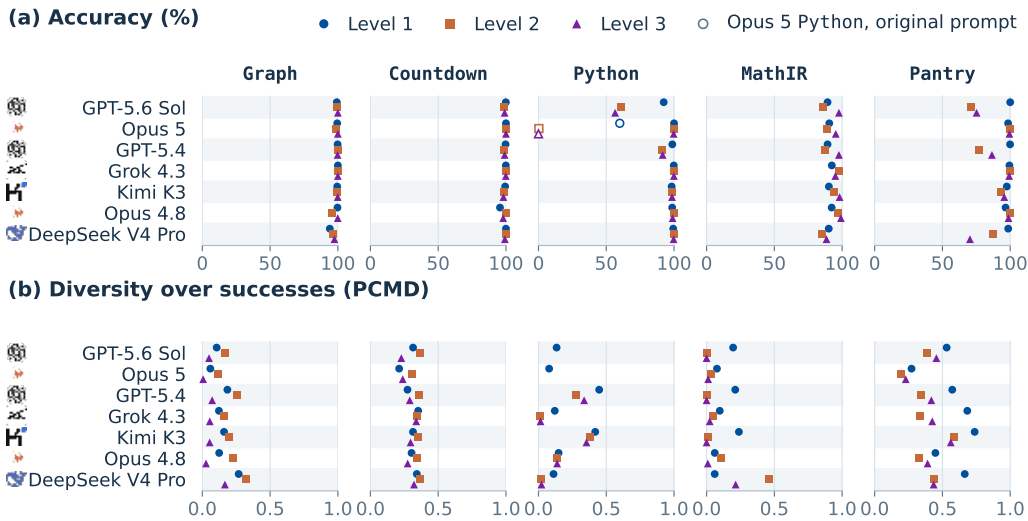


Figure 40: **Accurate hosted responses can still concentrate on few solution modes.** Columns show domains; colors and marker shapes distinguish Levels 1–3. Each deployment–domain–level cell uses 128 prompts and eight responses per prompt. Top: formatting-normalized response accuracy. Solid Opus 5 `Python` marks use revised wording without a system message; hollow marks show original wording. Bottom: promptwise `PCMD` under strict grading and original wording for every deployment, averaging over prompts with at least two verified draws. Cells with fewer than thirty eligible prompts are omitted. Thus the two rows use different grading and, for Opus 5 `Python`, different prompt conditions. All marks are point estimates. Table 55 gives per-level means; Table 56 gives normalized success and mode counts on the four domains with common wording.

**Comparison with common prompts.** The accuracy panels of Fig. 40 and the `pass@8/distinct@8` columns of Table 55 use normalized grading, with revised Python wording for Claude Opus 5. The lower figure panels and the table’s PCMD columns instead use strict grading and original wording. Table 56 compares normalized success and mode counts over Graph, Countdown, MathIR and Pantry, the four domains with identical prompts across deployments. Claude Opus 5 averages 1.42/1.45/1.35 verified modes at Levels 1–3, compared with 1.43/1.44/1.36 over five domains using revised Python wording and 1.36/1.17/1.08 using original Python wording. It has the lowest mean `distinct@8` at each level in all three comparisons. The revised condition increases its verified Python mode count, which also depends on response success; these means do not compare diversity at matched accuracy. The ordering of the other deployments depends on whether Python is included.

A separate Python comparison in App. K.4 uses original and neutral wording on three deployments, with 16 prompts and 64 responses per prompt at Levels 2 and 3. Pair-pooled diversity  $1 - C$  rises by +.024 to +.131 across its six deployment-level cells; the largest value under either wording is .369 (effective-mode transform 1.58). These values weight correct pairs equally, unlike promptwise PCMD. They describe that neutral-wording comparison and do not bound the effect of other wordings or the revised no-system condition on untested deployments.

| Model           | Level 1                 |         | Level 2                 |         | Level 3                 |         |
|-----------------|-------------------------|---------|-------------------------|---------|-------------------------|---------|
|                 | <code>pass@8</code> (%) | # modes | <code>pass@8</code> (%) | # modes | <code>pass@8</code> (%) | # modes |
| GPT-5.6 Sol     | 97.5                    | 1.90    | 98.2                    | 1.59    | 99.4                    | 1.54    |
| Claude Opus 5   | 97.9                    | 1.42    | 99.6                    | 1.45    | 99.8                    | 1.35    |
| GPT-5.4         | 97.7                    | 2.00    | 98.0                    | 1.68    | 99.4                    | 1.62    |
| Grok 4.3        | 99.4                    | 2.12    | 100.0                   | 1.70    | 100.0                   | 1.68    |
| Kimi K3         | 98.0                    | 2.26    | 100.0                   | 1.95    | 100.0                   | 1.79    |
| Claude Opus 4.8 | 98.4                    | 1.70    | 100.0                   | 1.75    | 99.8                    | 1.56    |
| DeepSeek V4 Pro | 98.0                    | 2.16    | 98.8                    | 2.04    | 99.6                    | 1.75    |

Table 56: **Few verified modes remain when all deployments share the same prompts.** Equal-domain means over Graph, Countdown, MathIR and Pantry, with 128 prompts per domain and level and eight responses per prompt, under normalized grading. Python is excluded for every deployment. `pass@8` is empirical prompt success; # modes is mean `distinct@8` over all prompts, including those with no verified response. Entries are point estimates; response accuracy and provider compute are not matched.

## O.5 THE REVISED PYTHON WORDING ON A SECOND DEPLOYMENT

The original and revised Python conditions of App. O.1 are evaluated on GPT-5.6 Sol through `api.openai.com`. Each condition uses the same 384 problems, with 128 prompts per level and eight responses per prompt, medium reasoning effort and an 8,192-token output cap. The original condition uses the benchmark request; the revised condition asks for a direct expression without a system message. Public task inputs, allowed operators and the verifier are unchanged, but wording and system-message presence change together. All responses report `gpt-5.6-sol`; build equality with the Azure deployment used in the hosted averages is not established. These are separate populations.

**Verified success and mode counts decrease under the revised condition.** There are no provider-declared refusals among the 6,144 responses in this comparison. The Azure GPT-5.6 Sol cohort also contains no provider-declared refusals; its missing strict Python diversity entries reflect too few verified draws. Under normalized grading, the revised condition lowers accuracy and mean `distinct@8` at each level (Table 57). Revised-minus-original differences use all 128 prompts per level, pairing problems across conditions and resampling whole eight-response groups: 20,000 replicates give pointwise 95% percentile intervals. Accuracy differences are fractions of responses; mode-count differences are modes per prompt: accuracy  $-.689 [-.745, -.631]$ ,  $-.420 [-.467, -.374]$ ,  $-.370 [-.418, -.320]$  at Levels 1–3; `distinct@8`  $-.945 [-1.078, -.820]$ ,  $-.703 [-.812, -.594]$ ,  $-.656 [-.766, -.547]$  respectively. The revised normalized cells have 28, 26, 25 prompts with at least two verified responses, all below the 30-prompt reporting threshold. Their success-conditional diversity is therefore omitted; these cells do not support a diversity contrast at the stated reporting threshold.

**Formatting failures accompany the lower scores.** The literal  $\text{\texttt{\LaTeX}}$  marker `\text{}` appears in 862 original-condition and 2,089 revised-condition responses (28% and 68%). The formatting normalizer does not remove this wrapper; none of these responses verifies. Among responses without the marker, normalized success is 94.9% and 59.3%, respectively. Absence of this marker does not guarantee parseability. These condition-dependent subsets are not matched semantic comparisons and provide no bound on a content or capability effect.

**Prompt-interface sensitivity differs across deployments.** The revised condition increases normalized success and verified mode counts for Opus 5 (App. O.1) but decreases both for this GPT-5.6 Sol deployment. This comparison does not isolate task wording from the system message, establish that either task is generally easier, or identify the effect on the remaining hosted deployments. The four-domain comparison in Table 56 holds the task prompts fixed across all seven deployments by excluding `Python`.

Table 57: **The revised `Python` condition lowers normalized success and verified mode counts on GPT-5.6 Sol.** Both conditions use the same 128 problems per level and eight responses per prompt through `api.openai.com`, with medium reasoning effort and an 8,192-token output cap. R counts provider-declared refusals; A is response accuracy (%); D is mean `distinct@8` over all prompts.  $1 - C$  is pair-pooled diversity: the fraction of correct pairs with different solution modes, weighting each correct pair equally across prompts. Dashes mark cells with fewer than 30 prompts containing at least two verified draws. Strict grading uses the original output; normalized grading applies the same typography rules to both conditions. Entries are point estimates; paired uncertainty for accuracy and mode-count differences is given in the text.

| Wording            | L | R | Strict |      |         | Normalized |      |         |
|--------------------|---|---|--------|------|---------|------------|------|---------|
|                    |   |   | A      | D    | $1 - C$ | A          | D    | $1 - C$ |
| Original           | 1 | 0 | 37.0   | 1.02 | .106    | 89.8       | 1.36 | .114    |
|                    | 2 | 0 | 10.4   | 0.14 | —       | 58.7       | 1.15 | .053    |
|                    | 3 | 0 | 13.0   | 0.16 | —       | 56.3       | 1.13 | .048    |
| Revised, no system | 1 | 0 | 18.7   | 0.27 | —       | 20.9       | 0.41 | —       |
|                    | 2 | 0 | 13.4   | 0.26 | —       | 16.7       | 0.45 | —       |
|                    | 3 | 0 | 15.5   | 0.25 | —       | 19.3       | 0.48 | —       |

| Model                                                                                               | Level 1    |         |      | Level 2    |         |      | Level 3    |         |      |
|-----------------------------------------------------------------------------------------------------|------------|---------|------|------------|---------|------|------------|---------|------|
|                                                                                                     | pass@8 (%) | # modes | PCMD | pass@8 (%) | # modes | PCMD | pass@8 (%) | # modes | PCMD |
|  GPT-5.6 Sol     | 82.5       | 1.31    | .226 | 66.9       | 0.76    | .063 | 76.2       | 0.88    | .095 |
|  GPT-5.4         | 82.5       | 1.67    | .347 | 71.2       | 0.92    | .134 | 70.0       | 0.94    | .160 |
|  Grok 4.3        | 82.5       | 1.64    | .493 | 86.9       | 1.41    | .296 | 90.0       | 1.45    | .314 |
|  Kimi K3         | 92.5       | 1.91    | .595 | 92.5       | 1.37    | .335 | 95.6       | 1.59    | .361 |
|  Claude Opus 4.8 | 91.2       | 1.46    | .216 | 100.0      | 1.62    | .226 | 100.0      | 1.51    | .185 |

Table 58: **Verified solution diversity varies across deployments with explicit reasoning disabled.** Each level contains 32 prompts per domain and eight responses per prompt, with formatting-normalized grading. `pass@8` is the fraction with any verified response; # modes is mean `distinct@8`, including zero-success prompts. These columns average five domains equally. PCMD instead weights equally the prompts with at least two verified responses in both this condition and the medium condition of Table 59. This selected population can differ between deployments and levels; it does not match overall accuracy. Entries are point estimates.

## O.6 MATCHED REASONING-CONTROL COMPARISON

Five deployments each answer the same 480 prompts under medium reasoning and disabled explicit reasoning. The sample uses 32 prompts in each of five domains and three levels, selected without reference to model outputs. Each condition has eight responses per prompt, or 3,840 responses per deployment. The system and user prompts are identical across conditions, including the original `Python` wording. These 32-prompt cells are a subset of the 128-prompt medium-reasoning evaluation; the Opus 5 `Python` wording change in Fig. 40 is outside this comparison.

Both conditions have an 8,192-token output cap. GPT-5.6 Sol and GPT-5.4 request `reasoning.effort=None`; Grok 4.3 and Kimi K3 request `reasoning_effort=None`. Claude Opus 4.8 requests `thinking.type=disabled`

with `output_config.effort=medium`. These settings disable explicit reasoning through the provider interface; they do not establish the absence of hidden computation or equal compute across conditions or providers. Both conditions use the same formatting normalization and canonical-key grader. Their responses were collected at different times. Temperature and `top_p` use provider defaults, which can change over time, and matching sample indices do not imply shared random seeds.

**Accuracy and conditional diversity use different populations.** `pass@8` and `distinct@8` use all 160 prompts at each level, with zero modes for unsolved prompts. Their overall means give every domain and level equal weight. PCMD gives each jointly eligible prompt equal weight: a prompt must contain at least two verified responses in both conditions. The level estimates pool such prompts across domains, and the *All* estimate pools them across domains and levels. Thus their weights depend on eligibility and are not equal-domain averages or means of the three level estimates. A reported PCMD requires at least thirty jointly eligible prompts in the displayed level or overall population. Joint eligibility makes the two conditions comparable on that selected population; it does not hold policy accuracy fixed or describe prompts that only one condition solves twice.

| Model                                                                                               | Level      | Medium reasoning        |         |      | Reasoning disabled      |         |      |
|-----------------------------------------------------------------------------------------------------|------------|-------------------------|---------|------|-------------------------|---------|------|
|                                                                                                     |            | <code>pass@8</code> (%) | # modes | PCMD | <code>pass@8</code> (%) | # modes | PCMD |
|  GPT-5.6 Sol       | 1          | 96.9                    | 1.81    | .311 | 82.5                    | 1.31    | .226 |
|                                                                                                     | 2          | 98.8                    | 1.54    | .117 | 66.9                    | 0.76    | .063 |
|                                                                                                     | 3          | 98.8                    | 1.51    | .195 | 76.2                    | 0.88    | .095 |
|                                                                                                     | <i>All</i> | 98.1                    | 1.62    | .226 | 75.2                    | 0.99    | .145 |
|                                                                                                     |            |                         |         |      |                         |         |      |
|  GPT-5.4           | 1          | 96.9                    | 2.06    | .338 | 82.5                    | 1.67    | .347 |
|                                                                                                     | 2          | 99.4                    | 1.68    | .162 | 71.2                    | 0.92    | .134 |
|                                                                                                     | 3          | 99.4                    | 1.82    | .247 | 70.0                    | 0.94    | .160 |
|                                                                                                     | <i>All</i> | 98.5                    | 1.85    | .255 | 74.6                    | 1.18    | .224 |
|                                                                                                     |            |                         |         |      |                         |         |      |
|  Grok 4.3        | 1          | 100.0                   | 1.93    | .211 | 82.5                    | 1.64    | .493 |
|                                                                                                     | 2          | 100.0                   | 1.62    | .136 | 86.9                    | 1.41    | .296 |
|                                                                                                     | 3          | 100.0                   | 1.54    | .121 | 90.0                    | 1.45    | .314 |
|                                                                                                     | <i>All</i> | 100.0                   | 1.70    | .154 | 86.5                    | 1.50    | .364 |
|                                                                                                     |            |                         |         |      |                         |         |      |
|  Kimi K3         | 1          | 96.9                    | 2.27    | .343 | 92.5                    | 1.91    | .595 |
|                                                                                                     | 2          | 100.0                   | 2.02    | .277 | 92.5                    | 1.37    | .335 |
|                                                                                                     | 3          | 100.0                   | 1.89    | .264 | 95.6                    | 1.59    | .361 |
|                                                                                                     | <i>All</i> | 99.0                    | 2.06    | .293 | 93.5                    | 1.62    | .426 |
|                                                                                                     |            |                         |         |      |                         |         |      |
|  Claude Opus 4.8 | 1          | 98.1                    | 1.61    | .202 | 91.2                    | 1.46    | .216 |
|                                                                                                     | 2          | 100.0                   | 1.67    | .218 | 100.0                   | 1.62    | .226 |
|                                                                                                     | 3          | 100.0                   | 1.57    | .185 | 100.0                   | 1.51    | .185 |
|                                                                                                     | <i>All</i> | 99.4                    | 1.62    | .202 | 97.1                    | 1.53    | .209 |
|                                                                                                     |            |                         |         |      |                         |         |      |

Table 59: **Lower success with disabled reasoning accompanies different changes in conditional diversity.** Formatting-normalized grading, eight responses per prompt. At each level, `pass@8` and # modes use all 160 prompts across five domains; *All* uses all 480 prompts. # modes is mean `distinct@8`. PCMD weights jointly eligible prompts equally, using 65–159 prompts per level and 245–458 overall. Each displayed population exceeds the thirty-prompt reporting threshold. Eligibility can vary by domain, level and deployment, so PCMD uses different weights from the other columns. Entries are descriptive point estimates without uncertainty intervals.

**Additional deployments.** For DeepSeek V4 Pro, strict grading gives PCMD .282 with medium reasoning and .369 with disabled reasoning, a difference of +.087 on 437 jointly eligible prompts. This comparison uses different grading from the normalized tables.

For Claude Opus 5, a response includes a thinking block and forty thinking tokens despite an explicit request to disable thinking. This violates the requested control, so its responses do not provide a validated reasoning-disabled comparison.

**Observed differences and limits.** All five tabulated deployments have lower overall `pass@8` and `distinct@8` under disabled reasoning. Changes in conditional diversity differ: overall

PCMD falls for both GPT deployments, rises for Kimi K3 and Grok 4.3, and changes from .202 to .209 for Opus 4.8. For Grok, PCMD rises from .154 to .364 while mean distinct modes falls. The common reduction in raw mode counts therefore does not imply a common increase in concentration among successful responses.

The level estimates also differ by deployment. GPT-5.6 Sol's PCMD falls at every level; Grok 4.3 and Kimi K3 rise at every level. GPT-5.4 rises by .009 at Level 1 and falls at Levels 2 and 3, so its change reverses sign. Opus 4.8's level changes range from approximately .0002 to .0142; these small positive estimates do not establish invariance. There are no uncertainty intervals for these comparisons. Eligibility changes the population being summarized, and collection time and provider defaults can confound the requested-control contrast. These observations describe output distributions under the tested configurations, not a causal effect of training or an isolated effect of internal deliberation.

## Part VI. Idealized analysis

### P MODE COLLAPSE AND VERIFIED REPLAY

For a fixed prompt, the binary-reward score estimators defined below have mean updates along the correctness gradient, while their sampling covariances can differ. With independently optimized categorical logits, these updates amplify differences among equally rewarded modes, and sampling can break an initially exact symmetry. Fixed-buffer replay supplies probability floors, restoration thresholds, and recovery bounds under the stated update assumptions. Neural retention requires additional conditions on complete-response likelihoods and local updates. These results do not certify the implemented AdamW/PPO trajectories. Table 2 defines the notation.

#### P.1 CATEGORICAL ABSTRACTION AND MEAN UPDATES

The categorical model assigns one independent logit to each outcome category. Its Euclidean mean flow and the subsequent finite-step, sampled, and neural updates have different assumptions; a neural solution mode need not have an independently optimized logit.

**Assumption P.1** (Categorical mean-flow model). Fix one isolated prompt and a finite category set  $\mathcal{A} = \mathcal{C} \dot{\cup} \mathcal{N}$  with  $m \geq 2$  correct solution modes and  $n \geq 1$  incorrect categories, and set  $p_a = \text{softmax}(z)_a$ ,  $P = \sum_{c \in \mathcal{C}} p_c$ , and  $q_c = p_c/P$ , so that  $P$  measures correctness while  $q$  describes how that correct probability is allocated among the individual correct modes.

- (A1) *Independent logits*. The policy carries one independently optimized logit for every category in  $\mathcal{A}$ , including one separate logit for each canonical correct mode.
- (A2) *Common length normalization*. Every response uses the same fixed length-normalization factor, absorbed into the positive coefficient of the group estimator.
- (A3) *Euclidean mean flow*. Updates are the infinitesimal expected updates in these logits, taken without optimizer preconditioning and without momentum carried between steps.
- (A4) *No competing regularizer*. Explicit reference KL is zero and no entropy term is present.
- (A5) *Nondegenerate start*. Logits are finite initially,  $0 < P(0) < 1$ , and the group size is  $G \geq 2$ .

Assumption P.1 includes all five conditions. AdamW’s preconditioning and momentum violate (A3); finite, repeated PPO updates are outside its infinitesimal on-policy limit. Response-specific length normalization violates (A2), while reference KL and entropy regularization violate (A4).

A language model induces probabilities over canonical outcomes, but those probabilities alone do not determine its update geometry. Aggregating several response strings into a mode does not generally commute with gradient flow in response logits. The independent-category model therefore requires an additional assumption beyond a neural policy and its validator equivalence classes.

Draw a group  $Z_1, \dots, Z_G \stackrel{\text{iid}}{\sim} p$  with  $G \geq 2$ , write  $R_i = \mathbf{1}[Z_i \in \mathcal{C}]$ , and let  $R = \sum_i R_i$ . At the on-policy point where the PPO ratio equals one, consider the common-length estimator

$$\hat{g}_G = \frac{1}{G} \sum_{i=1}^G w(R) \left( R_i - \frac{R}{G} \right) \nabla_z \log p_{Z_i}. \quad (13)$$

For the Dr.GRPO abstraction (Liu et al., 2025),  $w(R) = 1$  after absorbing the fixed positive length-normalization factor. Reward-standardized GRPO (Shao et al., 2024) with fixed common response-length normalization takes  $w(R)$  to be the reciprocal group reward standard deviation on mixed groups (with a fixed numerical stabilizer if used). Standard variable-length GRPO also weights each response by its own inverse length; that response-specific factor cannot generally be absorbed into  $w(R)$  and is outside this equivalence. In either included case,  $0 < w(\ell) < \infty$  for  $\ell \in \{1, \dots, G-1\}$ . The entire update is zero on all-equal groups, without evaluating an undefined reciprocal standard deviation; likewise,  $w(R)R(G-R)$  below is defined as zero when  $R = 0$  or  $R = G$ . All advantages and normalization weights are detached in the score estimator.

**Lemma P.2** (A group baseline does not remove frequency amplification). *For  $0 < P < 1$ , the expected update in Equation (13) is*

$$\mathbb{E}[\hat{g}_G] = c_G(P) \nabla_z P, \quad c_G(P) = \frac{\mathbb{E}[w(R)R(G-R)]}{G^2P(1-P)} > 0. \quad (14)$$

*For the Dr.GRPO abstraction,  $c_G(P) = (G-1)/G$ . Group centering and reward standardization rescale the expected gradient without changing its direction under these assumptions.*

*Proof.* Condition on the complete binary reward vector, whose sum is  $R$ . The score identities give

$$\mathbb{E}[\nabla_z \log p_{Z_i} \mid R_i = 1] = \nabla_z \log P, \quad \mathbb{E}[\nabla_z \log p_{Z_i} \mid R_i = 0] = \nabla_z \log(1 - P).$$

There are  $R$  correct rows with centered reward  $(G-R)/G$  and  $G-R$  incorrect rows with centered reward  $-R/G$ . Therefore the conditional expectation of the unaveraged sum in (13) is

$$w(R) \frac{R(G-R)}{G} (\nabla_z \log P - \nabla_z \log(1 - P)) = w(R) \frac{R(G-R)}{GP(1-P)} \nabla_z P.$$

The outer factor  $1/G$  and expectation over  $R \sim \text{Binomial}(G, P)$  yield (14). For  $w \equiv 1$ ,  $\mathbb{E}[R(G-R)] = G(G-1)P(1-P)$ , proving the Dr.GRPO value.  $\square$

The binomial identity also gives the denominator-free expression

$$c_G(P) = \frac{G-1}{G} \sum_{j=0}^{G-2} \binom{G-2}{j} w(j+1) P^j (1-P)^{G-2-j}.$$

The binomial weights sum to one, so  $c_G$  lies between  $(G-1) \min_j w(j+1)/G$  and  $(G-1) \max_j w(j+1)/G$ . Thus  $c_G$  extends smoothly and positively to  $P = 0, 1$ .

The practical MaxRL estimator has the same mean-gradient direction (Tajwar et al., 2026, Section 4.3 and App. D, Theorem 5). Its all-failure convention determines the finite-group coefficient.

**Lemma P.3** (Binary MaxRL has the same correctness-gradient direction). *Under Assumption P.1, let MaxRL assign  $A_i^{\text{MaxRL}} = GR_i/R - 1$  when  $R > 0$  and zero when  $R = 0$ , as in Section D.1. For  $\hat{g}_G^{\text{MaxRL}} = G^{-1} \sum_i A_i^{\text{MaxRL}} \nabla_z \log p_{Z_i}$  at the on-policy point, the expected update is*

$$\mathbb{E}[\hat{g}_G^{\text{MaxRL}}] = c_G^{\text{MaxRL}}(P) \nabla_z P, \quad c_G^{\text{MaxRL}}(P) = \frac{1 - (1-P)^{G-1}}{P} > 0, \quad (15)$$

*with continuous endpoint values  $c_G^{\text{MaxRL}}(0) = G-1$  and  $c_G^{\text{MaxRL}}(1) = 1$ .*

*Proof.* With  $N = G$  and pass rate  $P$ , the estimator in Tajwar et al. (2026, Theorem 5) equals  $\hat{g}_G^{\text{MaxRL}}$ . Its expected coefficient is  $\sum_{j=0}^{G-2} (1-P)^j$ , whose geometric sum gives Equation (15) and its endpoint values. The identity requires on-policy independent draws and the score identity.  $\square$

Integrating this coefficient gives the potential

$$\Psi_{\text{MaxRL}}(P) := \int_0^P c_G^{\text{MaxRL}}(s) ds = \sum_{k=1}^{G-1} \frac{1 - (1-P)^k}{k}. \quad (16)$$

In particular,  $\Psi_{\text{MaxRL}}(1) = \sum_{k=1}^{G-1} 1/k$ . For binary rewards, best@ $k$  equals pass@ $k$ , so this is a harmonic mixture of sampling objectives for  $k = 1, \dots, G-1$ . The upper limit is  $G-1$  because the implemented centered estimator sets the entire update to zero on all-failure groups. Subtracting an unconditional score baseline even on those groups instead gives the order- $G$  estimator. These are exact finite-group expectations at the on-policy point; subsequent clipping and adaptive optimization can alter the expected update.

## P.2 WINNER-TAKE-ALL COLLAPSE IN THE CATEGORICAL MEAN FLOW

Consider  $\dot{z} = c_G(P)\nabla_z P$ , with any of the smooth positive coefficients established above: Dr.GRPO, common-length reward-standardized GRPO, or the specified centered binary MaxRL estimator. This infinitesimal expected-update model has no sampling noise and uses independently optimized Euclidean logits. The conditional dynamics follow the frequency-dependent replicator system with fitness  $f_c(q) = q_c$  (Hofbauer & Sigmund, 1998; Harper, 2009). Related analyses of collapse among equally rewarded outcomes include Sinha et al. (2026); Lochab et al. (2026).

**Theorem P.4** (Winner-take-all collapse in the categorical mean flow). *Under Assumption P.1, if the conditional correct-mode distribution  $q(0)$  has a unique largest coordinate  $c^*$ , then*

$$P(t) \rightarrow 1, \quad q_{c^*}(t) \rightarrow 1, \quad q_c(t) \rightarrow 0 \quad (c \neq c^*).$$

*Thus these categorical mean flows converge to a single correct solution mode for generic initial conditions, even though every mode in  $\mathcal{C}$  has exactly the same reward.*

**Remark P.5.** Theorem P.4 is asymptotic and requires (A1)–(A5). Finite-time support loss and convergence of the stochastic neural optimizers used in the experiments are outside its scope.

*Proof.* The coefficient  $c_G$  is smooth and bounded on  $[0, 1]$ , and the softmax reward gradient is bounded. Thus the logit flow exists for all finite times and cannot develop infinite logits at finite time. For a correct mode  $c$  and incorrect category  $a$ ,

$$\dot{z}_c = c_G(P)p_c(1 - P), \quad \dot{z}_a = -c_G(P)p_aP.$$

Also  $\dot{P} = c_G(P)\|\nabla_z P\|_2^2 > 0$ . If  $P$  converged to an interior value, continuity and positivity of  $c_G$ , together with

$$\|\nabla_z P\|_2^2 \geq P^2(1 - P)^2 \left( \frac{1}{m} + \frac{1}{n} \right),$$

would keep  $\dot{P}$  bounded away from zero, a contradiction. Hence  $P \rightarrow 1$ .

Define the effective optimization time

$$\tau(t) = \int_0^t c_G(P(s))P(s)(1 - P(s)) ds.$$

If  $\tau(\infty) < \infty$ , every correct and incorrect logit would have finite total variation, because the absolute velocity of each is at most  $\dot{\tau}$ . All logits would then converge to finite values, contradicting  $P \rightarrow 1$ . Therefore  $\tau(t) \rightarrow \infty$ , leaving infinite effective time for the correct-simplex dynamics below.

On the correct simplex, changing from  $t$  to  $\tau$  gives the replicator flow

$$\frac{dq_c}{d\tau} = q_c \left( q_c - \sum_{d \in \mathcal{C}} q_d^2 \right), \quad \frac{d}{d\tau} \log \frac{q_c}{q_d} = q_c - q_d. \quad (17)$$

The initially unique maximum remains the unique maximum. For  $c \neq c^*$ , set  $\xi_c = q_c/q_{c^*}$ , so  $\xi_c(0) < 1$  and  $\xi_c$  is nonincreasing. Since  $q_{c^*} \geq 1/m$ ,

$$\frac{d \log \xi_c}{d\tau} = -q_{c^*}(1 - \xi_c) \leq -\frac{1 - \xi_c(0)}{m}.$$

Hence  $\xi_c(\tau) \leq \xi_c(0) \exp[-(1 - \xi_c(0))\tau/m] \rightarrow 0$ . Every competing mode vanishes relative to  $c^*$ ; normalization then yields  $q_{c^*} \rightarrow 1$ , establishing the claimed single-mode limit.  $\square$

If several modes tie for the largest initial probability, symmetry preserves the tie; the same ratio argument eliminates strictly smaller coordinates and leaves the tied maxima uniform. An exactly uniform correct distribution is therefore a special initial condition. The theorem concerns limiting concentration: finite softmax logits already give positive probabilities at every finite time, so finite-time positivity alone is not a retention result.

### P.3 CONDITIONAL TRAJECTORIES OF OUTCOME-BINARY MEAN FLOWS

Under Assumption P.1, every outcome-binary estimator with a positive correctness coefficient induces the same conditional trajectory in  $q$ . Its advantage changes only the rate at which that trajectory is traversed. Conditioning on correctness averages over all correct modes, so a reward-dependent advantage cannot distinguish their identities.

For one isolated prompt, let a finite categorical policy have  $m \geq 2$  correct modes and  $n \geq 1$  incorrect categories, with  $p = \text{softmax}(z)$ , correctness  $P = \sum_c p_c$ , and conditional mode probabilities  $q_c = p_c/P$ . Each category has an independent logit. Draw responses  $Z_1, \dots, Z_G$  independently with  $G \geq 2$ , with binary rewards  $R_i$  and  $R = \sum_i R_i$ . Assume finite initial logits, common response-length normalization, and on-policy Euclidean expected-gradient flow without KL, entropy, momentum, or preconditioning (Assumption P.1).

**Theorem P.6** (Conditional inertness of outcome-binary objectives). *Under Assumption P.1, let  $a(\cdot, \cdot)$  be any detached advantage measurable with respect to the group's binary reward vector, applied as  $\hat{g}_a = G^{-1} \sum_i a(R_i, R) \nabla_z \log p_{Z_i}$  at the on-policy point. Then*

$$\mathbb{E}[\hat{g}_a] = c_a(P) \nabla_z P, \quad c_a(P) = \frac{1}{G} \left( \frac{\mathbb{E}[R a(1, R)]}{P} - \frac{\mathbb{E}[(G - R) a(0, R)]}{1 - P} \right). \quad (18)$$

If  $c_a > 0$  on  $(0, 1)$ , then under the time change  $d\tau = c_a(P)P(1 - P) dt$  the conditional distribution  $q$  obeys Equation (17) exactly, for every such  $a$ . Dr.GRPO, reward-standardized GRPO, a leave-one-out baseline, the centered binary MaxRL estimator, and any reweighting shaped by a pass@ $k$  target are points in this class. The choice among them rescales optimization time along the correctness axis and leaves both the trajectory in  $q$  and the identity of the surviving mode exactly where they were.

*Proof of Theorem P.6.* The score identities used in Lemma P.2 give  $\mathbb{E}[\nabla_z \log p_{Z_i} \mid R_i = 1] = \nabla_z \log P$  and  $\mathbb{E}[\nabla_z \log p_{Z_i} \mid R_i = 0] = \nabla_z \log(1 - P)$ . Independence of the draws means that conditioning also on the other rows' rewards does not change either conditional mean. Since  $a(R_i, R)$  depends only on the binary rewards, conditioning on their complete vector and summing the  $R$  successful and  $G - R$  failed rows gives

$$\mathbb{E}[\hat{g}_a] = \frac{1}{G} \left( \mathbb{E}[R a(1, R)] \nabla_z \log P + \mathbb{E}[(G - R) a(0, R)] \nabla_z \log(1 - P) \right).$$

Substituting  $\nabla_z \log P = \nabla_z P/P$  and  $\nabla_z \log(1 - P) = -\nabla_z P/(1 - P)$  yields Equation (18). Hence  $\frac{d}{d\tau} \log(q_c/q_d) = \dot{z}_c - \dot{z}_d = c_a(P)P(1 - P)(q_c - q_d)$ . For  $c_a > 0$ , the positive time change  $d\tau = c_a(P)P(1 - P) dt$  gives the replicator dynamics in Equation (17).  $\square$

For a fixed reward-dependent advantage, the binomial identities  $R \binom{G}{R} = G \binom{G-1}{R-1}$  and  $(G - R) \binom{G}{R} = G \binom{G-1}{R}$  give

$$c_a(P) = \sum_{j=0}^{G-1} \binom{G-1}{j} P^j (1 - P)^{G-1-j} [a(1, j+1) - a(0, j)], \quad (19)$$

a polynomial in  $P$  of degree at most  $G - 1$ . Finite advantage values therefore give a smooth, bounded coefficient on  $[0, 1]$ .

The coefficients in this Bernstein expansion are

$$d_j := a(1, j+1) - a(0, j), \quad j = 0, \dots, G-1. \quad (20)$$

With the other  $G - 1$  rows fixed at  $j$  successes,  $d_j$  is the change in a row's advantage when its reward changes from failure to success. The Bernstein basis is nonnegative on  $[0, 1]$  and sums to one, so  $c_a(P)$  lies between  $\min_j d_j$  and  $\max_j d_j$ . Each basis function is strictly positive on  $(0, 1)$ ; hence nonnegative gaps, with at least one positive, imply  $c_a(P) > 0$  throughout the interior. This condition is sufficient: mixed-sign gaps alone do not determine the sign of  $c_a$ .

For Dr.GRPO, every  $d_j = (G - 1)/G$ , giving the constant coefficient in Lemma P.2. For the centered MaxRL estimator,  $d_0 = G - 1$  and  $d_j = G/(j + 1)$  for  $1 \leq j \leq G - 1$ . In particular,  $a(1, G) = 0$  and  $a(0, G - 1) = -1$  give  $d_{G-1} = 1$ . Substitution into Equation (19) gives  $[1 - (1 - P)^{G-1}]/P$ , as in Lemma P.3.

**Corollary P.7** (Collapse for correctness-improving outcome-binary mean flows). *Under Assumption P.1, let  $a(R_i, R)$  be a fixed, bounded advantage depending only on the binary rewards. Suppose the gaps in Equation (20) satisfy  $d_j \geq 0$  for every  $j$ , with at least one strictly positive gap. If  $q(0)$  has a unique largest coordinate  $c^*$ , then*

$$P(t) \longrightarrow 1, \quad q_{c^*}(t) \longrightarrow 1, \quad q_c(t) \longrightarrow 0 \quad (c \neq c^*).$$

*The surviving correct mode is the one with the largest initial probability; the choice of  $a$  changes the rate along the conditional trajectory.*

*Proof.* By Theorem P.6, the flow is  $\dot{z} = c_a(P)\nabla_z P$ . Equation (19) makes  $c_a$  smooth and bounded on  $[0, 1]$ . If  $d_{j_0} > 0$ , the nonnegative-gap hypothesis gives

$$c_a(P) \geq \binom{G-1}{j_0} P^{j_0} (1-P)^{G-1-j_0} d_{j_0} > 0 \quad (0 < P < 1).$$

These are the coefficient properties used in the proof of Theorem P.4: the logit flow exists for all finite times,  $P \rightarrow 1$ , and the effective time  $\tau \rightarrow \infty$ . The conditional replicator equation (17) then eliminates all coordinates strictly below the initial maximum. It contains neither the advantage coefficients nor the rollout group size, so the surviving mode depends only on  $q(0)$ .  $\square$

The nonnegative-gap condition is sufficient rather than necessary. An estimator with mixed-sign gaps also has the same limiting behavior if  $c_a(P) > 0$  throughout  $(0, 1)$ . An interior zero of  $c_a$  is a stationary correctness level for the mean flow, while a negative coefficient gives  $\dot{P} = c_a(P)\|\nabla_z P\|_2^2 < 0$  at an interior policy. The collapse conclusion requires positivity and an initial imbalance under Assumption P.1; it does not describe shared-prompt or sampled neural updates.

*Remark P.8* (Group size in the deterministic conditional flow). For each  $G \geq 2$ , group size enters the mean flow through  $c_a(P)$ . When this coefficient is positive, the time change in Theorem P.6 gives the same conditional replicator equation. Increasing  $G$  can change the speed of the mean flow but cannot change the surviving mode under the conditions of Corollary P.7. Group size also changes the probability of a mixed-reward group in Lemma P.19, which is small near both  $P = 0$  and  $P = 1$ , and the MaxRL coefficient in Corollary P.9. These properties concern total correctness  $P$ ; they do not provide a probability floor for an individual correct mode. Gradient covariance and finite-sample symmetry breaking can depend on  $G$ , so sampled updates need not share this equivalence.

**Corollary P.9** (MaxRL amplification of the correctness gradient). *For the coefficients of Lemmas P.2 and P.3,*

$$\frac{c_G^{\text{MaxRL}}(P)}{c_G^{\text{Dr.GRPO}}} = \frac{G[1 - (1-P)^{G-1}]}{(G-1)P}.$$

*This ratio is nonincreasing in  $P$ , with limits  $G$  as  $P \rightarrow 0$  and  $G/(G-1)$  as  $P \rightarrow 1$ . It is strictly decreasing for  $G > 2$  and constant at 2 for  $G = 2$ . At the same policy, MaxRL therefore multiplies the Dr.GRPO correctness-gradient coefficient by a factor approaching  $G$  when correctness is small. The probability of a nonzero fresh task component remains bounded by the mixed-group probability in Lemma P.19 for both estimators.*

*Proof.* Divide  $c_G^{\text{MaxRL}}(P) = [1 - (1-P)^{G-1}]/P$  by  $(G-1)/G$ . The finite-series identity  $c_G^{\text{MaxRL}}(P) = \sum_{j=0}^{G-2} (1-P)^j$  gives the endpoint limits and monotonicity. For  $G = 2$  the sum is the constant 1; for  $G > 2$  it contains the strictly decreasing term  $1 - P$ .  $\square$

#### P.4 NEURAL SCORE IDENTITIES AND THE LIMITS OF CONDITIONAL INERTNESS

Expected binary reward equals the total correct mass; conditional diversity depends on its allocation among solution modes. Sinha et al. (2026, Theorem 3.1, v1) derive the expected-return log-ratio dynamics for softmax outcome logits, and Lochab et al. (2026, Section 4.1, v1) identify binary reward’s indifference to allocation within the correct set. For differentiable neural policies, outcome-binary group estimators have per-prompt means proportional to the correctness gradient but can differ in covariance. Finite steps, shared prompts, and response-dependent weighting limit the resulting trajectory equivalence.

**Theorem P.10** (Per-prompt neural mean and covariance). *Fix a prompt and a differentiable policy  $\pi_\theta$  on a finite or countable response space with common positive support, where  $\theta$  lies in an open subset of  $\mathbb{R}^d$ . Let  $\mathcal{C}$  be a fixed verifier-positive event,  $P = \Pr_\theta(Y \in \mathcal{C}) \in (0, 1)$ , and  $s(Y) = \nabla_\theta \log \pi_\theta(Y)$ . Assume differentiation may be interchanged with the probability sums over the full support and over  $\mathcal{C}$ , and that  $\mathbb{E}\|s(Y)\|_2^2 < \infty$ . For  $G$  independent responses, write  $R_i = \mathbf{1}[Y_i \in \mathcal{C}]$ ,  $R = \sum_i R_i$ , and let  $a : \{0, 1\} \times \{0, \dots, G\} \rightarrow \mathbb{R}$  be any finite, detached advantage. Define*

$$\hat{g}_a = \frac{1}{G} \sum_i a(R_i, R) s(Y_i), \quad v = \nabla_\theta P, \quad \Sigma_r = \text{Cov}(s(Y) \mid \mathbf{1}[Y \in \mathcal{C}] = r).$$

With

$$\beta_r = \frac{1}{G} \left( \frac{r a(1, r)}{P} - \frac{(G - r) a(0, r)}{1 - P} \right), \quad c_a(P) = \mathbb{E}[\beta_R],$$

where  $R \sim \text{Binomial}(G, P)$ , the exact identities are

$$\mathbb{E}[\hat{g}_a] = c_a(P) v, \tag{21}$$

$$\text{Cov}(\hat{g}_a) = \text{Var}(\beta_R) v v^\top + \frac{1}{G^2} \mathbb{E}[R a(1, R)^2 \Sigma_1 + (G - R) a(0, R)^2 \Sigma_0]. \tag{22}$$

In particular,  $c_a$  is the coefficient in Equation (18), without an independent-logit assumption.

*Proof.* The conditional-score identity (Tajwar et al., 2026, Theorem 1) (also He et al., 2026, Lemma B.1) applied to the correct and incorrect events gives

$$\mathbb{E}[s(Y) \mid Y \in \mathcal{C}] = v/P, \quad \mathbb{E}[s(Y) \mid Y \notin \mathcal{C}] = -v/(1 - P).$$

Conditioned on the complete reward vector, the response scores remain independent. The conditional mean of  $\hat{g}_a$  is  $\beta_R v$  and its conditional covariance is  $G^{-2}[R a(1, R)^2 \Sigma_1 + (G - R) a(0, R)^2 \Sigma_0]$ . Taking expectations proves Equation (21); the law of total covariance proves Equation (22).  $\square$

For one prompt, a common locally Lipschitz preconditioner  $M(\theta)$  and positive coefficients  $c_a(P)$  give deterministic mean fields  $\dot{\theta} = c_a(P) M(\theta) \nabla_\theta P$  that trace the same orbit under a positive change of time, wherever the flows exist uniquely. This is a statement about the mean direction. Equation (22) shows that outcome-binary estimators can have different sampling covariance, including after their mean magnitudes are matched.

**Corollary P.11** (A finite-step comparison in neural parameters). *Under Theorem P.10, suppose  $c_a(P) > 0$  and take  $\theta^+ = \theta + \eta \hat{g}_a / c_a(P)$  with  $\eta > 0$ . Let  $f$  be a twice differentiable scalar observable with  $\|\nabla^2 f\|_{\text{op}} \leq L_f$  along every update segment, which is assumed to remain in its domain. Then, conditional on the current policy,*

$$|\mathbb{E}[f(\theta^+) - f(\theta)] - \eta \nabla f(\theta)^\top v| \leq \frac{L_f \eta^2}{2} \left( \|v\|_2^2 + \frac{\text{tr Cov}(\hat{g}_a)}{c_a(P)^2} \right). \tag{23}$$

*Proof.* Taylor’s theorem bounds the remainder after the linear term by  $L_f \eta^2 \|\hat{g}_a\|_2^2 / (2c_a(P)^2)$ . Take expectations and use Equation (21) together with  $\mathbb{E}\|\hat{g}_a\|_2^2 = c_a(P)^2 \|v\|_2^2 + \text{tr Cov}(\hat{g}_a)$ .  $\square$

For two estimators satisfying the corollary, division by their respective coefficients  $c_a(P)$  gives the same mean step  $\eta v$ ; the training updates do not use this normalization. Their expected changes in  $f$  then differ by at most the sum of the two second-order bounds. Conditional solution mode probabilities or PCMD may be used as  $f$  where the stated smoothness bounds hold. Mean-direction equivalence is therefore a first-order statement for stochastic finite updates; it does not determine their trajectories or eventual surviving solution modes. The bound assumes fresh on-policy score updates and does not cover Adam’s moment estimates or subsequent PPO epochs.

**Remark P.12** (Shared prompts need not follow the same trajectory). For a fixed prompt-sampling distribution  $\mu_x$  and shared parameters, the aggregate mean, assuming integrable updates, is

$$\mathbb{E}[\hat{g}_a] = \sum_x \mu_x c_a(P_x) \nabla_\theta P_x. \tag{24}$$

Different objectives generally assign different relative coefficients to the prompt gradients, so the per-prompt identity does not imply a common aggregate orbit. An explicit example also shows a

difference in conditional diversity. Let  $\theta = (u, v)$ ,  $\sigma(t) = (1 + e^{-t})^{-1}$ , and sample two prompts  $A, B$  equally. Set

$$P_A = \sigma(u + v), \quad P_B = \sigma(u - v), \quad q_A = (\sigma(v), 1 - \sigma(v)).$$

For prompt  $A$ , the probabilities of two correct responses and one incorrect response are  $(P_A\sigma(v), P_A[1 - \sigma(v)], 1 - P_A)$ ; for  $B$ , use  $(P_B/2, P_B/2, 1 - P_B)$ . At  $(u, v) = (0, \log 3)$ ,  $P_A = 3/4$ ,  $P_B = 1/4$ , and  $\nabla P_A = (3/16, 3/16)$ ,  $\nabla P_B = (3/16, -3/16)$ . For  $G = 3$ , Lemmas P.2 and P.3 give  $c_{Dr} = 2/3$  and  $c_{Max}(P) = 2 - P$ , hence

$$\dot{\theta}_{Dr} = (1/8, 0), \quad \dot{\theta}_{Max} = (9/32, -3/64).$$

Prompt  $A$  has conditional disagreement  $D_A = 1 - \sum_c q_{A,c}^2 = 2\sigma(v)[1 - \sigma(v)]$ , so direct differentiation gives

$$\dot{D}_{A,Dr} = 0, \quad \dot{D}_{A,Max} = 9/1024 > 0.$$

Both prompts' correctness increases under both mean fields, since  $\dot{P}_A = (3/16)(\dot{u} + \dot{v}) > 0$  and  $\dot{P}_B = (3/16)(\dot{u} - \dot{v}) > 0$  at this point. Changing a binary objective can therefore change conditional diversity through shared prompt gradients, despite the exact per-prompt identity.

**Proposition P.13** (The residual from response-dependent normalization). *Under Theorem P.10, multiply each response's score by a detached function  $\omega(Y)$  with finite conditional second moments; inverse response length is the case  $\omega(Y) = 1/L(Y)$  with  $L(Y) > 0$ . Define*

$$\hat{g}_{a,\omega} = \frac{1}{G} \sum_i a(R_i, R) \omega(Y_i) s(Y_i), \quad A_1 = \frac{\mathbb{E}[Ra(1, R)]}{G}, \quad A_0 = \frac{\mathbb{E}[(G - R)a(0, R)]}{G},$$

and, conditioning on  $\mathbf{1}[Y \in \mathcal{C}] = r$ , let  $\bar{\omega}_r = \mathbb{E}[\omega(Y) \mid r]$ ,  $K_r = \text{Cov}(\omega(Y), s(Y) \mid r)$ , and  $\sigma_{\omega,r}^2 = \text{Var}(\omega(Y) \mid r)$ . Then

$$\mathbb{E}[\hat{g}_{a,\omega}] = \tilde{c}_a \nabla_{\theta} P + A_1 K_1 + A_0 K_0, \quad \tilde{c}_a = \frac{A_1 \bar{\omega}_1}{P} - \frac{A_0 \bar{\omega}_0}{1 - P}, \quad (25)$$

$$\|A_1 K_1 + A_0 K_0\|_2 \leq \sum_{r=0}^1 |A_r| \sigma_{\omega,r} \sqrt{\text{tr } \Sigma_r}. \quad (26)$$

*Proof.* Conditioning on the reward vector gives  $\mathbb{E}[\hat{g}_{a,\omega}] = A_1 \mathbb{E}[\omega s \mid 1] + A_0 \mathbb{E}[\omega s \mid 0]$ . For each outcome,  $\mathbb{E}[\omega s \mid r] = \bar{\omega}_r \mathbb{E}[s \mid r] + K_r$ ; substituting the conditional score means proves Equation (25). Cauchy-Schwarz gives  $\|K_r\|_2 \leq \sigma_{\omega,r} \sqrt{\text{tr } \Sigma_r}$ ; summing these bounds with weights  $|A_r|$  proves Equation (26).  $\square$

The residual is a within-outcome covariance between response weighting and the policy score. For inverse-length weighting, a common response length makes it zero; equal mean lengths for correct and incorrect responses need not. Response-dependent weights need not be functions of binary outcomes alone. IPS uses outcome-frequency weights (Sinha et al., 2026, Section 4.4, v1), and UCPO adds a conditional-uniformity term (Lochab et al., 2026, Section 6, v1); these interventions are outside the outcome-binary class analyzed above.

The verified-key diversity metric is a specialization of SetPO's distributional functional. Its influence gives an exact metric gradient for shared neural parameters and fixed verified execution keys.

**Corollary P.14** (Verified-key specialization of SetPO). *Fix a prompt, a verifier-positive response event  $\mathcal{C}$ , and a deterministic key map  $V : \mathcal{C} \rightarrow \{1, \dots, m\}$ , all independent of  $\theta$ . Let  $\pi_{\theta}$  be differentiable with common positive support,  $P_{\theta} = \Pr_{\theta}(\mathcal{C}) > 0$ , and  $Q_{\theta} = \pi_{\theta}(\cdot \mid \mathcal{C})$ . Assume the score  $s_{\theta}(Y) = \nabla_{\theta} \log \pi_{\theta}(Y)$  is integrable under  $Q_{\theta}$  and differentiation commutes with the response sums over each key. Write  $q_c = Q_{\theta}(V(Y) = c)$  and  $S_2 = \sum_c q_c^2$ . For  $k(y, y') = \mathbf{1}[V(y) = V(y')]$  and  $g(u) = 1 - u$ , the functional  $\mathcal{F}(Q) = \mathbb{E}_{Y \sim Q}[g(\mathbb{E}_{Y' \sim Q} k(Y, Y'))]$  is exactly  $PCMD(q)$ . The influence formula of Li et al. (2026a, Theorem 4.2, v1) specializes, for  $y \in \mathcal{C}$ , to*

$$\left. \frac{d}{d\varepsilon} \mathcal{F}((1 - \varepsilon)Q_{\theta} + \varepsilon \delta_y) \right|_{\varepsilon=0} = 2(S_2 - q_{V(y)}), \quad (27)$$

$$\nabla_{\theta} P CMD(q) = 2 \mathbb{E}_{Y \sim Q_{\theta}}[(S_2 - q_{V(Y)}) s_{\theta}(Y)]. \quad (28)$$

*Proof.* The equality kernel and  $g$  satisfy SetPO’s Assumption 4.1, and its local mass is  $q_{V(y)}$ . Substitution into its influence formula proves the first identity. The influence has mean zero under  $Q_\theta$ , so the  $-\nabla_\theta \log P_\theta$  term in the conditional score cancels, giving the second identity.  $\square$

These identities permit shared neural parameters. Along an update, the sign of the diversity change depends on its alignment with this metric gradient; the influence formula alone supplies no monotonicity guarantee for a given training direction.

## P.5 DIVERSITY AT FINITE STEPS AND SAMPLING BUDGETS

Softmax outcome logits amplify probability differences among equally rewarded solutions (Sinha et al., 2026, Theorem 3.1 and App. A.1), and sampling imbalances can reinforce this effect (Lochab et al., 2026, Section 4.2). Under the categorical model, PCMD decreases from every nonuniform correct-mode distribution along both the mean flow and finite expected-gradient steps. Sampled updates can also break an initially uniform distribution’s symmetry. The bounds below relate concentration to the probability of observing a competing solution mode within a finite inference budget.

**Proposition P.15** (Exact diversity drift and matched correctness). *Under Assumption P.1, consider  $\dot{z} = c(P)\nabla_z P$  with  $c(P) > 0$ . Write  $D(q) = 1 - \sum_c q_c^2$ ,  $S_2 = \sum_c q_c^2$ , and  $V(q) = \sum_c q_c^3 - S_2^2 = \text{Var}_{C \sim q}(q_C)$ . For incorrect conditional probabilities  $r_a = p_a/(1 - P)$ , set  $T_2 = \sum_a r_a^2$ . In effective time  $d\tau = c(P)P(1 - P)dt$ ,*

$$\frac{dD}{d\tau} = -2V(q), \quad \frac{dD}{d \log[P/(1 - P)]} = -\frac{2V(q)}{S_2 + T_2}. \quad (29)$$

*Thus diversity strictly decreases at every nonuniform interior  $q$ . Positive-coefficient mean flows from the same initial logits follow the same  $D$ -versus- $P$  curve, independently of their chosen outcome-binary advantage or group size.*

*Proof.* For the replicator flow (17), the potential  $U(q) = S_2/2$  has fitness  $\partial U/\partial q_c = q_c$ . The fitness-variance identity of Harper (2009, Theorems 1–2) gives  $dU/d\tau = V(q)$ , hence  $dD/d\tau = -2V(q)$ . The variance vanishes exactly when the positive coordinates of  $q$  are equal. Also,

$$\dot{P} = c(P)\|\nabla_z P\|_2^2 = c(P)P^2(1 - P)^2(S_2 + T_2),$$

so  $d \log[P/(1 - P)]/d\tau = S_2 + T_2 > 0$ , proving the second identity. Finally, the full logit equation in effective time is  $dz_c/d\tau = q_c$  and  $dz_a/d\tau = -r_a$ ; it contains no  $c$ . The same initial logits therefore give the same full trajectory after the time change, and hence the same curve at matched correctness for every positive-coefficient outcome-binary estimator considered here.  $\square$

**Theorem P.16** (Collapse under finite expected-gradient steps). *Retain (A1), (A2), (A4), and (A5) of Assumption P.1, and replace (A3) by*

$$z_{k+1} = z_k + \eta_k c(P_k) \nabla_z P_k, \quad 0 < \eta_k < \infty, \quad \sum_{k=0}^{\infty} \eta_k = \infty, \quad (30)$$

*where  $c$  is continuous and positive on  $(0, 1)$ . Each step strictly decreases  $D(q_k)$  unless  $q_k$  is uniform. If  $q_0$  has a unique largest coordinate  $c^*$ , then  $P_k \rightarrow 1$  and  $q_{c^*,k} \rightarrow 1$ . No upper bound on the finite step sizes is required.*

*Proof.* Let  $h_k = \eta_k c(P_k) P_k (1 - P_k) > 0$ . The exact updates are  $z_c^+ = z_c + h_k q_c$  and  $z_a^+ = z_a - h_k r_a$ , so

$$q_c^+ = \frac{q_c e^{h_k q_c}}{\sum_d q_d e^{h_k q_d}}. \quad (31)$$

For fixed  $q$ , put  $u_c(s) = q_c e^{s q_c} / \sum_d q_d e^{s q_d}$ . Differentiation gives

$$\frac{d}{ds} \sum_c u_c(s)^2 = \sum_{c,d} u_c(s) u_d(s) (u_c(s) - u_d(s)) (q_c - q_d) \geq 0.$$

The ordering of  $u(s)$  is that of  $q$ , and the inequality is strict whenever  $q$  is nonuniform. Integrating from 0 to  $h_k$  proves the diversity claim. Equation (31) also preserves the leader and makes its conditional probability nondecreasing.

Every correct logit increases and every incorrect logit decreases, hence  $P_k$  increases. Jensen's inequality gives

$$\log \frac{P_{k+1}}{1 - P_{k+1}} - \log \frac{P_k}{1 - P_k} = \log \sum_c q_c e^{h_k q_c} - \log \sum_a r_a e^{-h_k r_a} \geq h_k \sum_c q_c^2 \geq h_k/m.$$

If  $P_k$  had an interior limit, continuity and positivity of  $c$  would imply  $\sum_k h_k = \infty$ , contradicting bounded correctness odds. Thus  $P_k \rightarrow 1$ . Moreover,  $\sum_k h_k = \infty$  is necessary: otherwise every logit would have finite total variation and converge to a finite value, contradicting  $P_k \rightarrow 1$ . Set  $\alpha = q_{c^*,0}$  and  $\xi_{c,k} = q_{c,k}/q_{c^*,k}$ . For  $c \neq c^*$ ,

$$\xi_{c,k+1} = \xi_{c,k} e^{-h_k(q_{c^*,k} - q_{c,k})}, \quad q_{c^*,k} - q_{c,k} \geq \alpha(1 - \xi_{c,0}) > 0.$$

The divergent sum of  $h_k$  forces every such ratio to zero, proving  $q_{c^*,k} \rightarrow 1$ .  $\square$

The update uses the exact expected gradient. Finite steps preserve the concentration conclusion but can change the trajectory across estimators. The identical diversity-versus-correctness curve in Proposition P.15 requires infinitesimal mean flow; it does not extend to these finite updates.

Finite multinomial counts can break a uniform policy's symmetry (Lochab et al., 2026, Proposition A.1). For the centered group estimator, the leading expected one-step loss in PCMD is quadratic in the learning rate, with a coefficient determined by group size and correctness.

**Proposition P.17** (Sampled updates break conditional symmetry). *Retain (A1), (A2), (A4), and (A5) of Assumption P.1, and replace (A3) by one sampled update  $z^+ = z + \eta \hat{g}_G$  from Equation (13), with  $\eta > 0$ . Suppose  $0 < P < 1$  and  $q_c = 1/m$  for every correct mode. Then*

$$\mathbb{E}[D(q^+)] < 1 - 1/m, \quad \mathbb{E}[q_c^+] = 1/m. \quad (32)$$

As  $\eta \rightarrow 0$ , for the fixed initial policy and estimator,

$$\mathbb{E}[D(q^+)] = 1 - \frac{1}{m} - \eta^2 \frac{m-1}{m^3} \mathbb{E} \left[ \frac{w(R)^2 R(G-R)^2}{G^4} \right] + O(\eta^3), \quad (33)$$

where the summand is zero on  $R = 0, G$ . For Dr.GRPO,  $w = 1$  and

$$\mathbb{E} \left[ \frac{R(G-R)^2}{G^4} \right] = \frac{G-1}{G^3} P(1-P) [1 + (G-2)(1-P)]. \quad (34)$$

*Proof.* Let  $N_c$  count occurrences of correct mode  $c$ . The centered advantages sum to zero, so the common softmax-score term cancels and  $\hat{g}_c = \kappa_R N_c$ , where  $\kappa_R = w(R)(G-R)/G^2$  on mixed groups and  $\kappa_R = 0$  otherwise. Conditional on  $R$ , the correct counts are  $N \sim \text{Multinomial}(R; 1/m, \dots, 1/m)$ , and

$$q_c^+ = \frac{e^{\eta \kappa_R N_c}}{\sum_d e^{\eta \kappa_R N_d}}.$$

Diversity is at most  $1 - 1/m$ , with equality only at uniform  $q^+$ . The event  $R = 1$  has positive probability; on this event one correct logit increases and the others do not, so  $q^+$  is nonuniform. This proves the strict inequality for every  $\eta > 0$ . Permutation symmetry of the correct labels gives  $\mathbb{E}[q_c^+] = 1/m$ .

For a fixed vector  $v$ , Taylor expansion about uniform logits gives

$$\sum_c \text{softmax}(\eta v)_c^2 = \frac{1}{m} + \frac{\eta^2}{m^2} \sum_c (v_c - \bar{v})^2 + O(\eta^3), \quad \bar{v} = \frac{1}{m} \sum_c v_c.$$

Use  $v_c = \kappa_R N_c$  and  $\mathbb{E}[\sum_c (N_c - R/m)^2 \mid R] = R(1 - 1/m)$ . There are finitely many possible groups and all mixed-group weights are finite, so the Taylor remainder is uniform over those groups. Averaging proves Equation (33). Finally,

$$\mathbb{E}[R(G-R)^2] = G(G-1)(G-2)P(1-P)^2 + G(G-1)P(1-P),$$

by the binomial factorial moments, yielding Equation (34).  $\square$

At fixed  $P \in (0, 1)$ , the Dr.GRPO coefficient in Equation (34) is  $P(1 - P)^2/G + O(G^{-2})$  as  $G \rightarrow \infty$ . The coefficient of the quadratic learning-rate term therefore decays asymptotically as  $1/G$ ; it need not decrease at every increment in finite group size. This sampling effect is absent from the uniform mean flow. The result concerns one step from a uniform correct-mode distribution and gives neither monotone expected diversity from an arbitrary initial policy nor stochastic-optimizer convergence.

**Corollary P.18** (Finite-budget disappearance of competing modes). *In the mean flow of Theorem P.4, let  $c^*$  be the unique initial correct leader and define*

$$\alpha = q_{c^*}(0), \quad \Gamma_0 = \sum_{c \neq c^*} \frac{q_c(0)}{\alpha - q_c(0)}.$$

At effective time  $\tau$ ,

$$1 - q_{c^*}(\tau) \leq \Gamma_0 e^{-\alpha\tau}, \quad D(q(\tau)) \leq 2\Gamma_0 e^{-\alpha\tau}. \quad (35)$$

For  $K \geq 1$  independent full-policy draws at that checkpoint,

$$\Pr(\text{any correct mode other than } c^* \text{ is observed}) \leq K\Gamma_0 e^{-\alpha\tau}. \quad (36)$$

In particular, if  $\tau \geq \max\{0, \alpha^{-1} \log(K\Gamma_0/\delta)\}$  for  $0 < \delta < 1$ , no competing correct mode is observed with probability at least  $1 - \delta$ , although every mode retains positive probability.

*Proof.* For  $c \neq c^*$ , write  $\xi_c = q_c/q_{c^*} < 1$ . The leader is nondecreasing, and Equation (17) gives

$$\frac{d}{d\tau} \log \frac{\xi_c}{1 - \xi_c} = -q_{c^*} \leq -\alpha.$$

Hence  $\xi_c(\tau) \leq [q_c(0)/(\alpha - q_c(0))]e^{-\alpha\tau}$ . Now  $1 - q_{c^*} = \sum_{c \neq c^*} \xi_c / (1 + \sum_{c \neq c^*} \xi_c)$  and  $D \leq 1 - q_{c^*}^2 \leq 2(1 - q_{c^*})$ , proving the first bounds. A single full-policy draw hits a competing correct mode with probability  $P(1 - q_{c^*}) \leq 1 - q_{c^*}$ ; a union bound over  $K$  draws completes the proof.  $\square$

The bound gives a finite-budget interpretation of the rare-output pruning mechanism discussed by Lochab et al. (2026, Section 4.2.5). It can also be indexed by correctness: Proposition P.15 and  $S_2 + T_2 \leq 2$  imply  $\tau \geq \frac{1}{2} \log\{[P/(1 - P)]/[P(0)/(1 - P(0))]\}$ . These are sufficient visibility bounds within the categorical model; they do not require literal support loss or identify an empirical collapse time for neural training.

## P.6 A REPLAY SIGNAL WHEN FRESH GROUPS SUPPLY NONE

**Lemma P.19** (Fresh-group starvation). *Under Assumption P.1 with correctness probability  $P$ , draw  $G$  fresh categories independently. The probability of a mixed-reward group is*

$$h_G(P) = 1 - P^G - (1 - P)^G \leq G \min\{P, 1 - P\}.$$

Every all-equal group has zero fresh task component for Equation (13) and for the specified centered MaxRL estimator, so the probability of a nonzero fresh task component is at most  $h_G(P)$ .

*Proof.* The all-correct and all-incorrect events are disjoint and have probabilities  $P^G$  and  $(1 - P)^G$ . A mixed group requires both a correct and an incorrect sample, so the union bound supplies each of the two stated bounds,  $GP$  and  $G(1 - P)$ , and the minimum of the pair is therefore also a bound.  $\square$

**Lemma P.20** (Replay ascent persists as fresh signal vanishes). *Under Assumption P.1, let  $\mathcal{B} \subseteq \mathcal{C}$  be a nonempty fixed verified buffer with normalized positive weights  $w_b$ , let  $R_w = -\sum_b w_b \log p_b$  with  $\rho_w > 0$ , and extend  $w$  by zero outside the buffer to obtain  $\bar{w}$ . The deterministic replay ascent vector is*

$$v^{\text{replay}} = -\rho_w \nabla R_w = \rho_w (\bar{w} - p),$$

which vanishes only at  $p = \bar{w}$ . If  $p_b \leq w_b/2$  for a buffered mode, then  $v_b^{\text{replay}} \geq \rho_w w_b/2$ , even when  $b$  is absent from the fresh group. For a uniform buffer of  $k \geq 2$  modes and fixed  $\rho_w > 0$ , as  $p \rightarrow \mathbf{e}_b$  at one buffered correct mode,  $h_G(P) \rightarrow 0$  whereas  $\|v^{\text{replay}}\|_2 \rightarrow \rho_w \sqrt{1 - 1/k} > 0$ .

*Proof.* For  $R_w = -\sum_b w_b \log p_b$ , the softmax score identity gives  $\partial R_w / \partial z_a = p_a - \bar{w}_a$ , proving the replay formula and coordinate bound. Finally,

$$\|\bar{w} - \mathbf{e}_b\|_2^2 = (1 - 1/k)^2 + (k-1)/k^2 = 1 - 1/k,$$

which gives the strictly positive limiting norm for  $k \geq 2$ .  $\square$

**Remark P.21.** A fixed verified buffer can supply a replay signal when the fresh task gradient is unavailable. Lemma P.19 bounds the probability of a nonzero fresh task component; it does not bound its magnitude or account for other loss terms, clipping, or momentum. The vector in Lemma P.20 averages the fixed buffer, so a stochastic exemplar draw realizes it only in expectation, and a nonzero logit coordinate can still be annihilated by a degenerate neural Jacobian. Replay also need not increase the next mixed-group probability, since  $h_G(P)$  is largest at  $P = 1/2$ .

## P.7 FIXED-BUFFER REPLAY AND QUALITATIVE RETENTION

Let  $\mathcal{B} \subseteq \mathcal{C}$  be a nonempty fixed buffer of  $k$  modes that the policy has produced and the validator has accepted. Uniform verified replay uses one exemplar per buffered mode and, in the categorical reduction, minimizes

$$R_{\mathcal{B}}(z) = -\frac{1}{k} \sum_{b \in \mathcal{B}} \log p_b. \quad (37)$$

For the uniform buffer target  $w$ , the likelihood-replay objective is  $R_{\mathcal{B}} = H(w) + \text{KL}(w||p)$  (Li et al., 2025, App. D.1, version 4). The target assigns mass  $1/k$  to each buffered mode and zero elsewhere. Decreasing this loss can raise total buffer mass as well as redistribute mass within the buffer; the buffer is not the conditioning event for  $p$ .

In an averaged continuous-time model, absorb a fixed round-robin replay frequency and loss scale into an effective dose  $\rho > 0$ . This assumes a fixed eligible-buffer set and constant relative weighting; it does not identify finite alternating updates with simultaneous gradient flow. For one isolated prompt, the categorical mean flow for any of the three fresh binary-reward estimators with verified replay is

$$\dot{z} = c_G(P) \nabla_z P - \rho \nabla_z R_{\mathcal{B}}(z). \quad (38)$$

Let  $\Psi(P) = \int_0^P c_G(s) ds$ . For the finite-group estimators in Lemmas P.2 and P.3,  $c_G$  has a finite positive extension to  $[0, 1]$ ; in particular, Dr.GRPO has  $c_G = (G-1)/G$  and MaxRL has  $c_G = [1 - (1-P)^{G-1}]/P$ . More generally, Theorem P.6 and Equation (19) give  $c_a(P) \nabla_z P$  for every fixed, bounded reward-measurable advantage, with  $c_a$  a polynomial. Its integral  $\Psi_a$  is also bounded on  $[0, 1]$ , so the retention argument applies even when the fresh coefficient changes sign. Define

$$F_{\mathcal{B}}(z) = \rho R_{\mathcal{B}}(z) - \Psi(P(z)). \quad (39)$$

Then (38) is exactly gradient flow  $\dot{z} = -\nabla_z F_{\mathcal{B}}$ .

**Theorem P.22** (Verified replay prevents buffered-mode extinction). *Under Assumption P.1, suppose logits are finite at time  $T$  and thereafter the nonempty verified buffer  $\mathcal{B}$  is fixed, fits within replay capacity, and receives a fixed positive effective replay dose  $\rho$ , the fresh objective being any bounded reward-measurable one in the sense of Theorem P.6. Set  $\Psi_{\max} = \max_{s \in [0, 1]} \Psi(s)$ , which is finite, and under (38) define*

$$C_T = R_{\mathcal{B}}(z(T)) + \frac{\Psi_{\max} - \Psi(P(T))}{\rho}.$$

Then for every  $b \in \mathcal{B}$  and every  $t \geq T$ ,

$$p_b(t) \geq \exp(-kC_T) > 0. \quad (40)$$

Thus each buffered probability is uniformly bounded away from zero for all  $t \geq T$ , including asymptotically, even when the buffer does not cover every correct mode. This is stronger than finite-time positivity, which already follows from finite softmax logits without replay.

*Proof.* The smooth vector field has bounded velocity because  $c_G$ ,  $\nabla_z P$ , and  $\nabla_z R_{\mathcal{B}}$  are bounded, so finite initial logits give a flow for every finite  $t \geq T$ . Along this flow,

$$\frac{d}{dt} F_{\mathcal{B}}(z(t)) = -\|\nabla_z F_{\mathcal{B}}(z(t))\|_2^2 \leq 0.$$

Since  $\Psi(P(t)) \leq \Psi_{\max}$ ,

$$R_{\mathcal{B}}(z(t)) \leq R_{\mathcal{B}}(z(T)) + \frac{\Psi_{\max} - \Psi(P(T))}{\rho} = C_T.$$

Every  $-\log p_b$  is nonnegative and their sum is  $kR_{\mathcal{B}}$ . Therefore  $-\log p_b(t) \leq kC_T$  for every buffered mode, which is exactly Equation (40) for every time after  $T$ .  $\square$

**A sharper certificate from normalization.** Write  $\text{kl}(a, r) = a \log(a/r) + (1-a) \log[(1-a)/(1-r)]$  with continuous endpoint conventions. For  $0 < a < 1$  and  $D \geq 0$ , let  $\ell_{-}(a, D)$  be the unique  $r \in (0, a]$  satisfying  $\text{kl}(a, r) = D$ ; set  $\ell_{-}(1, D) = e^{-D}$ . In particular,  $\ell_{-}(a, 0) = a$ .

**Corollary P.23** (Replay certificate by KL data processing). *Let  $w_b > 0$ ,  $\sum_{b \in \mathcal{B}} w_b = 1$ , and suppose a probability law  $p$  satisfies  $R_w(p) = -\sum_{b \in \mathcal{B}} w_b \log p_b \leq C < \infty$ . With  $D = C - H(w)$  and  $H(w) = -\sum_b w_b \log w_b$ , necessarily  $D \geq 0$ , and*

$$p_b \geq \ell_{-}(w_b, D) > 0, \quad p(\mathcal{B}) \geq e^{-D}, \quad p_b \geq \max\{0, w_b - \sqrt{D/2}\}. \quad (41)$$

*The binary-KL coordinate bound is sharp on the closed probability simplex when the alphabet has at least two categories. In Theorem P.22, take  $w_b = 1/k$  and  $C = C_T$  to obtain these bounds simultaneously for every  $t \geq T$ .*

*Proof.* Extend  $w$  by zero outside  $\mathcal{B}$ . Then  $\text{KL}(\bar{w} \| p) = R_w(p) - H(w) \leq D$ . Data processing under the partitions  $\{b\}, \{b\}^c$  and  $\mathcal{B}, \mathcal{B}^c$  gives  $\text{kl}(w_b, p_b) \leq D$  and  $-\log p(\mathcal{B}) \leq D$  (van Erven & Harremoës, 2014, Theorem 9). Binary inversion gives the first bound; Pinsker gives the third (van Erven & Harremoës, 2014, Theorem 31). For  $w_b < 1$ , equality in the first is attained by  $p_b = \ell_{-}(w_b, D)$  and  $p_j = (1 - p_b)w_j / (1 - w_b)$  on the remaining buffer, with zero mass elsewhere. For  $w_b = 1$ , put the remaining probability on another category. For  $D > 0$ , full-support laws approach the same bound by assigning vanishing mass off the buffer and slightly increasing the protected coordinate while keeping the divergence within the allowed budget.  $\square$

For illustration, a uniform buffer of 16 with  $C = \log 16 + 0.01$  gives  $p_b \geq 0.03397$ , whereas the termwise bound  $e^{-16C}$  is  $4.62 \times 10^{-20}$ . These are illustrative inputs, not measured training losses. A large excess loss  $D$  can still make the sharper certificate weak.

For the nonnegative coefficients of the named fresh estimators,  $\Psi_{\max} = \Psi(1)$ , recovering the simpler constant. Signed coefficients also satisfy the retention bound, but need not yield  $P \rightarrow 1$ . The elementary floor is qualitative and can be numerically vacuous. For example, at  $k = 16$  even a hypothetical  $C_T = 160$  gives  $e^{-2560}$ , far below any useful evaluation probability. The argument establishes a uniform-in-time lower bound in the stated ideal flow, not practical visibility or a prediction for the experimental replay loss. The effective dose also depends on relative fresh/replay frequencies, not the replay coefficient alone.

A common nonnegative, locally integrable rate  $\eta(t)$  multiplying both terms of (38) preserves the energy bound, since  $\dot{F}_{\mathcal{B}} = -\eta(t) \|\nabla F_{\mathcal{B}}\|_2^2 \leq 0$ . The convergence statements below additionally require  $\int_T^\infty \eta(t) dt = \infty$  and then follow by reparameterizing optimization time. Infinitely many replay visits alone establish neither this condition nor a fixed replay-to-fresh weighting. Different time-varying relative weights are outside the fixed-potential argument.

#### Full coverage and the categorical target.

**Corollary P.24** (Conditional full-coverage categorical limit). *Assume the conditions of Theorem P.22 and that by time  $T$  the buffer covers the full correct support,  $\mathcal{B} = \mathcal{C}$ . Assume additionally  $c_G(P) \geq 0$  on  $[0, 1]$ , as for the three named estimators. Let  $u_c = 1/m$  on  $\mathcal{C}$ . Then the categorical mean flow converges to*

$$P(t) \longrightarrow 1, \quad q(t) \longrightarrow u.$$

Consequently

$$\min_{c \in \mathcal{C}} q_c(t) \longrightarrow \frac{1}{m}, \quad \mathcal{H}(q(t)) \longrightarrow \log m,$$

*so asymptotic concentration onto a strict subset of the correct modes is impossible under full coverage. PCMD may nevertheless decline transiently before converging to its uniform maximum (Theorem P.26).*

*Proof.* For full coverage,

$$R_C = \log m - \log P + \text{KL}(u||q). \quad (42)$$

The potential is bounded below by  $-\Psi_{\max}$ , so its descent gives  $\int_T^\infty \|\nabla F_C\|_2^2 dt < \infty$ . The fixed reward-measurable advantage gives a smooth  $c_G$  on  $[0, 1]$ , and softmax derivatives and the cross-entropy gradient are bounded. Thus the Hessian of  $F_C$  and the velocity are bounded; the gradient along the trajectory is uniformly continuous. Square integrability then implies  $\nabla F_C \rightarrow 0$ : recurring gradients of fixed positive size would otherwise contribute energy over intervals of a uniform minimum length.

Every limiting probability vector therefore satisfies the correct-coordinate equations

$$0 = \rho(p_c - 1/m) - c_G(P)p_c(1 - P).$$

Summing them gives  $0 = -(1 - P)(\rho + c_G(P)P)$ , hence  $P = 1$ . Each correct coordinate is then  $1/m$  and every incorrect coordinate is zero. Compactness of the probability simplex and uniqueness of this limit establish convergence of the probability vector.  $\square$

**Corollary P.25** (Uniform allocation maximizes finite-budget coverage). *Fix correctness  $P \in (0, 1]$  and  $m \geq 2$  possible correct keys. Let  $D_K(P, q)$  count distinct verified keys in  $K$  independent draws with key probabilities  $Pq_1, \dots, Pq_m$  and failure probability  $1 - P$ . For  $u = (1/m, \dots, 1/m)$ , every  $K \geq 1$  and  $r = 1, \dots, m$  satisfy*

$$\Pr[D_K(P, u) \geq r] \geq \Pr[D_K(P, q) \geq r]. \quad (43)$$

*Thus uniform conditional allocation maximizes expected distinct@K at fixed correctness, as well as every coverage tail probability. This is the coupon-collection ordering of Anceaume et al. (2015, Theorem 3, v1), applied to independent draws from a fixed policy over a finite catalogue of verified execution keys.*

*Proof.* Map each correct key to a coupon of mass  $Pq_c$  and every failure to the null coupon of mass  $1 - P$ . The cited theorem makes the time to collect  $r$  distinct non-null coupons stochastically smallest when each has mass  $P/m$ . The event that this time is at most  $K$  is exactly  $D_K \geq r$ . Zero coordinates and  $P = 1$  follow by continuity of finite-horizon probabilities. Summing the tail inequalities over  $r = 1, \dots, m$  gives the same ordering of expected distinct counts for every finite sampling budget at the same correctness level and over the same catalogue of possible keys.  $\square$

This sampling optimum does not assert that a training algorithm reaches the uniform allocation.

**Positive weights and the implemented length normalization.** Uniformity of the target is separate from protection against extinction. For fixed weights  $w_b > 0$  with  $\sum_{b \in \mathcal{B}} w_b = 1$ , define  $R_w = -\sum_b w_b \log p_b$  and use replay coefficient  $\rho_w > 0$ . The same potential argument gives, for all  $t \geq T$ ,

$$p_b(t) \geq \exp(-C_{T,w}/w_b), \quad C_{T,w} = R_w(z(T)) + \frac{\Psi_{\max} - \Psi(P(T))}{\rho_w}.$$

Corollary P.23 sharpens this floor with  $C = C_{T,w}$  and the same normalized target  $w$ . For the following convergence statements assume  $c_G \geq 0$ . If  $\mathcal{B} = \mathcal{C}$ , then  $R_w = H(w) - \log P + \text{KL}(w||q)$ , and the independent-logit limit is  $P \rightarrow 1, q \rightarrow w$ . The preceding stationarity proof applies with  $1/m$  replaced by  $w_c$ : summing the correct-coordinate equations forces  $P = 1$ , then each equation forces  $p_c = w_c$ .

A partial buffer has a stronger limitation in this same ideal flow. For any fixed nonempty  $\mathcal{B} \subseteq \mathcal{C}$  and fixed positive weights summing to one, extend  $w$  by zero to  $\bar{w}$  on all categories. Then  $p(t) \rightarrow \bar{w}$ , so even correct modes outside the fixed buffer vanish asymptotically. For  $F = \rho_w R_w - \Psi(P)$ , the preceding energy and bounded-Hessian argument still gives  $\nabla F \rightarrow 0$ ; neither argument requires full buffer coverage. At every subsequential probability limit, the correct-coordinate equations are

$$0 = \rho_w(p_c - \bar{w}_c) - c_G(P)p_c(1 - P).$$

Summing over correct categories gives  $0 = -(1 - P)(\rho_w + c_G(P)P)$ , hence  $P = 1$ ; each equation then gives  $p_c = \bar{w}_c$ , and all incorrect mass is zero. Compactness and uniqueness of this limiting probability vector prove convergence. Thus protecting a fixed set of buffered

modes need not preserve unbuffered correct alternatives. This conclusion concerns the fixed-buffer categorical flow, not changing-buffer or stochastic neural training.

This distinction matters for the implemented mean-token exemplar loss. Even if each key were represented by exactly one categorical exemplar of fixed length  $L_b$ , its loss would be  $AR_w$ , where

$$A = \frac{1}{k} \sum_{b \in \mathcal{B}} \frac{1}{L_b}, \quad w_b = \frac{L_b^{-1}}{\sum_{j \in \mathcal{B}} L_j^{-1}}.$$

Its effective categorical replay coefficient would be  $\rho_w = \rho A$ . The full-coverage limit would therefore weight shorter exemplars more, and would be uniform only for equal lengths. For an actual language model,  $\pi_\theta(e_b \mid x)$  is only one contribution to the total key probability. Equal averaging of per-token exemplar scores does not imply uniform learned key probabilities; the categorical uniformity corollary is not that claim.

Full support is an additional premise, not a consequence of replay. In Python factors, evaluation prompts admit 16–1,512 correct execution modes (mean 229.44), while the buffer capacity is 16. The full-coverage corollary therefore cannot supply a domain-wide guarantee there. The retention theorem protects only buffered modes; it supplies no probability floor for an unbuffered key.

## P.8 WHEN REPLAY REVERSES CONCENTRATION, AND HOW FAST

The restoring logit field  $\rho(\bar{w} - p)$  is proportional to the ideal inverse probability scaling field of Sinha et al. (2026, Theorem 4.1, v1), normalized to a fixed target  $w$ . The uniform-correct target and its forward conditional-KL formulation also appear in Lochab et al. (2026, Sections 5.2 and 6.2, v1). For replay, the target is supported on the discovered buffer and the loss is evaluated from stored exemplars. The relative strengths of this restoring field and the fresh binary objective determine whether buffer diversity increases over a finite interval.

Fix a buffer  $\mathcal{B} \subseteq \mathcal{C}$  of  $k \geq 2$  modes, with uniform categorical weights, and consider Equation (38). Write  $c(P)$  for its fresh coefficient and define

$$M_{\mathcal{B}} = \sum_{b \in \mathcal{B}} p_b, \quad r_b = \frac{p_b}{M_{\mathcal{B}}}, \quad C_{\mathcal{B}} = \sum_{b \in \mathcal{B}} r_b^2, \quad (44)$$

$$a_\rho(t) = \rho - c(P(t))(1 - P(t)).$$

Thus  $1 - C_{\mathcal{B}}$  is PCMD conditional on the buffer. It equals the success-conditional PCMD when  $\mathcal{B} = \mathcal{C}$ ; otherwise it measures only allocation inside the buffer.

**Theorem P.26** (Exact replay threshold for buffer diversity). *Under the categorical mean-flow assumptions and a fixed uniform buffer,*

$$\frac{d}{dt} \log \frac{p_i}{p_j} = -a_\rho M_{\mathcal{B}}(r_i - r_j), \quad i, j \in \mathcal{B}, \quad (45)$$

$$\dot{r}_b = -a_\rho M_{\mathcal{B}} r_b(r_b - C_{\mathcal{B}}), \quad b \in \mathcal{B}, \quad (46)$$

$$\frac{d}{dt}(1 - C_{\mathcal{B}}) = 2a_\rho M_{\mathcal{B}} \left( \sum_{b \in \mathcal{B}} r_b^3 - C_{\mathcal{B}}^2 \right). \quad (47)$$

*For a nonuniform  $r$ , buffer-conditional diversity strictly increases when  $\rho > c(P)(1 - P)$ , strictly decreases when  $\rho < c(P)(1 - P)$ , and has zero instantaneous derivative at equality. These conclusions require no coverage of correct modes outside the buffer.*

*Proof.* For  $b \in \mathcal{B}$ , the score identities give  $\dot{z}_b = \rho/k - a_\rho p_b$ . Subtracting two such equations gives Equation (45). The distribution  $r$  is the softmax of the logits restricted to  $\mathcal{B}$ , so  $\dot{r}_b = r_b(\dot{z}_b - \sum_j r_j \dot{z}_j)$ , proving Equation (46). Differentiating  $C_{\mathcal{B}}$  gives Equation (47). Its parenthesis is  $\text{Var}_{b \sim r}(r_b)$ , which is strictly positive for a nonuniform distribution with positive coordinates.  $\square$

A positive dose therefore need not improve diversity immediately. The thresholds for Dr.GRPO and the specified MaxRL estimator are, respectively,

$$\rho > \frac{G-1}{G}(1-P), \quad \rho > \frac{(1-P)[1-(1-P)^{G-1}]}{P}.$$

The asymptotic result for every  $\rho > 0$  and this finite-time condition are compatible: the fresh concentration pressure vanishes as  $P \rightarrow 1$ .

**Theorem P.27** (Finite recovery from accumulated replay exposure). *In the setting of Theorem P.26, suppose  $a_\rho(u) \geq 0$  for  $u \in [T, t]$ , and define*

$$S(T, t) = \int_T^t a_\rho(u) M_{\mathcal{B}}(u) du, \quad D(t) = \log \frac{\max_{b \in \mathcal{B}} r_b(t)}{\min_{b \in \mathcal{B}} r_b(t)}.$$

Then

$$e^{D(t)} - 1 \leq (e^{D(T)} - 1)e^{-S(T, t)/k}, \quad \min_b r_b(t) \geq \frac{1}{1 + (k-1)e^{D(t)}}. \quad (48)$$

For  $0 < \delta < 1/k$ , put  $D_\delta = \log[(1/\delta - 1)/(k-1)]$ . If  $D(T) > D_\delta$ , the condition

$$S(T, t) \geq k \log \frac{e^{D(T)} - 1}{e^{D_\delta} - 1} \quad (49)$$

is sufficient for every  $r_b(t) \geq \delta$ . If  $D(T) \leq D_\delta$ , this conclusion already holds and is preserved on the interval. In particular, when  $a_\rho \geq a_0 > 0$  and  $M_{\mathcal{B}} \geq M_0 > 0$  on the interval, its required duration is at most the right-hand side of Equation (49) divided by  $a_0 M_0$ . At that endpoint every unconditional buffer probability is at least  $M_0 \delta$ .

*Proof.* Buffer-coordinate ordering is preserved by Equation (45): a difference of two buffer logits satisfies a scalar equation proportional to that difference, and ties persist. Writing  $r_{\max}$  and  $r_{\min}$  for the extremal coordinates,

$$\dot{D} = -a_\rho M_{\mathcal{B}}(r_{\max} - r_{\min}), \quad r_{\max} - r_{\min} = r_{\max}(1 - e^{-D}) \geq \frac{1 - e^{-D}}{k}.$$

Thus  $\frac{d}{dt}(e^D - 1) \leq -(a_\rho M_{\mathcal{B}}/k)(e^D - 1)$ ; integration proves the first bound. Since  $1 \leq r_{\min} + (k-1)r_{\max}$ , the second follows. Substituting  $D_\delta$  and solving the first bound for the exposure proves the remaining claims. When  $D(T) = 0$ , uniformity is preserved directly.  $\square$

The recovery bound assumes a nonnegative margin  $a_\rho$  throughout the interval. Buffer mass need not be monotone, and correctness need not stay above its value at  $T$ ; both contribute to  $S(T, t)$ . The stated unconditional visibility guarantee additionally uses the buffer-mass lower bound  $M_0$ .

**Proposition P.28** (Discrete exact-gradient contraction). *Replace the continuous update by simultaneous exact mean-gradient steps, while retaining independent logits and a fixed uniform buffer:*

$$z_{n+1} = z_n + \eta_n \{c(P_n) \nabla_z P_n + \rho(\bar{w} - p_n)\}, \quad \eta_n \geq 0.$$

Define  $D_n$  as above and  $\gamma_n = \eta_n M_{\mathcal{B}, n}[\rho - c(P_n)(1 - P_n)]$ . If  $0 \leq \gamma_n \leq 2$ , the order of all buffer probabilities is preserved at this step and

$$e^{D_{n+1}} - 1 \leq \left(1 - \frac{1 - e^{-\gamma_n}}{k}\right) (e^{D_n} - 1). \quad (50)$$

Consequently, if  $0 \leq \gamma_n \leq 1$  for  $n = 0, \dots, N-1$ , then

$$e^{D_N} - 1 \leq (e^{D_0} - 1) \exp\left(-\frac{1 - e^{-1}}{k} \sum_{n=0}^{N-1} \gamma_n\right).$$

*Proof.* For two buffer logits with difference  $d = z_i - z_j \geq 0$ ,

$$d_{\text{next}} = d - \gamma_n(r_i - r_j), \quad r_i - r_j = (r_i + r_j) \tanh(d/2) \leq d/2.$$

Hence  $d_{\text{next}} \geq 0$  when  $\gamma_n \leq 2$ . The same extremal indices therefore give  $D_{n+1} = D_n - \gamma_n(r_{\max} - r_{\min})$ . For  $x \in [0, 1]$ , convexity of  $e^{-\gamma_n x}$  yields  $1 - e^{-\gamma_n x} \geq (1 - e^{-\gamma_n})x$ . Set  $x = r_{\max} - r_{\min}$  and use  $e^{D_n} x \geq (e^{D_n} - 1)/k$  from the preceding proof. This gives Equation (50). Multiplication over steps,  $1 - y \leq e^{-y}$ , and  $1 - e^{-\gamma} \geq (1 - e^{-1})\gamma$  for  $0 \leq \gamma \leq 1$  prove the last bound.  $\square$

The discrete contraction holds for simultaneous exact mean-gradient steps. Clipping, sampling noise, optimizer state, and alternating replay visits are outside its assumptions, so it does not establish contraction for AdamW. Uniform categorical weights are also essential to the stated target. For fixed nonuniform weights  $w_b$ , the exact conditional equation is instead

$$\dot{r}_b = r_b \left\{ \rho \left( w_b - \sum_j r_j w_j \right) - a_\rho M_{\mathcal{B}}(r_b - C_{\mathcal{B}}) \right\}. \quad (51)$$

The inverse-length weights induced by the implemented mean-token loss belong to this weighted case. The uniform contraction theorem therefore describes an equal-length categorical specialization, and does not assert uniform key probabilities for neural exemplar replay.

## P.9 COMPLETE RESPONSES AND FINITE EVALUATION VISIBILITY

The learner scores a particular response string, while evaluation groups strings by execution key. A complete verified response is an event contained in the event of producing its execution key. This containment converts a complete-response probability floor into a key-probability floor while accounting for the loss’s length factor. Here  $p_\theta(b | x) = \Pr_{Y \sim \pi_\theta(\cdot | x)}[V(Y, s_x) = b]$  is the unconditional probability of producing key  $b$ , rather than the probability conditional on correctness.

**Lemma P.29** (Retention from a joint exemplar potential). *Fix finitely many verified complete-response exemplars  $(x_j, e_j)$ , each realizing key  $b_j$ , with lengths  $L_j > 0$  and fixed weights  $\alpha_j > 0$ . Let*

$$F(\theta) = \sum_j \alpha_j \frac{-\log \pi_\theta(e_j | x_j)}{L_j} - J(\theta), \quad J(\theta) \leq J_{\max} < \infty.$$

*Suppose  $F(\theta(T))$  is finite and  $F(\theta(t)) \leq F(\theta(T))$  for every  $t \geq T$ . With  $D = F(\theta(T)) + J_{\max}$ , every protected exemplar satisfies*

$$p_\theta(b_j | x_j) \geq \pi_\theta(e_j | x_j) \geq \exp\left(-\frac{L_j D}{\alpha_j}\right) > 0 \quad (t \geq T).$$

*Proof.* The weighted exemplar loss equals  $F(\theta(t)) + J(\theta(t)) \leq D$ . Each summand is nonnegative, so  $-\log \pi_\theta(e_j | x_j) \leq L_j D / \alpha_j$ . The complete-response event is a subset of the event producing key  $b_j$ , which proves the stated lower bound on every protected key’s probability.  $\square$

Exact descent of a smooth joint potential is sufficient for the energy inequality, even with shared parameters and multiple prompts. For example, the ideal categorical model permits  $J(\theta) = \sum_x \nu_x \Psi_x(P_x(\theta))$  with fixed nonnegative prompt weights and  $J_{\max} = \sum_x \nu_x \max_{s \in [0,1]} \Psi_x(s)$ . This descent condition is not established for stochastic, clipped, alternating AdamW updates, and it neither implies uniform neural key probabilities nor requires them. Its bound applies only to the listed exemplar-prompt pairs. Shared parameters do not extend it to held-out prompts; transfer to those prompts remains an empirical question.

The scored probability must describe the complete response, including its termination event or a fixed deterministic horizon, under the same prompt, temperature, masks, truncation rules, and verifier as the sampling law being bounded. A prefix score or a score under another decoding law does not supply this bound without a separate event or likelihood comparison. For a complete response of length  $L$  and mean-token log score  $s$ , its probability is  $e^{Ls}$ , not  $e^s$ . A change in this exemplar’s likelihood need not equal the change in its whole key’s probability.

**A certificate for an execution key.** For one fixed prompt  $x$ , take distinct complete verified responses  $e_j$  with lengths  $L_j$ , keys  $b_j$ , and coefficients  $\alpha_j > 0$ . Define

$$A_x = \sum_j \frac{\alpha_j}{L_j}, \quad w_j = \frac{\alpha_j / L_j}{A_x}, \quad W_b = \sum_{j: b_j = b} w_j.$$

If their mean-token replay loss is at most  $C_x$ , namely  $\sum_j (\alpha_j / L_j) [-\log \pi_\theta(e_j | x)] \leq C_x$ , then

$$\text{kl}(W_b, p_\theta(b | x)) \leq D_x := C_x / A_x - H(w), \quad p_\theta(b | x) \geq \ell_-(W_b, D_x) \quad (W_b > 0). \quad (52)$$

Indeed, the response-level divergence from the normalized exemplar target is at most  $D_x$ ; deterministic execution followed by the indicator of key  $b$  is data processing (van Erven & Harremoës, 2014, Theorem 9), followed by Corollary P.23. This uses the same complete response and decoding law as Lemma P.29. Normalize separately for each prompt; a bound on total replay loss also bounds each prompt’s nonnegative summand. A checkpoint certificate supplies no trajectory guarantee without a uniform loss bound.

**Allowing stochastic energy excursions.** Exact descent is sufficient but not necessary. The following is a direct application of Ville’s inequality (Howard et al., 2020, Lemma 1).

**Corollary P.30** (Retention from a conditional energy budget). *Use the fixed complete exemplars, weights and potential  $F$  of Lemma P.29. Let  $D_t = F(\theta_t) + J_{\max} \geq 0$  be adapted and integrable, with deterministic finite  $D_0$ . For  $N \in \mathbb{N} \cup \{\infty\}$ , suppose deterministic  $d_t \geq 0$  satisfy  $\mathbb{E}[D_{t+1} \mid \mathcal{F}_t] \leq D_t + d_t$  for  $t < N$ , with finite total budget  $B_N = \sum_{t < N} d_t$ . These hypotheses control expected upward energy changes without requiring every realized step to decrease the joint loss. For every  $0 < \delta < 1$ , with probability at least  $1 - \delta$ ,*

$$p_{\theta_t}(b_j \mid x_j) \geq \pi_{\theta_t}(e_j \mid x_j) \geq \exp\left[-\frac{L_j(D_0 + B_N)}{\alpha_j \delta}\right] \quad \text{for all } t \leq N \text{ and all } j. \quad (53)$$

*The sharper per-prompt certificate in Equation (52) also applies on this event. For  $N = \infty$ , every protected probability has an almost surely positive, potentially random, infimum over the iterations.*

*Proof.*  $X_t = D_t + \sum_{s=t}^{N-1} d_s$  is a nonnegative supermartingale. Ville’s inequality bounds its supremum by  $(D_0 + B_N)/\delta$  with probability at least  $1 - \delta$ . Every replay summand is nonnegative and bounded by  $D_t$ , so exponentiation and event containment give the claim without a union factor over exemplars. If  $D_0 + B_N = 0$ , then  $X_t = 0$  almost surely. For  $N = \infty$ , Ville’s inequality with thresholds tending to infinity gives  $\sup_t X_t < \infty$  almost surely.  $\square$

For example, the one-step estimate underlying Wang et al. (2017, Lemma 2.1, Eq. (2.13)) supplies this drift for the update  $\theta_{t+1} = \theta_t - \eta_t H_t g_t$  with an  $L$ -smooth  $F$ ,  $\mathbb{E}[g_t \mid \mathcal{F}_t] = \nabla F(\theta_t)$ ,  $\mathbb{E}[\|g_t - \nabla F(\theta_t)\|^2 \mid \mathcal{F}_t] \leq \sigma^2/m_t$ , and preconditioners measurable before the current gradient noise with  $\underline{\kappa}I \preceq H_t \preceq \bar{\kappa}I$ . Deterministic steps  $\eta_t \leq \underline{\kappa}/(L\bar{\kappa}^2)$  and batch sizes  $m_t$  permit  $d_t = L\eta_t^2 \bar{\kappa}^2 \sigma^2/(2m_t)$ . An infinite horizon additionally needs the displayed finite budget; constant steps and persistent noise generally supply only a finite-horizon bound. The current-gradient AdamW moments, clipping, changing buffers and cyclic replay schedule are not certified by these premises: their bias and energy effects require separate control. No such budget is established for the experimental runs.

**Retention and finite evaluation visibility.** At a fixed checkpoint, suppose  $k$  distinct buffered keys for one prompt have probability floors  $p_b \geq \delta_b > 0$ . For  $K$  independent draws from that same policy, the buffered distinct count  $D_{\mathcal{B},K}$  satisfies

$$\begin{aligned} \mathbb{E}[D_{\mathcal{B},K}] &\geq \sum_{b \in \mathcal{B}} [1 - (1 - \delta_b)^K], \\ \Pr(\text{some buffered key is unseen}) &\leq \sum_{b \in \mathcal{B}} (1 - \delta_b)^K \leq k e^{-K \delta_{\min}}, \end{aligned}$$

where  $\delta_{\min} = \min_b \delta_b$ . The first statement follows from indicator expectations and the second from the union bound. Thus  $K \geq \log(k/\varepsilon)/\delta_{\min}$  suffices to observe all buffered keys with probability at least  $1 - \varepsilon$ , for  $0 < \varepsilon < 1$ . The energy floors can make this budget impractically large; asymptotic retention does not imply visibility in an eight-draw evaluation.

Conversely, not observing a mode does not establish extinction. At a fixed checkpoint, zero sightings of a prespecified key in  $N$  independent draws give a one-sided  $(1 - \alpha)$  binomial upper bound  $1 - \alpha^{1/N}$  and lower bound zero (Clopper & Pearson, 1934). For  $N = 32$  and  $\alpha = .05$ , the upper bound is approximately 8.94%. These are single-key bounds; simultaneous statements need error allocation. Counts from changing checkpoints cannot be pooled to certify a current policy, and teacher-forced scores are not independent Bernoulli observations.

# P.10 WHEN A REPLAY GRADIENT IMPROVES A NEURAL EXEMPLAR

With shared parameters, a replay update’s effect on a particular exemplar depends on interference among score gradients and on the optimizer’s response. A lower bound on that exemplar’s change in log probability can be obtained from the realized parameter step, without assuming the global potential descent required by Lemma P.29.

**Proposition P.31** (Local replay response under shared parameters). *For a fixed buffer of  $k$  complete exemplars, define*

$$\ell_b(\theta) = \frac{\log \pi_\theta(e_b | x_b)}{L_b}, \quad h_b = \nabla_\theta \ell_b(\theta), \quad \bar{h} = \frac{1}{k} \sum_{j=1}^k h_j.$$

*Suppose a step has the form  $\Delta = \eta H(g + \lambda \bar{h}) + u$ , where  $\eta, \lambda \geq 0$ ,  $H$  is a fixed symmetric positive semidefinite matrix for this step,  $g$  is the fresh ascent direction, and  $u$  is the remaining update. Assume  $\ell_b$  is twice differentiable and its Hessian has operator norm at most  $B_b$  along  $\{\theta + s\Delta : 0 \leq s \leq 1\}$ . Set*

$$K_{bj} = h_b^\top H h_j, \quad \kappa_b = \frac{1}{k} \sum_j K_{bj}, \quad m_b = \eta h_b^\top H g + \eta \lambda \kappa_b + h_b^\top u - \frac{B_b}{2} \|\Delta\|_2^2.$$

*Then*

$$\ell_b(\theta + \Delta) - \ell_b(\theta) \geq m_b, \quad \pi_{\theta+\Delta}(e_b | x_b) \geq \pi_\theta(e_b | x_b) e^{L_b m_b}. \quad (54)$$

*In particular,  $m_b > 0$  guarantees that this update increases the complete exemplar’s probability, whether or not it appeared among the fresh samples.*

*Proof.* Taylor’s theorem with the Hessian bound gives  $\ell_b(\theta + \Delta) - \ell_b(\theta) \geq h_b^\top \Delta - (B_b/2) \|\Delta\|_2^2$ . Substitution of the stated step and  $h_b^\top H \bar{h} = k^{-1} \sum_j K_{bj}$  proves the first inequality. Multiplication by  $L_b$  and exponentiation prove the second.  $\square$

The required replay response is a row average  $\kappa_b$  of the score Gram matrix, not only its nonnegative diagonal entry  $K_{bb}$ . A sufficient interference condition is  $\kappa_b > 0$ , with replay strength satisfying

$$\eta \lambda \kappa_b > -\eta h_b^\top H g - h_b^\top u + \frac{B_b}{2} \|\Delta\|_2^2.$$

This inequality includes the step’s curvature cost, which itself depends on the chosen dose; arbitrarily increasing  $\lambda$  is not a sufficient rule.

The interference condition cannot be dropped. For example, consider three categorical logits sharing one scalar parameter, with slopes  $(1, -2, 0)$ . At a point with probabilities  $(s, s, 1 - 2s)$ , where  $0 < s < 1/4$ , let the first two categories be correct and buffered. Their unit-length log-score gradients are  $h_1 = 1 + s$  and  $h_2 = -2 + s$ . Uniform replay alone follows  $\bar{h} = s - 1/2$ , so

$$\left. \frac{d}{dt} \log p_1 \right|_{\theta=\bar{h}} = (1 + s)(s - 1/2) < 0.$$

Thus shared parameters can make replay initially reduce the probability of a rare stored mode while improving the average replay score.

For an AdamW step, replay can alter both moment estimates. The proposition therefore does not identify its  $H$  with an unchanged AdamW preconditioner. It applies to a specified decomposition of the realized step, with any remaining contribution included in  $u$ ; alternatively its Taylor bound can be evaluated directly from  $h_b^\top \Delta$ . Neither the interference margin nor a curvature bound is established for the experimental runs. The statement concerns complete stored responses under the sampling law specified in App. P.9. Raising one such probability strengthens that exemplar’s contribution to its key, but need not increase total key probability or transfer to a held-out prompt.

## P.11 REFERENCE-KL STATIONARY TARGETS AND RECOVERY

Assumption P.1(A4) sets the reference-KL coefficient to zero, as in the default training setting of Table 6. A positive fixed-reference coefficient changes the categorical target. For Dr.GRPO and centered MaxRL, the unique full-support stationary policy allocates its correct mass in the same proportions as the reference. Exact Dr.GRPO gradient steps converge to this policy under Corollary P.34. On a two-correct-mode face, reverse KL and replay also differ in recovery time: the reverse-KL restoring logit force vanishes with the depleted probability, while replay’s buffered-coordinate force has a positive limit. The measured KL comparison is in App. P.12.

For this analysis, replace (A4) by

(A4’) *Reference KL.* A fixed reference  $\mu$  with full support on the category set  $\mathcal{A}$  enters with a positive coefficient  $\beta$ , and no entropy term is present alongside it.

The KL estimator in GRPO has expectation  $\text{KL}(p \parallel \mu)$  under samples drawn from  $p$  (Shao et al., 2024). Unbiasedness of this value does not imply that differentiating it at fixed samples gives an unbiased gradient of the distributional KL. With the KL differentiated exactly, write  $D(z) = \text{KL}(p(z) \parallel \mu)$ . The prompt-isolated categorical mean flow is

$$\dot{z} = c_G(P) \nabla_z P - \beta \nabla_z D(z), \quad \nabla_{z_b} D = p_b (\log(p_b/\mu_b) - D), \quad (55)$$

which is the negative gradient flow of  $F_\beta(z) = -\Psi(P(z)) + \beta D(z)$ , with  $\Psi' = c_G$  as in Equation (39).

**Lemma P.32** (Reverse-KL sublevels and replay barriers). *For a full-support reference,  $D = \text{KL}(p \parallel \mu) \leq \log(1/\min_a \mu_a)$  on the simplex, while  $R_B \rightarrow \infty$  if a buffered coordinate vanishes. More precisely,*

$$\inf_{p: p_b=0} D(p) = \log \frac{1}{1 - \mu_b}.$$

*Thus every finite replay-loss sublevel bounds each protected probability away from zero. A reverse-KL sublevel  $\{D \leq B\}$  does so for coordinate  $b$  if  $B < -\log(1 - \mu_b)$ , but includes a point with  $p_b = 0$  if  $B \geq -\log(1 - \mu_b)$ .*

*Proof.* The global bound follows from  $\sum_a p_a \log p_a \leq 0$  and  $-\sum_a p_a \log \mu_a \leq \log(1/\min_a \mu_a)$ . On the face  $p_b = 0$ , write  $\tilde{\mu}_a = \mu_a/(1 - \mu_b)$  for  $a \neq b$ . Then  $D(p) = \text{KL}(p \parallel \tilde{\mu}) - \log(1 - \mu_b)$ , whose minimum is attained at  $\tilde{\mu}$ . Continuity and compactness exclude that face from any strictly lower sublevel and give a positive coordinate minimum. For replay,  $-\log p_b \leq k R_B$  gives the bound directly.  $\square$

Descent of  $F_\beta$  gives  $D(p(t)) \leq [F_\beta(z(T)) + \Psi_{\max}]/\beta$ . This energy bound excludes the face  $p_b = 0$  when its right side is below  $-\log(1 - \mu_b)$ ; an arbitrary initial energy need not satisfy that condition. A trajectory can have a positive probability floor even when this sublevel bound does not establish one, as for the exact Dr.GRPO gradient steps in Corollary P.34.

**Proposition P.33** (The retained conditional is the reference’s). *Under Assumption P.1 with (A4’), using the Dr.GRPO or centered MaxRL coefficient, the flow (55) has exactly one stationary policy of full support,*

$$p_a^* \propto \mu_a e^{\lambda R_a}, \quad \lambda = c_G(P^*)/\beta, \quad R_a = \mathbf{1}[a \in \mathcal{C}],$$

*with  $\lambda = (G - 1)/(G\beta)$  in closed form for the Dr.GRPO coefficient, and a unique  $\lambda$  for the MaxRL coefficient. Consequently*

$$q^* = \mu|_{\mathcal{C}}, \quad \mathcal{H}(q^*) = \mathcal{H}(\mu|_{\mathcal{C}}), \quad \text{logit } P^* = \text{logit } \mu(\mathcal{C}) + \lambda.$$

*This stationary conditional does not depend on  $\beta$ : the coefficient sets the correctness level there and leaves the allocation among correct modes at the reference’s.*

*Proof.* Stationarity of (55) in a coordinate with  $p_b > 0$  reads  $c_G(P)(R_b - P) = \beta(\log(p_b/\mu_b) - D)$ , so  $\log(p_b/\mu_b) = D + (c_G(P)/\beta)(R_b - P)$  and the  $b$ -independent terms are absorbed by

normalization, giving the stated form with  $\lambda = c_G(P)/\beta$ . Conversely each  $\lambda \geq 0$  determines  $P(\lambda) = \mu(\mathcal{C})e^\lambda / (\mu(\mathcal{C})e^\lambda + 1 - \mu(\mathcal{C}))$ , strictly increasing from  $\mu(\mathcal{C})$  to 1. The consistency requirement  $\lambda = c_G(P(\lambda))/\beta$  has a solution by the intermediate value theorem, since the left side increases from 0 without bound while the right side is positive and bounded; it is unique whenever  $c_G$  is non-increasing, which covers the constant Dr.GRPO coefficient and  $c_G^{\text{MaxRL}}(P) = [1 - (1 - P)^{G-1}]/P$ , nonincreasing from  $G - 1$  to 1 and strictly decreasing for  $G > 2$ . Two correct modes give  $\log(p_c/\mu_c) = \log(p_d/\mu_d)$ , hence  $q^* \propto \mu$  on  $\mathcal{C}$ , which is  $\beta$ -free. The logit identity is  $P^*/(1 - P^*) = [\mu(\mathcal{C})/(1 - \mu(\mathcal{C}))]e^\lambda$ .  $\square$

Dr.GRPO with exact reference KL is an entropy-regularized softmax bandit, so Mei et al. (2020, Theorem 5 and Lemma 13) give convergence and a positive trajectory-wide probability floor. Its Gibbs target is the regularized maximizer of Geist et al. (2019, Proposition 1).

**Corollary P.34** (Exact Dr.GRPO with reference KL converges). *Fix one prompt, a finite category set  $\mathcal{A} = \mathcal{C} \dot{\cup} \mathcal{N}$  with both parts nonempty, and independently optimized logits  $p_n = \text{softmax}(z_n)$ . Let  $z_1$  be finite, let  $\mu$  be a reference distribution with  $\mu_a > 0$  for every category, and take  $\beta > 0$  and  $\alpha = (G - 1)/G$  for  $G \geq 2$ . Define*

$$\mathcal{J}_\beta(p) = \alpha P(p) - \beta \text{KL}(p \parallel \mu).$$

*For constant  $0 < \eta \leq 1/\beta$ , consider the exact gradient updates*

$$z_{n+1} = z_n + \eta \nabla_z \mathcal{J}_\beta(p_n). \quad (56)$$

*Then  $p_n \rightarrow p^*$ , where*

$$p_a^* = \frac{\mu_a \exp(\alpha R_a / \beta)}{\sum_b \mu_b \exp(\alpha R_b / \beta)}, \quad R_a = \mathbf{1}[a \in \mathcal{C}], \quad q^* = \mu|_{\mathcal{C}}.$$

*There are constants  $\underline{p} > 0$  and  $C < \infty$ , depending only on  $|\mathcal{A}|, z_1, \mu, \alpha, \beta$ , such that for every  $n \geq 1$ ,*

$$\min_a p_{n,a} \geq \underline{p}, \quad \mathcal{J}_\beta(p^*) - \mathcal{J}_\beta(p_n) = \beta \text{KL}(p_n \parallel p^*) \leq C e^{-2\beta \eta \underline{p}(n-1)}. \quad (57)$$

*Proof.* Set  $\tilde{r}_a = \alpha R_a + \beta \log \mu_a$ . Then  $\mathcal{J}_\beta(p) = \langle p, \tilde{r} \rangle + \beta H(p)$ . For  $B = \max\{1, \max_a \tilde{r}_a - \min_a \tilde{r}_a\}$ , define  $r'_a = (\tilde{r}_a - \min_b \tilde{r}_b)/B \in [0, 1]$ ,  $\tau = \beta/B$ , and  $\eta' = \eta B$ . Equation (56) is exactly the entropy-regularized bandit update of Mei et al. (2020) with reward  $r'$ , temperature  $\tau$ , and step  $\eta' \leq 1/\tau$ . Their Lemma 13 supplies  $\underline{p}$ , and their Theorem 5 supplies the displayed rate after multiplying the objective by  $B$ . The optimum softmax( $r'/\tau$ ) is  $p^*$ ; substitution gives the KL gap identity. The gap tends to zero, so Pinsker's inequality also gives convergence of the full categorical probability law and its conditional allocation among the correct categories.  $\square$

This positive trajectory floor depends on initialization and the reference; it need not provide useful finite-sample visibility. It is compatible with the vanishing coordinate force below and with increasingly slow recovery as the initial probability decreases. The result concerns exact discrete gradients with a fixed reference and the stated step size, not sampled KL gradients, PPO epochs, or AdamW. For centered MaxRL with  $G \geq 3$ ,  $c_G(P)$  is nonconstant, so its potential cannot be absorbed into the fixed reward vector used in this reduction; Proposition P.33 continues to specify its stationary target without asserting convergence.

The reference-conditional target differs from full-coverage uniform replay: Corollary P.24 gives  $q \rightarrow u$ , whereas Corollary P.34 gives  $q_n \rightarrow \mu|_{\mathcal{C}}$  for exact Dr.GRPO steps. A broader reference changes that allocation; replay constructs its target from verified discoveries. For MaxRL, Proposition P.33 identifies the stationary conditional without a convergence claim. Neither result bounds transient diversity by the reference's diversity or certifies a neural optimizer trajectory.

**Remark P.35.** The correctness prediction also depends on reference correctness  $s_\mu = \mu(\mathcal{C})$ :  $1 - P^* = (1 - s_\mu)/(1 - s_\mu + s_\mu e^\lambda)$ . For Dr.GRPO with  $G = 16$  and  $\beta \leq .1$ ,  $\lambda \geq 9.375$ ; this amplifies the reference correctness odds by at least  $e^{9.375}$  but does not give a reference-independent bound on the correctness deficit.

**Lemma P.36** (Reference KL has no probability-independent boundary force). *Under (A4') the logit velocity contributed by the reference-KL term is*

$$\dot{z}_b^{\text{KL}} = \beta p_b (\log(\mu_b/p_b) + D) \xrightarrow{p_b \rightarrow 0} 0, \quad |\dot{z}_b^{\text{KL}}| \leq \beta \left( \frac{\mu_b}{e} + \log \frac{1}{\min_a \mu_a} \right),$$

Table 60: **Replay recovery time grows approximately linearly with log-odds depletion.** Time to reach  $p_b = 0.05$  under the exact Dr.GRPO categorical mean flow, with  $m = 4$  correct and three incorrect categories and  $G = 16$ . Columns give initial odds  $d = p_b(0)/p_c(0)$ , with  $z_c = 0$ ,  $z_b = \log d$ , and all other logits  $-40$ . KL uses a uniform reference over all categories; replay uses a uniform target over correct categories. Each tenfold decrease in  $d$  multiplies KL recovery time by 7.3–8.2, while replay adds approximately  $\log(10)/\rho$ . Times are numerical integrations in the flow’s own units; the two-mode asymptotics are in Corollary P.37.

| Regularizer                    | $d = 10^{-3}$ | $10^{-4}$ | $10^{-5}$ | $10^{-6}$ |
|--------------------------------|---------------|-----------|-----------|-----------|
| Reference KL, $\beta = 0.04$   | 2.12e+03      | 1.55e+04  | 1.20e+05  | 9.83e+05  |
| Reference KL, $\beta = 0.1$    | 8.49e+02      | 6.19e+03  | 4.81e+04  | 3.93e+05  |
| Verified replay, $\rho = 0.01$ | 4.07e+02      | 6.37e+02  | 8.68e+02  | 1.10e+03  |
| Verified replay, $\rho = 0.1$  | 4.07e+01      | 6.37e+01  | 8.68e+01  | 1.10e+02  |

whereas the verified-replay term contributes  $\rho(w_b - p_b) \rightarrow \rho w_b > 0$  at the same limit. This is a property of the exactly evaluated penalty. The corresponding score multiplier contains an unbounded log ratio, so Lemma P.39’s uniformly bounded-multiplier premise does not apply. The exact KL force nevertheless vanishes at this boundary.

*Proof.* The displayed velocity is the softmax score identity used above. For the bound,  $\sup_{0 < s \leq 1} s \log(\mu_b/s) = \mu_b/e$  at  $s = \mu_b/e$ , the same expression is bounded below by  $\log \mu_b$  at  $s = 1$ , and  $0 \leq D \leq \log(1/\min_a \mu_a)$  by Lemma P.32. The replay expression is Lemma P.20.  $\square$

**Corollary P.37** (Recovery on a two-correct-mode face). *Consider the continuous extension of the categorical probability flow to the face supported on two correct modes  $b, c$ , with  $p_b = s$ ,  $p_c = 1 - s$ . This boundary reduction replaces the full-support start in (A5); here  $P = 1$  and the fresh correctness gradient is zero. For reference KL take fixed  $\beta > 0$  and  $\mu_b, \mu_c > 0$ . For replay take the uniform buffer  $\{b, c\}$  and fixed  $\rho > 0$ . Fix  $0 < \delta < \min\{1/2, \mu_b/(\mu_b + \mu_c)\}$ . The times to reach  $p_b = \delta$  from  $s \downarrow 0$  satisfy*

$$T_{\text{KL}}(s) = \frac{1 + o(1)}{2\beta s \log(1/s)}, \quad T_{\text{replay}}(s) = \frac{\log(\delta/s)}{\rho} + O(1).$$

*In particular their ratio diverges. The KL order is  $\Theta(1/[s \log(1/s)])$ , and the replay order is  $\Theta(\log(1/s))$ .*

*Proof.* Write  $u = p_b$  along this face. The softmax identity  $\dot{u} = u(1 - u)(\dot{z}_b - \dot{z}_c)$  gives exactly

$$\dot{u}_{\text{KL}} = 2\beta u^2(1 - u)^2 \log \frac{(1 - u)\mu_b}{u\mu_c}, \quad \dot{u}_{\text{replay}} = \rho u(1 - u)(1 - 2u).$$

Both velocities are positive up to  $\delta$  by its stated restriction. Separating variables expresses each hitting time as the integral from  $s$  to  $\delta$  of the reciprocal velocity. For KL the integrand is  $(1 + o(1))/(2\beta u^2 \log(1/u))$  as  $u \downarrow 0$ , and its integral is asymptotic to  $1/(2\beta s \log(1/s))$  (equivalently, apply l’Hopital’s rule to the two divergent functions). For replay the integrand is  $1/(\rho u) + O(1)$ , giving the second expression for any fixed admissible target  $\delta$ .  $\square$

Table 60 integrates the exact Dr.GRPO mean field in a system with 4 correct and three incorrect categories, with  $G = 16$ . The initial logits are  $z_c = 0$ ,  $z_b = \log d$ , and  $-40$  for every other category; thus  $d = p_b(0)/p_c(0)$  is the initial odds ratio, rather than  $p_b(0)$  itself. The KL reference is uniform over all seven categories, and the replay target is uniform over the four correct categories. Recovery ends at  $p_b = 0.05$ . Each tenfold decrease in  $d$  multiplies the KL time by 7.3–8.2 and adds approximately  $\log(10)/\rho$  to the replay time. These finite-depth integrations illustrate the recovery contrast; the asymptotic constants in Corollary P.37 apply to its specified two-mode face.

**Forward KL and cross-entropy barriers.** Forward KL to the same reference satisfies  $\text{KL}(\mu \| p) = -\sum_a \mu_a \log p_a - \mathcal{H}(\mu)$ . It differs from the weighted cross-entropy  $R_\mu$  in App. P.7 by the constant  $-\mathcal{H}(\mu)$  and diverges when any coordinate vanishes. For coefficient  $\beta > 0$ , descent of  $-\Psi(P) + \beta R_\mu$  therefore gives  $p_a(t) \geq \exp(-C_{T,\mu}/\mu_a)$ , where  $C_{T,\mu} = R_\mu(z(T)) + [\Psi_{\max} - \Psi(P(T))]/\beta$ . This is the weighted replay bound with target  $w = \mu$ .

The target distinguishes forward reference KL from verified replay. A small reference mass  $\mu_a$  weakens the coordinate floor through its exponent  $1/\mu_a$ , while a uniform buffer gives every stored solution mode equal weight. A full-support reference protects incorrect categories as well as correct ones; a verified buffer assigns weight only to correct categories. Estimating the forward expectation with draws from  $p$  uses importance weights  $\mu_a/p_a$ , while sampling from  $\mu$  or a stored target gives the corresponding cross-entropy expectation directly. Verified replay uses the latter construction with a target supported on discovered verified modes.

**Correctness at the analyzed targets.** For Dr.GRPO and MaxRL with a full-support reference, the interior KL stationary point has  $P^* < 1$ , because the reference assigns positive mass to incorrect categories. Cross-entropy over a fixed nonempty verified buffer has the limit  $P \rightarrow 1$  under the replay convergence assumptions with  $c_G \geq 0$ . These statements concern the identified stationary point and convergence theorem, respectively. They do not exclude boundary stationary points of the continuously extended reverse-KL probability flow or establish a universal correctness cost for neural KL training.

**Stationary targets and finite training budgets.** Full-coverage uniform replay has the same categorical limit for every  $\rho > 0$ , but this does not make dose selection irrelevant. The finite-time analysis above compares replay with the instantaneous fresh concentration pressure  $c_G(P)(1 - P)$ . Small doses can allow diversity to decline before the asymptotic target is approached.

For reference KL, increasing  $\beta$  changes stationary correctness through  $\text{logit } P^* = \text{logit } \mu(\mathcal{C}) + c_G(P^*)/\beta$ . Its boundary recovery time also depends on  $\beta$ ; on the two-mode face the leading scale is  $1/[2\beta s \log(1/s)]$ . Both coefficients therefore matter at a finite horizon. The replay-dose dependence here is a prediction of the specified categorical dynamics; the reported replay experiments do not estimate a dose-response curve.

**Scope of the comparison.** The stationary and rate statements concern independently optimized categorical logits. Their assumptions exclude schedules, changing references, momentum, and shared neural parameters. The implemented sampled KL term also differs from an exactly differentiated distributional KL: an absent response supplies no direct sampled score, although normalization and shared parameters can still change its probability. Reference KL has guarantees for regularized control and proximity to a reference (Geist et al., 2019; Vieillard et al., 2020). The results here distinguish its stationary target and boundary force from verified replay; they do not exclude useful finite-time diversity preservation by KL.

**Corollary P.38** (Two limits and a KL stationary value in the diversity metric). *For the conditional solution mode distribution,  $PCMD = 1 - C(q) = 1 - \sum_{c \in \mathcal{C}} q_c^2$ . In the categorical mean flow, a bounded outcome-binary objective satisfying the positivity and unique-maximum hypotheses of Corollary P.7 sends  $PCMD \rightarrow 0$ , and adding full-coverage uniform verified replay sends  $PCMD \rightarrow 1 - 1/m$  for every dose  $\rho > 0$  (Corollary P.24). With the Dr.GRPO or centered MaxRL coefficient, adding a reference KL instead gives a single interior stationary point, at which*

$$PCMD = 1 - \sum_{c \in \mathcal{C}} \left( \frac{\mu_c}{\mu(\mathcal{C})} \right)^2 \quad \text{for every } \beta > 0,$$

*the fixed reference's success-conditional diversity (Proposition P.33).*

*Proof.* Substitute  $\mathbf{e}_{c^*}$ , the uniform  $u$  on  $\mathcal{C}$ , and  $\mu|_{\mathcal{C}}$  into  $1 - \sum_c q_c^2$ , using continuity of  $C$  on the simplex.  $\square$

The first two statements are convergence results under their stated hypotheses. The third also holds as a limit for the exact Dr.GRPO steps of Corollary P.34; for centered MaxRL with  $G \geq 3$ , it identifies the stationary value without a convergence claim. Lemma P.32 separately describes which KL sublevels exclude a boundary face. Neither the stationary value nor the two-mode recovery rate is a bound on transient neural diversity. These targets specify asymptotic or stationary allocations under the respective categorical assumptions.

**Scarce fresh successes and depleted discovered modes.** As  $P \rightarrow 0$ , fresh verified responses become rare. For the centered estimators in Lemma P.19, only mixed reward groups supply a fresh

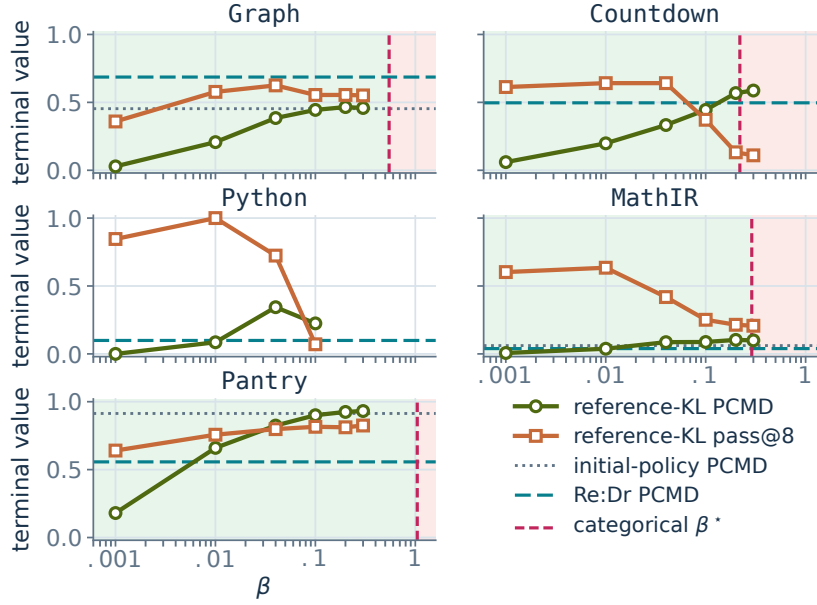


Figure 41: **KL coefficient effects vary by domain.** Circles show terminal PCMD and squares show pass@8 for Qwen2.5-0.5B at Level 1; lines connect seed means without uncertainty bands. Horizontal rules mark the initial policy’s reportable PCMD (dotted) and Re:Dr’s terminal PCMD (dashed). Vertical rules mark the categorical plug-in threshold  $\beta^*$ ; green and red indicate stationary single-draw correctness above and below one half under that calculation. This threshold does not predict the observed pass@8 knee.

task update, and their probability is at most  $GP$ . Corollary P.9 gives MaxRL a mean-gradient magnitude ratio approaching  $G$  relative to Dr.GRPO in this limit. Its potential is a harmonic mixture of pass@ $k$  for  $k = 1, \dots, G - 1$  (Equation (16)). This coefficient increase does not change the probability of sampling a verified response at a fixed policy. A previously populated buffer can still supply verified targets when current correctness is low.

At  $p_b \rightarrow 0$  for a discovered correct mode, the bounded sampled-coordinate force vanishes, while verified replay contributes  $\rho(w_b - p_b)$ . The reference-KL force also vanishes, at a different rate. These are coordinate-level comparisons in the stated geometry: they do not exclude retention through parameter sharing or other update rules.

The fresh and replay terms compose into  $F = \rho R_B - \Psi_{\text{MaxRL}}(P)$ . Its descent gives the fixed-buffer retention guarantee, and the finite-time analysis quantifies when replay overcomes concentration. Their gradients can oppose one another, so composition does not imply an absence of interference. Replay also requires stored targets and additional gradient computation, quantified in Equation (58) and App. Q.

#### P.12 MEASURED KL TRADEOFFS AND REPLAY OCCUPANCY

Table 61 compares terminal PCMD at Qwen2.5-0.5B, Level 1, after 3,072 updates. Dr.GRPO with reference KL uses 6 coefficients,  $\beta \in \{0.001, 0.01, 0.04, 0.1, 0.2, 0.3\}$ , across 5 domains. These arms use the zero-replay-derivative control configuration with a nonzero KL coefficient and the initial policy as the reference. The comparisons match optimizer updates and fresh-rollout budgets. They do not equalize total computation: reference-policy evaluation and replay processing incur different costs, and the control does not spend those costs on additional effective updates or fresh rollouts.

**Initial concentration and coefficient dependence.** Dr.GRPO without KL reaches PCMD at or below .004 in all five domains. MaxRL without replay reaches .148 in Graph and at or below .022 elsewhere. These low values are consistent with concentration, but the neural runs do not satisfy the independent-logit assumptions of Corollary P.7. KL produces larger conditional-diversity estimates at several coefficients in every domain. The response is not uniformly monotone: Graph changes from .465 to .458 and MathIR from .103 to .099 between  $\beta = .2$  and

Table 61: **KL and replay have different domain-level tradeoffs.** Terminal PCMD seed means for Qwen2.5-0.5B, Level 1. Superscripts give counts below five seeds; dashes indicate unavailable cells. The four zero-KL arms use 32 disjoint evaluation streams; KL arms use their own disjoint draws with the same estimator. Reference summaries are initial-policy PCMD and  $1 - 1/\bar{k}$  for each replay arm, where  $\bar{k}$  is the mean number of keys replayed per measured update. These do not bound neural diversity. The final column computes categorical  $\beta^*$  from mean initial single-draw correctness.

| Domain    | without replay |       | + reference KL, by $\beta$ |       |       |                    |       |       | + replay |        | reference summaries |       |        |           |
|-----------|----------------|-------|----------------------------|-------|-------|--------------------|-------|-------|----------|--------|---------------------|-------|--------|-----------|
|           | Dr.GRPO        | MaxRL | .001                       | .01   | .04   | .1                 | .2    | .3    | Re:Dr    | Re:Max | Initial             | Re:Dr | Re:Max | $\beta^*$ |
| Graph     | 0.003          | 0.148 | 0.029                      | 0.207 | 0.385 | 0.444              | 0.465 | 0.458 | 0.686    | 0.625  | 0.45                | 0.75  | 0.70   | 0.55      |
| Countdown | 0.004          | 0.022 | 0.060                      | 0.198 | 0.333 | 0.444              | 0.568 | 0.587 | 0.497    | 0.531  | —                   | 0.56  | 0.52   | 0.22      |
| Python    | 0.000          | 0.000 | 0.000                      | 0.086 | 0.344 | 0.225 <sup>3</sup> | —     | —     | 0.099    | 0.239  | —                   | 0.15  | 0.42   | —         |
| MathIR    | 0.003          | 0.008 | 0.007                      | 0.038 | 0.086 | 0.088              | 0.103 | 0.099 | 0.039    | 0.018  | 0.06                | 0.15  | 0.12   | 0.29      |
| Pantry    | 0.000          | 0.000 | 0.180                      | 0.660 | 0.824 | 0.901              | 0.925 | 0.931 | 0.557    | 0.556  | 0.91                | 0.85  | 0.85   | 1.05      |

.3, while Python changes from .344 to .225 between  $\beta = .04$  and .1. These are comparisons of point estimates, without a seed-level uncertainty analysis for their differences.

**Stationary predictions and measured checkpoints.** Proposition P.33 identifies the categorical interior stationary conditional as  $q^* = \mu|_C$ , independently of  $\beta$ . It does not bound transient diversity or establish that a finite neural checkpoint reaches this conditional. The measured PCMD varies with  $\beta$  and can exceed the initial estimate: at  $\beta = .2$ , Graph reaches .465 against .454 initially, MathIR .103 against .061, and Pantry .925 against .913. These estimates also condition on each policy’s own verified responses, rather than on a common set of solved prompts.

For a fixed prompt with  $0 < \mu(C) < 1/2$ , the Dr.GRPO formula gives

$$\beta^* = \frac{(G-1)/G}{-\text{logit } \mu(C)}, \quad P^*(\beta^*) = \frac{1}{2}.$$

The displayed values substitute each domain’s mean initial correctness and  $G = 16$ , giving 0.22–1.05. This aggregation is descriptive: substituting a mean into the nonlinear stationary formula does not give the mean of its promptwise predictions. Moreover,  $P^*$  is single-draw correctness, whereas the plotted pass@8 measures success within eight responses. Declines in that metric already occur below the displayed threshold: MathIR falls from .634 at  $\beta = .01$  to .418 at .04, well below its  $\beta^* \simeq .29$ . Countdown similarly falls from .642 at .04 to .372 at .1, below .22. The thresholds therefore do not locate the measured knees. Python has zero observed initial successes, so no finite plug-in threshold or reportable initial PCMD is shown; this does not establish zero population correctness.

Finite training, shared parameters, prompt sampling, and the adaptive optimizer can all separate the measured neural trajectory from the categorical stationary calculation. The comparison does not identify their individual effects. The boundary-force identities and the two-mode  $\Theta(1/[s \log(1/s)])$  versus  $\Theta(\log(1/s))$  recovery result characterize their stated categorical models, not measured neural recovery rates.

**What buffer occupancy measures.** For a nonempty fixed uniform categorical buffer with  $k$  keys, the limiting diversity is  $1 - 1/k$ . Across such buffers,  $\mathbb{E}[1 - 1/k] \leq 1 - 1/\mathbb{E}[k]$  by Jensen’s inequality. The table instead uses  $\bar{k}$ , the number of keys replayed per measured update, averaged within runs and then across seeds. Buffers change during training, and updates without replay can contribute zero keys. Thus  $1 - 1/\bar{k}$  is a descriptive transform of training occupancy, not an estimate of the nonempty fixed-buffer expectation in that inequality. Neither quantity bounds held-out neural PCMD.

Python illustrates a possible discovery difference. Re:Max averages 1.73 replayed keys per update against Re:Dr’s 1.18, while terminal PCMD is .239 versus .099. Corollary P.9 provides a categorical motivation for stronger fresh learning when correctness is scarce, but these measurements do not isolate discovery from prompt weighting, sampling, or optimization. Occupancy also does not order all outcomes: in Countdown, Re:Max has fewer replayed keys per update (2.10 versus 2.28) but higher PCMD (.531 versus .497). In Pantry both arms average about 6.7 keys and reach about .557 PCMD. The capacity of 16 keys would allow a uniform fixed-buffer value of  $1 - 1/16 = .9375$ ; the occupancy summary near .85 is not that capacity limit and does not prove why KL performs better there.

**Observed correctness–diversity comparisons.** In Python and Pantry, at least one measured KL coefficient exceeds both replay arms on PCMD and has no lower pass@8 point es-

timate. In Countdown and MathIR, coefficients that exceed both replay arms on PCMD have lower pass@8 than both. In Graph, all measured coefficients are below both replay arms on both metrics. These comparisons concern seed means; they establish neither statistical dominance nor a ranking at equal total compute.

**Relative and absolute readings of the Re:Max effect.** Table 61 carries both arms of the Re:Max versus Dr.GRPO comparison on one evaluation, so the effect can be read either way. Equally weighting the five domains, Dr.GRPO reaches pass@8 .405 and Re:Max .772, a relative increase of 90%; averaging the five per-domain relative increases instead gives 124%, because Python rises from a small base. A relative reading of PCMD is not available: terminal Dr.GRPO PCMD is .000–.004 across these domains and exactly .000 in Python and PantryPlan, so two of the five ratios are undefined and the rest are dominated by the denominator. The count-valued reading is well defined instead. Re:Max reaches  $N_{\text{eff}}^{(2)}$  of 2.66, 2.13, 1.32, 1.02 and 2.25 on Graph, Countdown, Python, MathIR and PantryPlan, against 1.00–1.004 for Dr.GRPO: a mean ratio of 1.87 effective verified modes for each one the control retains. Paired per-seed PCMD differences on a common eligible population, which weight prompts and seeds differently, are reported in Table 1.

Figure 11 averages Graph, MathIR, and Pantry, the three domains with reportable initial PCMD, across all six coefficients. Between  $\beta = .2$  and  $.3$ , mean PCMD changes from .498 to .496, while pass@8 changes from .527 to .528.

### P.13 REFERENCE QUALITY, DISCOVERY, AND THE SCOPE OF THE COMPARISON

**Reference quality can favor KL.** A full-support fixed reference yields the full-support categorical stationary policy in Proposition P.33. Reference KL uses that policy rather than a replay buffer; it requires reference-policy evaluation but no rule for admitting exemplars. In Pantry at  $\beta = 0.3$ , the 5-seed mean reaches PCMD = 0.931 and pass@8 0.825. The replay arm with larger PCMD reaches 0.557 and 0.797, respectively. Pantry’s initial PCMD is 0.91, consistent with a useful reference conditional. The measurements support reference quality as one possible explanation; they do not show that replay’s capacity or discovery alone causes the difference.

**Reference diversity varies across evaluation populations.** App. N.1 reports 7 hosted deployments with macro PCMD spanning 0.14–0.31; Kimi K3 is highest at 0.31. Of 101 reportable model–domain–level cells, 7 reach .50 and the maximum is 0.74. These estimates are lower than Pantry’s initial 0.91 in the local comparison, but the models, evaluation populations, and reportability sets differ. The hosted cohort therefore describes additional candidate reference distributions; it does not establish how training with those references would perform. Under the categorical proposition, each reference determines a stationary conditional, not a ceiling on a retrained neural policy. Replay instead targets the solution modes represented by its discovered buffer.

**Correctness costs depend on the domain and coefficient.** The largest observed pass@8 reduction among KL cells that exceed both replay arms on PCMD occurs in MathIR at  $\beta = 0.3$ . Its 5-seed mean is 0.099 PCMD and 0.207 pass@8, compared with 0.039 and 0.816 for the replay arm with larger PCMD. This finite-checkpoint comparison supports a domain-dependent correctness–diversity tradeoff, without identifying the categorical threshold as its cause. Replay’s positive dose does not change its full-coverage categorical limit under Corollary P.24, but it changes restoration time in Theorem P.27. A measured replay-dose sweep would be needed to compare coefficient sensitivity.

**Limits of the evidence.** Reference PCMD requires at least 30 defined prompts in an initial evaluation. This eligibility condition affects which policy conditionals can be compared. Python’s zero observed initial correctness cannot support an inheritance explanation for its measured KL gains. MathIR’s support certificate counts derivations and does not imply that a policy will produce every certified solution mode.

The categorical results separate a fixed-reference target from a discovered-buffer target and characterize restoration under their stated assumptions. The neural measurements compare finite checkpoints. They do not establish a universal preference between KL and replay, a discovery cause for every replay shortfall, or an advantage under equal total compute.

## P.14 SCOPE RELATIVE TO ENTROPY, KL, ADMISSION, AND NEURAL OPTIMIZATION

**Dependence on optimization geometry.** For a smooth parameterization  $z = z(\theta)$  with Jacobian  $J_\theta = \partial z / \partial \theta$ , Euclidean parameter flow instead induces  $\dot{z} = c_G(P) J_\theta J_\theta^\top \nabla_z P$ . The independently trained logit theorem uses  $J_\theta J_\theta^\top = I$  and does not automatically extend to arbitrary shared parameters or another optimization metric. Equal binary reward alone does not force this winner selection. The natural policy gradient update of Agarwal et al. (2021, Lemma 15), specialized to this bandit, is  $p_a^+ \propto p_a \exp(\eta c_G(P) R_a)$  and preserves every ratio of correct probabilities at each finite step. Its continuous version has  $\dot{z}_a = c_G(P)(R_a - P)$ , with  $R_a = \mathbf{1}[a \in \mathcal{C}]$ . The distinction between Euclidean and natural updates also underlies the tabular entropy identities of Cui et al. (2025, Theorems 1–2); entropy and verified-mode diversity remain different observables.

The mean-gradient identity has broader scope than the collapse conclusion. For one prompt and a common autonomous preconditioner  $M(\theta)$ , positive coefficients give flows  $c_a(P)M(\theta)\nabla_\theta P$  that follow the same orbit after a change of time. Outcome-dependent covariance, response-length weights, shared prompts, and optimizer memory can break the corresponding training-trajectory equivalence, as the preceding neural analysis shows. Parameter count alone does not identify this geometry. The empirical scale comparisons change pretrained models and schedules; they do not establish the cause of cross-scale differences or the training history of a hosted model.

With uniformly bounded detached advantages, the expected sampled-logit update vanishes as the category probability tends to zero, even when advantages depend on the group. Exact entropy gradients have unbounded surprisal weights and therefore require a separate boundary analysis.

**Lemma P.39** (Bounded sampled scores have no probability-independent boundary force). *Under Assumption P.1, condition on the training history, draw  $Z_1, \dots, Z_G$  independently from  $p$ , and let  $a_i$  be any detached, possibly group-dependent advantages with  $|a_i| \leq B$ . To interpret the bound uniformly as  $p_b \rightarrow 0$ , assume a finite  $B$  independent of that limit. At the on-policy point define*

$$\hat{g} = \frac{1}{G} \sum_{i=1}^G a_i (e_{Z_i} - p).$$

*For every category  $b$ ,*

$$|\mathbb{E}[\hat{g}_b]| \leq \mathbb{E}[\hat{g}_b] \leq 2Bp_b(1 - p_b).$$

*On a group with no draw of  $b$ , exactly  $\hat{g}_b = -p_b G^{-1} \sum_i a_i$ ; it is zero if the applied advantages sum to zero. By comparison, the verified-replay component for a fixed target  $w$  has logit velocity  $\rho(w_b - p_b)$ , which approaches  $\rho w_b > 0$  for a buffered category as  $p_b \rightarrow 0$ .*

*Proof.* Use the triangle inequality and  $|a_i| \leq B$ . Each marginal draw satisfies  $\mathbb{E}[\mathbf{1}\{Z_i = b\} - p_b] = 2p_b(1 - p_b)$ , irrespective of the dependence of  $a_i$  on the group. On the absence event all indicator terms are zero. The replay formula follows by differentiating  $-\sum_b w_b \log p_b$ .  $\square$

The bound compares the isolated logit-gradient components. It does not prove inevitable collapse, rule out retention by a bounded estimator, or guarantee that replay dominates the full update. An unsampled mode can still change through softmax normalization or shared parameters. Exact entropy is outside the uniformly bounded-score premise because its surprisal is unbounded near zero; the lemma is not an impossibility result for entropy regularization. Nor does the absence of a mode from a finite sample establish zero probability.

**Exact entropy is a different comparator.** For a known full correct support and uniform  $u$ , the elementary identities

$$H(q) = \log m - \text{KL}(q||u), \quad R_u(p) = \log m - \log P + \text{KL}(u||q)$$

show that exact conditional entropy and categorical replay both favor a uniform conditional target. With strictly increasing correctness potential  $\Psi$ , both  $\Psi(P) + \beta H(q)$  and  $\Psi(P) - \rho R_u(p)$  have the unique global optimum  $P = 1, q = u$  in probability space on the simplex with  $P > 0$ , for positive coefficients. This is a boundary limit for finite-logit softmax policies. The optimum follows from nonnegativity of KL and increasing  $\Psi$ ; it does not assert convergence of either neural update. The KL direction distinguishes their loss level sets (Pereyra et al., 2017;

Agarwal et al., 2021): replay loss diverges when a protected coordinate vanishes, whereas conditional entropy can remain  $\log(m - 1)$  with one missing mode.

Exact full-output entropy is a third objective and also regularizes incorrect mass. Exact tabular entropy-regularized policy gradient has its own positive probability and convergence guarantees under specified assumptions (Mei et al., 2020); our collapse theorem contains no entropy term. The sampled semantic scores in the experiments are not exact gradients of full-output or conditional verified-key entropy. Neither their empirical comparison nor the bounded-score lemma is an impossibility result for entropy.

**Discovery and optimization remain separate requirements.** Retention starts after a verified key enters active memory. Actual insertion depends on generation, validation, capacity, and eligibility; a positive sampling probability alone does not establish eventual admission. Once a non-evicting capacity-16 buffer fills, an unbuffered key can remain ineligible. Changing membership or exemplar weights changes the loss being bounded, so fixed-buffer descent cannot simply be carried across those changes. In particular, the fixed-weight argument does not cover a target whose weights continue to change with fresh observation counts.

For a fixed joint potential, a discrete bound  $F(\theta_{n+1}) \leq F(\theta_n) + \epsilon_n$  with  $\epsilon_n \geq 0$  and  $\sum_{n \geq n_0} \epsilon_n = E < \infty$  would extend Lemma P.29, using  $D = F(\theta_{n_0}) + J_{\max} + E$  for steps  $n \geq n_0$ . This follows by telescoping and bounds retention, not convergence. Corollary P.30 instead permits conditional expected drift with a controlled accumulated error budget, and yields a simultaneous probability statement over iterations and protected exemplars. An optimizer-specific argument is still needed to establish either form of energy control. A positive replay coefficient or small nominal learning rate does not establish it. Adam moments, clipping, momentum, prompt scheduling, and buffer changes are not covered by the exact-flow proofs. The experiments do not establish the required joint-energy or update-error assumptions. The categorical and conditional-exemplar guarantees therefore remain distinct from the observed neural training behavior and measured exemplar likelihoods.

#### P.15 ASSUMPTIONS AND THE EVIDENCE THAT TESTS THEM

The results in this part are exact for the flows they assume, and the preceding subsection states where those assumptions stop. Each result also implies a sign, an ordering, or a directly countable quantity that does not depend on the exact-flow assumptions. Those implications are measured elsewhere in this supplement. The table collects them, together with the case in which the idealized statement does not carry over to neural training.

| Idealized claim                                                                                  | Implication for neural training                                                                                                                           | Measured in      | What the measurement shows                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lemma P.39: a group whose rewards agree contributes no fresh task gradient                       | Under sparse binary verification many updates carry no fresh task gradient, and replay raises the fraction of mixed groups                                | App. H.4         | Mixed-group fractions span .015–.364 under Dr.GRPO and .185–.954 under Re:Dr; 98.5% of Python control updates carry zero fresh task gradient                                                                                                     |
| Theorem P.4: winner-take-all among correct modes                                                 | Training under binary verification retains fewer correct modes than it began with                                                                         | Fig. 8, App. H.3 | Fewer solution modes after training in 14 of 15 comparisons; every Python control seed reaches byte-identical completions on all 128 prompts by steps 288–480                                                                                    |
| Uniform rehearsal severs rehearsal frequency from discovery frequency (Eq. 3)                    | Rehearsing in proportion to discovery restores the self-reinforcement that uniform weighting removes, so uniform weighting retains at least as many modes | App. F.7         | + .318 extra modes per prompt ([+.272, +.358]), with positive within-seed domain averages in all five seeds, driven by Graph, Countdown and PantryPlan; the paired $P_8$ effect +.010 ([−.119, +.139]) identifies neither a gain nor equivalence |
| Theorem P.22 and Corollary P.24: a fixed buffer holds every retained mode above a positive floor | With shared parameters the floor is not exact for an individual exemplar; typical exemplars rise while a minority falls                                   | App. H.6         | Median change +.038 nat/token ([+.025, +.048]), tenth percentile −.256, and 2.61% of exemplars lose more than .5 nat/token ([1.54, 3.77]%)                                                                                                       |

The first three rows measure a quantity the idealized statement names, and each carries the predicted sign. The fourth row is where the categorical guarantee does not carry over: a per-exemplar floor is exact in the categorical flow, while under shared parameters a measurable minority of protected exemplars falls. Proposition P.31 is the statement that applies in that setting, and it is conditional on a local response term rather than categorical. The  $P_8$  interval in the third row likewise leaves correctness unidentified, so the ordering it supports concerns retained modes rather than accuracy.

#### P.16 MEASURED SIGNATURES OF THE CATEGORICAL PREDICTIONS

The categorical results predict an ordering among coexisting correct modes and a direction for which mode is lost. Both are signs rather than trajectories, so both survive the move to AdamW, clipping and shared parameters, and both are measurable in the runs already reported. This section measures them on the five-domain Qwen2.5-0.5B verified-replay pairs and their matched controls (five seeds per domain, thirty-three evaluated checkpoints per run, thirty-two sampled draws per prompt). Canonical keys are recorded for incorrect draws as well, so only draws carrying reward one contribute.

Two estimator choices matter. Ranking a pair of modes and measuring their starting log-odds from the same draws biases the ordering statistic below one half, because a mode over-sampled at a checkpoint is both more likely to be ranked ahead and more likely to fall back at the next one; pooled over  $1.1 \times 10^5$  pairs that artifact alone reads .48 under both objectives. The draws at each checkpoint are therefore split by index parity: one half fixes which mode leads, the disjoint half fixes the starting log-odds, and the next checkpoint supplies the outcome. Counts are Laplace-smoothed so that a mode reaching zero is a defined move rather than a dropped observation; discarding those pairs removes precisely the outcome the prediction concerns.

| Domain     | Arm     | Ordering |                   | Loss              |  | Coexisting               |
|------------|---------|----------|-------------------|-------------------|--|--------------------------|
|            |         | pairs    | Pr[leader gains]  | Pr[rarer lost]    |  | first $\rightarrow$ last |
| Graph      | control | 947      | .693 [.663, .721] | .911 [.881, .934] |  | 21.0 $\rightarrow$ 0.2   |
|            | replay  | 36,488   | .551 [.546, .557] | .720 [.711, .728] |  | 21.0 $\rightarrow$ 108.2 |
| Countdown  | control | 1,014    | .546 [.516, .577] | .759 [.720, .794] |  | 1.4 $\rightarrow$ 1.0    |
|            | replay  | 16,613   | .461 [.454, .469] | .836 [.821, .851] |  | 1.4 $\rightarrow$ 81.2   |
| PantryPlan | control | 6,081    | .460 [.448, .473] | .719 [.704, .734] |  | 98.2 $\rightarrow$ 0.0   |
|            | replay  | 29,802   | .522 [.516, .528] | .824 [.816, .831] |  | 98.2 $\rightarrow$ 65.0  |
| MathIR     | control | 103      | .447 [.354, .543] | .667 [.516, .790] |  | 1.8 $\rightarrow$ 0.2    |
|            | replay  | 292      | .555 [.497, .611] | .734 [.664, .794] |  | 1.8 $\rightarrow$ 5.2    |
| Python     | replay  | 718      | .655 [.619, .688] | .938 [.852, .976] |  | 0.0 $\rightarrow$ 5.8    |
| All five   | control | 8,157    | .499 [.488, .509] | .744 [.731, .756] |  | 24.5 $\rightarrow$ 0.3   |
|            | replay  | 83,913   | .524 [.521, .527] | .779 [.773, .784] |  | 24.5 $\rightarrow$ 53.1  |

Intervals are Wilson intervals at the nominal 95% level over pairs, which are not independent across checkpoints or prompts; they describe precision, not a seed-level test. `Python` control contributes twelve pairs in total and is omitted, since control coexistence there is zero throughout.

Read together, the two statistics identify which route concentration takes. The categorical flow concentrates by reweighting: a leading correct mode draws probability from its rivals while both remain present. That is not what the runs show in general. Among modes that both persist across a checkpoint, the leader gains in .499 of control transitions, indistinguishable from chance. Among pairs in which exactly one mode is absent at the next checkpoint, the missing one is the rarer member in .744 of control transitions and .779 of replay transitions, and that direction holds in every domain, reaching .911 in `Graph` control. Concentration under binary verification therefore proceeds mainly by truncation of the rare tail rather than by drift among the modes that survive, and it reaches the conclusion of Theorem P.4 by a route the categorical flow does not describe. The distinction matters for the mechanism: replay acts on exactly the event that carries the effect, since rehearsing a stored exemplar prevents the disappearance that truncation depends on, and it does so without needing to win a reweighting contest.

Drift is present where modes are few and well separated, and absent where they are many and close. `Graph` control sits at .693 and `Countdown` control at .546, both above one half, while `PantryPlan` control runs below it at .460; `MathIR` is uninformative at this sample size. Pooling conceals this entirely, because `PantryPlan` supplies 6,081 of the 8,157 control pairs and drags the aggregate to chance. The categorical ordering claim thus describes `Graph` and `Countdown`, and does not describe `PantryPlan`, whose correct modes are numerous and close in probability. In `Graph` both routes operate at once, which is also the domain where replay retains the most modes.

Coexistence is what both statistics condition on, and it moves in opposite directions under the two objectives: an average of 24.5 prompts per run carry at least two correct modes at the first checkpoint, falling to 0.3 under the control and rising to 53.1 under replay, with `Graph` rising from 21.0 to 108.2. Because the control retains almost no such prompts late in training, its ordering and loss statistics are computed on a shrinking and atypical set of prompts; the within-domain statements above are the ones the design supports, and the cross-arm difference in those two statistics is not like-for-like. Finally, verified diversity dips below both of its endpoints before recovering in 11 of 25 replay runs against 3 of 25 controls, and in all five seeds of both `MathIR` and `PantryPlan`, the transient decline permitted by Theorem P.26.

## Q WHAT VERIFIED REPLAY COSTS

### Q.1 WHAT THE CONTROL MATCHES, AND WHAT IT DOES NOT

The core factorial matches initialization, data, optimizer settings, 3,072 optimizer updates, and  $G = 16$  fresh rollouts per visited prompt. Both arms maintain buffers and use the same configured replay traversal; the control replaces the replay score derivative with exact zeros. Replay gradients join the fresh-task gradients before the existing optimizer step. Activating replay therefore does not add an optimizer step or a fresh rollout, and the control already incurs replay computation.

This design isolates the effect of applying the replay learning signal under the common schedule. It does not match total training compute: policies can produce different rollout lengths and populate different buffers, changing whether replay executes and how many tokens it processes. Equal code paths and update counts do not imply equal FLOPs or wall-clock time. More fundamentally, the control’s replay work contributes no learning signal and is not reassigned to useful baseline updates. We do not report a control that spends an equivalent budget on extra task-gradient steps, extra fresh rollouts, or a longer training horizon. The results therefore do not establish that replay is the best use of an equal total-compute budget.

The uniform-versus-frequency weighting ablation addresses a different question: how to allocate replay weight among stored modes. It retains the same buffer construction, scheduling, and update rule, but its policies can also develop different workloads. It supports a contribution from balancing modes; it is not a comparison of performance at equal total compute.

## Q.2 TOKEN-BASED COST ACCOUNTING AND ITS LIMITS

Cost estimates use diagnostics from 75 replay runs: five seeds in each of five domains at Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B. Table 62 compares replay storage and a likelihood-pass subtotal with the corresponding components of a hypothetical reference-KL implementation. Those are component estimates; the runtime difference between replay and its control is measured separately below.

**Resident state.** Under full fine-tuning, a separate fixed reference copy requires  $\Theta(|\theta|)$  parameter bytes. A replay buffer stores prompt and exemplar token identifiers, whose cost depends on the number of prompts, retained keys, and token lengths rather than directly on parameter count. The estimated mean buffer footprint is at most 444 kB across these families, while the reference copy grows from 0.99 to 6.17 GB. The corresponding reference-to-buffer ratios are 2,554–15,697. These are component storage estimates, not peak accelerator-memory measurements: buffer bytes assume four bytes per token and exclude bookkeeping, optimizer state, and activations. A reference recovered by disabling an adapter need not incur a separate resident parameter copy.

**A likelihood-pass subtotal, not total training FLOPs.** Let  $T_{\text{replay}}$  include the prompt and response tokens processed by a selected buffer, and  $T_{\text{fresh}}$  the corresponding fresh-group token estimate. Approximating a forward pass by  $2|\theta|T$  and its backward by  $4|\theta|T$ , the table prices one differentiable replay forward/backward at  $6|\theta|T_{\text{replay}}$  and one reference forward at  $2|\theta|T_{\text{fresh}}$ . Thus its reported ratio is

$$\frac{\text{replay likelihood-pass FLOPs}}{\text{reference-forward FLOPs}} \simeq \frac{3T_{\text{replay}}}{T_{\text{fresh}}} \approx \frac{3\mathbb{E}|\mathcal{B}|}{G}, \quad (58)$$

where the final approximation additionally assumes comparable fresh and stored sequence lengths. Stored exemplars and current rollouts need not have the same lengths, so occupancy alone does not determine the ratio. The token-based cell means range from 0.19 to 1.64; the subtotal is below one in 12 of 15 cells. The three exceptions are `PantryPlan`. This denominator is a reference forward, not the baseline’s complete training cost.

The training implementation also makes a detached scoring forward before the differentiable replay pass. Under the same approximation, these passes together would cost about  $8|\theta|T_{\text{replay}}$ , before activation recomputation and other system costs. The table’s  $6|\theta|T$  subtotal omits that scoring pass and cannot estimate total training FLOPs. Both the active and exact-zero control paths retain scoring and backward computation; this subtotal must not be interpreted as their measured cost difference.

**Measurement scope.** Workload estimates are averaged within runs and then equally across seeds in each model–domain cell. Fresh-group token counts use response lengths and prompt-length estimates. These averages describe the selected likelihood passes, not cumulative computation through the matched training horizon. They exclude rollout generation, padding, attention-specific costs, activation recomputation, verification, buffer maintenance, optimizer work, communication, and evaluation. They therefore do not establish an end-to-end speedup or an advantage at equal total compute.

**Measured runtime against the matched control.** The estimates above price components. What an equal-compute argument needs is the whole difference, and four cohorts on this panel supply it directly: the control, Re:Dr, capacity-one replay and the twelve-pass control all ran on the same

Table 62: **Replay buffers use less estimated storage than separate reference models.** The first row per family gives mean selected-buffer occupancy, reference parameter bytes, estimated buffer token bytes, and their storage ratio. The second row is the likelihood-pass/reference-forward FLOP ratio of Equation (58); it excludes the detached scoring pass and is not a replay/control overhead ratio. Diagnostics are averaged within each run, then across five seeds per cell (75 runs overall).

| Family       | Graph | Countd. | Python | MathIR | Pantry | Ref (GB) | Buffer (kB) | Ratio  |
|--------------|-------|---------|--------|--------|--------|----------|-------------|--------|
| Qwen2.5-0.5B | 3.98  | 2.28    | 1.18   | 1.17   | 6.74   | 0.99     | 387         | 2,554  |
| ratio        | 0.75  | 0.43    | 0.22   | 0.22   | 1.26   |          |             |        |
| Falcon3-1B   | 3.83  | 1.88    | 1.24   | 1.20   | 8.76   | 3.34     | 444         | 7,520  |
| ratio        | 0.72  | 0.35    | 0.23   | 0.23   | 1.64   |          |             |        |
| Qwen2.5-3B   | 3.69  | 1.50    | 1.27   | 1.03   | 6.46   | 6.17     | 393         | 15,697 |
| ratio        | 0.69  | 0.28    | 0.24   | 0.19   | 1.21   |          |             |        |

two `cs` A5000 nodes, on the same prompts and seeds, under the same runtime. Pairing by domain and seed over 25 cells gives Re:Dr’s overhead as +2.4% of total learner compute and +1.1% of wall clock (Tab. 63). The two measures fail differently — the first is a median per-update learner time times the update count, reproducible from disk and insensitive to a noisy neighbour; the second is whole-job elapsed, which also carries evaluation, checkpointing and I/O but is sensitive to node sharing — and they agree to about a point, which is the reason to trust either. Capacity-one replay is cheaper than the control on both, as a one-slot buffer should be.

This is what makes the next subsection quantitative rather than asserted. The twelve-pass control received +44.9% more total learner compute and +44.7% more wall clock, so it was handed roughly 19 times the marginal compute replay actually consumes on the conservative measure, and 39 times on the other. These are runtime measurements on fixed hardware, not FLOP counts, and they inherit whatever the scheduler and the colocated vLLM instance contribute; the margin is wide enough that the conclusion does not turn on the difference.

Table 63: **Replay’s runtime overhead, and what the equal-compute control was given.** Paired by domain and seed over 25 cells per cohort, Qwen2.5-0.5B at Level 1, all on `cs` A5000 node203/node204. Learner hours are a median per-update time times the update count; wall-clock hours are whole-job elapsed. Percentages are paired ratios against the control in the first row.

| Arm                               | learner |            | wall clock |            |
|-----------------------------------|---------|------------|------------|------------|
|                                   | hours   | vs control | hours      | vs control |
| control, eight passes             | 2.80    | —          | 4.90       | —          |
| Re:Dr replay, eight passes        | 2.85    | +2.4%      | 4.95       | +1.1%      |
| capacity-one replay, eight passes | 2.68    | -4.5%      | 4.76       | -2.8%      |
| control, twelve passes            | 4.05    | +44.9%     | 7.09       | +44.7%     |

### Q.3 THE CONTROL AT A LONGER HORIZON

An equal-compute comparison requires a baseline that reallocates replay’s work to learning. Additional task-gradient passes over existing fresh groups, larger fresh-rollout groups, and more fresh-sampling updates beyond the eight-pass horizon are distinct alternatives. None is represented by a zero replay derivative. A comparison of their correctness and PCMD would require common cumulative budgets, initialization, task splits, seeds, evaluation requests, and hardware.

One of those alternatives is measured. The extended-horizon cohort runs the control objective with only the horizon moved, 12 prompt passes in place of 8: 4,608 optimizer updates against 3,072, with 50% more fresh rollouts, over 25 cells on the same prompts, seeds, placement, runtime and evaluator. The horizon is deliberately generous rather than matched, and by a measured margin: Re:Dr’s own overhead over this control is +2.4% of total learner compute and +1.1% of wall clock (App. Q.2), while this arm was given +44.9% and +44.7% respectively — roughly 19 times the marginal compute replay consumes. The resulting comparison is one-sided in the control’s favour, and is read as one-sided.

Spending that budget on more learning does not move the endpoints toward replay. Averaging the five domains, terminal `pass@8` goes from .444 at 8 passes to .434 at 12, a change of -.010, against .748 for Re:Dr at 8 passes. The longer control is below Re:Dr in 5 of five domains, and the gap it leaves is .314. Diversity behaves likewise: mean PCMD is .013 at the longer horizon and .004 at the shorter

one, against .376 for Re:Dr, which in effective verified modes is 1.013 and 1.004 against 1.601. Both horizons sit within a hundredth of a mode of answering every held-out prompt the same way.

Python is the sharpest case, and it moves the wrong way. It is the domain where 98.5% of control updates carry no fresh task gradient (App. H.4), and the additional 1,536 updates take terminal `pass@8` from .338 to .182, a shift of -.155. A task gradient that is identically zero is not made nonzero by granting more of them. Its PCMD column carries no estimate at this horizon: 0 of five cells reach the thirty-prompt threshold, at a mean of 23.4 defined prompts.

Two qualifications belong with the result. The registered prediction for diversity was that the longer control would concentrate further or hold; the measured direction is instead a small rise, higher in 3 of five domains, at a magnitude that leaves both horizons indistinguishable from a single verified mode. And cumulative FLOPs or timing on controlled hardware would have to include rollout generation and every learner pass, with consistent treatment of evaluation and of any budget overshoot from the final update; those quantities are outside this measurement. What the cohort establishes is narrower and one-directional: handed strictly more additional computation than replay consumes, the control closes neither the correctness gap nor the diversity gap.

**Table 64: A longer control horizon closes neither gap.** Terminal endpoints for Qwen2.5-0.5B at Level 1, five seeds per cell. The control is shown at its 8-pass horizon and at 12 passes with only that horizon moved; Re:Dr is at 8 passes. Means weight the five domains equally. Python PCMD at 12 passes rests on fewer than thirty defining prompts per cell and is reported for completeness rather than as a reportable estimate.

| Domain     | control, 8 passes |      | control, 12 passes |      | Re:Dr, 8 passes |      |
|------------|-------------------|------|--------------------|------|-----------------|------|
|            | pass@8            | PCMD | pass@8             | PCMD | pass@8          | PCMD |
| Graph      | .325              | .003 | .316               | .000 | .926            | .686 |
| Countdown  | .589              | .017 | .530               | .011 | .674            | .497 |
| Python     | .338              | .000 | .182               | .006 | .525            | .099 |
| MathIR     | .441              | .000 | .582               | .001 | .816            | .039 |
| PantryPlan | .527              | .000 | .559               | .047 | .797            | .557 |
| Mean       | .444              | .004 | .434               | .013 | .748            | .376 |

#### Q.4 REPLAY WITHOUT MODE IDENTITY

Re:Dr adds two things to Dr.GRPO at once: a supervised likelihood term on the policy’s own past verified outputs, and the rule that the term is keyed by canonical mode and rehearsed uniformly across the modes a prompt has discovered. A reader can accept the second and still ask whether the first alone explains the result. The question has force in the sparse-gradient regime: on Python, 98.5% of control updates carry no fresh task gradient (App. H.4), so almost any likelihood term on a past success would move the policy.

The single-slot cohort separates them. It is the Re:Dr objective with buffer capacity set to one, over 25 cells on the matched control’s prompts, seeds, placement and evaluator. The same replay loss, the same coefficient, the same one scheduled buffer per update, the same admission rule and schedule are held; the buffer simply retains the first canonical mode each prompt discovers and no other. Mode identity therefore cannot enter the objective, and the replay term reduces to a length-normalized likelihood on exactly one stored success per prompt. The capacity, the retained-mode count and the live replay derivative are logged at every optimizer step of every cell and hold at one throughout, so the arm is a single-slot arm by measurement rather than by configuration.

Most of the diversity effect survives the removal. Averaging the five domains, PCMD is .218 at capacity one against .004 for the matched control and .372 for Re:Dr: the single-slot arm recovers 58% of the Re:Dr effect over its control while never seeing a second key. Correctness moves with it, from .444 to .707 against .796 for Re:Dr. The registered expectation was that capacity-one PCMD would not differ systematically from its control, on the reasoning that the mechanism credited for breadth is absent by construction. That expectation is not borne out, and the claim it was registered against is stated accordingly in Sec. 5: retaining and rehearsing a verified success is what produces most of the breadth, and balancing across retained modes strengthens it.

Balancing is nonetheless what separates the arms, in every domain. The how much it adds is an interval, not a count of domains. Paired seed by seed, full Re:Dr minus capacity one is  $+155$  over 25 cells, 95% CI  $[+.071, +.239]$ : keying adds diversity overall. Per domain it separates on Graph ( $+330$ , CI  $[+.290, +.371]$ ) and Countdown ( $+384$ , CI  $[+.256, +.512]$ ) and is indistinguishable from zero on the other three. That includes `PantryPlan`, at  $-.063$  with CI  $[-.252, +.126]$ . An earlier reading counted the domains whose capacity-one point estimate fell below Re:Dr's and called `PantryPlan` an exception where one slot beats the full buffer. The count was the wrong instrument: that domain's interval spans zero, so what the data supports is that keying adds nothing detectable there, not that removing it helps.

The per-domain recovered fraction is the more honest statistic than the mean, because it ranges from 1% to 113%. It is reported for every domain next to the effect it was divided by, and no magnitude threshold decides which domains count. That distinction matters: the Re:Dr effect itself is present everywhere. Paired within cohort against its own control, seed by seed, it is positive in 24 of the 25 cells, zero in 1, and negative in 0, and its five-domain mean is positive in 5 of five. `MathIR` carries the smallest effect at  $+0.18$  and still shows it in 5 of five seeds against a control that is identically zero. What varies across domains is the magnitude of the effect, not whether there is one.

Magnitude at Level 1 is also not a property of the domain. `Python`'s Re:Dr effect here is  $+111$ , but at Level 3 replay takes it from zero to  $.418$  terminal PCMD over five seeds in both arms, the largest gain of any domain at that level — though also the gain most sensitive to the choice of canonical key (App. L.3).

Placing the intermediate arm on the same axis shows why. Re:Dr differs from its control in three separable ways — it retains a verified success at all, it retains several, and it rehearses them uniformly rather than in proportion to how often each was discovered — and a third cohort isolates the middle step by holding one exemplar per discovered mode and weighting the buffer by discovery frequency (App. F.7). That cohort reports `distinct@8`, the superseded breadth metric, so its PCMD is measured here from its own terminal mode-coverage draws with the estimator used throughout. The resulting ladder is *not* monotonic (Tab. 66): retaining many modes under frequency weighting scores  $.160$ , below the  $.218$  of retaining exactly one. Reading the four rungs as steps, retention supplies  $+214$ , adding frequency-weighted breadth to it costs  $-.057$ , and making the rehearsal uniform supplies  $+212$ .

The reason is that a frequency-weighted buffer concentrates its rehearsal on the mode that was already common, which is close to what the control does unaided; a one-slot buffer at least rehearses its single mode against whatever else the policy samples. Uniformity is therefore not a bonus applied on top of multi-mode retention. It is what makes multi-mode retention pay at all, and a larger buffer without it is worse than no choice of buffer contents.

Every number above is within cohort. An earlier reading of this comparison was not: the uniform-rehearsal arm was the published Re:Dr cohort, whose cells span an A100 and an A5000 node, while the control and the capacity-one arm came from this one. That seam is now closed by running Re:Dr on this panel, 25 cells on the same prompts, seeds, placement, runtime and evaluator as the control, at Re:Dr's own buffer capacity with the replay derivative live at every logged step (Tab. 67). It was worth closing on the correctness axis and worth checking on the diversity axis. Matched Re:Dr reaches  $.372$  mean terminal PCMD against the published cohort's  $.376$ , a difference of  $.003$ , so every PCMD comparison made against the published arm stood. Correctness moves further:  $.796$  against its control's  $.444$ , where the published arm reported a lower figure off a different control, which is why the accuracy comparison needed the matched cohort and the diversity one did not.

Three qualifications bound the reading. `Python` and `MathIR` carry PCMD near zero in all three arms, so the five-domain separation count rests on the three domains with a measurable effect. `Python`'s correctness at capacity one is bimodal across seeds rather than centred on its mean, reaching  $.612$  on average from per-seed values spanning almost the whole interval, and is reported as a finding about supervision in sparse-reward regimes rather than as a stable endpoint. And 2 of the 25 cells, both in `Python`, never discovered a second mode at all, so the single slot withheld nothing there and the ablation is empty for those two cells.

Table 65: **Replay without mode identity recovers most of the breadth effect.** Terminal endpoints for Qwen2.5-0.5B at Level 1, five seeds per cell, all arms at eight passes. Capacity one retains a single verified success per prompt, so the objective never sees a second key; Re:Dr retains up to sixteen and rehearses them uniformly. Means weight the five domains equally. Three capacity-one cells rest on fewer than thirty defining prompts and are included for completeness.

| Domain     | control |      | capacity one |      | Re:Dr  |      |
|------------|---------|------|--------------|------|--------|------|
|            | pass@8  | PCMD | pass@8       | PCMD | pass@8 | PCMD |
| Graph      | .325    | .003 | .798         | .380 | .950   | .711 |
| Countdown  | .589    | .017 | .552         | .155 | .663   | .538 |
| Python     | .338    | .000 | .612         | .001 | .842   | .111 |
| MathIR     | .441    | .000 | .794         | .005 | .786   | .018 |
| PantryPlan | .527    | .000 | .779         | .547 | .738   | .484 |
| Mean       | .444    | .004 | .707         | .218 | .796   | .372 |

Table 66: **The mechanism ladder is not monotonic.** Terminal PCMD for Qwen2.5-0.5B at Level 1, five seeds per cell, all arms at eight passes and all measured with the same estimator. Retaining many modes under discovery-frequency weighting scores below retaining exactly one; uniform rehearsal is what converts multi-mode retention into breadth. The recovered fraction is the capacity-one effect over the Re:Dr effect; the effect it is divided by is given beside it, since on Python and MathIR that denominator is small at this level and the ratio is correspondingly unstable.

| Domain     | control | capacity one | frequency | uniform | Re:Dr effect | recovered |
|------------|---------|--------------|-----------|---------|--------------|-----------|
| Graph      | .003    | .380         | .383      | .711    | .708         | 53%       |
| Countdown  | .017    | .155         | .169      | .538    | .521         | 26%       |
| Python     | .000    | .001         | .002      | .111    | .111         | 1%        |
| MathIR     | .000    | .005         | .005      | .018    | .018         | 26%       |
| PantryPlan | .000    | .547         | .243      | .484    | .484         | 113%      |
| Mean       | .004    | .218         | .160      | .372    | .368         |           |

Table 67: **Re:Dr on the control’s own panel, paired seed by seed.** Terminal PCMD for Qwen2.5-0.5B at Level 1, 25 cells on the same prompts, seeds, placement and evaluator as the control. *Effect* is the within-cohort paired difference and *seeds* counts how many of the five are strictly positive. The published column is the cross-cohort arm this cohort replaces; its five-domain mean differs by .003. The effect is positive in 24 of 25 cells and negative in 0.

| Domain     | control | Re:Dr | effect | seeds | published |
|------------|---------|-------|--------|-------|-----------|
| Graph      | .003    | .711  | +.708  | 5/5   | .686      |
| Countdown  | .017    | .538  | +.521  | 5/5   | .497      |
| Python     | .000    | .111  | +.111  | 4/5   | .099      |
| MathIR     | .000    | .018  | +.018  | 5/5   | .039      |
| PantryPlan | .000    | .484  | +.484  | 5/5   | .557      |
| Mean       | .004    | .372  | +.368  | 24/25 | .376      |

## Part VII. Extended comparison

### R DIVERSITY-PRESERVING RLVR: EXTENDED COMPARISON

Diversity-preserving RLVR methods differ in their measurement, training signals, and target distributions. Cui et al. (2025) study token-level entropy decline and its relation to policy-update covariance, and report empirical entropy–performance curves. Clip-Cov removes selected token updates with positive covariance; KL-Cov applies a KL penalty to tokens with large covariance. Lochab et al. (2026) analyze binary reward’s indifference to allocation within the correct set and add a forward conditional-KL term that favors uniform correct-response probabilities. DMPO (Li et al., 2026b) uses group-level distribution matching motivated by forward KL. Its implemented penalty is mean squared error between a Boltzmann reward target and normalized mean-token likelihoods, added to GRPO. These mechanisms address different aspects of concentration.

**Measurement and evaluation populations.** Token entropy averages uncertainty over generated prefixes and the next-token distribution. It does not isolate variation among correct solution modes: incorrect responses and differences between strings with the same execution key also contribute. UCPO reports  $\text{pass}@K$  and the fraction of a correct response’s extracted equations absent from the other correct responses for that prompt (Lochab et al., 2026, App. F). DMPO reports feasibility and solution quality on NP-Bench and MM-NP-Bench; its quality ratio assigns zero to invalid responses (Li et al., 2026b, Section 4.2). These measures describe different response properties.

At a fixed prompt and sampling law,  $\text{pass}@K$  depends only on correctness  $P$ , while  $\text{distinct}@K$  depends on both  $P$  and the conditional allocation  $q$  (Equation (1)). Across the 375 untrained measurements in App. B.5,  $\text{distinct}@8$  correlates .936 with  $\text{pass}@8$ . The population quantity  $\text{PCMD}(q) = 1 - \sum_c q_c^2$  is unchanged when  $P$  changes at fixed  $q$ . Its finite-sample estimate still requires verified responses, and a change in which prompts meet the eligibility threshold can change an aggregate. Section 5.1 therefore uses prompts eligible at both checkpoints for its longitudinal contrasts. MODEBENCH measures sameness with domain-specific canonical execution keys returned by its validators; it does not infer key equivalence from a learned judge or an embedding threshold.

**Fresh samples and stored exemplars.** Clip-Cov and KL-Cov modify token updates on current rollouts; UCPO redistributes advantage among sampled correct responses; DMPO matches probabilities within the sampled group. None supplies a likelihood term on a stored response absent from that group. Absence does not imply an unchanged mode probability: softmax normalization and shared neural parameters can change unsampled outputs.

Lemma P.39 quantifies one categorical comparison. For an on-policy estimator  $G^{-1} \sum_i a_i \nabla_z \log p_{z_i}$  with detached row advantages bounded by a finite  $B$  independently of  $p_b$ ,  $|\mathbb{E}[\hat{g}_b]| \leq 2Bp_b(1 - p_b)$ . If  $b$  is absent from the group, its logit component is exactly  $-p_b G^{-1} \sum_i a_i$ , which is zero when the advantages sum to zero. This bound concerns the independent-logit abstraction and the specified bounded component; it does not automatically cover an entire token-level objective, KL penalty, or neural optimizer.

In the UCPO implementation, let  $G^+ > 0$  count active verifier-positive rows and let  $A^+$  be their common base advantage. The estimate  $\hat{q}_i$  is obtained by normalizing rollout-policy *sequence likelihoods* over these rows. For  $\omega \in [0, 1]$ , set  $v_i = 1/\hat{q}_i$  and  $A_i = G^+ A^+ ((1 - \omega)/G^+ + \omega v_i / \sum_j v_j)$ . Thus  $|A_i| \leq G^+ |A^+|$  and the total positive advantage is conserved. The weights are not estimates of canonical-key counts. With fixed  $G$  and bounded base advantages, this redistribution satisfies the row-score bound under the lemma’s categorical assumptions. Verified replay instead scores a fixed stored target: its categorical component is  $\rho(w_b - p_b)$ , which approaches  $\rho w_b > 0$  for a buffered key as  $p_b \rightarrow 0$ . Neither comparison proves inevitable collapse or neural retention. Exact entropy has unbounded surprisal near zero and is outside the bounded-score premise (App. P.14).

**Target distributions and replay weighting.** For a known full correct category set, uniform categorical replay and exact conditional entropy favor the same conditional target, with penalties  $\text{KL}(u||q)$  and  $\text{KL}(q||u)$ , respectively (App. P.14). UCPO also targets a uniform distribution over correct responses. These targets coincide only when they refer to the same categories.

Uniformity over response strings need not be uniformity over execution keys. A partial buffer covers only discovered keys, and the implemented mean-token exemplar loss adds length dependence; neither implies uniform neural key probabilities (App. P.7).

DMPO’s target for sampled row  $i$  is proportional to  $\exp(r_i/\alpha)$ , with temperature  $\alpha > 0$  (Li et al., 2026b, Eq. 5). Under binary rewards, correct rows have equal target weights and incorrect rows retain positive target mass. Conditional on the correct rows, repeated occurrences of a key therefore receive target mass proportional to their count. This within-group weighting is distinct from replaying a buffer with weights proportional to cumulative fresh-rollout key counts. The buffer-weighting ablation is not an evaluation of DMPO’s distribution-matching objective.

The replay ablation holds buffer construction and the update rule fixed while changing uniform versus fresh-frequency weights; subsequent buffer contents can differ across policies. On Qwen2.5-0.5B, uniform weighting increases  $B_8 = \text{distinct@8} - \text{pass@8}$  by +.318 extra verified modes per prompt ([+.272, +.358]), averaging the five domains within each of five paired seeds. The  $\text{pass@8}$  contrast is +.010 ([−.119, +.139]); these nominal paired-seed bootstrap intervals establish neither equal accuracy nor an accuracy gain. Extra-mode counts remain coupled to correctness. The result supports a benefit from uniform weighting within this replay procedure (App. F.7).

**Evaluated comparators.** The UCPO implementation follows its Equations 16, 47, and 51 and is compared with Dr.GRPO using matched training seeds at Qwen2.5-0.5B and Falcon3-1B (App. F.5). In the common-prompt analysis, Qwen2.5-0.5B Countdown  $\text{pass@8}$  improves by +.133 ([+.005, +.262]), while its PCMD contrast is inconclusive: +.002 ([−.018, +.023]). MathIR effects are also inconclusive. Intervals are nominal paired 95% Student- $t$  intervals over five training seeds. The diversity contrasts average promptwise differences where both arms have at least two verified responses, with at least 30 common eligible prompts per seed. These populations differ from comparisons using separate per-arm eligibility thresholds.

Fixed Semantic MaxEnt uses a bounded surprisal bonus on freshly sampled verified outcomes, based on accumulated and peer counts (App. E). It is not an implementation of Clip-Cov or KL-Cov and does not provide an exact entropy gradient or an individual-mode retention guarantee. Sparse RLEP-Dr adapts verified-trajectory replay (Zhang et al., 2025) using an offline response pool and a mixed fresh/replay advantage. It preserves response frequency rather than balancing canonical keys; its collection and update differ from the online buffer procedure.

Self-imitation learning (Oh et al., 2018) and rejection-sampling fine-tuning (Dong et al., 2023) likewise revisit the learner’s own past successes, and verified replay belongs to that family. They rank stored experience by return or reward and carry no notion of solution identity, so eight copies of one verified mode and eight distinct ones are interchangeable targets. Under a binary verifier every correct response earns the same return, so the ranking reduces to how often a mode was discovered. What distinguishes verified replay is therefore not off-policy reuse but the key: one exemplar per canonical execution key, rehearsed uniformly. The uniform-versus-frequency ablation of App. F.7 isolates exactly that difference, holding the buffer construction and update rule fixed. Its MathIR correctness gain in App. F.5 is consistent with reuse helping performance without uniform key weighting. Additional fresh-group comparisons with GAPO and SetPO appear in App. F.6.

**Scope of the comparison.** UCPO and sparse RLEP-Dr cover the two smaller model scales on MODEBENCH. Clip-Cov, KL-Cov, and DMPO are literature comparisons, without direct runs in this evaluation. MODEBENCH’s executable domains, verified-key metric, and eligibility rules differ from the mathematical reasoning and combinatorial optimization evaluations in those papers. The comparisons therefore assess the specified mechanisms and benchmark conditions, without ranking the methods on their original tasks. The categorical boundary calculation and the buffer-weighting ablation answer separate questions: whether a stored target can supply a signal without a fresh occurrence, and how weighting stored keys affects measured performance.

## Part VIII. Data and reproducibility

### S DATA AND REPRODUCIBILITY

#### S.1 EVALUATION COMPARISONS

Paired method comparisons match model, domain, and seed at the specified training endpoint. Initial references follow each study’s protocol. Historical saved-output comparisons use each method’s own initial evaluation. In the fresh 64-response evaluation (App. H.1), each training seed is paired with an initial sampling replica shared across the four methods; all five replicas use the same initial model weights. Terminal replay effects compare trained checkpoints. Each result states its measured population, sample size, and conditional-diversity eligibility rule.

#### S.2 DATASETS AND MODELS

The ModeBench dataset provides benchmark tasks by level and domain, with documented split roles. The loading guide and evaluation guide describe their use and distinguish auxiliary data. The model repository contains terminal checkpoints organized by model, domain, method, and seed, together with configurations, tokenizers, and applicable base-model licenses. Model links identify the version used for each export.

#### S.3 ANALYSIS RESOURCES

The experiment index connects the scale comparisons, MaxRL factorial, Level-2 factorial, replay-weighting ablation, and supporting controls to their model families. The analysis-input index and model-loading guide provide numerical inputs and instructions for using the released models. Figure data and rendering scripts are listed in `paper/FIGURE_MANIFEST.md`.